

Contents

1	Introduction	5
1.1	Basics	5
1.1.1	Probability	6
1.2	Independence	10
1.2.1	Conditional Probability	15
1.3	Random Variable	17
1.3.1	Distribution of a Random Variable	18
1.3.2	Distribution Function	20
1.3.3	Multidimensional Distributions	28
1.4	Expectation	32
1.5	Moments & Inequalities	38
1.5.1	Markov & Chebyshev inequality	40
1.5.2	Jensen's inequality	42
1.5.3	Holder's inequality	44
1.5.4	Minkowski's inequality	45
1.6	Independence of Random Variables	46
1.7	Supplementary Exercises	52
2	Generating function	55
2.1	Definition	55
2.2	Properties	57
2.3	Computation of Generating function	61
2.3.1	Recurrence Relation	63
2.4	Convolution	71
2.4.1	Random Sum	74

2.5	Fixed point of Generating Function	75
2.6	Supplementary Exercises	77
3	Branching Process	81
3.1	Introduction	81
3.2	Extinction	85
3.3	Extinction time	89
3.3.1	The subcritical case	90
3.3.2	The critical case	91
3.3.3	The supercritical case	93
3.4	Supplementary Exercises	96
4	Random walk	101
4.1	Definition and Basic Example	101
4.1.1	The Gambler's Ruin Problem	102
4.2	Simple symmetric random walk in one dimension	102
4.2.1	Path Combinatorics	103
4.2.2	Return Probabilities	104
4.2.3	First Return	105
4.2.4	Renewal Equations	107
4.3	Reflection Principle and Ballot Theorem	108
4.3.1	Ballot Theorem	109
4.3.2	Probability of No Return	110
4.3.3	Time of Last Return	113
4.3.4	Long Leads	114
4.3.5	Distribution of Maxima	116
4.4	Advanced Topics and Applications	118
4.5	Supplementary Problems	119
5	Markov chain	121
5.1	Introduction	121
5.2	Examples of Markov Chains	125
5.2.1	Random walks	127
5.2.2	Queueing model	138
5.2.3	Renewal event chains	139

5.2.4	Genetics	140
5.2.5	Miscellaneous examples	142
5.3	State Transition Diagrams	147
5.4	Supplementary Exercises	147
6	Higher Order Transitions	155
6.1	Joint distribution of Markov chain	155
6.2	Higher order transition probabilities	158
6.2.1	Chapman Kolmogorov Equations	159
7	Recurrence and transience	163
7.1	Recurrence and transience	163
7.1.1	Computation of ever returning	166
7.2	Basic Renewal Equation	167
7.3	Classification of states	172
7.4	Periodicity of states	181
8	First step analysis	185
8.1	Entering a closed set	185
8.2	Probability of visiting a state	197
8.3	Recurrence of irreducible chains	200
8.4	Visiting a state before another	203
9	Stationary distribution	207
9.1	Stationary probability distribution	207
9.1.1	Reversible chains	211
9.2	Construction of a stationary distribution	214
9.3	Uniqueness of the stationary distribution	218
9.3.1	Stationary distribution for arbitrary chains	225
10	Limit theory of Markov chains	229
10.1	Absolute probabilities	229
10.2	Limiting probabilities	230
10.3	Periodic chains	237
10.4	Finite state chains	237

11 Strong Markov Property	239
11.1 Some measure theoretic concepts	239
11.2 Stopping time	240
11.3 Strong Markov Property	242
11.4 Applications of strong Markov property	243
11.5 Regeneration	244
11.6 Time average and Stationary distribution	245
12 Martingale methods in Markov chains	251
12.1 Introduction to Martingale	251
12.2 Examples	253
12.3 Martingale & stopping times	257
12.4 Convergence of Martingale	262
12.5 Sufficient Criteria for Recurrence	262
13 Simulation	267
13.1 Simulation Basics	267
13.2 Simulate Markov chains	269
13.3 Computation on Markov chain	270
13.4 Python Code	270

Chapter 1

Introduction

1.1 Basics

Let Ω be the collection of all possible outcomes of some random experiment, like tossing a coin (once or multiple times), rolling a dice (again once or multiple times) etc. We call Ω the sample space. Generally we will be interested in describing the occurrence of certain phenomenon during the experiment. In other words, we want to know whether during the experiment, a given subset of the sample space is observed or not. These subsets will be called events. More generally, We will be interested in studying collections of events of the sample space which we will denote by $\mathcal{F}$. There are some required properties of the collection $\mathcal{F}$, namely,

- a) $\emptyset, \Omega \in \mathcal{F}$;
- b) if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;
- c) if $\{A_n : n \geq 1\}$ are such that $A_n \in \mathcal{F}$, then $\cup_{n=1}^{\infty} A_n \in \mathcal{F}$.

Such a collection is said to be a σ -algebra on Ω .

Observe that many σ -algebras may be defined on the same sample space. Note $\mathcal{F}_1 = \{\emptyset, \Omega\}$ is the smallest such collection which satisfy the conditions above and $\mathcal{F}_2 = \mathcal{P}(\Omega)$, the collection of all possible subsets of Ω is another such collection and is the largest such collection. For any subset A , the collection $\{\emptyset, A, A^c, \Omega\}$ is a σ -algebra.

If a random experiment can have only finitely many outcomes, say $1, 2, \dots, n$, it is often enough to take $\Omega = \{1, 2, \dots, n\}$ and $\mathcal{P}(\Omega)$ as the σ -algebra. This will be useful for random experiments which will take finitely values. For experiments which can possibly

result in countably many values, we can take $\mathbb{N} := \{1, 2, \dots\}$, $\mathbb{N}^+ := \{0, 1, 2, \dots\}$, $\mathbb{Z} := \{0, \pm 1, \pm 2, \dots\}$, $\mathbb{Z}^d = \{(i_1, i_2, \dots, i_d) : i_k \in \mathbb{Z}, \text{ for } k = 1, 2, \dots, d\}$ or suitable subsets of them as the sample space. In these cases also, the $\mathcal{P}(\Omega)$ can be taken as the σ -algebra.

For experiments which results in all possible values in a continuous range, we often take $(0, 1]$, $\mathbb{R}$ or $\mathbb{R}^d$ as Ω . We can define sigma algebras, called the Borel σ -algebra (denoted by $\mathcal{B}(0, 1]$, $\mathcal{B}(\mathbb{R})$ or $\mathcal{B}(\mathbb{R}^d)$ respectively) which includes all intervals of various types (and other sets such as open sets, closed sets, compact sets etc.), and satisfy the required conditions of the definition of σ -algebra. This is also the smallest σ -algebra generated by each of the above collections. For most of our study we will not be requiring more details of these concepts. For advanced topics, such as the strong Markov property (see Chapter 11) or martingale methods (see Chapter 12), we will describe the necessary details.

1.1.1 Probability

Our aim is to attach a number in $[0, 1]$ to the events which represents the chance of the event occurring. We will view probability from a axiomatic point of view. We think of it as a mapping from the collection of events to the closed unit interval. More formally,

Definition 1.1. A probability is mapping from such a collection of sets described above to the closed interval $[0, 1]$, i.e., $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ such that

- a) $\mathbb{P}(\Omega) = 1$;
- b) if $\{A_n : n \geq 1\}$ are pairwise disjoint sets in $\mathcal{F}$, i.e., $A_i \cap A_j = \emptyset$ for $i, j \geq 1$ and $i \neq j$, then

$$\mathbb{P}(\cup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mathbb{P}(A_n). \quad (1.1)$$

Note, in general cases, it is required to set that $\mathbb{P}(\emptyset) = 0$. However, in this case, this can be derived from b). Note that $\emptyset = \cup_{i=1}^{\infty} A_i$ where $A_i = \emptyset$. Since $A_i \cap A_j = \emptyset$ for any $i, j \geq 1$, we must have that $\mathbb{P}(\emptyset) = \sum_{i=1}^{\infty} \mathbb{P}(\emptyset)$ which can hold if and only if $\mathbb{P}(\emptyset) = 0$.

The condition c) in equation (1.1) above implies that if $A_1, A_2, \dots, A_n$ are finitely many pairwise disjoint sets in $\mathcal{F}$, i.e., $A_i \cap A_j = \emptyset$ for $1 \leq i, j \leq n$ and $i \neq j$, taking $A_{n+k} = \emptyset$ for $k \geq 1$ and using $\mathbb{P}(\emptyset) = 0$, we have

$$\mathbb{P}(\cup_{k=1}^n A_k) = \sum_{k=1}^n \mathbb{P}(A_k). \quad (1.2)$$

Example 1.2. Consider a discrete set Ω and the power set $\mathcal{P}(\Omega)$ as $\mathcal{F}$. Now consider non-negative real numbers $\{p(\omega) : \omega \in \Omega\}$ such that $\sum_{\omega \in \Omega} p(\omega) = 1$. Then, for any set $A \in \mathcal{F}$, we set

$$\mathbb{P}(A) = \sum_{\omega \in A} p(\omega). \quad (1.3)$$

It is easy to check that $\mathbb{P}$ is a probability on $(\Omega, \mathcal{F})$. Note here that we are not specifying any conditions on the values of $p(\omega)$ at this moment. The conditions will arise from the problem at hand.

Typically we choose $\mathbb{N}, \mathbb{N}^+, \mathbb{Z}, \mathbb{Z}^d$ or a subset of $\mathbb{Z}$ or $\mathbb{Z}^d$ as Ω . For example, for a coin tossing example, we may take $\Omega = \{0, 1\}$ and $p(0) = q = 1 - p(1) = 1 - p$, with 0 representing the failure and 1 representing the success. For a fair coin, we will further assume $p = q = 1/2$. For a dice rolling example one may take $\Omega = \{1, 2, 3, 4, 5, 6\}$ and $p(1), p(2), p(3), p(4), p(5), p(6) \geq 0$ such that $p(1) + p(2) + p(3) + p(4) + p(5) + p(6) = 1$. If we assume that the dice is fair, then we set $p(i) = 1/6$ for $1 \leq i \leq 6$. Other examples may also be constructed in a similar way.

It is also possible to consider other discrete subsets. For example we may consider $\Omega = \{0, 1\}^2 = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$. We may set $p((0, 0)) = p_{0,0}, p((0, 1)) = p_{0,1}, p((1, 0)) = p_{1,0}$ and $p((1, 1)) = p_{1,1}$ where $p_{0,0} + p_{0,1} + p_{1,0} + p_{1,1} = 1$. We can consider this sample space for an experiment of tossing two coins. Similar spaces may be considered for multiple coin tosses or rolling dice multiple times. ■

For any event A , we have that $A \cup A^c = \Omega$ and hence using (1.1), we have $\mathbb{P}(A) + \mathbb{P}(A^c) = 1$ which implies

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A). \quad (1.4)$$

For any two events A and B , we have that $A = (A \cap B^c) \cup (A \cap B)$ and $(A \cap B^c) \cap (A \cap B) = \emptyset$. Hence again using equation (1.2), we have

$$\mathbb{P}(A) = \mathbb{P}(A \cap B^c) + \mathbb{P}(A \cap B).$$

If we have $B \subseteq A$, then $A \cap B = B$. Thus, we have $\mathbb{P}(A) = \mathbb{P}(A \cap B^c) + \mathbb{P}(A \cap B) = \mathbb{P}(A \cap B^c) + \mathbb{P}(B)$ which implies

$$\mathbb{P}(A) \geq \mathbb{P}(B). \quad (1.5)$$

For any two events A and B , we can write $A \cup B$ as disjoint union of $(A \cap B^c), (B \cap A^c)$

and $(A \cap B)$. Therefore, we have

$$\begin{aligned}\mathbb{P}(A \cup B) &= \mathbb{P}(A \cap B^c) + \mathbb{P}(B \cap A^c) + \mathbb{P}(A \cap B) \\ &= [\mathbb{P}(A \cap B^c) + \mathbb{P}(A \cap B)] + [\mathbb{P}(B \cap A^c) + \mathbb{P}(A \cap B)] - \mathbb{P}(A \cap B) \\ &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).\end{aligned}\tag{1.6}$$

Using the fact that the probability of any event is non-negative, we have

$$\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B).$$

Exercise 1.3. Let A and B be events with probabilities $\mathbb{P}(A) = 3/4$ and $\mathbb{P}(B) = 1/3$. Show that $\frac{1}{12} \leq \mathbb{P}(A \cap B) \leq 1/3$, and give examples to show that both extremes are possible. Find corresponding bounds for $\mathbb{P}(A \cup B)$. ◀

Exercise 1.4. For any two events A and B , show that

$$\max\{\mathbb{P}(A), \mathbb{P}(B)\} \leq \mathbb{P}(A \cup B) \leq 2 \max\{\mathbb{P}(A), \mathbb{P}(B)\}.$$

and

$$\mathbb{P}(A \cap B)\mathbb{P}(A \cup B) \leq \mathbb{P}(A)\mathbb{P}(B).$$

When can one claim that the last relation is actually an identity? ◀

Exercise 1.5. Let $A_1, A_2, \dots, A_n$ any events.

a) Prove, by induction,

$$\begin{aligned}\mathbb{P}(\cup_{i=1}^n A_i) &= \sum_i \mathbb{P}(A_i) - \sum_{i < j} \mathbb{P}(A_i \cap A_j) + \sum_{i < j < k} \mathbb{P}(A_i \cap A_j \cap A_k) \\ &\quad - \dots + (-1)^{n-1} \mathbb{P}(A_1 \cap A_2 \cap \dots \cap A_n).\end{aligned}$$

Also, show that

$$\mathbb{P}(\cup_{i=1}^n A_i) \leq \sum_{i=1}^n \mathbb{P}(A_i).$$

b) **Bonferroni's inequality:** Show that

$$\mathbb{P}(\cup_{i=1}^n A_i) \geq \sum_{i=1}^n \mathbb{P}(A_i) - \sum_{i < j} \mathbb{P}(A_i \cap A_j).$$

c) **Kounias's inequality:** Show that

$$\mathbb{P}(\cup_{i=1}^n A_i) \leq \min \left\{ \sum_{i=1}^n \mathbb{P}(A_i) - \sum_{k \neq i} \mathbb{P}(A_i \cap A_k) : 1 \leq k \leq n \right\}. \quad \blacktriangleleft$$

Proposition 1.6. Let $\{A_n : n \geq 1\}$ be a sequence of events.

a) Suppose that A_n 's are increasing, i.e., $A_n \subseteq A_{n+1}$ for all $n \geq 1$. Then

$$\lim_{n \rightarrow \infty} \mathbb{P}(A_n) = \mathbb{P}(\cup_{n=1}^{\infty} A_n).$$

b) Suppose that A_n 's are decreasing, i.e., $A_{n+1} \subseteq A_n$ for all $n \geq 1$. Then

$$\lim_{n \rightarrow \infty} \mathbb{P}(A_n) = \mathbb{P}(\cap_{n=1}^{\infty} A_n).$$

To prove the above we use a method which is very common in probability. Given any sequence of events, we may *disjointify* these events. More formally, we define a new set of events, which are pairwise disjoint, but when taken union both yield the same result.

Given any sequence of events, we set

$$B_1 = A_1 \text{ and for } n \geq 2, B_n = A_n \setminus (\cup_{i=1}^{n-1} A_i).$$

It is easy to observe, the sequence of events $\{B_n : n \geq 1\}$ are pairwise disjoint, i.e., $B_i \cap B_j = \emptyset$ for $i \neq j$. Furthermore, for any $n \geq 1$, we have

$$\cup_{i=1}^n B_i = \cup_{i=1}^n A_i \text{ and } \cup_{i=1}^{\infty} B_i = \cup_{i=1}^{\infty} A_i.$$

Proof : We prove part a). The part b) follows from part a) and the fact that if A_n 's are decreasing, then A_n^c 's are increasing along with the equation (1.4).

Suppose $\{A_n : n \geq 1\}$ are a sequence of increasing events. Let us consider the disjointification of the events A_n as defined above. Then, we have

$$\begin{aligned} \mathbb{P}(\cup_{i=1}^{\infty} A_i) &= \mathbb{P}(\cup_{i=1}^{\infty} B_i) = \sum_{i=1}^{\infty} \mathbb{P}(B_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{P}(B_i) \\ &= \lim_{n \rightarrow \infty} \mathbb{P}(\cup_{i=1}^n B_i) = \lim_{n \rightarrow \infty} \mathbb{P}(\cup_{i=1}^n A_i) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n). \quad \blacksquare \end{aligned}$$

Exercise 1.7. Let $\{A_n : n \geq 1\}$ be any sequence of events.

a) Suppose that $A_{n+1} \subseteq A_n$ and $\cap_{n=1}^{\infty} A_n = \emptyset$. Show that

$$\lim_{n \rightarrow \infty} \mathbb{P}(A_n) = 0.$$

b) **Boole's inequality:** show that

$$\mathbb{P}(\cup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

c) Suppose that $\mathbb{P}(A_n) = 1$ for all $n \geq 1$. Show that $\mathbb{P}(\cap_{n=1}^{\infty} A_n) = 1$. ◀

1.2 Independence

Independence is a fundamental concept in probability. In plain terms it implies that occurrence of an event has no bearing in the occurrence of another event.

Definition 1.8. Events A and B are called independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B). \quad (1.7)$$

A finite collection $\{A_1, \dots, A_n\}$ of events is independent if

$$\mathbb{P}(A_{k_1} \cap \dots \cap A_{k_j}) = \mathbb{P}(A_{k_1}) \dots \mathbb{P}(A_{k_j}) \quad (1.8)$$

for $2 \leq j \leq n$ and $1 \leq k_1 < \dots < k_j \leq n$. An infinite (perhaps uncountable) collection of events is defined to be independent if each of its finite sub-collection is independent.

It is obvious from equation (1.8) that reordering the sets clearly has no effect on the condition for independence. Therefore, if any collection of events are independent, then any reordering of that collection is also independent. Further, for any sub-collection of independent events, the condition (1.8) holds trivially and hence is also independent.

We should observe that the sets $\emptyset$ and the whole sample space Ω are independent of any event A since $\emptyset \cap A = \emptyset$ and $\Omega \cap A = A$ so that $\mathbb{P}(\emptyset \cap A) = \mathbb{P}(\emptyset) = 0 = \mathbb{P}(\emptyset)\mathbb{P}(A)$. Similarly $\mathbb{P}(\Omega \cap A) = \mathbb{P}(A) = \mathbb{P}(\Omega)\mathbb{P}(A)$.

Exercise 1.9. Let $\Omega = \{1, 2, \dots, p\}$ where p is prime. With notations as in Example 1.2, consider $p(i) = 1/p$. Show that, the events A and B are independent if and only if at least one of A and B is trivial (the empty set or the whole sample space). ◀

Exercise 1.10. Suppose that a fair dice is rolled. Find the number of all non-trivial events A and B such that the events A and B are independent and $\mathbb{P}(A) = 1/2$. What is the above count when $\mathbb{P}(A) = \mathbb{P}(B) = 1/2$? If A represents the event that the toss results in an even number, what is the above count? ◀

If $n = 3$, (1.8) imposes for $j = 2$ the three constraints

$$\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_1)\mathbb{P}(A_2), \mathbb{P}(A_1 \cap A_3) = \mathbb{P}(A_1)\mathbb{P}(A_3) \text{ and } \mathbb{P}(A_2 \cap A_3) = \mathbb{P}(A_2)\mathbb{P}(A_3)$$

and for $j = 3$ the single constraint

$$\mathbb{P}(A_1 \cap A_2 \cap A_3) = \mathbb{P}(A_1)\mathbb{P}(A_2)\mathbb{P}(A_3).$$

For a collection of events $\{A_1, A_2, \dots, A_k\}$, if all pairs of events $\{A_i, A_j\}$ is independent, they are often called pairwise independent events.

Independence requires that (1.8) hold for each $j = 2, \dots, n$ and each choice of $k_1, \dots, k_j$, a total of $\sum_{j=2}^n \binom{n}{j} = 2^n - 1 - n$ constraints. This set of constraints may be stated in a different way: if $B_1, \dots, B_n$ are sets such that for each $i = 1, \dots, n$ either we have $B_i = A_i$ or $B_i = \Omega$, then

$$\mathbb{P}(B_1 \cap B_2 \cap \dots \cap B_n) = \mathbb{P}(B_1)\mathbb{P}(B_2) \dots \mathbb{P}(B_n). \quad (1.9)$$

The point is that if $B_i = \Omega$, then B_i can be ignored in the intersection on the left and the factor $\mathbb{P}(B_i) = 1$ can be ignored in the product on the right.

Example 1.11. For a fair dice rolling experiment, consider the events $A_1 = \{1, 2, 3, 4\}$ and $A_2 = A_3 = \{4, 5, 6\}$, then

$$\begin{aligned} \mathbb{P}(A_1 \cap A_2 \cap A_3) &= \mathbb{P}(\{4\}) = 1/6; \\ \mathbb{P}(A_1) &= 4/6, \mathbb{P}(A_2) = \mathbb{P}(A_3) = 3/6; \\ \mathbb{P}(A_1)\mathbb{P}(A_2)\mathbb{P}(A_3) &= 4/6 \times 3/6 \times 3/6 = 1/6. \end{aligned}$$

Thus, the last condition above holds but check that none of the others hold. Hence, the events are not independent. ■

Exercise 1.12. Suppose a fair dice is rolled. Find three events that are not independent even though each is independent of the intersection of the other two. ◀

Example 1.13. Let $\Omega_1, \Omega_2, \dots, \Omega_n$ be any collection of discrete sets for some $n \geq 2$. Let us consider $\mathcal{F}_i = \mathcal{P}(\Omega_i)$ and $\mathbb{P}_i$ any probability on Ω_i for $i = 1, 2, \dots, n$ (see Example 1.2). We now consider the product of the Ω_i 's as the sample space, i.e., $\Omega = \prod_{i=1}^n \Omega_i = \Omega_1 \times \Omega_2 \times \dots \times \Omega_n = \{(\omega_1, \omega_2, \dots, \omega_n) : \omega_1 \in \Omega_1, \omega_2 \in \Omega_2, \dots, \omega_n \in \Omega_n\}$. Consider the power set on Ω as $\mathcal{F}$. Now, consider a probability on $(\Omega, \mathcal{F})$ by setting

$$\mathbb{P}(\{(\omega_1, \omega_2, \dots, \omega_n)\}) = \prod_{i=1}^n \mathbb{P}_i(\{\omega_i\}) \quad (1.10)$$

and

$$\mathbb{P}(A) = \sum_{\omega=(\omega_1, \dots, \omega_n) \in A} \mathbb{P}(\{(\omega_1, \omega_2, \dots, \omega_n)\}). \quad (1.11)$$

It is easy to check that $\mathbb{P}$ is actually a probability. Also we observe that for any set A of the form $A = \prod_{i=1}^n A_i = A_1 \times A_2 \cdots \times A_n$, where $A_i \in \mathcal{F}_i$ for $i = 1, 2, \dots, n$, we have

$$\mathbb{P}(A) = \mathbb{P}\left(\prod_{i=1}^n A_i\right) = \prod_{i=1}^n \mathbb{P}_i(A_i).$$

Let us now consider events $B_i = \Omega_1 \times \cdots \times \Omega_{i-1} \times C_i \times \Omega_{i+1} \times \cdots \times \Omega_n$ where $C_i \subseteq \Omega_i$. Clearly, we have $\mathbb{P}(B_i) = \mathbb{P}_i(C_i)$ for $i = 1, 2, \dots, n$. Also, we have

$$B_{i_1} \cap B_{i_2} \cdots \cap B_{i_k} = \{(\omega_1, \dots, \omega_n) : \omega_{i_j} \in C_{i_j} \text{ for } j = 1, 2, \dots, k\} = \prod_{i=1}^n D_i$$

where $D_i = \Omega_i$ if $i \notin \{i_1, i_2, \dots, i_k\}$ and $D_{i_j} = C_{i_j}$ for $j = 1, 2, \dots, k$. Thus, we have

$$\mathbb{P}(B_{i_1} \cap B_{i_2} \cdots \cap B_{i_k}) = \mathbb{P}\left(\prod_{i=1}^n D_i\right) = \prod_{i=1}^n \mathbb{P}_i(D_i) = \prod_{j=1}^k \mathbb{P}_{i_j}(C_{i_j}) = \prod_{j=1}^k \mathbb{P}(B_{i_j}).$$

Thus, the events $\{B_1, B_2, \dots, B_n\}$ is a collection of independent events.

Returning to the Example 1.2, in the last part of tossing two coins, we observe that if $B_1 = \{(1, 0), (1, 1)\} = \{1\} \times \{0, 1\} = \{\text{the first toss is a success}\}$ and $B_2 = \{(0, 1), (1, 1)\} = \{0, 1\} \times \{1\} = \{\text{the second toss is a success}\}$, then the events are independent if and only if

$$(p_{1,0} + p_{1,1})(p_{0,1} + p_{1,1}) = p_{1,1}.$$

If we assume that $p_1 = p_{1,0} + p_{1,1}$ (getting success in the first toss) and, $p_2 = p_{0,1} + p_{1,1}$ (getting success in the second toss), for independence of B_1 and B_2 , we must have

$$p_{1,1} = p_1 p_2.$$

This is a generic way to construct sample space in which a outcomes of n experiments are independent. ■

Exercise 1.14. A fair dice is rolled $n \geq 3$ times. Let A_{ij} be the event that the i -th and j -th rolls produce the same number. Show that the events $\{A_{ij} : 1 \leq i < j \leq n\}$ are pairwise independent but not independent. ◀

Example 1.15. Let A and B be independent events. Then we have

$$\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A) - \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A)[1 - \mathbb{P}(B)] = \mathbb{P}(A)\mathbb{P}(B^c).$$

Therefore, the events A and B^c are independent. Similar arguments can be carried out to prove the pairs $\{A^c, B\}$ and $\{A^c, B^c\}$ are also independent.

We may generalize the above to any number of event. Let $\{A_1, A_2, \dots, A_n\}$ be a collection of independent events. For each $i \geq 1$, set $B_i = A_i$ or A_i^c . We show that the collection $\{B_1, B_2, \dots, B_n\}$ is independent.

We can proceed by induction on the number of events B_i which equal A_i^c , i.e., $m = |\{i : B_i = A_i^c\}|$ where $|A|$ denotes the number of members in a set A . If $m = 0$, there is nothing to prove. For $m = 1$, by renaming the events we may assume that $B_1 = A_1^c$ and $B_i = A_i$ for all $i \geq 2$. Now consider any finite sub-collection $B_{i_1}, B_{i_2}, \dots, B_{i_k}$. If the indices $\{i_1, i_2, \dots, i_k\}$ does not contain 1, again there is nothing to prove. If $\{i_1, i_2, \dots, i_k\}$ contains 1, we may assume that $B_{i_1} = A_{i_1}^c$ and $B_{i_j} = A_{i_j}$ for $j \geq 2$. Thus, we have

$$\begin{aligned} \mathbb{P}(B_{i_1} \cap B_{i_2} \cdots \cap B_{i_k}) &= \mathbb{P}(A_1^c \cap A_{i_2} \cdots \cap A_{i_k}) \\ &= \mathbb{P}(A_{i_2} \cdots \cap A_{i_k}) - \mathbb{P}(A_1 \cap A_{i_2} \cdots \cap A_{i_k}) \\ &= \mathbb{P}(A_{i_2}) \cdots \mathbb{P}(A_{i_k}) - \mathbb{P}(A_1)\mathbb{P}(A_{i_2}) \cdots \mathbb{P}(A_{i_k}) \\ &= (1 - \mathbb{P}(A_1))\mathbb{P}(A_{i_2}) \cdots \mathbb{P}(A_{i_k}) = \mathbb{P}(B_{i_1})\mathbb{P}(B_{i_2}) \cdots \mathbb{P}(B_{i_k}). \end{aligned}$$

Now we assume the result holds for $1 \leq m < n$. Now, we want to prove the result for $m + 1$. Again, by renaming the events we may assume that $B_1 = A_1^c$. Now consider any finite sub-collection $B_{i_1}, B_{i_2}, \dots, B_{i_k}$. If the indices $\{i_1, i_2, \dots, i_k\}$ does not contain 1, again there is nothing to prove, as the sub-collection may have at most m many B_i 's which equal A_i^c . If $\{i_1, i_2, \dots, i_k\}$ contains 1, then, using the induction hypothesis, we have

$$\begin{aligned} \mathbb{P}(B_{i_1} \cap B_{i_2} \cdots \cap B_{i_k}) &= \mathbb{P}(A_1^c \cap B_{i_2} \cdots \cap B_{i_k}) \\ &= \mathbb{P}(B_{i_2} \cap \cdots \cap B_{i_k}) - \mathbb{P}(A_1 \cap B_{i_2} \cdots \cap B_{i_k}) \\ &= \mathbb{P}(B_{i_2}) \cdots \mathbb{P}(B_{i_k}) - \mathbb{P}(A_1)\mathbb{P}(B_{i_2}) \cdots \mathbb{P}(B_{i_k}) \\ &= (1 - \mathbb{P}(A_1))\mathbb{P}(B_{i_2}) \cdots \mathbb{P}(B_{i_k}) = \mathbb{P}(B_{i_1})\mathbb{P}(B_{i_2}) \cdots \mathbb{P}(B_{i_k}). \end{aligned}$$

This proves the result. ■

Exercise 1.16. Show, by an example, even if the collections $\{A, B\}$ and $\{A, C\}$ are independent, the collection of events $\{A, B \cap C\}$ or the collection $\{A, B \cup C\}$ may not be independent. ◀

Example 1.17. Suppose $\{A, B, C, D\}$ be independent events. We consider the events $E_1 = A \cup B$ and $E_2 = C^c \cup D$. We want to show that E_1 and E_2 are independent. Using first part of of Example 1.15, it is enough to that $E_1^c = A^c \cap B^c$ and $E_2^c = C \cap D^c$ are independent. Now, using the second part the collection of events $\{A^c, B^c, C, D^c\}$ is independent. Therefore, we have

$$\begin{aligned}\mathbb{P}(E_1^c \cap E_2^c) &= \mathbb{P}(A^c \cap B^c \cap C \cap D^c) = \mathbb{P}(A^c)\mathbb{P}(B^c)\mathbb{P}(C)\mathbb{P}(D^c) \\ &= \mathbb{P}(A^c \cap B^c)\mathbb{P}(C \cap D^c) = \mathbb{P}(E_1^c)\mathbb{P}(E_2^c).\end{aligned}$$

Therefore, the events E_1^c and E_2^c are independent.

We may also consider more complicated events. For example consider $E_3 = A^c$ and $E_4 = (B \cup C^c) \cup (B \cap D)$. Let us denote $F_1 = B \cup C^c$ and $F_2 = (B \cap D)$ so that $E_4 = F_1 \cup F_2$. Same arguments as above shows that the collections $\{E_3, F_1\}$ and $\{E_3, F_2\}$ are independent. Now using Example 1.15, we have

$$\begin{aligned}\mathbb{P}(E_3 \cap (F_1 \cap F_2)) &= \mathbb{P}((E_3 \cap F_1) \cup (E_3 \cap F_2)) \\ &= \mathbb{P}(E_3 \cap F_1) + \mathbb{P}(E_3 \cap F_2) - \mathbb{P}(E_3 \cap F_1 \cap F_2) \\ &= \mathbb{P}(E_3)\mathbb{P}(F_1) + \mathbb{P}(E_3)\mathbb{P}(F_2) - \mathbb{P}(A^c \cap (B \cap C^c) \cap (B \cap D)) \\ &= \mathbb{P}(E_3)\mathbb{P}(F_1) + \mathbb{P}(E_3)\mathbb{P}(F_2) - \mathbb{P}(A^c \cap B \cap C^c \cap D) \\ &= \mathbb{P}(E_3)\mathbb{P}(F_1) + \mathbb{P}(E_3)\mathbb{P}(F_2) - \mathbb{P}(A^c)\mathbb{P}(B \cap C^c \cap D) \\ &= \mathbb{P}(E_3)\mathbb{P}(F_1) + \mathbb{P}(E_3)\mathbb{P}(F_2) - \mathbb{P}(E_3)\mathbb{P}(F_1 \cap F_2) \\ &= \mathbb{P}(E_3)[\mathbb{P}(F_1) + \mathbb{P}(F_2) - \mathbb{P}(F_1 \cap F_2)] = \mathbb{P}(E_3)\mathbb{P}(F_1 \cup F_2)\end{aligned}$$

This above argument can be repeated for any number of times. Thus, we will obtain the following result:

Proposition 1.18. Suppose that $\{A_1, A_2, \dots, A_n\}$ is a family of independent events. Consider the sub-collections $\mathcal{A}_1 = \{A_1, \dots, A_{k_1}\}, \mathcal{A}_2 = \{A_{k_1+1}, \dots, A_{k_2}\}, \dots, \mathcal{A}_m = \{A_{k_{m-1}+1}, \dots, A_n\}$. Suppose for each sub-collections $\mathcal{A}_i$, we construct events E_i using any combination of union, intersection and taking complement for $i = 1, 2, \dots, m$. Then the family $\{E_1, E_2, \dots, E_m\}$ is independent.

Note that we have taken here only a finite collection. However, we do not need to restrict ourselves to only finite collections, as long as each sub-collection have no common index.

1.2.1 Conditional Probability

Often we will be dealing with events which are not independent. Therefore, we will be interested in finding the probability of an event when we know that some other event has already occurred.

Definition 1.19. For a event A with $\mathbb{P}(A) > 0$, we define a new probability $\mathbb{P}(\cdot|A)$, called the *conditional probability given the event A* , as a probability on the $\mathcal{F}$ by

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} \quad (1.12)$$

for $B \in \mathcal{F}$.

It is easy to check that $\mathbb{P}(\cdot|A)$ is a probability of $\mathcal{F}$ in the sense that it satisfies the conditions as given in Definition 1.1. The probability $\mathbb{P}(B|A)$ is referred as the conditional probability of B given A . The interpretation is that we know that the event A has occurred, then with that knowledge what is the probability of the event B occurring. Clearly, the equation (1.12) can be rewritten as

$$\mathbb{P}(A \cap B) = \mathbb{P}(B|A)\mathbb{P}(A). \quad (1.13)$$

It should be noted that $\mathbb{P}(A|A) = 1$, i.e., the set A^c does not contribute anything to the probability.

For a finite collection of events $\{A_1, A_2, \dots, A_n\}$, we can now generalize the equation (1.13) as

$$\mathbb{P}(A_1 \cap A_2 \cap \dots \cap A_n) = \mathbb{P}(A_1)\mathbb{P}(A_2|A_1)\mathbb{P}(A_3|A_1 \cap A_2) \dots \mathbb{P}(A_n|A_1 \cap A_2 \cap \dots \cap A_{n-1}). \quad (1.14)$$

Example 1.20. Urn model : An urn contains b blue and r red balls. A ball is drawn at random. It is replaced and, moreover, c balls of the color drawn are added to the urn. A new random drawing is made from the urn (now containing $r + b + c$ balls), and this procedure is repeated. Let us consider the event E_i be the event that the i -th ball drawn is red.

We now observe that $\mathbb{P}(E_1) = r/(r+b)$. To compute the probability E_2 , we need to compute the joint probabilities of the events $E_1 \cap E_2$ and $E_1^c \cap E_2$. Now, we have

$$\mathbb{P}(E_1 \cap E_2) = \mathbb{P}(E_1) \times \mathbb{P}(E_2 \mid E_1).$$

Observe that, given the first draw results in red, i.e., E_1 happens, then we will have $(r+c)$ red balls and b blue balls in the urn before the second draw. Therefore, we have $\mathbb{P}(E_2 \mid E_1) = (r+c)/(r+b+c)$. Similar arguments will show that $\mathbb{P}(E_2 \mid E_1^c) = r/(r+b+c)$. Therefore, combining the above calculations, we obtain

$$\begin{aligned} \mathbb{P}(E_2) &= \mathbb{P}(E_1) \times \mathbb{P}(E_2 \mid E_1) + \mathbb{P}(E_1^c) \times \mathbb{P}(E_2 \mid E_1^c) \\ &= \frac{r}{r+b} \times \frac{r+c}{r+b+c} + \frac{b}{r+b} \times \frac{r}{r+b+c} = \frac{r}{r+b}. \end{aligned}$$

Exercise 1.21. With notations as in Example 1.20, consider a events $F_1, F_2, \dots, F_n$ where each F_i is either E_i or E_i^c . Then, show that

$$\mathbb{P}(\cap_{i=1}^n F_i) = \frac{r(r+c) \cdots (r+(k-1)c) \times b(b+c) \cdots (b+(n-k-1)c)}{(r+b)(r+b+c) \cdots (r+b+(n-1)c)}$$

where $k = |\{i : F_i = E_i\}|$. ◀

We end with a famous result on the conditional probability.

Proposition 1.22. (Bayes Theorem) Suppose that $\{A_n : n \geq 1\}$ is a partition of Ω , i.e., $A_i \cap A_j = \emptyset$ for $i \neq j$ and $\cup_{n \geq 1} A_n = \Omega$ and that $\mathbb{P}(A_n) > 0$ for each $n \geq 1$, $\mathbb{P}(B) > 0$. Then, we have

$$\mathbb{P}(A_n \mid B) = \frac{\mathbb{P}(B \mid A_n) \mathbb{P}(A_n)}{\sum_{j=1}^{\infty} \mathbb{P}(B \mid A_j) \mathbb{P}(A_j)}. \quad (1.15)$$

Proof : We have

$$\begin{aligned} \mathbb{P}(B) &= \mathbb{P}(\Omega \cap B) = \mathbb{P}((\cup_{j \geq 1} A_j) \cap B) = \mathbb{P}(\cup_{j \geq 1} (A_j \cap B)) \\ &= \sum_{j=1}^{\infty} \mathbb{P}(A_j \cap B) = \sum_{j=1}^{\infty} \mathbb{P}(B \mid A_j) \mathbb{P}(A_j). \end{aligned}$$

Now, we have

$$\mathbb{P}(A_n \mid B) = \frac{\mathbb{P}(A_n \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B \mid A_n) \mathbb{P}(A_n)}{\sum_{j=1}^{\infty} \mathbb{P}(B \mid A_j) \mathbb{P}(A_j)}.$$

This proves the result. ■

1.3 Random Variable

The result of a random experiment needs to be encoded in a numeric form in order to make further studies. Often a random or a family of random variables will serve this purpose. A random variable will be understood as a summary of the result of the experiment. Formally speaking it is a mapping from the sample space to real line or higher dimensional Euclidean space (for random vectors). We require also a technical condition, which is given below. But for our study we will be assuming that and not be bothered with it.

Definition 1.23. A random variable on a sample space Ω is a real-valued function $X = X(\omega)$ from $\Omega \rightarrow \mathbb{R}$ such that for any set $B \in \mathcal{B}$, the Borel σ -algebra, we must have $\sigma(X) := \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}$.

Note that, we always assume that in the background a sample space Ω and a collection of events $\mathcal{F}$, suitable for our purpose, is given so that requirement that $\{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}$ is satisfied. We remark here that one may reduce the requirement by restricting to smaller class of subsets such as the open sets, closed sets, compact sets or intervals of various types, rather than working with all set in the Borel σ -algebra.

For illustration, we demonstrate how to construct random variables which we will be using mostly. The simplest of random variables are those which take a finitely many values, say $\{x_1, x_2, \dots, x_n\}$. We may consider a sample space $\Omega = \{1, 2, \dots, n\}$ and $\mathcal{F} = \mathcal{P}(\Omega)$, i.e., the collection of all subsets of Ω . Finally, set

$$X(\omega) = x_i \text{ if } \omega = i.$$

It is clear that $\{\omega : X(\omega) \in B\}$ must be a subset of Ω and hence will belong to $\mathcal{P}(\Omega)$. Therefore, X is a random variable. Such a random variable is known as the *simple random variable*.

We may now consider random variables which takes values $\{x_1, x_2, \dots\}$ and construct it the same way. Consider $\Omega = \{1, 2, \dots\} = \mathbb{N}$ and again take $\mathcal{F} = \mathcal{P}(\Omega)$ and the same way, set

$$X(\omega) = x_i \text{ if } \omega = i.$$

Again here

Let f be a function from $\mathbb{R} \rightarrow \mathbb{R}$ such that $f^{-1}(B) = \{u : f(u) \in B\} \in \mathcal{B}$ for all $B \in \mathcal{B}$. Again this conditions may be reduced to check for a much smaller set, such as

open sets, closed sets, intervals of various types etc. All continuous functions, monotone functions etc. satisfy the above condition (enough to check for open intervals). In fact it is really difficult (but possible) to construct an example which do not satisfy the above condition.

Suppose X be a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$. Let us consider the map $Y : \Omega \rightarrow \mathbb{R}$ by setting $Y(\omega) = f(X(\omega))$. Now note that for any set $B \in \mathcal{B}$, we have $\{\omega : Y(\omega) \in B\} = \{\omega : f(X(\omega)) \in B\} = \{\omega : X(\omega) \in f^{-1}(B)\} \in \mathcal{B}$ as $f^{-1}(B) \in \mathcal{B}$ and X is a random variable. Thus, $Y = f(X)$ is a random variable. We also observe that

$$\begin{aligned}\sigma(Y) &= \{Y^{-1}(B) : B \in \mathcal{B}\} = \{X^{-1}(f^{-1}(B)) : B \in \mathcal{B}\} \\ &\subseteq \{X^{-1}(B) : B \in \mathcal{B}\} = \sigma(X)\end{aligned}$$

since $f^{-1}(B) \in \mathcal{B}$ for all $B \in \mathcal{B}$.

1.3.1 Distribution of a Random Variable

Definition 1.24. For a random variable X on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, we define a probability on the line, μ_X , defined by the following: for $B \in \mathcal{B} \rightarrow [0, 1]$

$$\mu_X(B) = \mathbb{P}(\{\omega : X(\omega) \in B\})$$

where $\mathcal{B}$ is Borel σ -algebra.

It is easy to check that μ_X is indeed a probability on the Borel σ -algebra. This is known as the *distribution of the random variable*.

Example 1.25. Let Ω be any non-empty set and $X : \Omega \rightarrow \mathbb{R}$ be defined by $X(\omega) = x_0 \in \mathbb{R}$. Clearly, for any Borel set B , we have

$$\{\omega \in \Omega : X(\omega) \in B\} = \begin{cases} \emptyset & \text{if } x_0 \notin B \\ \Omega & \text{if } x_0 \in B. \end{cases}$$

Therefore, X is a random variable, known as the degenerate random variable at x_0 , having distribution $\mu_X(B) = \mathbb{I}_B(x_0)$ where

$$\mathbb{I}_A(x) = \begin{cases} 0 & \text{if } x \notin A \\ 1 & \text{if } x \in A. \end{cases}$$

Such a function is called *indicator* function. For a random variable X taking finitely many values, $\{x_1, x_2, \dots, x_n\}$, we have the distribution of X given by

$$\mu_X(B) = \sum_{i=1}^k \mathbb{P}(X = x_i) \mathbb{I}_B(x_i). \quad \blacksquare$$

Example 1.26. Let $\{p_k : k = 0, \pm 1, \pm 2, \dots\}$ be a positive numbers such that $\sum_{k \in \mathbb{Z}} p_k = 1$. Now, define $\Omega = \mathbb{Z}$ with $\mathcal{F} = \mathcal{P}(\mathbb{Z})$. Now, define $X : \Omega \rightarrow \mathbb{R}$ by $X(\omega) = \omega$. It is easy to check that X is random variable. Note that, for any $B \in \mathcal{B}$, let us define

$$\mu_X(B) = \sum_{k: k \in \mathbb{Z} \cap B} p_k = \sum_{k: k \in \mathbb{Z}} p_k \mathbb{I}_B(k).$$

This is a probability on $\mathcal{B}$. We note that for particular sets $\{k\}$ where $k \in \mathbb{Z}$, we have $\mu_X(\{k\}) = \mathbb{P}(X = k) = p_k$. It is easy to check by setting different values of $\{p_k : k \in \mathbb{Z}\}$ we can define various random variables. $\blacksquare$

Example 1.27. Let us consider $\Omega = \{0, 1, 2, 3\}$ with $\mathcal{F} = \mathcal{P}(\Omega)$. Let $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ be defined by $\mathbb{P}(A) = \sum_{i=0}^3 \frac{1}{4} \mathbb{I}_A(i)$. Now, we define two maps on Ω . Let X, Y be defined as follows:

$$X(\omega) = \begin{cases} 0 & \text{if } \omega = 0, 1 \\ 1 & \text{if } \omega = 2, 3 \end{cases} \quad \text{and} \quad Y(\omega) = \begin{cases} 0 & \text{if } \omega = 0, 2 \\ 1 & \text{if } \omega = 1, 3. \end{cases}$$

Clearly, the maps X and Y are different. However, we observe that

$$\mu_X(A) = \frac{1}{2} \mathbb{I}_A(0) + \frac{1}{2} \mathbb{I}_A(1) \quad \text{and} \quad \mu_Y(A) = \frac{1}{2} \mathbb{I}_A(0) + \frac{1}{2} \mathbb{I}_A(1).$$

Therefore, the distributions of X and Y are identical. So, it possible that two distinct random variables having the same (induced) distribution on $\mathbb{R}$.

Further note that $\sigma(X) = \{\phi, \{0, 1\}, \{2, 3\}, \Omega\}$ and $\sigma(Y) = \{\phi, \{0, 2\}, \{1, 3\}, \Omega\}$ and for $A \in \sigma(X), B \in \sigma(Y)$, we have $\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B)$. Therefore, the σ -algebras $\sigma(X)$ and $\sigma(Y)$ are independent. These shows that we are able to construct independent and identically distributed random variables, called *Bernoulli* random variables (see Example 1.34). $\blacksquare$

We have so far described random variables which take values in a discrete set. However, the most important example of random variables which take values in continuous region.

Now, consider the function $U : \Omega \rightarrow \mathbb{R}$ defined by

$$U(\omega) = \omega. \quad (1.16)$$

Clearly, for any set $B \in \mathcal{B}((0, 1])$, we note $\{\omega \in \Omega : U(\omega) \in B\} = B \in \mathcal{B}((0, 1]) = \mathcal{F}$. Thus, this is a random variable. We know it as the *uniform random variable* and will denote this $U \sim U(0, 1)$. This shows that we can construct uniform random variables. We will return to construction of more general random variables later.

1.3.2 Distribution Function

Though the distribution of a random variable, captures all the information, we will rarely deal with it. Instead we look for other properties through which we may get information about the distribution of a random variable.

Definition 1.28. A function $F : \mathbb{R} \rightarrow [0, 1]$ is called a distribution function if satisfies the following conditions:

- a) F is increasing, i.e., for $x < y$, we have $F(x) \leq F(y)$.
- b) F converges to 1 as $x \rightarrow \infty$, i.e., $\lim_{x \rightarrow \infty} F(x) = 1$.
- c) F converges to 0 as $x \rightarrow -\infty$, i.e., $\lim_{x \rightarrow -\infty} F(x) = 0$.
- d) F is right continuous for any $x_0 \in \mathbb{R}$, i.e., $\lim_{x \downarrow x_0} F(x) = F(x_0)$.

The following exercise provides some useful properties of any distribution function.

Exercise 1.29. Let F be any monotone function.

1. Show that the both the left hand limit and the right limits exist. In fact, if F is increasing, the left hand limit $F(x_0-) = \lim_{x \uparrow x_0} F(x) = \sup\{F(x) : x < x_0\}$ and $F(x_0+) = \lim_{x \downarrow x_0} F(x) = \inf\{F(x) : x > x_0\}$ for $x_0 \in \mathbb{R}$. Similarly, if F decreasing, we have $F(x_0-) = \lim_{x \uparrow x_0} F(x) = \inf\{F(x) : x < x_0\}$ and $F(x_0+) = \lim_{x \downarrow x_0} F(x) = \sup\{F(x) : x > x_0\}$ for $x_0 \in \mathbb{R}$.
2. Show that the set of discontinuities of any monotone function is at most countable. Therefore, the set of continuity points is dense in $\mathbb{R}$.

3. For any function F , let D_F be the set of discontinuities of F . Show that, for a monotone function,

$$\begin{aligned} D_F &= \{x_0 \in \mathbb{R} : F(x_0-) \neq F(x_0) \text{ or } F(x_0+) \neq F(x_0)\} \\ &= \{x_0 \in \mathbb{R} : F(x_0+) \neq F(x_0-)\}. \end{aligned}$$

If F is a distribution function, then

$$D_F = \{x_0 \in \mathbb{R} : F(x_0) > F(x_0-)\}.$$

Conclude that D_F is either empty or finite or countable. Produce examples of each type. ◀

We now associate a *distribution function* with any random variable.

Definition 1.30. For any random variable X , we will associate a non-decreasing function $F_X : \mathbb{R} \rightarrow [0, 1]$ in the following way: for $u \in \mathbb{R}$,

$$F_X(u) = \mathbb{P}(X \in (-\infty, u]) = \mathbb{P}(X \leq u).$$

We call this the (*cumulative*) *distribution function of the random variable X* (we will drop the word cumulative) and denote by F_X . If there is no chance of ambiguity, we will also drop the suffix X . Note here that the event $\{X \in (-\infty, u]\}$ belongs to the class of events that we are considering.

We now verify that the (cumulative) distribution function of a random variable X indeed satisfies the properties of a distribution function. We will later that every distribution function is actually a distribution function of a random variable.

- a) F is non-decreasing, i.e., if $u < v$, the $F(u) \leq F(v)$ since $\{X \in (-\infty, u]\} \subseteq \{X \in (-\infty, v]\}$ and monotonicity of the probability (see equation 1.5).
- b) We have $\lim_{x \rightarrow \infty} F(x) = 1$. For any sequence $x_n \uparrow \infty$, $(-\infty, x_n] \uparrow \mathbb{R}$. Hence, $\{\omega : X(\omega) \leq x_n\} \uparrow \{\omega : X(\omega) \in \mathbb{R}\} = \Omega$. Now, using monotonicity of the distribution function and Proposition 1.6, we conclude the claim.
- c) We have $\lim_{x \rightarrow -\infty} F(x) = 0$. For any sequence $x_n \downarrow -\infty$, $(-\infty, x_n] \downarrow \emptyset$ and hence $\{X \leq x_n\} \downarrow \emptyset$. Again using using monotonicity of the distribution function and Proposition 1.6, we conclude the result.

- d) F is right continuous, i.e., $\lim_{x \downarrow x_0} F(x) = F(x_0)$. For any sequence $x_n \downarrow x_0$, $(-\infty, x_n] \downarrow (-\infty, x_0]$ and hence $\{X \leq x_n\} \downarrow \{X \leq x_0\}$. Again using monotonicity of the distribution function and Proposition 1.6, we conclude the result.

From Exercise 1.29, F has left limits as F is monotone increasing and $F(x_0-) = \mathbb{P}(X \in (-\infty, x_0))$ and hence we have

$$\begin{aligned} D_{F_X} &= (D_F) = \{x_0 \in \mathbb{R} : F(x_0) > F(x_0-)\} \\ &= \{x_0 \in \mathbb{R} : \mathbb{P}(X \in (-\infty, x_0]) > \mathbb{P}(X \in (-\infty, x_0))\} \\ &= \{x_0 \in \mathbb{R} : \mathbb{P}(X = x_0) > 0\}. \end{aligned}$$

It is therefore the case that each discontinuity point of F are precisely the points where the random variable gives positive mass.

Next we state a result (without proof) that the distribution function of a random variable uniquely determines the (induced) distribution of a random variable. Therefore, it is enough to study the distribution function of a random variable.

Theorem 1.31. *Let X and Y be two random variables have the same distribution function, i.e.,*

$$F_X(u) = F_Y(u) \text{ for all } u \in \mathbb{R}.$$

Then their (induced) distribution are same, i.e.,

$$\mu_X(B) = \mathbb{P}(\{\omega : X(\omega) \in B\}) = \mathbb{P}(\{\omega : Y(\omega) \in B\}) = \mu_Y(B) \text{ for all } B \in \mathcal{B}.$$

Remark. Since the distribution function is right continuous, it is enough to verify the above equality of distribution functions for a dense subset of $\mathbb{R}$.

Example 1.32. Let X be a random variable.

- For the degenerate random variable at x_0 , the distribution function $F : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F(u) = \begin{cases} 0 & \text{if } u < x_0; \\ 1 & \text{if } u \geq x_0. \end{cases} \quad (1.17)$$

Observe that is a step function with discontinuity only at x_0 .

- For a random variable X taking values in $\mathbb{Z}$ with $\mathbb{P}(X = k) = p_k$ for $k \in \mathbb{Z}$, we have the distribution function is given by

$$F(u) = \sum_{k=-\infty}^{\lfloor u \rfloor} p_k \quad (1.18)$$

where $\lfloor x \rfloor$ is largest integer smaller or equal to x . Observe that the discontinuity points of F are precisely the points in the set $\{k \in \mathbb{Z} : p_k > 0\}$.

- For the *uniform* random variable, we observe that

$$F(u) = \begin{cases} 0 & \text{if } u < 0; \\ u & \text{if } 0 \leq u < 1; \\ 1 & \text{if } 1 \leq u. \end{cases} \quad (1.19)$$

We observe that the distribution function is continuous, i.e., the set of discontinuities is an empty set, i.e., $D_F = \emptyset$.

- Let $f : \mathbb{R} \rightarrow [0, \infty)$ be a function such that $\int_{-\infty}^{\infty} f(x)dx = 1$. Now, we set, for $u \in \mathbb{R}$,

$$F(u) = \int_{-\infty}^u f(x)dx. \quad (1.20)$$

It is easy check that the function F is a distribution function and further it is continuous. If a random variable X has the distribution function F which is given by equation (1.20) for some f , then we will call f as the *density function* of the random variable X . It should be noted that there is no uniqueness of the density function as we may change the density function at finitely many points with changing the integral(s). ■

A random variable is called *discrete* if there exists a discrete set S (either finite or countable) such that $\sum_{s \in S} \mathbb{P}(X = s) = \sum_{s \in S} F(s) - F(s-) = 1$. We call a random variable continuous if the distribution function of the random variable is continuous. We should note that there may be random variables which are neither discrete nor continuous (see Example 1.33).

Example 1.33. Let us consider the following function $F : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F(x) = \begin{cases} 0 & \text{if } u < 1; \\ 1 - \frac{1}{2} \exp(-(u - 1)) & \text{if } u \geq 1. \end{cases}$$

It is easy to check that this is a distribution function. Note that $D_F = \{1\}$ and $\mathbb{P}(X = 1) = F(1) - F(1-) = \frac{1}{2}$. Therefore, this is neither a discrete distribution nor a continuous distribution. ■

Example 1.34. (Named Discrete Distribution) Let X be a non-negative integer valued random variable.

- a) We say X is a *Bernoulli* random variable (number of success in a single trial of a coin) with parameter $0 \leq p \leq 1$ (denoted subsequently by $X \sim \text{Ber}(p)$), if

$$\mathbb{P}(X = 1) = p = 1 - q = \mathbb{P}(X = 0).$$

- b) We say X is a *Binomial* random variable (number of successes in n independent trials of a coin) with parameter $n \geq 1$ and $0 \leq p \leq 1$ (denoted subsequently by $X \sim \text{Bin}(n, p)$), if

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

for $k = 0, 1, \dots, n$.

- c) We say X is a *discrete uniform* random variable on $\{1, 2, \dots, N\}$ for $N \geq 1$, if

$$\mathbb{P}(X = k) = \frac{1}{N}$$

for $k = 1, \dots, N$.

- d) We say that X is a *Poisson* random variable with parameter $\lambda > 0$ (denoted subsequently by $X \sim \text{Poi}(\lambda)$), if

$$\mathbb{P}(X = k) = \exp(-\lambda) \frac{\lambda^k}{k!}$$

for $k = 0, 1, \dots$.

- e) We say that X is *geometric* random variable (number of failures before the first success in an independent coin tossing experiment) with parameter $0 < p < 1$ (denoted subsequently by $X \sim \text{Geo}(p)$), if

$$\mathbb{P}(X = k) = pq^k$$

for $k = 0, 1, \dots$.

- f) We say that X is a negative Binomial random variable (number of failures before the r^{th} success in an independent coin tossing experiment) with parameters $r \geq 1$ and $0 < p < 1$ (denoted subsequently by $X \sim \text{NB}(r, p)$), if

$$\mathbb{P}(X = k) = \binom{r+k-1}{k-1} p^r q^k$$

for $k = 0, 1, \dots$. ■

Exercise 1.35. Consider a random variable X taking values in $\mathbb{N}^+$ such that for all $n, m \geq 0$, we have

$$\mathbb{P}(X \geq n+m) = \mathbb{P}(X \geq n)\mathbb{P}(X \geq m).$$

Show that X is a geometric random variable with parameter $p = 1 - \mathbb{P}(X > 0)$. ◀

Example 1.36. (Named Continuous Distribution) Let X be a random variable with a continuous distribution function F .

- We say X is an *uniform* random variable on (a, b) if the distribution function is given by

$$F(u) = \begin{cases} 0 & \text{if } u \leq a \\ \frac{u-a}{b-a} & \text{if } a < u < b \\ 1 & \text{if } u \geq b. \end{cases}$$

In such a case, the density can be taken as

$$f(u) = \begin{cases} \frac{1}{b-a} & \text{if } a < u < b \\ 0 & \text{otherwise.} \end{cases}$$

- We say X is an *exponential* random variable with parameter $\lambda > 0$ if the distribution function is given by

$$F(u) = \begin{cases} 0 & \text{if } u \leq 0 \\ 1 - \exp(-\lambda u) & \text{if } u > 0. \end{cases}$$

In such a case, the density can be taken as

$$f(u) = \begin{cases} \lambda \exp(-\lambda u) & u > 0 \\ 0 & \text{otherwise.} \end{cases}$$

- We say X is an *double exponential* random variable with parameter $\lambda > 0$ if the distribution function is given by

$$F(u) = \begin{cases} \frac{1}{2} \exp(\lambda u) & \text{if } u \leq 0 \\ 1 - \frac{1}{2} \exp(-\lambda u) & \text{if } u > 0. \end{cases}$$

In such a case, the density can be taken as

$$f(u) = \frac{\lambda}{2} \exp(-\lambda|u|)$$

for $u \in \mathbb{R}$.

- We say X is an *Gamma* random variable with parameter $\alpha > 0, \beta > 0$ if the distribution function is given by

$$F(u) = \begin{cases} 0 & \text{if } u \leq 0 \\ \int_0^u \frac{1}{\Gamma(\alpha)} x^{\alpha-1} \exp(-\beta x) \beta^\alpha dx & \text{if } u > 0 \end{cases}$$

where $\Gamma(\alpha)$ is the *gamma* function. In such a case, the density can be taken as

$$f(u) = \begin{cases} \frac{1}{\Gamma(\alpha)} u^{\alpha-1} \exp(-\beta u) \beta^\alpha & u > 0 \\ 0 & \text{otherwise.} \end{cases}$$

We should note that if $\alpha < 1$, $f(u) \rightarrow \infty$ as $u \downarrow 0$.

- We say X is a *Beta* random variable with parameter $\alpha > 0, \beta > 0$ if the distribution function is given by

$$F(u) = \begin{cases} 0 & \text{if } u \leq 0 \\ \int_0^u \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1} dx & \text{if } 0 < u < 1 \\ 1 & \text{if } u \geq 1 \end{cases}$$

where $B(\alpha, \beta)$ is the *beta* function. In such a case, the density can be taken as

$$f(u) = \begin{cases} \frac{1}{B(\alpha, \beta)} u^{\alpha-1} (1-u)^{\beta-1} & 0 < u < 1 \\ 0 & \text{otherwise.} \end{cases}$$

We should observe that if $\alpha < 1$, the density function diverges to ∞ as $u \downarrow 0$.

Similarly if $\beta < 1$, the density function diverges to ∞ as $u \uparrow 1$.

- We say X is an *Normal* random variable with parameter $\mu \in \mathbb{R}, \sigma > 0$ if the distribution function is given by

$$F(u) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^u \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx.$$

In such a case, the density can be taken as

$$f(u) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(u-\mu)^2}{2\sigma^2}\right)$$

for $u \in \mathbb{R}$. When $\mu = 0, \sigma = 1$, we call this the standard Normal density and the distribution is called the standard Normal distribution.

- We say X is an *Cauchy* random variable with parameter $\mu \in \mathbb{R}, \gamma > 0$ if the distribution function is given by

$$F(u) = \frac{1}{\pi} \tan^{-1}\left(\frac{u-\mu}{\gamma}\right) + \frac{1}{2}.$$

In such a case, the density can be taken as

$$f(u) = \frac{1}{\pi} \left[\frac{\gamma}{(u-\mu)^2 + \gamma^2} \right].$$

for $u \in \mathbb{R}$. ■

Suppose that F is a continuously differentiable with $F' = f$. Then for any $a < b, a, b \in \mathbb{R}$, by the fundamental theorem of calculus, we have

$$F(b) - F(a) = \int_a^b f(u) du.$$

Since the distribution of X is determined by the intervals $\{(a, b] : a < b, a, b \in \mathbb{R}\}$, this shows that $F' = f$ can be taken as a density function for the random variable.

Exercise 1.37. Let X be a random variable with a continuous distribution function F . Find expressions for the distribution functions of the following random variables:

- a) X^2 , b) $\sqrt{X}$ for X taking positive values
c) $\sin(X)$, d) $F(X)$, e) $G^{-1}(X)$, f) $G^{-1}(F(X))$

where G is a continuous and strictly increasing function. ◀

Exercise 1.38. Express the distribution functions of

$$X^+ = \max\{0, X\}, X^- = \max\{0, -X\} = -\min\{0, X\}, |X| = X^+ + X^-, -X$$

in terms of the distribution function F of the random variable X . ◀

Exercise 1.39. Show that if F and G are distribution functions and $0 \leq \lambda \leq 1$ then $\lambda F + (1 - \lambda)G$ is a distribution function. Is the product FG a distribution function? ◀

Exercise 1.40. Let F be a distribution function and r a positive integer. Show that the following are distribution functions (convention $0 \cdot \log(0) = 0$):

$$a) G_1(x) = (F(x))^r; \quad b) G_2(x) = 1 - (1 - F(x))^r;$$

$$c) G_3(x) = F(x) + (1 - F(x)) \log(1 - F(x));$$

$$d) G_4(x) = \exp(1 - F(x)) - (1 - F(x))e. \quad \blacktriangleleft$$

Exercise 1.41. Let X be a random variable with distribution function F , and let $a < b$. Sketch the distribution functions of the *truncated* random variables $Y_{(a)}^{(b)}$ and $Z_{(a)}^{(b)}$ given by

$$Y_{(a)}^{(b)}(\omega) = \begin{cases} a & \text{if } X(\omega) < a, \\ X(\omega) & \text{if } a \leq X(\omega) \leq b, \\ b & \text{if } X(\omega) > b; \end{cases} \quad \text{and} \quad Z_{(a)}^{(b)}(\omega) = \begin{cases} X(\omega) & \text{if } a \leq X(\omega) \leq b, \\ 0 & \text{otherwise.} \end{cases}$$

Describe the behaviour of the distribution functions of $Y_{(a)}^{(b)}$ and $Z_{(a)}^{(b)}$ when $a \rightarrow -\infty, b \rightarrow \infty$. ◀

We have so far constructed only the uniform random variable. The question now remains that how do we construct other continuous random variables. Towards this end, we introduce the concept of quantile function.

1.3.3 Multidimensional Distributions

For a k -dimensional random vector $\mathbf{X} = (X_1, X_2, \dots, X_k)$ the (induced) distribution is a probability on the k -dimensional Borel field on $\mathbb{R}^k$. The (joint) distribution function F is a real-valued function on $\mathbb{R}^k$, defined by

$$\begin{aligned} F_{\mathbf{X}}(\mathbf{x}) &= \mathbb{P}\left((X_1, X_2, \dots, X_k) \in (-\infty, x_1] \times (-\infty, x_2] \times \dots \times (-\infty, x_k]\right) \\ &= \mathbb{P}(X_1 \leq x_1, X_2 \leq x_2, \dots, X_k \leq x_k) \end{aligned}$$

where $\mathbf{x} = (x_1, x_2, \dots, x_k) \in \mathbb{R}^k$. Taking $S_{\mathbf{x}}$ as the (infinite)-rectangle with its *southwest* point as $\mathbf{x}$, consisting of all points $\{(y_1, y_2, \dots, y_k) : y_i \leq x_i, \text{ for all } i = 1, 2, \dots, k\}$, we have

$$F(\mathbf{x}) = \mathbb{P}(\{\omega : (X_1(\omega), X_2(\omega), \dots, X_k(\omega)) \in S_{\mathbf{x}}\}) = \mathbb{P}(\mathbf{X} \in S_{\mathbf{x}}).$$

Note, F is increasing in each variable, and for bounded rectangles $A = (a_1, b_1] \times (a_2, b_2] \times \dots \times (a_k, b_k]$, we have

$$\mathbb{P}((X_1, X_2, \dots, X_k) \in A) = \sum_{s_i \in \{a_i, b_i\}} (-1)^{\#\{i: s_i = a_i\}} F(s_1, s_2, \dots, s_k) \geq 0. \quad (1.21)$$

We should note that for any $\mathbf{h} = (h_1, h_2, \dots, h_k)$ with $h_i \geq 0$, using the the distribution function is monotone in each variable, we have

$$\begin{aligned} & \mathbb{P}((X_1, X_2, \dots, X_k) \in (-\infty, x_1 + h_{\min}] \times (-\infty, x_2 + h_{\min}] \times \dots \times (-\infty, x_k + h_{\min}]) \\ & \leq \mathbb{P}((X_1, X_2, \dots, X_k) \in (-\infty, x_1 + h_1] \times (-\infty, x_2 + h_2] \times \dots \times (-\infty, x_k + h_k]) \\ & \leq \mathbb{P}((X_1, X_2, \dots, X_k) \in (-\infty, x_1 + h_{\max}] \times (-\infty, x_2 + h_{\max}] \times \dots \times (-\infty, x_k + h_{\max}]) \end{aligned}$$

where $h_{\min} = \min\{h_1, h_2, \dots, h_k\}$ and $h_{\max} = \max\{h_1, h_2, \dots, h_k\}$. Further, for any $h > 0$ and $\mathbf{x} = (x_1, x_2, \dots, x_k)$, the rectangle $S_{\mathbf{x}+h}$ decreases to $S_{\mathbf{x}}$ as h decreases to 0 where $\mathbf{x} + h = (x_1 + h, x_2 + h, \dots, x_k + h)$. Therefore, using the above, we conclude that F is continuous from above.

For any $i \geq 1$, the function $x \mapsto F(x_1, x_2, \dots, x_{i-1}, x, x_{i+1}, \dots, x_k) \rightarrow 0$ if $x \rightarrow -\infty$ where all the other coordinates are held fixed. It is also easy to observe that $F(x_1, x_2, \dots, x_k) \rightarrow 1$ if all $x_i \rightarrow \infty$.

There is a theorem, equivalent to Theorem 13.4, for multidimensional distribution functions. Indeed, for any F with these properties, there exists a random vector having this distribution.

Since, for $h > 0$ and h decreases to 0, observe that $S_{\mathbf{x}-h}$ increases to the interior $S_{\mathbf{x}^\circ} = \{y : y_i < x_i, i = 1, 2, \dots, k\}$. Thus, again using the above arguments, we have

$$\lim_{h \downarrow 0} F(\mathbf{x} - h) = \mathbb{P}(\mathbf{X} \in S_{\mathbf{x}^\circ}).$$

Since F is non-decreasing in each variable, it is continuous at $\mathbf{x}$ if and only if it is continuous from below there in the sense that this last limit coincides with $F(\mathbf{x})$. Thus F is continuous at $\mathbf{x}$ if and only if $F(\mathbf{x}) = \mathbb{P}(\mathbf{X} \in S_{\mathbf{x}}) = \mathbb{P}(\mathbf{X} \in S_{\mathbf{x}^\circ})$ which holds if and

only if the boundary $\partial(S_{\mathbf{x}}) = S_{\mathbf{x}} \setminus S_{\mathbf{x}^o}$ (the set of points $\mathbf{y} = (y_1, y_2, \dots, y_k)$ where $y_i \leq x_i$ for all $i = 1, 2, \dots, k$ and for some $i_0 = 1, 2, \dots, k, y_{i_0} = x_{i_0}$) satisfies $\mathbb{P}(\mathbf{X} \in \partial(S_{\mathbf{x}})) = 0$.

There are a few distinct differences in the multi-dimensional case from the one-dimensional case. We know that in the one-dimensional the discontinuity points are at most countable and each discontinuity will have positive mass. However, for $k > 1$, F can have discontinuity points even if there are no point masses (a point mass is a point $\mathbf{x}_0$ such that $\mathbb{P}(\mathbf{X} = \mathbf{x}_0) > 0$).

Example 1.42. Consider the distribution function

$$F(x, y) = \begin{cases} 0 & \text{if } x < 0 \text{ or } y < 0 \\ y & \text{if } x \geq 0, 0 \leq y < 1 \\ 1 & \text{if } x \geq 0, y \geq 1. \end{cases}$$

It is easily verified that F is a distribution function. Consider the set of points $D = \{(0, y) : 0 < y \leq 1\}$. For $(0, y_0) \in D$, we have $F(0, y_0) = y_0 > 0$ but $\lim_{x \uparrow 0} F(x, y_0) = \lim_{x \uparrow 0} 0 = 0$. Therefore, the points in set D are discontinuity points of F . Hence the discontinuity points may be uncountable for multidimensional distribution functions.

It should be noted that no point of D does not contain any point mass. Indeed for $(0, y_0) \in D$, we have

$$\begin{aligned} \mathbb{P}((X, Y) = (0, y_0)) &= \mathbb{P}(\cap_{n \geq 1} \{X \in (-\frac{1}{n}, 0], Y \in (y_0 - \frac{1}{n}, y_0]\}) \\ &= \lim_{n \rightarrow \infty} \mathbb{P}(X \in (-\frac{1}{n}, 0], Y \in (y_0 - \frac{1}{n}, y_0]) \\ &= \lim_{n \rightarrow \infty} \mathbb{P}((X, Y) \in (-\frac{1}{n}, 0] \times (y_0 - \frac{1}{n}, y_0]) \\ &= \lim_{n \rightarrow \infty} [F(0, y_0) - F(-\frac{1}{n}, y_0) - F(0, y_0 - \frac{1}{n}) + F(-\frac{1}{n}, y_0 - \frac{1}{n})] \\ &= \lim_{n \rightarrow \infty} (y_0 - 0 - (y_0 - \frac{1}{n}) + 0) = 0. \end{aligned} \quad \blacksquare$$

We should further note that for any $\mathbf{x} = (x_1, x_2, \dots, x_k)$, the boundaries of $S_{\mathbf{x}+h}$ for distinct $h > 0$ are disjoint. Therefore, only a countably many such sets can have positive probability. This implies that for any $\mathbf{x}$, uncountably many vertices $\mathbf{x}+h$ are in the continuity points of F . Clearly, we can choose a sequence of such continuity points which converge to $\mathbf{x}$. Thus, continuity points are dense in $\mathbb{R}^k$.

We should observe that the distribution function of a subset of $(X_1, X_2, \dots, X_k)$ can be easily obtained from the distribution function of the whole set by letting rest of

co-ordinates to $+\infty$. In other words, we have

$$\begin{aligned} F_{(X_1, X_2, \dots, X_{k-1})}(x_1, x_2, \dots, x_{k-1}) &= \mathbb{P}(X_1 \leq x_1, X_2 \leq x_2, \dots, X_{k-1} \leq x_{k-1}) \\ &= \lim_{x \rightarrow \infty} \mathbb{P}(X_1 \leq x_1, X_2 \leq x_2, \dots, X_{k-1} \leq x_{k-1}, X_k \leq x) \\ &= \lim_{x \rightarrow \infty} F_{\mathbf{X}}(x_1, x_2, \dots, x_{k-1}, x). \end{aligned}$$

Returning to the Example 1.42, we observe that the marginal distribution function of X is given by

$$F_X(x) = \lim_{y \rightarrow \infty} F(x, y) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0. \end{cases}$$

Thus, X is a discrete random variable taking value 1 with probability 1. The marginal distribution function of Y is given by

$$F_Y(y) = \lim_{x \rightarrow \infty} F(x, y) = \begin{cases} 0 & \text{if } x < 0 \\ y & \text{if } 0 \leq y < 1 \\ 1 & \text{if } y \geq 1. \end{cases}$$

Thus, Y is a continuous random variable. In fact it is a uniform random variable on $(0, 1]$.

If we have a function $f : \mathbb{R}^k \rightarrow [0, \infty)$ such that for any $\mathbf{x} = (x_1, x_2, \dots, x_k)$ we have

$$F(\mathbf{x}) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} \cdots \int_{-\infty}^{x_k} f(u_1, u_2, \dots, u_k) du_1 du_2 \cdots du_k$$

we will call f to be a density of the distribution. We should further note, from above that a density of a subset of $(X_1, X_2, \dots, X_k)$ can be obtained from the density function of the whole set by integrating over the rest of the co-ordinate. Therefore, we have

$$f_{(X_1, X_2, \dots, X_{k-1})}(x_1, x_2, \dots, x_{k-1}) = \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_{k-1}, u) du$$

is a density function for the vector $(X_1, X_2, \dots, X_{k-1})$.

Suppose that $g : U \rightarrow V$ is an one-to-one function where U, V are open subsets of $\mathbb{R}^k$. We will also assume that $T = g^{-1}$ is continuously differentiable and let $J(x)$ be the Jacobian of T . Now, assume that X is random variable which admits a density f and f vanishes on U^c . Then, we have, for $A \subseteq V$

$$\mathbb{P}(g(X) \in A) = \mathbb{P}(X \in T(A)) = \int_{T(A)} f(u) du = \int_A f(T(x)) |J(x)| dx$$

by the change of variable formula. Therefore, we have the density of $g(X)$ given by $f(T(x))|J(x)|$ on V and 0 outside.

Example 1.43. Suppose that we consider (X_1, X_2) on $\mathbb{R}^2$ with density

$$f(x_1, x_2) = (2\pi)^{-1} \exp\left(-\frac{1}{2}(x_1^2 + x_2^2)\right)$$

for $(x_1, x_2) \in \mathbb{R}^2$. Suppose $g : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the polar co-ordinate transformation, i.e., $g(x_1, x_2) = (r, \theta)$ where $r = \sqrt{x_1^2 + x_2^2}$ and $\theta = \tan^{-1}\left(\frac{x_1}{x_2}\right)$. Let R and Θ be the co-ordinates of $g(X_1, X_2)$. We see that (R, Θ) has the joint density on $(0, \infty) \times (0, 2\pi)$ defined by

$$h(r, \theta) = \frac{1}{2\pi} \times r \exp(-r^2/2).$$

Therefore, the marginal distribution (by integrating over the other co-ordinate) of R and Θ are given by $r \exp(-r^2/2)$ on $(0, \infty)$ and $(2\pi)^{-1}$ on $(0, 2\pi)$ respectively (and 0 in rest of the region). ■

1.4 Expectation

Definition 1.44. For a simple random variable X , taking values $\{x_1, x_2, \dots, x_n\}$, the expectation of X is defined by

$$\mathbb{E}(X) = \sum_{i=1}^n x_i \mathbb{P}(X = x_i). \quad (1.22)$$

For any two random variables X, Y of this type, it is easy to check that for any $\alpha, \beta \in \mathbb{R}$, we have $\mathbb{E}(\alpha X + \beta Y) = \alpha \mathbb{E}(X) + \beta \mathbb{E}(Y)$.

Next our aim is to define the expectation for non-negative random variables. For any non-negative random variable X , let Y be a non-negative random variable taking finitely many values such that $0 \leq Y \leq X$. Then, the expectation of X is defined by

$$\mathbb{E}(X) = \sup\{\mathbb{E}(Y) : 0 \leq Y \leq X, Y \text{ takes finitely many values}\}.$$

It should be note here that for any non-negative random variable, the expectation is defined, but it may take value $+\infty$.

Exercise 1.45. Let X and Y be any two random variables such that $0 \leq X \leq Y$. Show that $\mathbb{E}(X) \leq \mathbb{E}(Y)$. ◀

For general random variable, X , consider the random variables $X^+ = \max\{0, X\}$ and $X^- = \max\{0, -X\}$, (see Exercise 1.38 also), and define the expectation as

$$\mathbb{E}(X) = \mathbb{E}(X^+) - \mathbb{E}(X^-)$$

as long as the right hand side is not of the type $\infty - \infty$. Note that $|X| = X^+ + X^-$ and $\mathbb{E}(|X|) = \mathbb{E}(X^+) + \mathbb{E}(X^-) < \infty$ if and only if $\mathbb{E}(X^+) < \infty$ and $\mathbb{E}(X^-) < \infty$. In such a case, $|\mathbb{E}(X)| < \infty$. Further, check that that $|\mathbb{E}(X)| \leq \mathbb{E}(|X|)$ in all cases when it is defined.

A useful result for non-negative random variables is called the Monotone Convergence Theorem.

Theorem 1.46. (*Monotone Convergence Theorem - MCT*) Let $\{X_n : n \geq 1\}$ be a sequence of random variables such that $0 \leq X_n \leq X_{n+1}$ and $X_n \rightarrow X$. Then,

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = \mathbb{E}(X). \quad (1.23)$$

Note that we are not assuming that $\mathbb{E}(X)$ is finite. Indeed, if the limit is infinite, then $\mathbb{E}(X) = +\infty$ and vice versa. Using MCT, it is also easy to check that, for non-negative random variable X, Y and for any $\alpha, \beta \geq 0$, $\alpha X + \beta Y$ is non-negative random variable and we have $\mathbb{E}(\alpha X + \beta Y) = \alpha \mathbb{E}(X) + \beta \mathbb{E}(Y)$.

One result that is often useful for general random variables is the Dominated Convergence Theorem.

Theorem 1.47. (*Dominated Convergence Theorem - DCT*) Let X_n and X be random variables such that $X_n \rightarrow X$. Suppose that there is a non-negative random variable Y such that $|X_n| \leq Y$ for any $n \geq 1$ and $\mathbb{E}(Y) < \infty$. Then,

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = \mathbb{E}(X). \quad (1.24)$$

It should be note that when $\mathbb{E}(|X|), \mathbb{E}(|Y|) < \infty$, for any $\alpha, \beta \in \mathbb{R}$, we have $\mathbb{E}(\alpha X + \beta Y) = \alpha \mathbb{E}(X) + \beta \mathbb{E}(Y)$.

Exercise 1.48. Let X and Y be any two random variables.

1. If $0 \leq X \leq M$, then show that $0 \leq \mathbb{E}(X) \leq M$.
2. If $X \geq 0$ and $\mathbb{E}(X) = 0$, then show that $\mathbb{P}(X = 0) = 1$.

3. If $X \leq Y$ and $\mathbb{E}(Y) < \infty$, then show that $\mathbb{E}(X) \leq \mathbb{E}(Y) < \infty$. ◀

Example 1.49. The case of random variables taking values in $\{0, 1, 2, \dots\}$ is useful for us. Suppose X takes values $k \in \mathbb{N}^+$. We want to show

$$\mathbb{E}(X) = \sum_{k=0}^{\infty} k\mathbb{P}(X = k).$$

Let Y_n be the random variable defined by

$$X_n(\omega) = X(\omega)\mathbb{I}_{\{0,1,\dots,n\}}(X(\omega)) = \begin{cases} X(\omega) & \text{if } X(\omega) \in \{0, 1, \dots, n\} \\ 0 & \text{otherwise.} \end{cases}$$

Clearly, we have $0 \leq X_n \leq X_{n+1}$ and $X_n \rightarrow X$. Thus, we have

$$\mathbb{E}(X_n) = \sum_{i=1}^n i\mathbb{P}(X = i) + 0 \times (\mathbb{P}(X = 0) + \mathbb{P}(X > n)) = \sum_{i=1}^n i\mathbb{P}(X = i).$$

Using MCT, we now conclude that $\mathbb{E}(X) = \lim_{n \rightarrow \infty} \mathbb{E}(X_n) = \lim_{n \rightarrow \infty} \sum_{i=1}^n i\mathbb{P}(X = i) = \sum_{i=1}^{\infty} i\mathbb{P}(X = i) = \sum_{i=0}^{\infty} i\mathbb{P}(X = i)$.

We should note here that we are not assuming that the infinite series $\sum_{i=0}^{\infty} i\mathbb{P}(X = i) < \infty$. The result just states that if the series $\sum_{i=0}^{\infty} i\mathbb{P}(X = i) = +\infty$, then the expectation is $+\infty$. On the other hand, if the infinite series is convergent, then the expected value is finite. ■

Exercise 1.50. For each discrete random variables in Example 1.34, compute the expected value. ◀

Exercise 1.51. Consider a discrete random variable taking values in $\mathbb{N}$ with

$$\mathbb{P}(X = i) = \frac{1}{i(i+1)}$$

for $i \geq 1$. Show that $\mathbb{E}(X) = +\infty$. ◀

For more general random variables, we may consider the following functions: for any $n \geq 1$,

$$\psi_n(x) = \begin{cases} (i-1)2^{-n} & \text{if } (i-1)2^{-n} < x \leq i2^{-n}, \text{ for } i = 1, \dots, n2^n \\ 0 & \text{otherwise.} \end{cases} \quad (1.25)$$

For any non-negative random variable X , note that $0 \leq X_n = \psi_n(X) \uparrow X$. We may use MCT (Theorem 1.46) now to calculate expectation.

Example 1.52. For the uniform random variable U , applying MCT, we obtain

$$\begin{aligned}\mathbb{E}(U) &= \lim_{n \rightarrow \infty} \mathbb{E}(\psi_n(U)) \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^{n2^n} \frac{i-1}{2^n} \mathbb{P}(U \in (\frac{i-1}{2^n}, \frac{i}{2^n}]) = \lim_{n \rightarrow \infty} \sum_{i=1}^{2^n} \frac{i-1}{2^n} \mathbb{P}(U \in (\frac{i-1}{2^n}, \frac{i}{2^n}]) \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^{2^n} \frac{i-1}{2^n} \frac{1}{2^n} = \lim_{n \rightarrow \infty} \frac{1}{2^{2n}} \frac{(2^n - 1)(2^n)}{2} = \frac{1}{2}.\end{aligned}$$

Example 1.53. Suppose X is a random variable with density function g such that g is positive only on $[0, 1]$ and $\int_0^1 xg(x)dx$ exists. Now, taking ψ_n as in equation (1.25) above, we have

$$\begin{aligned}\mathbb{E}(\psi_n(X)) &= \sum_{i=1}^{n2^n} \frac{i-1}{2^n} \mathbb{P}(X \in (\frac{i-1}{2^n}, \frac{i}{2^n}]) = \sum_{i=1}^{n2^n} \frac{i-1}{2^n} (F(\frac{i}{2^n}) - F(\frac{i-1}{2^n})) \\ &= \sum_{i=1}^{2^n} \frac{i-1}{2^n} (F(\frac{i}{2^n}) - F(\frac{i-1}{2^n})) = \sum_{i=1}^{2^n} \frac{i-1}{2^n} \int_{\frac{i-1}{2^n}}^{\frac{i}{2^n}} g(u) du \\ &= \sum_{i=1}^{2^n} \int_{\frac{i-1}{2^n}}^{\frac{i}{2^n}} u g(u) du - \sum_{i=1}^{2^n} \int_{\frac{i-1}{2^n}}^{\frac{i}{2^n}} (u - \frac{i-1}{2^n}) g(u) du \\ &= \int_0^1 u g(u) du - \sum_{i=1}^{2^n} \int_{\frac{i-1}{2^n}}^{\frac{i}{2^n}} (u - \frac{i-1}{2^n}) g(u) du.\end{aligned}$$

Here we have used the fact that $F(x) = 1$ for all $x \geq 1$, will imply that for all $i > 2^n$, $\mathbb{P}(X \in (\frac{i-1}{2^n}, \frac{i}{2^n}]) = 0$.

Thus, we have

$$\begin{aligned}|\mathbb{E}(\psi_n(X)) - \int_0^1 u g(u) du| &\leq |\sum_{i=1}^{2^n} \int_{\frac{i-1}{2^n}}^{\frac{i}{2^n}} (u - \frac{i-1}{2^n}) g(u) du| \\ &\leq \sum_{i=1}^{2^n} \int_{\frac{i-1}{2^n}}^{\frac{i}{2^n}} \left| (u - \frac{i-1}{2^n}) \right| g(u) du \leq \frac{1}{2^n} \sum_{i=1}^{2^n} \int_{\frac{i-1}{2^n}}^{\frac{i}{2^n}} g(u) du \leq \frac{1}{2^n} \rightarrow 0\end{aligned}$$

as $n \rightarrow \infty$. Thus, by MCT, we have $\mathbb{E}(X) = \int_0^1 u g(u) du$.

This argument extends to any density on $[0, M]$ in a straightforward way. Now, suppose that g is defined on $[0, \infty)$ and $\int_0^\infty xg(x)dx < \infty$. By the definition of infinite Riemann integral we must have that $\int_0^n xg(x)dx \rightarrow \int_0^\infty xg(x)dx$ as $n \rightarrow \infty$. Now, taking ψ_n as before, we have, using the same calculations,

$$\mathbb{E}(\psi_n(X)) = \sum_{i=1}^{n2^n} \frac{i-1}{2^n} \mathbb{P}(X \in (\frac{i-1}{2^n}, \frac{i}{2^n}]) = \int_0^n u g(u) du - \sum_{i=1}^{n2^n} \int_{\frac{i-1}{2^n}}^{\frac{i}{2^n}} (u - \frac{i-1}{2^n}) g(u) du.$$

Again, we have

$$|\mathbb{E}(\psi_n(X)) - \int_0^n ug(u)du| \leq \frac{1}{2^n} \rightarrow 0$$

as $n \rightarrow \infty$. Thus, again by MCT, we conclude that $\mathbb{E}(X) = \int_0^\infty xg(x)dx$. Furthermore, if $\int_0^n ug(u)du \rightarrow \infty$, we conclude that $\mathbb{E}(X) = \lim_{n \rightarrow \infty} \int_0^n ug(u)du = +\infty$. ■

Exercise 1.54. For discrete random variables in Example 1.36, compute the expected value (when defined). ◀

Example 1.55. Let us now return to the Example 1.33. Here again we may consider the sequence of random variables $\psi_n(X)$. Then we have

$$\begin{aligned} \mathbb{E}(\psi_n(X)) &= \sum_{i=1}^{n2^n} \frac{i-1}{2^n} \mathbb{P}(X \in (\frac{i-1}{2^n}, \frac{i}{2^n}]) = \sum_{i=1}^{n2^n} \frac{i-1}{2^n} (F(\frac{i}{2^n}) - F(\frac{i-1}{2^n})) \\ &= (1 - \frac{1}{2^n})(F(1) - F(1 - \frac{1}{2^n})) + \sum_{i=2^n+1}^{n2^n} \frac{i-1}{2^n} \frac{1}{2} \left(\exp(-\frac{i-1-2^n}{2^n}) - \exp(-\frac{i-2^n}{2^n}) \right) \\ &= (1 - \frac{1}{2^n}) \frac{1}{2} + \frac{1}{2} \sum_{i=1}^{(n-1)2^n} \left(1 + \frac{i-1}{2^n} \right) \left(\exp(-\frac{i-1}{2^n}) - \exp(-\frac{i}{2^n}) \right) \\ &= (1 - \frac{1}{2^n}) \frac{1}{2} + \frac{1}{2} (1 - \exp(-(n-1))) + \sum_{i=1}^{(n-1)2^n} \frac{i-1}{2^n} \left(\exp(-\frac{i-1}{2^n}) - \exp(-\frac{i}{2^n}) \right). \end{aligned}$$

Now we observe that

$$\left(\exp(-\frac{i-1}{2^n}) - \exp(-\frac{i}{2^n}) \right) = \int_{\frac{i-1}{2^n}}^{\frac{i}{2^n}} \exp(-u)du.$$

Therefore, following the same argument as in Example 1.53, we conclude that

$$\begin{aligned} &\left| \mathbb{E}(\psi_n(X)) - 1 - \frac{1}{2} \int_0^{n-1} u \exp(-u)du \right| \\ &\leq \frac{1}{2} \left[\frac{1}{2^n} + \exp(-(n-1)) + \frac{1}{2^n} (1 - \exp(-(n-1))) \right] \rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$. Therefore, we conclude that $\mathbb{E}(\psi_n(X)) \rightarrow 1 + \frac{1}{2} \int_0^\infty u \exp(-u)du = 3/2$. Hence, by MCT, we have $\mathbb{E}(X) = 3/2$. ■

Next we prove a result which provides a generic formula for computing the expectation of any non-negative random variable.

Proposition 1.56. *For any non-negative random variable X , we have*

$$\mathbb{E}(X) = \int_0^\infty \mathbb{P}(X \geq t) dt = \int_0^\infty (1 - F(t-)) dt. \quad (1.26)$$

We should note from the formula above that the expectation is actually determined by the distribution function of the random variable. Therefore, if two random variables have the same distribution function, they will have the same expectation, provided it exists. We will often write the expectation in the form of an integral $\int u dF_X(u)$, where it should be interpreted as a sum or an integral or a combination of both as the case may be.

Proof : We will now use another approximation from below to compute the expectation. For $n \geq 1$, let $\chi_n : \mathbb{R} \rightarrow [0, \infty)$ be defined as follows:

$$\chi_n(x) = \begin{cases} \frac{i-1}{2^n} & \text{if } x \in [\frac{i-1}{2^n}, \frac{i}{2^n}), i = 1, 2, \dots, n2^n \\ n & \text{if } x \geq n \\ 0 & \text{otherwise.} \end{cases}$$

Again, it is easy to check that for all $x \geq 0$, $\chi_n(x) \uparrow x$.

Consider $X_n = \chi_n(X)$. Note that $\mathbb{E}(X_n) \rightarrow \mathbb{E}(X)$ by MCT. Now, we have

$$\begin{aligned} \mathbb{E}(X_n) &= \sum_{i=1}^{n2^n} \frac{i-1}{2^n} \mathbb{P}(X \in [\frac{i-1}{2^n}, \frac{i}{2^n})) + n\mathbb{P}(X \geq n) \\ &= \sum_{i=1}^{n2^n} \frac{i-1}{2^n} [\mathbb{P}(X \geq \frac{i-1}{2^n}) - \mathbb{P}(X \geq \frac{i}{2^n})] + n\mathbb{P}(X \geq n) \\ &= \sum_{i=0}^{n2^n-1} \frac{i}{2^n} \mathbb{P}(X \geq \frac{i}{2^n}) - \sum_{i=1}^{n2^n} \frac{i-1}{2^n} \mathbb{P}(X \geq \frac{i}{2^n}) + n\mathbb{P}(X \geq n) \\ &= \sum_{i=0}^{n2^n-1} \frac{i}{2^n} \mathbb{P}(X \geq \frac{i}{2^n}) - \sum_{i=1}^{n2^n} \frac{i}{2^n} \mathbb{P}(X \geq \frac{i}{2^n}) + \sum_{i=1}^{n2^n} \mathbb{P}(X \geq \frac{i}{2^n}) + n\mathbb{P}(X \geq n) \\ &= \sum_{i=1}^{n2^n} \frac{1}{2^n} \mathbb{P}(X \geq \frac{i}{2^n}). \end{aligned}$$

Now, for $i \geq 1$, we have

$$\begin{aligned} \frac{1}{2^n} \mathbb{P}(X \geq \frac{i}{2^n}) &= \int_{\frac{i-1}{2^n}}^{\frac{i}{2^n}} \mathbb{P}(X \geq \frac{i}{2^n}) dt \\ &= \int_{\frac{i-1}{2^n}}^{\frac{i}{2^n}} \mathbb{P}(X \geq t) dt + \int_{\frac{i-1}{2^n}}^{\frac{i}{2^n}} [\mathbb{P}(X \geq \frac{i}{2^n}) - \mathbb{P}(X \geq t)] dt. \end{aligned}$$

Thus, using the fact that F is monotone, we have

$$\begin{aligned} \left| \frac{1}{2^n} \mathbb{P}(X \geq \frac{i}{2^n}) - \int_{\frac{i-1}{2^n}}^{\frac{i}{2^n}} \mathbb{P}(X \geq t) dt \right| &\leq \int_{\frac{i-1}{2^n}}^{\frac{i}{2^n}} |\mathbb{P}(X \geq \frac{i}{2^n}) - \mathbb{P}(X \geq t)| dt \\ &\leq \int_{\frac{i-1}{2^n}}^{\frac{i}{2^n}} [\mathbb{P}(X \geq \frac{i-1}{2^n}) - \mathbb{P}(X \geq \frac{i}{2^n})] dt = \frac{1}{2^n} [\mathbb{P}(X \geq \frac{i-1}{2^n}) - \mathbb{P}(X \geq \frac{i}{2^n})]. \end{aligned}$$

Therefore, we obtain that

$$\begin{aligned} \left| \mathbb{E}(Y_n) - \int_0^n \mathbb{P}(X \geq t) dt \right| &= \left| \sum_{i=1}^{n2^n} \frac{1}{2^n} \mathbb{P}(X \geq \frac{i}{2^n}) - \sum_{i=1}^{n2^n} \int_{\frac{i-1}{2^n}}^{\frac{i}{2^n}} \mathbb{P}(X \geq t) dt \right| \\ &\leq \sum_{i=1}^{n2^n} \frac{1}{2^n} [\mathbb{P}(X \geq \frac{i-1}{2^n}) - \mathbb{P}(X \geq \frac{i}{2^n})] \leq \frac{1}{2^n} [1 - \mathbb{P}(X \geq n)] \rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$. Now, taking limit as $n \rightarrow \infty$, we conclude that $\mathbb{E}(X) = \int_0^\infty \mathbb{P}(X \geq t) dt = \int_0^\infty (1 - F(t-)) dt$. ■

Exercise 1.57. For any non-negative random variable X , show that

$$\mathbb{E}(X) = \int_0^\infty \mathbb{P}(X > t) dt = \int_0^\infty (1 - F(t)) dt.$$

[Hint : modify the function χ_n suitably.] ■

Exercise 1.58. For any random variable X and $\lambda \geq 0$, show that

$$\mathbb{E}(X \mathbb{I}\{X \geq \lambda\}) = \lambda(1 - F(\lambda-)) + \int_\lambda^\infty (1 - F(t)) dt. \quad \blacksquare$$

1.5 Moments & Inequalities

For a function $f : \mathbb{R} \rightarrow \mathbb{R}$, let us define the random variable $Y = f(X)$. Then, the expectation of Y can be computed from the distribution of X , i.e., without computing the distribution of $f(X)$. Indeed, let X be a random variable taking values in $\mathbb{Z}$ and f is a non-negative function. Then, we have

$$\mathbb{E}(f(X)) = \sum_{i=-\infty}^{\infty} f(i) \mathbb{P}(X = i). \quad (1.27)$$

If X has a density g , and f is a non-negative function, we have

$$\mathbb{E}(f(X)) = \int_{-\infty}^{\infty} f(x) g(x) dx. \quad (1.28)$$

If general functions f , whenever the sums or the integrals of f^+ and f^- are finite, we may write the expectation as the sum or the integral given by the right hand side of the above equations (1.27) or (1.28). We observe that $\mathbb{E}(f(X)) < \infty$ for any bounded continuous function f .

Exercise 1.59. The distribution of random variables is determined by the expectations of bounded continuous functions. In other words, for random variables X and Y , if $\mathbb{E}(f(X)) = \mathbb{E}(f(Y))$ for all bounded continuous functions, then

$$X \stackrel{d}{=} Y, \text{ i.e., for any } u \in \mathbb{R}, \mathbb{P}(X \leq u) = \mathbb{P}(Y \leq u). \quad \blacktriangleleft$$

If we specialize to the function $x \mapsto x^k$, we can now define higher moments of a random variable. Indeed, for any k , the k -th order moment of a non-negative random variable is defined by

$$\mathbb{E}(X^k) = \int x^k dF(x) \text{ which equals } \sum_{i=-\infty}^{\infty} i^k \mathbb{P}(X = i) \text{ or } \int_{-\infty}^{\infty} x^k g(x) dx \quad (1.29)$$

or a combination of both as the case might be. The variance of a random variable is given by

$$\text{Var}(X) = \mathbb{E}((X - \mathbb{E}(X))^2) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2. \quad (1.30)$$

Note the variance is defined only when $\mathbb{E}(|X|) < \infty$.

Exercise 1.60. If $\mathbb{E}(|X|^r) < \infty$, then $\mathbb{E}(|X|^s) < \infty$ for all $s < r$. $\blacktriangleleft$

Exercise 1.61. For any non-negative random variable X , show that

$$\sum_{n=1}^{\infty} n \mathbb{P}(X > n) \leq \mathbb{E}(X^2) \leq 1 + 4 \sum_{n=1}^{\infty} n \mathbb{P}(X > n).$$

Hence prove that $\mathbb{E}(X^2) < \infty$ if and only if $\sum_{n=1}^{\infty} n \mathbb{P}(|X| > n) < \infty$. $\blacksquare$

Exercise 1.62. For any random variable X and real number $r > 0$, show that

1. $\mathbb{E}(|X|^r) = \int_0^{\infty} r t^{r-1} \mathbb{P}(|X| \geq t) dt = \int_0^{\infty} r t^{r-1} \mathbb{P}(|X| > t) dt;$
2. $\mathbb{E}(|X|) < \infty$, if and only if $\sum_{n=0}^{\infty} c^n \mathbb{P}(|X| > c^n) < \infty$ for any $c > 1$. $\blacksquare$

Exercise 1.63. Suppose that the non-negative random variable X is such that for any $\alpha > 1$, we have

$$\frac{\mathbb{P}(X \geq \alpha n)}{\mathbb{P}(X \geq n)} \rightarrow 0$$

as $n \rightarrow \infty$. Prove that then X admits finite moments of all orders.

1.5.1 Markov & Chebyshev inequality

For any $\lambda > 0$ and non-negative random variable X , we have that $\lambda \mathbb{I}\{X \geq \lambda\} \leq X$. Therefore, we have $\lambda \mathbb{P}(X \geq \lambda) \leq \mathbb{E}(X)$ so that we have

$$\mathbb{P}(X \geq \lambda) = (1 - F(\lambda-)) \leq \frac{1}{\lambda} \mathbb{E}(X). \quad (1.31)$$

Furthermore if we assume that $\mathbb{E}(|X|) < \infty$, we have $|X| \mathbb{I}\{|X| \geq \lambda\} \rightarrow 0$ as $\lambda \uparrow \infty$ and $|X| \mathbb{I}\{|X| \geq \lambda\} \leq |X|$. Therefore, by DCT, $\mathbb{E}(|X| \mathbb{I}\{|X| \geq \lambda\}) \rightarrow 0$ as $\lambda \uparrow \infty$. Hence,

$$\lambda \mathbb{P}(|X| \geq \lambda) \leq \mathbb{E}(|X| \mathbb{I}\{|X| \geq \lambda\}) \rightarrow 0$$

as $\lambda \rightarrow \infty$.

Exercise 1.64. If $\mathbb{E}(|X|^r) < \infty$, then show that $\lambda^r \mathbb{P}(|X| > \lambda) \rightarrow 0$ as $\lambda \rightarrow \infty$. ◀

For non-negative random variable X and for any $k > 0$, we have

$$\mathbb{P}(X \geq \lambda) = \mathbb{P}(X^k \geq \lambda^k) \leq \frac{1}{\lambda^k} \mathbb{E}(X^k). \quad (1.32)$$

This is known as Markov Inequality. For any random variable X , with $\mu = \mathbb{E}(X) < \infty$, we apply the above inequality on $|X - \mu|$ with $k = 2$ which yields Chebyshev's inequality

$$\mathbb{P}(|X - \mu| \geq \lambda) \leq \frac{1}{\lambda^2} \mathbb{E}(|X - \mu|^2) = \frac{1}{\lambda^2} \text{Var}(X). \quad (1.33)$$

Exercise 1.65. A random variable X is a constant with probability 1 if and only if $\text{Var}(X) = 0$. ◀

Exercise 1.66. Let X be a random variable with $\mathbb{E}(X) = \mu$ and variance $\text{Var}(X) = \sigma^2 < \infty$.

(a) Prove Cantelli's inequality : for $\lambda \geq 0$

$$\mathbb{P}(X - \mu \geq \lambda) \leq \frac{\sigma^2}{\sigma^2 + \lambda^2}.$$

(b) Show that

$$\mathbb{P}(|X - \mu| \geq \lambda) \leq \frac{2\sigma^2}{\sigma^2 + \lambda^2}.$$

When is this better than Chebyshev's inequality?

(c) By considering a random variable assuming two values, show that Cantelli's inequality is sharp.

Exercise 1.67. For a random variable X on $[0, 1]$ with $F(1) = 1$ and $F(x) < 1$ for all $x < 1$, show that

$$[\mathbb{E}(|X|^r)]^{\frac{1}{r}} \rightarrow 1$$

as $r \rightarrow \infty$. If F is such that $F(x) < 1$ for all $x \in \mathbb{R}$ and all moments exist, then show that

$$[\mathbb{E}(|X|^r)]^{\frac{1}{r}} \rightarrow +\infty$$

as $r \rightarrow \infty$. ◀

Exercise 1.68. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function. Define the **Bernstein** polynomial $B_n^f : \mathbb{R} \rightarrow \mathbb{R}$ as follows

$$B_n^f(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}.$$

Show that, for any $\delta > 0$

$$\sup_{x \in [0,1]} |f(x) - B_n^f(x)| \leq w_f(\delta) + \frac{M}{2n\delta^2}$$

where $w_f(\delta) = \sup\{|f(u) - f(v)| : u, v \in [0, 1], |u - v| \leq \delta\}$ and $M = \sup\{|f(x)| : x \in [0, 1]\}$. Conclude that any continuous function on $[0, 1]$ can be uniformly approximated by polynomials.

[Hint: Note that, if X is a Binomial (n, p) random variable, $\mathbb{E}(X) = np$ and $\mathbb{V}\text{ar}(X) = np(1-p)$.] ■

Exercise 1.69. The distribution of a bounded random variable is determined by its moments. More precisely, for random variables $0 \leq X, Y \leq 1$, if $\mathbb{E}(X^m) = \mathbb{E}(Y^m)$ for all $m \geq 1$, then $X \stackrel{d}{=} Y$ (see also Exercise 1.59.) ■

Exercise 1.70. For non-negative random variable X , we observe that the transformed random variable $\exp(-\lambda X)$ is a bounded random variable for any $\lambda \geq 0$ and hence $\mathbb{E}(\exp(-\lambda X)) < \infty$. The Laplace transform, for a non-negative random variable is defined by, for $\lambda \geq 0$.

$$L_X(\lambda) := \mathbb{E}(\exp(-\lambda X)). \quad (1.34)$$

Observe that $L_X(0) = 1$ and L_X is a non-decreasing function in $\lambda \geq 0$. The distribution of a non-negative random variable is determined by its Laplace transform. More precisely, for random variables $X, Y \geq 0$, if $L_X(\lambda) = L_Y(\lambda)$ for all $\lambda \geq 0$, then $X \stackrel{d}{=} Y$ (see also Exercises 1.69 and 1.59.) ■

1.5.2 Jensen's inequality

Definition 1.71. A function ϕ on an open interval I (bounded or unbounded) is called convex if for $x_1, x_2 \in I$ and $0 \leq \lambda \leq 1$,

$$\phi(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda\phi(x_1) + (1 - \lambda)\phi(x_2). \quad (1.35)$$

By induction, it follows that

$$\phi\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i \phi(x_i)$$

for all $x_i \in I$ and $\lambda_i \geq 0$ with $\sum_{i=1}^n \lambda_i = 1$.

If ϕ has a continuous, non-decreasing derivative ϕ' on I , then ϕ is convex. Indeed, for $x_1 < x_2$ and $\lambda \in (0, 1)$, set $x = \lambda x_1 + (1 - \lambda)x_2$. Then, we have

$$\frac{\phi(x) - \phi(x_1)}{x - x_1} = \frac{1}{x - x_1} \int_{x_1}^x \phi'(t) dt \leq \phi'(x) \leq \frac{1}{x_2 - x} \int_x^{x_2} \phi'(t) dt = \frac{\phi(x_2) - \phi(x)}{x_2 - x}.$$

This inequality is equivalent to equation (1.35) with the identification that $\lambda = (x_2 - x)/(x_2 - x_1)$. An useful corollary of the above is that if ϕ is twice differentiable with $\phi''(x) \geq 0$ then ϕ is convex. A direct application shows that $x \mapsto x^p$ is convex on $(0, \infty)$ for $p \geq 1$ or $x \mapsto \exp(x)$ is convex on $\mathbb{R}$. However, it is not necessary that a convex function is differentiable (it is continuous though). It is easy to check $x \mapsto |x|$ is a convex function, even though the function is not differentiable at 0. Also, it is straightforward to show that the composition of a non-decreasing convex function with a convex function yields a convex function. Using this we can prove that $x \mapsto |x|^p$ is convex on $\mathbb{R}$ for $p \geq 1$.

The following exercise will be helpful in proving the result.

Exercise 1.72. Suppose that $\phi : I \rightarrow \mathbb{R}$ is a convex function where I is an interval. For $x_1, x_2 \in I$, set

$$R(x_1, x_2) = \frac{\phi(x_1) - \phi(x_2)}{x_1 - x_2}.$$

Note that $R(x_1, x_2)$ represents the slope of the line joining $(x_1, \phi(x_1))$ and $(x_2, \phi(x_2))$. It is also clear that R is symmetric in x_1, x_2 , i.e., interchanging their roles do not change the value of R .

- a) Show that ϕ is convex if and only if $R(x_1, x_2)$ is monotonically increasing in x_1 for every fixed x_2 or monotonically increasing in x_2 for every fixed x_1 .

b) Show that ϕ is continuous.

c) Show that for any $x \in I^\circ$ (the interior of the interval I), the left derivative $\phi'(x-)$ and the right derivative $\phi'(x+)$ exist. In fact,

$$\begin{aligned}\phi'(x+) &= \lim_{y \downarrow x} \frac{\phi(y) - \phi(x)}{y - x} = \inf_{y \in I: x < y} \frac{\phi(y) - \phi(x)}{y - x} = \inf_{y \in I: x < y} R(y, x) \\ \phi'(x-) &= \lim_{y \uparrow x} \frac{\phi(y) - \phi(x)}{y - x} = \sup_{y \in I: x > y} \frac{\phi(y) - \phi(x)}{y - x} = \sup_{y \in I: x > y} R(y, x).\end{aligned}$$

Further, conclude from (a) that $\phi'(x+) \geq \phi'(x-)$.

d) Show that for any $x \in I$,

$$\phi(y) \geq \phi(x) + m(y - x) \text{ for all } y \in I$$

for any $m \in [\phi'(x-), \phi'(x+)]$. Conclude that the equality holds if and only if ϕ is linear.

Theorem 1.73. Suppose $\phi : I \rightarrow \mathbb{R}$ be a convex function. Let X be random variable taking values in I . Suppose that $\mathbb{E}(|X|) < \infty$. Then $\mathbb{E}(\phi(X))$ is defined and

$$\phi(\mathbb{E}(X)) \leq \mathbb{E}(\phi(X)). \quad (1.36)$$

Proof : Observe that $|\mathbb{E}(X)| \leq \mathbb{E}(|X|) < \infty$. Thus, we must have $x_0 = \mathbb{E}(X) \in I$. Then, by part c) of Exercise 1.72, we have for any $x \in I$,

$$\phi(x) \geq \phi(x_0) + m(x - x_0)$$

for all $x \in I$ where $m \in [\phi'(x_0-), \phi'(x_0+)]$.

Now, set $Y = \phi(x_0) + m(X - x_0)$. From above relation, we obtain,

$$\phi(X) \geq Y. \quad (1.37)$$

Observe that $|Y| \leq |\phi(x_0)| + |m|(|X| + |x_0|)$. Hence, $\mathbb{E}(|Y|) < \infty$. Since $\phi(X) \geq Y$, we have $Y^- = \max(-Y, 0) \geq \max(-\phi(X), 0) = \phi(X)^-$. Thus, $\mathbb{E}(\phi(X)^-) \leq \mathbb{E}(Y^-) \leq \mathbb{E}(|Y|) < \infty$. Thus, $\mathbb{E}(\phi(X))$ is defined and $\mathbb{E}(\phi(X)) > -\infty$. Now, if $\mathbb{E}(\phi(X)) = +\infty$, we have nothing to prove. So, we may assume $\mathbb{E}(\phi(X)) < +\infty$. Thus, we may take expectation of both sides of the equation (1.37), we conclude that

$$\mathbb{E}(\phi(X)) \geq \mathbb{E}(Y) = \phi(x_0) + m(\mathbb{E}(X) - x_0) = \phi(x_0) = \phi(\mathbb{E}(X))$$

using the fact that $x_0 = \mathbb{E}(X_0)$. This completes the proof. ■

Note that the equality holds in equation (1.36) if either X is a constant with probability 1 or ϕ is linear.

Exercise 1.74. For a positive random variable, i.e., $\mathbb{P}(X > 0) = 1$, show that $\mathbb{E}(1/X) \geq 1/\mathbb{E}(X)$. When is $\mathbb{E}(1/X) = 1/\mathbb{E}(X)$? ◀

Exercise 1.75. For $0 < r < s < \infty$, show that

$$[\mathbb{E}(|X|^r)]^{\frac{1}{r}} \leq [\mathbb{E}(|X|^s)]^{\frac{1}{s}}. \quad \blacktriangleleft$$

Exercise 1.76. For a random variable X with $\mathbb{E}(X) = 0$, $\mathbb{E}(X^2) = \sigma^2$ and $x > 0$, show that

$$\mathbb{E}(|x + X|) \leq x + \frac{\sigma^2}{2x}. \quad \blacktriangleleft$$

Exercise 1.77. Let $\{p_k : k \geq 0\}$ and $\{q_k : k \geq 0\}$ be two sequences of positive reals with $\sum_{k=0}^{\infty} p_k = \sum_{k=0}^{\infty} q_k = 1$. Show that

$$\sum_{k=0}^{\infty} p_k \log q_k \leq \sum_{k=0}^{\infty} p_k \log p_k. \quad \blacktriangleleft$$

1.5.3 Holder's inequality

Theorem 1.78. For two random variables X, Y and $p, q > 1$ with

$$\frac{1}{p} + \frac{1}{q} = 1$$

we have

$$\mathbb{E}(|XY|) \leq [\mathbb{E}(|X|^p)]^{\frac{1}{p}} [\mathbb{E}(|Y|^q)]^{\frac{1}{q}}.$$

A special case of this is when $p = q = 2$, known as **Cauchy-Schwartz inequality**

$$\mathbb{E}(|XY|) \leq \sqrt{\mathbb{E}(|X|^2)\mathbb{E}(|Y|^2)}.$$

Exercise 1.79. Let $X \geq 0$ be a random variable with $\mathbb{E}(X^2) < \infty$. For any $\lambda \in (0, 1)$ show that

$$\mathbb{P}(X > \lambda \mathbb{E}(X)) \geq (1 - \lambda)^2 \frac{\mathbb{E}[X]^2}{\mathbb{E}[X^2]}.$$

Conclude that

$$\mathbb{P}(X > 0) \geq \frac{\mathbb{E}[X]^2}{\mathbb{E}[X^2]}. \quad \blacktriangleleft$$

Exercise 1.80. Let X and Y be two random variables with mean $\mathbb{E}(X) = \mathbb{E}(Y) = 0$, $\mathbb{E}(X^2) = \mathbb{E}(Y^2) = 1$, and $\mathbb{E}(XY) = \rho$. Show that

$$\mathbb{E}(\max\{X^2, Y^2\}) \leq 1 + \sqrt{1 - \rho^2}. \quad \blacktriangleleft$$

The proof of this uses Young's inequality for positive reals.

Proposition 1.81. (Young's Inequality) For $a, b \in \mathbb{R}$ and $p, q \geq 1$ such that $1/p + 1/q = 1$,

$$|ab| \leq \frac{|a|^p}{p} + \frac{|b|^q}{q}.$$

Proof : For $a, b \in \mathbb{R}$, $|a|, |b| \geq 0$ and we can choose $s, t \in \mathbb{R}$ such that $|a| = \exp(s/p)$ and $|b| = \exp(t/q)$ so that $|a|^p = \exp(s)$ and $|b|^q = \exp(t)$. Now, we have

$$\begin{aligned} |ab| &= |a||b| = \exp(s/p) \exp(t/q) = \exp\left[\frac{s}{p} + \frac{t}{q}\right] \\ &\leq \frac{1}{p} \exp(s) + \frac{1}{q} \exp(t) = \frac{|a|^p}{p} + \frac{|b|^q}{q}. \end{aligned}$$

In the above we have used the fact that $x \rightarrow \exp(x)$ is a convex function and $1/p + 1/q = 1$. ■

Proof : (Holder's Inequality) Note that if $\mathbb{E}(|X|^p) = 0$, $X = 0$ with probability 1, so that $XY = 0$ with probability 1, thus the inequality holds. Similarly, for $\mathbb{E}(|Y|^q) = 0$. If either $\mathbb{E}(|X|^p) = \infty$ or $\mathbb{E}(|Y|^q) = \infty$, the inequality holds trivially. So, we assume that $0 < \mathbb{E}(|X|^p) < \infty$ and $0 < \mathbb{E}(|Y|^q) < \infty$. Consider the random variables

$$X' = \frac{|X|}{[\mathbb{E}(|X|^p)]^{\frac{1}{p}}} \text{ and } Y' = \frac{|Y|}{[\mathbb{E}(|Y|^q)]^{\frac{1}{q}}}.$$

Then, we observe that $\mathbb{E}[(X')^p] = 1 = \mathbb{E}[(Y')^q]$. Hence, using Young's inequality, we have

$$\frac{\mathbb{E}[|XY|]}{[\mathbb{E}(|X|^p)]^{\frac{1}{p}} [\mathbb{E}(|Y|^q)]^{\frac{1}{q}}} = \mathbb{E}[|X'Y'|] \leq \frac{\mathbb{E}[(X')^p]}{p} + \frac{\mathbb{E}[(Y')^q]}{q} = \frac{1}{p} + \frac{1}{q} = 1.$$

This completes the proof. ■

1.5.4 Minkowski's inequality

Theorem 1.82. For $p \geq 1$, we have

$$[\mathbb{E}(|X + Y|^p)]^{\frac{1}{p}} \leq [\mathbb{E}(|X|^p)]^{\frac{1}{p}} + [\mathbb{E}(|Y|^p)]^{\frac{1}{p}}.$$

Proof : We observe that if $p = 1$, it is trivial. So, consider $p > 1$. If $\mathbb{E}(|X + Y|^p) = 0$, then the equality holds trivially. Now, using convexity of $|x|^p$, we have

$$\left|\frac{1}{2}x + \frac{1}{2}y\right|^p \leq \frac{1}{2}|x|^p + \frac{1}{2}|y|^p$$

so that we have

$$|x + y|^p \leq 2^{p-1} [|x|^p + |y|^p].$$

Hence, if $\mathbb{E}(|X + Y|^p) = \infty$, we must have $\mathbb{E}[|X|^p + |Y|^p] = \infty$ so that at least one of $\mathbb{E}(|X|^p) = \infty$ or $\mathbb{E}(|Y|^p) = \infty$. Thus, in such a case also, the inequality holds. Now, we assume that $0 < \mathbb{E}(|X + Y|^p) < \infty$. Take $q \geq 1$ so that $1/p + 1/q = 1$. Hence, $p = q(p-1)$. Now, we have

$$\begin{aligned} \mathbb{E}(|X + Y|^p) &= \mathbb{E}(|X + Y| \cdot |X + Y|^{p-1}) \leq \mathbb{E}[(|X| + |Y|)|X + Y|^{p-1}] \\ &= \mathbb{E}[|X| \cdot |X + Y|^{p-1}] + \mathbb{E}[|Y| \cdot |X + Y|^{p-1}] \\ &\leq \mathbb{E}[|X|^p]^{\frac{1}{p}} \mathbb{E}[|X + Y|^{q(p-1)}]^{\frac{1}{q}} + \mathbb{E}[|Y|^p]^{\frac{1}{p}} \mathbb{E}[|X + Y|^{q(p-1)}]^{\frac{1}{q}} \\ &\leq \left[\mathbb{E}[|X|^p]^{\frac{1}{p}} + \mathbb{E}[|Y|^p]^{\frac{1}{p}} \right] \mathbb{E}[|X + Y|^{q(p-1)}]^{\frac{1}{q}} \\ &\leq \left[\mathbb{E}[|X|^p]^{\frac{1}{p}} + \mathbb{E}[|Y|^p]^{\frac{1}{p}} \right] \mathbb{E}[|X + Y|^p]^{\frac{1}{q}}. \end{aligned}$$

Now, dividing both sides by $\mathbb{E}[|X + Y|^p]^{\frac{1}{q}}$, we have

$$\mathbb{E}[|X + Y|^p]^{\frac{1}{p}} = \mathbb{E}[|X + Y|^p]^{1-\frac{1}{q}} \leq \mathbb{E}[|X|^p]^{\frac{1}{p}} + \mathbb{E}[|Y|^p]^{\frac{1}{p}}.$$

This completes the proof. ■

1.6 Independence of Random Variables

We have already introduced independent events and σ -algebras. Now, we extend these for random variables.

Definition 1.83. We will say $\{X_1, X_2, \dots, X_n\}$ are independent if the family of σ -algebras $\{\sigma(X_1), \sigma(X_2), \dots, \sigma(X_n)\}$ is independent. Any family $\{\sigma(X_\lambda) : \lambda \in \Lambda\}$ is independent if any finite sub-family $\{\sigma(X_{\lambda_1}), \sigma(X_{\lambda_2}), \dots, \sigma(X_{\lambda_n}) : \lambda_1, \lambda_2, \dots, \lambda_n \in \Lambda\}$ is independent.

We know that $\sigma(X) = \{X^{-1}(B) : B \in \mathcal{B}\}$. Thus, a finite family of random variables $\{X_1, X_2, \dots, X_n\}$ will be independent if for any finite sets $B_1, B_2, \dots, B_n \subseteq \mathbb{R}$ (Borel

sets in $\mathbb{R}$), the family of events $\{A_i = \{X_i \in B_i\} : i = 1, 2, \dots, n\}$ is independent. In other words, we must have

$$\mathbb{P}(X_i \in B_i \text{ for } i = 1, 2, \dots, n) = \mathbb{P}(X_1 \in B_1)\mathbb{P}(X_2 \in B_2) \cdots \mathbb{P}(X_n \in B_n). \quad (1.38)$$

Since the Borel σ -algebra is smallest σ -algebra containing intervals $\{(-\infty, x] : x \in \mathbb{R}\}$, it is enough to check the independence only for sets $\{X_1, X_2, \dots, X_n\}$ are independent if and only if

$$\mathbb{P}(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n) = \mathbb{P}(X_1 \leq x_1)\mathbb{P}(X_2 \leq x_2) \cdots \mathbb{P}(X_n \leq x_n) \quad (1.39)$$

for all $x_1, x_2, \dots, x_n \in \mathbb{R}$. Thus, the joint distribution function of $(X_1, X_2, \dots, X_n)$ at $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ is the product of the individual distribution function.

Remark. It is, in fact, enough to check for any class of subsets $\mathcal{A}$ which is closed under intersection and any interval of type $(-\infty, x]$ may be obtained by unions or intersections of these sets. ►

Let $X_1, X_2, \dots, X_n$ be a family of independent random variables having densities $f_1, f_2, \dots, f_n$ respectively. Then, for $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$, from equation (1.38), we have

$$\begin{aligned} & \mathbb{P}(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n) \\ &= \int_{-\infty}^{x_1} f_1(u_1) du_1 \int_{-\infty}^{x_2} f_2(u_2) du_2 \cdots \int_{-\infty}^{x_n} f_n(u_n) du_n \\ &= \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} \cdots \int_{-\infty}^{x_n} f_1(u_1) f_2(u_2) \cdots f_n(u_n) du_n \dots du_2 du_1 \\ &= \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} \cdots \int_{-\infty}^{x_n} f(\mathbf{u}) du_n \dots du_2 du_1 \end{aligned}$$

where, for $\mathbf{u} = (u_1, u_2, \dots, u_n) \in \mathbb{R}^n$,

$$f(\mathbf{u}) = f_1(u_1) f_2(u_2) \cdots f_n(u_n). \quad (1.40)$$

Thus, $f(\mathbf{u})$ is a density of the joint distribution of $(X_1, X_2, \dots, X_n)$.

On the other hand, if the density of $(X_1, X_2, \dots, X_n)$ splits into product of functions as in equation (1.40), then the family $\{X_1, X_2, \dots, X_n\}$ is independent. To establish

this, we note that

$$\begin{aligned}
 1 &= \mathbb{P}((X_1, X_2, \dots, X_n) \in \mathbb{R}^n) \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_1(u_1) f_2(u_2) \cdots f_n(u_n) du_n \cdots du_2 du_1 \\
 &= \prod_{i=1}^n \int_{-\infty}^{\infty} f_i(u_i) du_i.
 \end{aligned}$$

Therefore, for each $i \geq 1$, we must have $0 < \int_{-\infty}^{\infty} f_i(u_i) du_i < \infty$. Now, for $i \geq 1$ and $u \in \mathbb{R}$, set

$$g_i(u) = \frac{f_i(u)}{\int_{-\infty}^{\infty} f_i(u_i) du_i}.$$

so that

$$\int_{-\infty}^{\infty} g_i(u) du = 1$$

and for $\mathbf{u} \in \mathbb{R}^n$,

$$f(\mathbf{u}) = f_1(u_1) f_2(u_2) \cdots f_n(u_n) = g_1(u_1) g_2(u_2) \cdots g_n(u_n).$$

Now, we set, for $x \in \mathbb{R}$,

$$F_i(x) = \int_{-\infty}^x g_i(u) du$$

Therefore, F_i is a distribution function for each $i \geq 1$. Finally, we observe that

$$\begin{aligned}
 &\mathbb{P}(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n) \\
 &= \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} \cdots \int_{-\infty}^{x_n} g_1(x_1) g_2(x_2) \cdots g_n(x_n) dx_n \cdots dx_2 dx_1 \\
 &= \int_{-\infty}^{x_1} g_1(x_1) dx_1 \int_{-\infty}^{x_2} g_2(x_2) dx_2 \cdots \int_{-\infty}^{x_n} g_n(x_n) dx_n \\
 &= F_1(x_1) F_2(x_2) \cdots F_n(x_n).
 \end{aligned}$$

By taking limit as $x_1, x_2, \dots, x_{n-1}$ tend to ∞ in the above, we obtain that F_n must be distribution function of X_n . Therefore, we can take g_n can be taken as a density function of X_n . This holds for any $i \geq 1$. Thus, we have verified the equation (1.39) which implies that the family $\{X_1, X_2, \dots, X_n\}$ is independent.

Example 1.84. We return to Example 1.43 to observe that the joint density of (X_1, X_2) can be written as the product of two standard normal densities. Therefore, we conclude that X_1, X_2 are independent and identically distributed standard random variables. We further observe that the polar transformed random variables R and Θ are also independent, however they are not identically distributed. ■

We now state an extension of Theorem 1.18.

Theorem 1.85. *Suppose that $\{X_1, X_2, \dots, X_n\}$ is a collection of independent random variables. Assume that $\mathcal{F}_1 = \sigma(\{X_1, \dots, X_{k_1}\})$, $\mathcal{F}_2 = \sigma(\{X_{k_1+1}, \dots, X_{k_2}\})$, $\dots$, $\mathcal{F}_m = \sigma(\{X_{k_{m-1}+1}, \dots, X_n\})$. Then, the family of σ -algebras $\{\mathcal{F}_1, \mathcal{F}_2, \dots, \mathcal{F}_m\}$ is independent.*

With the above notations as in Theorem 1.85, let us consider measurable functions $f_i : \mathbb{R}^{k_{i+1}-k_i} \rightarrow \mathbb{R}^{d_i}$ for $i = 1, 2, \dots, m$ (with notations $k_0 = 0, k_m = n$). If $\{X_1, X_2, \dots, X_n\}$ is a collection of independent random variables, then $\{Y_i = f_i(X_{k_{i-1}+1}, \dots, X_{k_i}) : i = 1, 2, \dots, m\}$ is a collection of independent random vectors.

If we have a family of independent and identically distributed random variables $\{X_n : n \geq 1\}$, then it is easy to observe that, for any $i_1, i_2, \dots, i_k$, the distribution function of $(X_{i_1}, X_{i_2}, \dots, X_{i_k})$ is identical to $(X_1, X_2, \dots, X_k)$ for any distinct $i_1, i_2, \dots, i_k \geq 1, k \geq 1$. Therefore, the joint distribution of $(X_{i_1}, X_{i_2}, \dots, X_{i_k})$ is identical to $(X_1, X_2, \dots, X_k)$. This implies that

$$\mathbb{P}((X_{i_1}, X_{i_2}, \dots, X_{i_k}) \in B) = \mathbb{P}((X_1, X_2, \dots, X_k) \in B) \quad (1.41)$$

for any set $B \in \mathcal{B}(\mathbb{R}^k)$. We will be using this later. Note that, this further implies that for any function $f : \mathbb{R}^k \rightarrow \mathbb{R}^m$, we also must have that the distribution of $f(X_{i_1}, X_{i_2}, \dots, X_{i_k})$ must be identical to $f(X_1, X_2, \dots, X_k)$.

Exercise 1.86. (Exercise 1.40 revisited) Let X be a random variable with F as distribution function and r a positive integer. Construct random variables having the following distribution functions :

$$a) G_1(x) = (F(x))^r; \quad b) G_2(x) = 1 - (1 - F(x))^r. \quad \blacktriangleleft$$

Exercise 1.87. For a non-negative random variable X with distribution function F , show that

$$\int_0^\infty (1 - F(t))^2 dt \leq (\mathbb{E}(\sqrt{X}))^2. \quad \blacktriangleleft$$

Suppose that $\{X_i : i = 1, 2, \dots, n\}$ is a finite family of independent random variables. Let f_i be function on $\mathbb{R}$ for $i = 1, 2, \dots, n$. Let

$$Y_i = f_i(X_i)$$

for $i = 1, 2, \dots, n$. For $B_1, B_2, \dots, B_n \subset \mathbb{R}$, set

$$A_i = \{\omega : Y_i(\omega) \in B_i\} = \{\omega : f_i(X_i(\omega)) \in B_i\} = \{\omega : X_i(\omega) \in f_i^{-1}(B_i)\}$$

where $f_i^{-1}(A) = \{x \in \mathbb{R} : f_i(x) \in A\} \subseteq \mathbb{R}$. Since the family $\{X_1, X_2, \dots, X_n\}$ is independent, we have that the events $\{A_1, A_2, \dots, A_n\}$ are independent. Therefore, the family $\{X_1, X_2, \dots, X_n\}$ is also independent.

In a similar way, let $\{X_1, X_2, \dots, X_{i_1}, X_{i_1+1}, X_{i_1+2}, \dots, X_{i_1+i_2}\}$ be a family of independent random variables. Let $f : \mathbb{R}^{i_1} \rightarrow \mathbb{R}$ and $g : \mathbb{R}^{i_2} \rightarrow \mathbb{R}$ and $X = f(X_1, X_2, \dots, X_{i_1})$ and $Y = g(X_{i_1+1}, X_{i_1+2}, \dots, X_{i_1+i_2})$. Then, for $B_1, B_2 \subseteq \mathbb{R}$, we have

$$\begin{aligned} A = \{\omega : X(\omega) \in B_1\} &= \{\omega : f(X_1(\omega), \dots, X_{i_1}(\omega)) \in B_1\} \\ &= \{\omega : X_1(\omega), \dots, X_{i_1}(\omega) \in f^{-1}(B_1)\} \end{aligned}$$

and

$$\begin{aligned} B = \{\omega : Y(\omega) \in B_2\} &= \{\omega : g(X_{i_1+1}(\omega), \dots, X_{i_1+i_2}(\omega)) \in B_2\} \\ &= \{\omega : X_{i_1+1}(\omega), \dots, X_{i_1+i_2}(\omega) \in g^{-1}(B_2)\}. \end{aligned}$$

where $f^{-1}(B_1) = \{(x_1, \dots, x_{i_1}) \in \mathbb{R}^{i_1} : f(x) \in B_1\} \subseteq \mathbb{R}^{i_1}$ and $g^{-1}(B_2) = \{(x_1, \dots, x_{i_2}) \in \mathbb{R}^{i_2} : g(x) \in B_2\} \subseteq \mathbb{R}^{i_2}$. Since the family $\{X_1, X_2, \dots, X_{i_1}, X_{i_1+1}, X_{i_1+2}, \dots, X_{i_1+i_2}\}$ is independent, we have that the events A and B are independent. Hence, the random variables X and Y are independent. This may be extended for any finitely many set of random variables. Compare this result with Theorem 1.18.

We will end this section with a final result.

Proposition 1.88. *If X, Y are two independent random variables with $\mathbb{E}(|X|) < \infty$ and $\mathbb{E}(|Y|) < \infty$. Then, $\mathbb{E}(|XY|) < \infty$ and*

$$\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y). \quad (1.42)$$

Proof : First we prove the result when both X, Y take finitely many values. Suppose X assumes the values $\{x_1, x_2, \dots, x_n\}$ and Y assumes the values $\{y_1, y_2, \dots, y_m\}$. Hence we may write

$$X = \sum_{i=1}^n x_i \mathbb{I}_{A_i} \text{ and } Y = \sum_{j=1}^m y_j \mathbb{I}_{B_j}$$

where $\{A_i = \{\omega : X(\omega) = x_i\} : 1 \leq i \leq n\}$ and $\{B_j = \{\omega : Y(\omega) = y_j\} : 1 \leq j \leq m\}$ are partitions of Ω . Thus, we have

$$XY = \sum_{i=1}^n \sum_{j=1}^m x_i y_j \mathbb{I}_{A_i} \mathbb{I}_{B_j}.$$

Clearly, the product XY also takes only finitely many values. Thus, we must have $\mathbb{E}(|XY|) < \infty$. Hence, we have

$$\begin{aligned}\mathbb{E}(XY) &= \sum_{i=1}^n \sum_{j=1}^m x_i y_j \mathbb{P}(A_i \cap B_j) = \sum_{i=1}^n \sum_{j=1}^m x_i y_j \mathbb{P}(X = x_i, Y = y_j) \\ &= \sum_{i=1}^n \sum_{j=1}^m x_i y_j \mathbb{P}(X = x_i) \mathbb{P}(Y = y_j) = \sum_{i=1}^n x_i \mathbb{P}(X = x_i) \sum_{j=1}^m y_j \mathbb{P}(Y = y_j) \\ &= \mathbb{E}(X) \mathbb{E}(Y).\end{aligned}$$

Now, we consider both X, Y non-negative random variables. Fix $n \geq 1$ and consider the function $\psi_n : [0, \infty) \cup \{+\infty\} \rightarrow [0, \infty)$ in the following way:

$$\psi_n(x) = \begin{cases} \frac{(k-1)}{2^n} & \text{if } \frac{(k-1)}{2^n} \leq x < \frac{k}{2^n}, k = 1, 2, \dots, n2^n \\ n & \text{if } x \geq n. \end{cases}$$

Let us denote $X_n = \psi_n(X), Y_n = \psi_n(Y)$. Note that X_n is also non-negative and $0 \leq X_n \leq X, 0 \leq Y_n \leq Y$ and $X_n \uparrow X, Y_n \uparrow Y$. Further, X_n, Y_n are both random variables taking only a finitely many values.

Since X and Y are independent, then $X_n = \psi_n(X)$ and $Y_n = \psi_n(Y)$ are independent. Thus, we must have

$$\mathbb{E}(X_n Y_n) = \mathbb{E}(X_n) \mathbb{E}(Y_n).$$

Now, using MCT, $\mathbb{E}(X_n) \rightarrow \mathbb{E}(X)$ and $\mathbb{E}(Y_n) \rightarrow \mathbb{E}(Y)$. Finally, it is also easy to check that $X_n Y_n \uparrow XY$. Therefore, we must have $\mathbb{E}(X_n Y_n) \rightarrow \mathbb{E}(XY)$. Thus, taking limit in the above equation, we conclude the result for non-negative random variables.

For general X, Y , the families $\{X^+, Y^+\}$, $\{X^+, Y^-\}$, $\{X^-, Y^+\}$ and $\{X^-, Y^-\}$ are all independent. Therefore, we must have

$$\begin{aligned}\mathbb{E}(X^+ Y^+) &= \mathbb{E}(X^+) \mathbb{E}(Y^+), & \mathbb{E}(X^+ Y^-) &= \mathbb{E}(X^+) \mathbb{E}(Y^-) \\ \text{and } \mathbb{E}(X^- Y^+) &= \mathbb{E}(X^-) \mathbb{E}(Y^+), & \mathbb{E}(X^- Y^-) &= \mathbb{E}(X^-) \mathbb{E}(Y^-).\end{aligned}$$

Now, we note that $(XY)^+ = X^+ Y^+ + X^- Y^-$ and $(XY)^- = X^+ Y^- + X^- Y^+$. Now, we have, using the fact that $|X|$ and $|Y|$ are functions of X and $|Y|$ respectively, we have

$$\mathbb{E}(|XY|) = \mathbb{E}(|X||Y|) = \mathbb{E}(|X|) \mathbb{E}(|Y|) < \infty.$$

Also, we have

$$\begin{aligned}
 \mathbb{E}(XY) &= \mathbb{E}((XY)^+) - \mathbb{E}((XY)^-) = \mathbb{E}(X^+Y^+ + X^-Y^-) - \mathbb{E}(X^+Y^- + X^-Y^+) \\
 &= \mathbb{E}(X^+Y^+) + \mathbb{E}(X^-Y^-) - \mathbb{E}(X^+Y^-) - \mathbb{E}(X^-Y^+) \\
 &= \mathbb{E}(X^+)\mathbb{E}(Y^+) + \mathbb{E}(X^-)\mathbb{E}(Y^-) + \mathbb{E}(X^+)\mathbb{E}(Y^-) + \mathbb{E}(X^-)\mathbb{E}(Y^+) \\
 &= (\mathbb{E}(X^+) - \mathbb{E}(X^-))(\mathbb{E}(Y^+) - \mathbb{E}(Y^-)) = \mathbb{E}(X)\mathbb{E}(Y).
 \end{aligned}$$

This completes the proof. ■

Exercise 1.89. Show that X, Y are independent if and only if for all bounded and continuous functions f, g on $\mathbb{R}$, we have

$$\mathbb{E}(f(X)g(Y)) = \mathbb{E}(f(X))\mathbb{E}(g(Y)).$$

If we further need f, g to be differentiable, does the above condition imply independence of X and Y . ◀

Exercise 1.90. Show that, if $0 \leq X, Y \leq 1$ be random variables such that for any $m, n \geq 1$, we have

$$\mathbb{E}(X^m Y^n) = \mathbb{E}(X^m)\mathbb{E}(Y^n),$$

then X and Y are independent. ◀

1.7 Supplementary Exercises

Exercise 1.91. For any sequence of events $\{A_n : n \geq 1\}$, set

$$B_n = \cup_{m=n}^{\infty} A_m \text{ and } C_n = \cap_{m=n}^{\infty} A_m.$$

Observe that $C_n \subseteq A_n \subseteq B_n$ and the sequence of events $\{B_n : n \geq 1\}$ and $\{C_n : n \geq 1\}$ are decreasing and increasing respectively. Set

$$\begin{aligned}
 \limsup_{n \rightarrow \infty} A_n &= \cap_{n \geq 1} B_n = \cap_{n \geq 1} \cup_{m=n}^{\infty} A_m \\
 \liminf_{n \rightarrow \infty} A_n &= \cup_{n \geq 1} C_n = \cup_{n \geq 1} \cap_{m=n}^{\infty} A_m.
 \end{aligned}$$

Show that

- a) $\limsup_{n \rightarrow \infty} A_n = \{\omega \in \Omega : \omega \in A_n \text{ for infinitely many values of } n\}.$

1. $\liminf_{n \rightarrow \infty} A_n = \{\omega \in \Omega : \omega \in A_n \text{ for all but finitely many values of } n\}$.

b) Show that

$$\left(\liminf_{n \rightarrow \infty} A_n\right)^c = \limsup_{n \rightarrow \infty} (A_n)^c \text{ and } \left(\limsup_{n \rightarrow \infty} A_n\right)^c = \liminf_{n \rightarrow \infty} (A_n)^c.$$

c) Show that

$$\cap_{n \geq 1} A_n \subseteq \liminf_{n \rightarrow \infty} A_n \subseteq \limsup_{n \rightarrow \infty} A_n \subseteq \cup_{n \geq 1} A_n.$$

d) Show that

$$\limsup_{n \rightarrow \infty} A_n \setminus \liminf_{n \rightarrow \infty} A_n = \limsup_{n \rightarrow \infty} (A_n \cap A_{n+1}^c) = \limsup_{n \rightarrow \infty} (A_n^c \cap A_{n+1}).$$

e) If $\{A_n : n \geq 1\}$ is a sequence of increasing events, i.e., $A_n \subseteq A_{n+1}$ for $n \geq 1$, show that that

$$\limsup_{n \rightarrow \infty} A_n = \liminf_{n \rightarrow \infty} A_n = \cup_{n=1}^{\infty} A_n.$$

f) If $\{A_n : n \geq 1\}$ is a sequence of decreasing events, i.e., $A_{n+1} \subseteq A_n$ for $n \geq 1$, show that that

$$\limsup_{n \rightarrow \infty} A_n = \liminf_{n \rightarrow \infty} A_n = \cap_{n=1}^{\infty} A_n.$$

g) For any sequence $\{A_n\}$, show that the sets $\limsup_{n \rightarrow \infty} A_n$ and $\liminf_{n \rightarrow \infty} A_n$ are in $\mathcal{F}$ and we have

$$\mathbb{P}(\liminf_{n \rightarrow \infty} A_n) \leq \liminf_{n \rightarrow \infty} \mathbb{P}(A_n) \leq \limsup_{n \rightarrow \infty} \mathbb{P}(A_n) \leq \mathbb{P}(\limsup_{n \rightarrow \infty} A_n).$$

h) We say that the sequence $\{A_n : n \geq 1\}$ converges to a limit set A if

$$\limsup_{n \rightarrow \infty} A_n = A = \liminf_{n \rightarrow \infty} A_n.$$

In this case, we write, $\lim_{n \rightarrow \infty} A_n = A$. Note that if A_n 's are increasing, then $\lim_{n \rightarrow \infty} A_n = \cup_{n \geq 1} A_n$. If A_n 's are decreasing, then $\lim_{n \rightarrow \infty} A_n = \cap_{n \geq 1} A_n$.

If $A_n \rightarrow A$, then show that $\mathbb{P}(A_n) \rightarrow \mathbb{P}(A)$. ◀

Exercise 1.92. Let $\{A_n : n \geq 1\}$ and $\{B_n : n \geq 1\}$ to two sequences of events. Show that

a) $\limsup_{n \rightarrow \infty} A_n \cap \limsup_{n \rightarrow \infty} B_n \supseteq \limsup_{n \rightarrow \infty} (A_n \cap B_n)$;

b) $\limsup_{n \rightarrow \infty} A_n \cup \limsup_{n \rightarrow \infty} B_n = \limsup_{n \rightarrow \infty} (A_n \cup B_n);$

c) $\liminf_{n \rightarrow \infty} A_n \cap \liminf_{n \rightarrow \infty} B_n = \liminf_{n \rightarrow \infty} (A_n \cap B_n);$

d) $\liminf_{n \rightarrow \infty} A_n \cup \liminf_{n \rightarrow \infty} B_n \subseteq \liminf_{n \rightarrow \infty} (A_n \cup B_n).$

Show by examples that the containment in the above may be strict. ◀

Exercise 1.93. Suppose that an event A is independent of itself, then $\mathbb{P}(A) = 0$ or 1 . ◀

Exercise 1.94. ★ Suppose the family of events $\{A_n : n \geq 1\}$ are independent. Then $\mathbb{P}(\limsup_{n \rightarrow \infty} A_n)$ is either 0 or 1 . What can you say about $\mathbb{P}(\liminf_{n \rightarrow \infty} A_n)$? ◀

Chapter 2

Generating function

2.1 Definition

Definition 2.1. Let $\{a_k : k \geq 0\}$ be a sequence of real numbers. Define

$$A(s) = \sum_{k=0}^{\infty} a_k s^k. \quad (2.1)$$

If the formal power series converges for some interval $(-R, R)$ for some $R > 0$, then $A(s)$ will be called the generating function of the sequence $\{a_k : k \geq 0\}$.

From the theory of power series, this formal power series will have a radius of convergence $R = [\limsup_{n \rightarrow \infty} |a_n|^{1/n}]^{-1}$. Clearly, if the sequence is bounded above, i.e., $|a_n| \leq M$ for all $n \geq 0$, for some $M > 0$, we have

$$|A(s)| = \left| \sum_{k=0}^{\infty} a_k s^k \right| \leq \sum_{k=0}^{\infty} |a_k| |s|^k \leq M \sum_{k=0}^{\infty} |s|^k < \infty$$

if $|s| < 1$. Hence, $R \geq 1$.

If X is a non-negative integer valued random variable and let $p_k = \mathbb{P}(X = k)$ for $k \geq 0$. Then, clearly $|p_k| = p_k \leq 1$ and therefore $P(s) = \sum_{k=0}^{\infty} p_k s^k$ converges absolutely for $|s| < 1$. In this case, $P(s)$ is referred as the the *probability generating function* (pgf for short) of X . Note that, $P(s)$ can be written as an expectation in the following manner:

$$P(s) = \mathbb{E}(s^X). \quad (2.2)$$

Further, for $s = 1$, $P(1) = \sum_{k=0}^{\infty} p_k = 1$.

Example 2.2. Let $\{a_k : k \geq 0\}$ be real numbers.

- a) If $a_k = 1$ for all $k \geq 0$, then $A(s) = \sum_{k=0}^{\infty} a_k s^k = \sum_{k=0}^{\infty} s^k = 1/(1-s)$. Note that the radius of convergence is 1 here.
- b) If $a_k = k$ for all $k \geq 0$, then $A(s) = \sum_{k=0}^{\infty} a_k s^k = \sum_{k=1}^{\infty} k s^k = s/(1-s)^2$. Here again the radius of convergence is 1.
- c) If $a_k = 1/k!$ for all $k \geq 1$, then $A(s) = \sum_{k=0}^{\infty} a_k s^k = \sum_{k=0}^{\infty} 1/k! s^k = \exp(s)$. The radius of convergence is ∞ here.
- d) If $a_k = \alpha^k$ for all $k \geq 1$ for some $\alpha \in \mathbb{R}$, then $A(s) = \sum_{k=0}^{\infty} a_k s^k = \sum_{k=0}^{\infty} (\alpha s)^k = (1 - \alpha s)^{-1}$ for $|\alpha||s| < 1$. ■
- e) If $a_k = \binom{2k}{k} \alpha^k$ for all $k \geq 1$ and $\alpha \in \mathbb{R}$, then $A(s) = \sum_{k=0}^{\infty} a_k s^k = \sum_{k=0}^{\infty} \binom{2k}{k} (\alpha s)^k = (1 - 4\alpha s)^{-1/2}$ for $4|\alpha||s| < 1$. ■

Exercise 2.3. Find the generating functions of the sequence $\{a_k : k \geq 0\}$ where a) $a_k = (-1)^k$, b) $a_k = k^2$, c) $a_k = k^3$ and d) $a_k = (-1)^k k^2$. ◀

Example 2.4. Let X be a non-negative integer valued random variable.

- a) If X is a Bernoulli random variable with parameter p , (see Example 1.34 a)) then $P(s) = 1 - p + ps$.
- b) If X is Binomial random variable with parameter $n \geq 1$ and $0 \leq p \leq 1$ (see Example 1.34 b)), then $P(s) = \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} s^k = \sum_{k=0}^n \binom{n}{k} (ps)^k (1-p)^{n-k} = (1-p+ps)^n$.
- c) If X is the discrete uniform on $1, 2, \dots, N$, then $P(s) = \sum_{k=1}^N \frac{1}{N} s^k = \frac{s(1-s)^N}{N(1-s)}$.
- d) If X is a Poisson random variable with parameter $\lambda > 0$ (see Example 1.34 d)), then $P(s) = \sum_{k=0}^{\infty} \frac{\exp(-\lambda) \lambda^k}{k!} s^k = \exp(-\lambda(1-s))$. If X is a Poisson random variable with parameter $\lambda > 0$, truncated below 1, i.e., $\mathbb{P}(X=0) = 0$ and for $k \geq 1$,

$$\mathbb{P}(X=k) = \frac{\exp(-\lambda) \lambda^k}{(1 - \exp(-\lambda)) k!}$$

$$\text{then } P(s) = \frac{\exp(-\lambda)}{1 - \exp(-\lambda)} (\exp(\lambda s) - 1).$$

- e) If X is a geometric random variable with parameter $0 < p < 1$ (see Example 1.34 e)), then $P(s) = \sum_{k=0}^{\infty} p(1-p)^k s^k = p/(1 - (1-p)s)$.

- e) If X is a negative Binomial random variable with parameters $r \geq 1$ and $0 < p < 1$ (see Example 1.34 f)), then $P(s) = \sum_{k=0}^{\infty} \binom{r+k-1}{k} p^r (1-p)^k s^k = p^r / (1 - (1-p)s)^r$.
- f) If X taking values $2, 3, \dots$, with probabilities $\frac{k-1}{2^k}$, then $P(s) = \sum_{k=2}^{\infty} \frac{(k-1)s^k}{2^k} = \frac{s^2}{(2-s)^2}$. ■

2.2 Properties

Some simple properties of the probability generating function are easy to read off from the definition.

- a) The generating function, being a power series, is infinitely differentiable on $|s| < R$ where $R > 0$ is the radius of convergence. Note that the k -th derivative of the generating function is given by

$$A^{(k)}(s) = \sum_{n=k}^{\infty} n(n-1) \cdots (n-k+1) a_n s^{n-k} \quad (2.3)$$

for $|s| < R$. Therefore, for a sequence of non-negative numbers, we have $A^{(k)}(s) \geq 0$ for all $0 \leq s < R$.

- b) The generating function determines the sequence uniquely. Indeed, for any $k \geq 0$, we have from (2.3)

$$a_k = \frac{1}{k!} A^{(k)}(0).$$

It is therefore the case that the probability generating function (on $|s| < 1$) of a non-negative integer valued random variable determines the distribution uniquely. Note that pgf may be thought as a special case of the Laplace transform (see Exercise ??) for non-negative random variables.

- c) For the probability generating function $\lim_{s \uparrow 1} P(s) = P(1) = 1$ (see also Abel's Lemma (Lemma 2.5)).
- d) The probability generating function $P(s) : [0, 1] \rightarrow [0, 1]$ is non-decreasing, as every term is non-negative. If $p_0 < 1$, then $P(s)$ is strictly increasing.
- e) The second derivative of the probability generating function is given by $P^{(2)}(s) = \sum_{n=2}^{\infty} n(n-1) p_n s^{n-2} \geq 0$ for $s \geq 0$. Thus the probability generating function

is convex on $[0, 1]$. Further, if $p_0 + p_1 < 1$, then the second derivative is strictly positive. Hence $P(s)$ is strictly convex on $[0, 1]$.

Lemma 2.5. (Abel's Lemma) *Let $\{a_n : n \geq 0\}$ be a sequence of real numbers with $A(s)$ as the generating function with the radius of convergence of the power series being at least 1.*

(a) *If $\alpha = \sum_{n=0}^{\infty} a_n < \infty$, then*

$$\lim_{s \uparrow 1} A(s) = \sum_{n=0}^{\infty} a_n.$$

(b) *Suppose $a_n \geq 0$ for all $n \geq 0$. Let $\lim_{s \uparrow 1} A(s) = \lim_{s \uparrow 1} \sum_{n=0}^{\infty} a_n s^n = a \leq \infty$. Then*

$$\sum_{n=0}^{\infty} a_n = \lim_{N \rightarrow \infty} \sum_{n=0}^N a_n = a.$$

Proof : (a) Let us set $s_{-1} = 0$ and for $n \geq 0$, $s_n = \sum_{k=0}^n a_k$. Note that,

$$\begin{aligned} \sum_{k=0}^n a_k s^k &= \sum_{k=0}^n (s_k - s_{k-1}) s^k = \sum_{k=0}^n s_k s^k - \sum_{k=0}^n s_{k-1} s^k \\ &= s_n s^n + \sum_{k=0}^{n-1} s_k s^k - s \sum_{k=0}^{n-1} s_k s^k = s_n s^n + (1-s) \sum_{k=0}^{n-1} s_k s^k. \end{aligned}$$

Letting $n \rightarrow \infty$, and for $|s| < 1$, $s_n s^n \rightarrow 0$, we have

$$A(s) = \sum_{n=0}^{\infty} a_n s^n = \lim_{n \rightarrow \infty} \sum_{k=0}^n a_k s^k = (1-s) \sum_{n=0}^{\infty} s_n s^n$$

for $|s| < 1$. Note here that s_n converges to $\alpha = \sum_{n=0}^{\infty} a_n < \infty$. Hence s_n is bounded for all $n \geq 0$ and the power series $\sum_{n=0}^{\infty} s_n s^n$ is convergent for $|s| < 1$.

Since $s_n \rightarrow \alpha$, we can choose M so that $|s_n - \alpha| \leq M$ for any $n \geq 0$. Now given $\epsilon > 0$, we choose N so large that $|s_n - \alpha| < \epsilon/2$ for all $n > N$. Using the fact that $(1-s) \sum_{n=0}^{\infty} s^n = 1$ for any $|s| < 1$, we may write

$$\begin{aligned} |A(s) - \alpha| &= \left| (1-s) \sum_{n=0}^{\infty} s_n s^n - (1-s) \sum_{n=0}^{\infty} \alpha s^n \right| = \left| (1-s) \sum_{n=0}^{\infty} (s_n - \alpha) s^n \right| \\ &\leq (1-s) \sum_{n=0}^N |s_n - \alpha| |s|^n + (1-s) \sum_{n=N+1}^{\infty} |s_n - \alpha| |s|^n \\ &\leq (1-s) N M + \frac{1-s}{2} \sum_{n=N+1}^{\infty} \epsilon |s|^n \\ &\leq \frac{\epsilon}{2} + \frac{|s|^{N+1} (1-s) \epsilon}{2(1-|s|)} \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \end{aligned}$$

for $0 < 1 - s \leq \epsilon/(2NM)$ and using the fact that $|s| \leq 1$ and $(1 - |s|) = (1 - s)$ for $s > 0$. This proves the result.

(b) Since $a_n \geq 0$, the sequence of partial sums $\sum_{n=0}^N a_n$ is non-decreasing. So, $\sum_{n=0}^{\infty} a_n = \lim_{N \rightarrow \infty} \sum_{n=0}^N a_n$ with the provision that $\sum_{n=0}^{\infty} a_n$ may be infinite. Now, for any $s < 1$, we have that $A(s) = \sum_{n=0}^{\infty} a_n s^n \leq \sum_{n=0}^{\infty} a_n$. Thus, we must have

$$a \leq \sum_{n=0}^{\infty} a_n. \quad (2.4)$$

If $a = \infty$, the result is obvious from above, i.e., $\sum_{n=0}^{\infty} a_n \geq a = \infty$.

Now assume that $a < \infty$. Then by our hypothesis, we have $A(s) \leq a$ for any $0 < s < 1$. Hence, for any $n \geq 1$,

$$\sum_{k=0}^n a_n s^k \leq A(s) \leq a$$

as all the terms are non-negative. So, letting $s \uparrow 1$, we have, for any $n \geq 0$, $\sum_{k=0}^n a_n \leq a$. Thus, letting $n \rightarrow \infty$, we have $\sum_{k=0}^{\infty} a_k \leq a$. Combining with (2.4), we now conclude the result. ■

The following theorem shows how to compute moments using the pgf and its derivatives.

Theorem 2.6. (Factorial moments) *Suppose that X is a random variable with pgf $P(s)$. Then*

$$\begin{aligned} a) \quad \mathbb{E}(X) &= P'(1) := \lim_{s \uparrow 1} P'(s) \\ b) \quad \mathbb{E}[X(X-1) \cdots (X-k+1)] &= P^{(k)}(1) := \lim_{s \uparrow 1} P^{(k)}(s) \end{aligned}$$

for any $k \geq 1$.

Proof : For any $k \geq 1$, since P is a power series having radius at least 1, we have

$$P^{(k)}(s) = \sum_{n=0}^{\infty} n(n-1) \cdots (n-k+1) p_n s^{n-k}$$

for any $|s| < 1$. Since the terms are non-negative we can apply Abel's Lemma to obtain

$$\mathbb{E}[X(X-1) \cdots (X-k+1)] = \sum_{n=0}^{\infty} n(n-1) \cdots (n-k+1) p_n = \lim_{s \uparrow 1} P^{(k)}(s). \quad \blacksquare$$

Proposition 2.7. *Let X be a non-negative integer valued random variable. Define for each $k \geq 0$,*

$$q_k = \mathbb{P}(X > k) = p_{k+1} + p_{k+2} + \cdots .$$

Then the generating function of $\{q_k : k \geq 0\}$ is given by

$$Q(s) = \frac{1 - P(s)}{1 - s} \quad \text{for } |s| < 1. \quad (2.5)$$

Proof : We have, for each $k \geq 0$,

$$q_k = \sum_{j=k+1}^{\infty} p_j = p_{k+1} + \sum_{j=k+2}^{\infty} p_j = p_{k+1} + q_{k+1}.$$

Therefore, multiplying both sides by s^{k+1} and summing over $k = 0, 1, 2, \dots$, we have

$$\begin{aligned} \sum_{k=0}^{\infty} q_k s^{k+1} &= \sum_{k=0}^{\infty} p_{k+1} s^{k+1} + \sum_{k=0}^{\infty} q_{k+1} s^{k+1} = \sum_{k=1}^{\infty} p_k s^k + \sum_{k=1}^{\infty} q_k s^k \\ &= P(s) - p_0 + Q(s) - q_0 = P(s) + Q(s) - 1 \end{aligned}$$

since $q_0 = \sum_{k=1}^{\infty} p_k = 1 - p_0$. Thus, we have,

$$P(s) + Q(s) - 1 = \sum_{k=0}^{\infty} q_k s^{k+1} = s \sum_{k=0}^{\infty} q_k s^k = sQ(s).$$

Therefore, we have,

$$Q(s) = \frac{1 - P(s)}{1 - s}. \quad \blacksquare$$

Applying mean value theorem in equation (2.5), we note that $Q(s) = P'(\sigma)$ for some $s < \sigma < 1$. Hence, if $s \rightarrow 1$, then $\sigma \rightarrow 1$. Thus, using Abel's Lemma (part b)) again,

$$\sum_{k=0}^{\infty} \mathbb{P}(X > k) = \sum_{k=0}^{\infty} q_k = Q(1) = \lim_{s \uparrow 1} Q(s) = \lim_{s \uparrow 1} P'(s) = \mathbb{E}(X). \quad (2.6)$$

This shows that the expectation can be expressed as the sum of the tail probabilities. We may continue this again. Observe that

$$P''(s) = Q(s) - (1 - s)Q'(s).$$

Letting $s \uparrow 1$ and using Theorem 2.6, we conclude the following proposition.

Proposition 2.8. Let X be a non-negative integer valued random variable. Then, we have

$$a) \mathbb{E}(X) = Q(1) = \sum_{k=0}^{\infty} \mathbb{P}(X > k); \quad b) \mathbb{E}(X(X-1)) = 2Q'(1) = 2 \sum_{k=1}^{\infty} k \mathbb{P}(X > k).$$

In case, one side of the equality is infinite, then both sides become infinite.

Exercise 2.9. Suppose $P(s) = (q + ps)^n$. Use the pgf to verify the well-known formulas for $\mathbb{E}[X]$ and $\text{Var}(X)$ for a Binomial(n, p) random variable.

Exercise 2.10. Express the variance $\text{Var}(X)$ solely in terms of $P'(1)$ and $P''(1)$. Find the mean and variance of the random variables in Example 2.4. ◀

Exercise 2.11. The *factorial moment generating function* is defined as $F(t) = G_X(1+t)$. Find $\mathbb{E}[X]$ and $\mathbb{E}[X(X-1)]$ in terms of derivatives of $F(t)$ evaluated at $t = 0$.

Exercise 2.12. Find the third moment of a random variable in terms of its pgf ◀

2.3 Computation of Generating function

In this section, we evaluate generating functions for a number of cases. If one knows the pgf of a random variable X , it is often possible to express the pgfs of other functions of X in terms of the pgf X .

Example 2.13. Let X be non-negative random variable with generating function $P(s)$. For $Y = X + 1$, let $Q(s)$ be the generating function of Y , then

$$Q(s) = \mathbb{E}(s^Y) = \mathbb{E}(ss^X) = s\mathbb{E}(s^X) = sP(s).$$

For $Z = 2X$, let $R(s)$ be the generating function of Z , then

$$R(s) = \mathbb{E}(s^Z) = \mathbb{E}(s^{2X}) = \mathbb{E}[(s^2)^X] = P(s^2). \quad \blacksquare$$

Example 2.14. Let X be non-negative, integer valued random variable. The generating functions of the sequences of various probabilities (i) $\mathbb{P}(X \leq n)$, (ii) $\mathbb{P}(X < n)$, (iii) $\mathbb{P}(X \geq n)$, (iv) $\mathbb{P}(X > n+1)$ and (v) $\mathbb{P}(X = 2n)$ can be obtained from the probability generating function of X . Indeed, we have

(i) We have, $\mathbb{P}(X \leq k) = 1 - \mathbb{P}(X > k) = 1 - q_k$. Thus

$$F_1(s) = \sum_{k=0}^{\infty} \mathbb{P}(X \leq k) s^k = \sum_{k=0}^{\infty} (1 - q_k) s^k = \frac{1}{1-s} - \frac{1 - P(s)}{1-s} = \frac{P(s)}{1-s}.$$

(ii) Note that, $\mathbb{P}(X < 0) = 0$ and for all $k \geq 1$, $\mathbb{P}(X < k) = \mathbb{P}(X \leq k-1)$. Thus,

$$\begin{aligned} F_2(s) &= \sum_{k=0}^{\infty} \mathbb{P}(X < k) s^k = \sum_{k=1}^{\infty} \mathbb{P}(X < k) s^k = \sum_{k=1}^{\infty} \mathbb{P}(X \leq k-1) s^k \\ &= s \sum_{k=1}^{\infty} \mathbb{P}(X \leq k-1) s^{k-1} = s \sum_{k=0}^{\infty} \mathbb{P}(X \leq k) s^k = s F_1(s) = \frac{s P(s)}{1-s}. \end{aligned}$$

(iii) For all $k \geq 0$, $\mathbb{P}(X \geq k) = 1 - \mathbb{P}(X < k)$. Thus, we have

$$F_3(s) = \sum_{k=0}^{\infty} [1 - \mathbb{P}(X < k)] s^k = \frac{1}{1-s} - F_2(s) = \frac{1 - s P(s)}{1-s}.$$

(iv) We have for all $k > 0$, $\mathbb{P}(X > k+1) = q_{k+1}$. Thus,

$$\begin{aligned} F_4(s) &= \sum_{k=0}^{\infty} q_{k+1} s^k = \frac{1}{s} \sum_{k=0}^{\infty} q_{k+1} s^{k+1} = \frac{1}{s} [Q(s) - \mathbb{P}(X > 0)] \\ &= \frac{1}{s} \left[\frac{1 - P(s)}{1-s} - 1 + p_0 \right] = \frac{(s - P(s)) - p_0(1-s)}{s(1-s)}. \end{aligned}$$

(v) We have

$$\begin{aligned} F_5(s) &= \sum_{k=0}^{\infty} \mathbb{P}(X = 2k) s^k = \sum_{k=0}^{\infty} \mathbb{P}(X = 2k) (\sqrt{s})^{2k} \\ &= \frac{1}{2} \left[\sum_{k=0}^{\infty} \mathbb{P}(X = k) (\sqrt{s})^k + \sum_{k=0}^{\infty} \mathbb{P}(X = k) (-\sqrt{s})^k \right] \\ &= (P(\sqrt{s}) + P(-\sqrt{s}))/2. \end{aligned}$$

Exercise 2.15. Let X be non-negative, integer valued random variable. Find the generating function of (a) $a_n = \mathbb{P}(X = 2n) + \mathbb{P}(X = 2n+1)$ for $n \geq 0$, (b) $a_n = \mathbb{P}(X = 3n) + \mathbb{P}(X = 3n+1) + \mathbb{P}(X = 3n+2)$ for $n \geq 0$, (c) $a_0 = 0$ and $a_n = \mathbb{P}(n \leq X < 2n)$ for $n \geq 1$, in terms of the generating function of X . ◀

Example 2.16. Suppose that a coin is tossed n times. We assume that the probability of getting a HEAD at a given toss is p where $0 < p < 1$. Unless otherwise specified, we

will continue to assume it. Let p_n be the probability that the number of heads obtained is even. Define $P(s)$ be the generating function of $\{p_k : k \geq 1\}$. Then, $p_0 = 1$ and for $k \geq 1$, we have

$$p_k = \sum_{i: \text{even}} \binom{k}{2j} p^{2j} (1-p)^{k-2j}.$$

Now,

$$\begin{aligned} 1 &= \sum_{i=0}^k \binom{k}{i} p^i (1-p)^{k-i} = \sum_{i: \text{even}} \binom{k}{i} p^i (1-p)^{k-i} + \sum_{i: \text{odd}} \binom{k}{i} p^i (1-p)^{k-i} \\ &= p_k + \sum_{i: \text{odd}} \binom{k}{i} p^i (1-p)^{k-i} \end{aligned}$$

and

$$\begin{aligned} (1-2p)^k &= ((1-p) - p)^k = \sum_{i=0}^k \binom{k}{i} (-p)^i (1-p)^{k-i} \\ &= \sum_{i: \text{even}} \binom{k}{i} p^i (1-p)^{k-i} - \sum_{i: \text{odd}} \binom{k}{i} p^i (1-p)^{k-i} \\ &= p_k - \sum_{i: \text{odd}} \binom{k}{i} p^i (1-p)^{k-i}. \end{aligned}$$

Hence, $p_k = [1 + (1-2p)^k]/2$ and therefore,

$$P(s) = \sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{\infty} \frac{1}{2} s^k + \sum_{k=0}^{\infty} \frac{1}{2} (1-2p)^k s^k = \frac{1}{2(1-s)} + \frac{1}{2(1-(1-2p)s)}. \quad \blacksquare$$

2.3.1 Recurrence Relation

The most common method of evaluation of the generating function of a sequence is to find a recurrence relation between terms of the sequence. In Example 2.16, we were able to find an exact expression for p_k . However, it is not always possible to find the exact formula. In that case, we try to find a recursion relation.

Example 2.17. We revisit Example 2.16, but this time we try to find a recurrence relation. Let us denote, for $i \geq 1$,

$$\epsilon_i = \begin{cases} 1 & \text{if } i\text{-th toss is HEAD} \\ 0 & \text{if } i\text{-th toss is TAIL.} \end{cases}$$

Denote, $S_0 = 0$ and $S_k = \sum_{i=1}^k \epsilon_i$ for $k \geq 1$. Then, we have, for $k \geq 0$,

$$p_k = \mathbb{P}(S_k \text{ is even}).$$

Thus, $p_0 = 1$. Now, fix $k \geq 1$. We have,

$$\begin{aligned} \mathbb{P}(S_k \text{ is even}) &= \mathbb{P}(S_k \text{ is even}, \epsilon_1 = 1) + \mathbb{P}(S_k \text{ is even}, \epsilon_1 = 0) \\ &= \mathbb{P}(S_k - S_1 \text{ is odd}, \epsilon_1 = 1) + \mathbb{P}(S_k - S_1 \text{ is even}, \epsilon_1 = 0) \\ &= \mathbb{P}(S_k - S_1 \text{ is odd})\mathbb{P}(\epsilon_1 = 1) + \mathbb{P}(S_k - S_1 \text{ is even})\mathbb{P}(\epsilon_1 = 0) \end{aligned}$$

since $S_k - S_1 = \epsilon_2 + \epsilon_3 + \dots + \epsilon_k$ depends only on $\{\epsilon_2, \epsilon_3, \dots, \epsilon_k\}$ and, therefore, is independent of ϵ_1 .

Since all the coin tosses are **iid**, the joint distribution of $\{\epsilon_1, \epsilon_2, \dots, \epsilon_{k-1}\}$ is identical to the joint distribution of $\{\epsilon_2, \epsilon_3, \dots, \epsilon_k\}$ (see equation (1.41) also). Hence

$$\begin{aligned} \mathbb{P}(S_k - S_1 \text{ is even}) &= \mathbb{P}(\epsilon_2 + \epsilon_3 + \dots + \epsilon_k \text{ is even}) \\ &= \mathbb{P}(\epsilon_1 + \epsilon_2 + \dots + \epsilon_{k-1} \text{ is even}) = \mathbb{P}(S_{k-1} \text{ is even}) = p_{k-1}. \end{aligned}$$

Similarly, $\mathbb{P}(S_k - S_1 \text{ is odd}) = \mathbb{P}(S_{k-1} \text{ is odd}) = 1 - p_{k-1}$. Thus we have the recurrence relation

$$p_k = qp_{k-1} + p(1 - p_{k-1}) \text{ for } k \geq 1.$$

Hence, for any $|s| < 1$, we have

$$\begin{aligned} P(s) &= \sum_{k=0}^{\infty} p_k s^k = 1 + \sum_{k=1}^{\infty} p_k s^k = 1 + \sum_{k=1}^{\infty} [qp_{k-1} + p(1 - p_{k-1})] s^k \\ &= 1 + qs \sum_{k=0}^{\infty} p_k s^k + ps \sum_{k=0}^{\infty} s^k - ps \sum_{k=0}^{\infty} p_k s^k \\ &= 1 + qsP(s) + \frac{ps}{(1-s)} - psP(s). \end{aligned}$$

Therefore, solving, we have

$$P(s) = \frac{1}{2(1 - (q-p)s)} + \frac{1}{2(1-s)}.$$

■

Remark. Observe that $p_n \rightarrow \frac{1}{2}$ as $n \rightarrow \infty$.

►

Sometime, it is possible to obtain a recurrence relation which is not linear. Nevertheless, it is possible to solve that equation to obtain expression for the sequence as well as other properties. We illustrate this with a couple of examples.

Example 2.18. (Fibonacci sequence) Consider the Fibonacci sequence, i.e., let $a_0 = 0, a_1 = 1$ and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$. Let $A(s)$ be the formal power series defined by $A(s) = \sum_{k=1}^{\infty} a_k s^k$. Then,

$$\begin{aligned} A(s) &= \sum_{k=0}^{\infty} a_k s^k = a_1 s + \sum_{k=2}^{\infty} a_k s^k = s + \sum_{k=2}^{\infty} (a_{k-1} + a_{k-2}) s^k \\ &= s + \sum_{k=1}^{\infty} a_k s^{k+1} + \sum_{k=0}^{\infty} a_k s^{k+2} = s + sA(s) + s^2 A(s). \end{aligned}$$

Thus, we have, $A(s)(1 - s - s^2) = s$, so that

$$A(s) = \frac{s}{1 - s - s^2}.$$

We can now obtain the expression of the n -th term. Indeed, we have

$$\begin{aligned} A(s) &= \frac{s}{\left[1 - \frac{s(\sqrt{5}-1)}{2}\right] \left[1 + \frac{s(\sqrt{5}+1)}{2}\right]} = \frac{1}{\sqrt{5}} \left[\frac{1}{1 - \frac{s(\sqrt{5}-1)}{2}} - \frac{1}{1 + \frac{s(\sqrt{5}-1)}{2}} \right] \\ &= \frac{1}{\sqrt{5}} \left[\sum_{k=0}^{\infty} \left(\frac{s(\sqrt{5}+1)}{2} \right)^k - \sum_{k=0}^{\infty} \left(\frac{-s(\sqrt{5}-1)}{2} \right)^k \right] \\ &= \frac{1}{\sqrt{5}} \sum_{k=0}^{\infty} \left[\left(\frac{\sqrt{5}+1}{2} \right)^k - (-1)^k \left(\frac{\sqrt{5}-1}{2} \right)^k \right] s^k. \end{aligned}$$

Therefore,

$$a_k = \frac{1}{\sqrt{5}} \left[\left(\frac{\sqrt{5}+1}{2} \right)^k - (-1)^k \left(\frac{\sqrt{5}-1}{2} \right)^k \right]. \quad \blacksquare$$

Remark. In the Fibonacci sequence example, it is straightforward to show that $a_n \leq 2^n$ and hence the power series has radius of convergence at least $1/2$ (see also Example 2.2 d)). Therefore, we can expand the power series in the region $|s| < 1/2$. Once we have found the expression of the n -th term, we can now get the exact radius of convergence, namely $(\sqrt{5} - 1)/2$. ►

Exercise 2.19. Find the generating functions for the following sequences defined by the recurrence relations and try to obtain the expression of the general term:

- (a) $a_0 = 1, a_1 = 4, a_n = 4a_{n-1} - 3a_{n-2}$ for $n \geq 2$;
- (b) $a_0 = 0, a_n = 2a_{n-1} + 1$ for $n \geq 1$. (Note, this sequence identifies the number of moves needed to solve the Tower of Hanoi puzzle). ◀

Example 2.20. Suppose a coin is tossed until two consecutive Heads (HH) are observed for the first time. Let p_k be the probability that the pattern is observed for the first time after the k -th toss is done, i.e., $(k-1)$ -th and k -th both results in HEAD (H) but up to the tosses $(k-2)$ two consecutive Heads (HH) did not appear.

Define A_k as the event that the pattern is observed for the first time after k tosses. Let us define a more general event $B_k^{(i)}$ that starting from the $(i+1)$ -th coin toss, the sequence HH appears for the first time after k tosses. Then, it is clear that $A_k = B_k^{(0)}$ for any $k \geq 1$. We note that while considering $B_k^{(i)}$, we ignore the results of first i tosses. This implies that we could have had both the i -th toss and the $(i+1)$ -th toss as H but that will not imply that the event $B_1^{(i)}$ occurs.

Let ϵ_i be as defined in Example 2.17. Further, using the fact that $\{\epsilon_1, \epsilon_2, \dots, \epsilon_k\}$ is a sequence of **iid** random variables, the distribution of $\{\epsilon_{i+1}, \epsilon_{i+2}, \dots, \epsilon_k\}$ is identical to that of $\{\epsilon_1, \epsilon_2, \dots, \epsilon_{k-i}\}$. Therefore, for any $0 \leq i < k$, we also have

$$\mathbb{P}(A_{k-i}) = \mathbb{P}(B_k^{(i)}).$$

We should further note that, since $B_k^{(i)}$ depends only on the coin tosses after time $i+1$, it is independent of any event which involves only the random variables $\{\epsilon_1, \dots, \epsilon_i\}$.

We note that $p_0 = p_1 = 0$ and $p_2 = p^2$. Now, fix $k \geq 3$

$$p_k = \mathbb{P}(A_k) = \mathbb{P}(A_k, \epsilon_1 = 1) + \mathbb{P}(A_k, \epsilon_1 = 0)$$

We will consider the second event first. We note that for any $\omega \in A_k \cap \{\epsilon_1 = 0\}$, we clearly have $\omega \in B_k^{(1)} \cap \{\epsilon_1 = 0\}$. Conversely, if $\omega \in B_k^{(1)} \cap \{\epsilon_1 = 0\}$, since the first toss has resulted in TAIL, if we start looking from the first toss itself, the pattern HH appears after k tosses for the first time. Thus, we have $\omega \in A_k \cap \{\epsilon_1 = 0\}$. Also, from above discussion, we have that $B_k^{(1)}$ and $\{\epsilon_1 = 0\}$ are independent. Thus,

$$\mathbb{P}(A_k, \epsilon_1 = 0) = \mathbb{P}(B_k^{(1)}, \epsilon_1 = 0) = \mathbb{P}(B_k^{(1)})\mathbb{P}(\epsilon_1 = 0) = q\mathbb{P}(A_{k-1}) = qp_{k-1}.$$

For the first term, however we are unable to argue the first part. Indeed, when we start looking from time 2, the first coin toss itself may result in H and therefore it will mean that the event A_2 will occur. So, we need to look at what happens at time 2.

$$\mathbb{P}(A_k, \epsilon_1 = 1) = \mathbb{P}(A_k, \epsilon_1 = \epsilon_2 = 1) + \mathbb{P}(A_k, \epsilon_1 = 1, \epsilon_2 = 0).$$

Now, we observe that if both first and second tosses result in H, then the pattern HH occurs for $k = 2$, whereas we have assumed that $k \geq 3$. Therefore, $\mathbb{P}(A_k, \epsilon_1 = \epsilon_2 =$

1) = 0. For the other part, again a similar argument as above can be made to show that $A_k \cap \{\epsilon_1 = 1, \epsilon_2 = 0\} = B_k^{(2)} \cap \{\epsilon_1 = 1, \epsilon_2 = 0\}$. Hence, we have

$$\begin{aligned}\mathbb{P}(A_k, \epsilon_1 = 1, \epsilon_2 = 0) &= \mathbb{P}(B_k^{(2)}, \epsilon_1 = 1, \epsilon_2 = 0) \\ &= \mathbb{P}(B_k^{(2)})\mathbb{P}(\epsilon_1 = 1, \epsilon_2 = 0) = pq\mathbb{P}(A_{k-2}) = pqp_{k-2}.\end{aligned}$$

Therefore, for $k \geq 3$, we have

$$p_k = qp_{k-1} + pqp_{k-2}.$$

Now,

$$\begin{aligned}P(s) &= \sum_{k=0}^{\infty} p_k s^k = p^2 s^2 + \sum_{k=3}^{\infty} p_k s^k \quad \text{since } p_0 = p_1 = 0 \text{ and } p_2 = p^2 \\ &= p^2 s^2 + \sum_{k=3}^{\infty} (qp_{k-1} + pqp_{k-2}) s^k = p^2 s^2 + qs \sum_{k=3}^{\infty} p_{k-1} s^{k-1} + pqs^2 \sum_{k=3}^{\infty} p_{k-2} s^{k-2} \\ &= p^2 s^2 + qs \sum_{k=2}^{\infty} p_k s^k + pqs^2 \sum_{k=1}^{\infty} p_k s^k = p^2 s^2 + qsP(s) + pqs^2 P(s) \\ &= p^2 s^2 + P(s)(qs + pqs^2)\end{aligned}$$

where we have used the fact $p_0 = p_1 = 0$. Thus,

$$P(s) = \frac{p^2 s^2}{1 - qs - pqs^2}.$$

We note here that, using Abel's lemma,

$$\sum_{k=0}^{\infty} p_k = \lim_{s \uparrow 1} P(s) = P(1) = \frac{p^2}{(1 - q - pq)} = 1.$$

Hence, this experiment will stop in finite time with probability 1. Thus we can define T_{HH} to be the random time when the experiment stops. This is a well-defined random variable and $\mathbb{P}(T_{\text{HH}} = n) = p_n$. The expectation and variance of T can be computed using Proposition 2.7 and 2.8. For example,

$$\mathbb{E}(T_{\text{HH}}) = P'(1) = \frac{2p^2}{1 - q - pq} + \frac{p^2(q + 2pq)}{(1 - q - pq)^2} = 2 + \frac{q(1 + 2p)}{p^2} = \frac{1 + p}{p^2}. \quad \blacksquare$$

Remark. When we say that a random variable X is finite with probability 1, we mean that $\mathbb{P}(X = +\infty) = \mathbb{P}(X = -\infty) = 0$. This happens if $\mathbb{P}(|X| \geq n) \rightarrow 0$ or equivalently $\mathbb{P}(|X| \leq n) \rightarrow 1$. In course of this book, we will come across cases when such conditions are not satisfied. ►

Sometimes it is possible to get a recurrence relation involving the given sequence and another known sequence. In the following example, we illustrate this.

Example 2.21. Suppose that a coin is tossed until the pattern HT is obtained for the first time. Let p_k be the probability that the experiment stops after k tosses. Our aim is to derive a recurrence relation and finally compute the generating function of $\{p_k : k \geq 0\}$.

Define A_k as the event that the pattern is observed after k tosses for the first time. Let ϵ_i be as defined in Example 2.17. Again we define the event $B_k^{(i)}$ that starting from the $(i+1)$ -th toss, the pattern HT is observed after k -tosses for the first time. Again, we will have that $\mathbb{P}(B_k^{(i)}) = \mathbb{P}(A_{k-i}) = p_{k-i}$ and $B_k^{(i)}$ is independent of the random variables $\{\epsilon_1, \dots, \epsilon_i\}$.

We note that $p_0 = p_1 = 0$ and for $k \geq 2$

$$p_k = \mathbb{P}(A_k) = \mathbb{P}(A_k, \epsilon_1 = 1) + \mathbb{P}(A_k, \epsilon_1 = 0).$$

We can now argue, as before, that $A_k \cap \{\epsilon_1 = 0\} = B_k^{(1)} \cap \{\epsilon_1 = 0\}$, so that, we have

$$\mathbb{P}(A_k, \epsilon_1 = 0) = \mathbb{P}(B_k^{(1)}, \epsilon_1 = 0) = \mathbb{P}(B_k^{(1)})\mathbb{P}(\epsilon_1 = 0) = q\mathbb{P}(A_{k-1}) = qp_{k-1}.$$

For the first term, we note that if the second toss results in T ($\epsilon_2 = 0$), we will have the pattern HT at the second trial, which is not possible unless $k = 2$, thus $\epsilon_2 = 1$. Continuing this argument, we note that if T appears before the k^{th} coin toss, we will have the pattern HT before k tosses. Thus, we have the only possibility as that all subsequent tosses $2, 3, \dots, k-1$ must result in H and the k -th coin toss must result in T. Therefore, we have

$$\mathbb{P}(A_k, \epsilon_1 = 1) = \mathbb{P}(\epsilon_i = 1, \text{ for } 1 \leq i \leq k-1, \epsilon_k = 0) = qp^{k-2}.$$

Therefore, for $k \geq 2$, we have

$$p_k = qp^{k-1} + qp_{k-1}.$$

Hence, we have

$$\begin{aligned} P(s) &= \sum_{k=0}^{\infty} p_k s^k = \sum_{k=2}^{\infty} p_k s^k = \sum_{k=2}^{\infty} qp^{k-1} s^k + \sum_{k=2}^{\infty} qp_{k-1} s^k \\ &= pqs^2 \sum_{k=2}^{\infty} p^{k-2} s^{k-2} + qs \sum_{k=2}^{\infty} p_{k-1} s^{k-1} = pqs^2 \sum_{k=0}^{\infty} p^k s^k + qs \sum_{k=1}^{\infty} p_k s^k \\ &= \frac{pqs^2}{(1-ps)} + qsP(s) \end{aligned}$$

where we have used $p_0 = p_1 = 0$. Thus,

$$P(s) = \frac{pqs^2}{(1-ps)(1-qs)}.$$

Here also we see that

$$\sum_{k=0}^{\infty} p_k = \lim_{s \uparrow 1} P(s) = P(1) = 1.$$

Here also, we can define a random time T_{HT} when we observe the pattern HT for the first time. From the arguments above we have now shown that T_{HT} is finite with probability 1. Also we have $\mathbb{P}(T_{\text{HT}} = k) = p_k$ for $k \geq 0$.

The exact formula for p_k can be easily obtained from here by expanding the generating function as a partial fraction as follows: for $p \neq q$,

$$\begin{aligned} P(s) &= \frac{pqs^2}{p-q} \left[\frac{p}{1-ps} - \frac{q}{1-qs} \right] = \frac{pqs^2}{p-q} \left[p \sum_{k=0}^{\infty} (ps)^k - q \sum_{k=0}^{\infty} (qs)^k \right] \\ &= \frac{pq}{p-q} \left[\sum_{k=0}^{\infty} p^{k+1} s^{k+2} - \sum_{k=0}^{\infty} q^{k+1} s^{k+2} \right] = \frac{pq}{p-q} \left[\sum_{k=2}^{\infty} p^{k-1} s^k - \sum_{k=2}^{\infty} q^{k-1} s^k \right]. \end{aligned}$$

So, we have

$$p_k = \frac{pq(p^{k-1} - q^{k-1})}{(p-q)} \quad \text{for } k \geq 2.$$

If $p = q = 1/2$, we have

$$P(s) = \frac{s^2}{4(1-s/2)^2} = \frac{s^2}{4} \sum_{k=0}^{\infty} \frac{(k+1)s^k}{2^k} = \sum_{k=0}^{\infty} \frac{(k+1)s^{k+2}}{2^{k+2}} = \sum_{k=2}^{\infty} \frac{(k-1)s^k}{2^k}.$$

Hence,

$$p_k = \frac{k-1}{2^k} \quad \text{for } k \geq 2.$$

We can compute the expected time for this above experiment to stop. Indeed, we observe that

$$\mathbb{E}(T_{\text{HT}}) = \begin{cases} \frac{1-p^2q^2}{pq} & \text{if } p \neq q \\ 4 & \text{if } p = q = 1/2. \end{cases} \quad \blacksquare$$

Remark. For any pattern of length r , say, we break the coin tosses into blocks of length r . For any of the block B_i , define

$$I_i = \begin{cases} 1 & \text{if the given pattern occurs in the block} \\ 0 & \text{otherwise.} \end{cases}$$

Note I_i 's are **iid** Bernoulli random variables with $\mathbb{P}(I_i = 1) =$ the probability of the given pattern, i.e.,

$$\mathbb{P}(I_i = 1) = p^{\#\{\text{No of H in pattern}\}}(1 - p)^{\#\{\text{No of T in pattern}\}}.$$

Let G denotes the total number of blocks required to get the first block which the given pattern, i.e., $G = \inf\{i \geq 1 : I_i = 1\}$. Thus, $G - 1$ is a geometric random variable (see Example 2.4). Further, it is clear that the first occurrence of the pattern must happen before time rG , i.e., $T_{\text{pattern}} \leq rG$. It is therefore the case that $\mathbb{P}(T_{\text{pattern}} < \infty) = 1$ as $G < \infty$ with probability 1. Further, all moments of T_{pattern} must also exist since all moments of G exists. Therefore to compute moments in the problems we need not evaluate the probability generating function and then differentiate it at 1.

We illustrate the above in the case of Example 2.20. We have the relation

$$p_0 = p_1 = 0, p_2 = p^2 \text{ and } p_k = qp_{k-1} + pqp_{k-2} \text{ for } k \geq 3$$

where $p_n = \mathbb{P}(T_{\text{HH}} = n)$. Thus, multiplying by k and summing over $k \geq 3$, we obtain,

$$\begin{aligned} \mathbb{E}(T_{\text{HH}}) - 2p^2 &= \sum_{k=3}^{\infty} kp_k = \sum_{k=3}^{\infty} kqp_{k-1} + \sum_{k=3}^{\infty} kqp_{k-2} \\ &= q \left[\sum_{k=3}^{\infty} (k-1)p_{k-1} + \sum_{k=3}^{\infty} p_{k-1} \right] + pq \left[\sum_{k=3}^{\infty} (k-2)p_{k-2} + \sum_{k=3}^{\infty} 2p_{k-2} \right] \\ &= q \left[\sum_{k=2}^{\infty} kp_k + \sum_{k=2}^{\infty} p_k \right] + pq \left[\sum_{k=1}^{\infty} kp_k + 2 \sum_{k=1}^{\infty} p_k \right] \\ &= q[\mathbb{E}(T_{\text{HH}}) + 1] + pq[\mathbb{E}(T_{\text{HH}}) + 2]. \end{aligned}$$

Now, solving the linear equation, we may obtain the expression for the expectation. ►

Exercise 2.22. (a) Let p_k be the probability that the pattern Head, Tail and Head (HTH) is obtained for the first time after the k^{th} toss of a coin. Find the generating function of $\{p_k : k \geq 0\}$. Compute the expected time for this to happen.

(b) Do the above for the appearance of three consecutive Heads (HHH). Compare the expected time in both cases for the fair coin, i.e., $p = 1/2$. ◀

Exercise 2.23. Let p_k be the probability that the total obtained in k throws of a dice (not necessarily fair) is divisible by 3. Find the generating function of $\{p_k : k \geq 0\}$. Find $\lim_{k \rightarrow \infty} p_k$. ◀

2.4 Convolution

Definition 2.24. Let $\{a_k : k \geq 0\}$ and $\{b_k : k \geq 0\}$ be two sequences of real numbers. Define for each $k \geq 0$,

$$c_k = a_0 b_k + a_1 b_{k-1} + \cdots + a_{k-1} b_1 + a_k b_0 = \sum_{j=0}^k a_j b_{k-j}. \quad (2.7)$$

Then, the sequence $\{c_k : k \geq 0\}$ is called the convolution of the sequences $\{a_k : k \geq 0\}$ and $\{b_k : k \geq 0\}$ and is denoted by

$$\{c_k\} = \{a_k\} \star \{b_k\}.$$

It is easy to note that the definition does not depend on the order of $\{a_k\}$ and $\{b_k\}$, i.e.,

$$\{a_k\} \star \{b_k\} = \{b_k\} \star \{a_k\}.$$

Also, it should be further noted that if we have three sequences $\{a_k\}$, $\{b_k\}$ and $\{c_k\}$, then we may define $\{\{a_k\} \star \{b_k\}\} \star \{c_k\}$ as well as $\{a_k\} \star \{\{b_k\} \star \{c_k\}\}$. But, easy calculations show that both the convolutions give the same sequence. Therefore, we may define both the above convolutions as $\{a_k\} \star \{b_k\} \star \{c_k\}$. So, while defining self convolution, we can define

$$\{a_k\}^{2\star} = \{a_k\} \star \{a_k\} \quad \{a_k\}^{3\star} = \{a_k\}^{2\star} \star \{a_k\}$$

and in general,

$$\{a_k\}^{n\star} = \{a_k\}^{(n-1)\star} \star \{a_k\}.$$

Example 2.25. Let $\{a_k : k \geq 0\}$ and $\{b_k : k \geq 0\}$ be two sequences of real numbers.

a) If $a_k = b_k = 1$ for all $k \geq 0$, then

$$c_k = k + 1.$$

b) If $a_k = k$ and $b_k = 1$ for all $k \geq 0$, then

$$c_k = 1 + 2 + \cdots + k = k(k+1)/2.$$

c) If $a_0 = a_1 = 1$ and $a_k = 0$ for all $k \geq 2$, then $c_0 = b_0$ and for $k \geq 1$,

$$c_k = b_k + b_{k-1}. \quad \blacksquare$$

If X and Y are two non-negative integer-valued random variables with $a_k = \mathbb{P}(X = k)$ and $b_k = \mathbb{P}(Y = k)$ for all $k \geq 0$. Now, if X and Y are independent, then

$$\begin{aligned}\mathbb{P}(X + Y = k) &= \mathbb{P}(\cup_{j=0}^k \{X = j, Y = k - j\}) = \sum_{j=0}^k \mathbb{P}(X = j, Y = k - j) \\ &= \sum_{j=0}^k \mathbb{P}(X = j) \mathbb{P}(Y = k - j) = \sum_{j=0}^k a_j b_{k-j}.\end{aligned}$$

Thus, the distribution of $X + Y$ is given by the convolution of $\{a_k\}$ and $\{b_k\}$.

Theorem 2.26. Suppose that $\{a_k\}$ and $\{b_k\}$ be two sequences with generating functions $A(s)$ and $B(s)$ respectively and let $\{c_k\}$ be the convolution of $\{a_k\}$ and $\{b_k\}$ with the generating function $C(s) = \sum_{k=0}^{\infty} c_k s^k$. Assume that the radius of convergence of $A(s)$ and $B(s)$ is at least R . Then, radius of convergence of $C(s)$ at least R and is given by

$$C(s) = A(s)B(s). \quad (2.8)$$

Proof : Since the radius of convergence of $A(s)$ and $B(s)$ is at least R , for any $\epsilon > 0$, we must have

$$|a_n| \leq A(R^{-1} + \epsilon)^n \quad \text{and} \quad |b_n| \leq B(R^{-1} + \epsilon)^n$$

for all $n \geq 0$ where $A, B > 0$. Thus, we have

$$|c_n| \leq \sum_{k=0}^n |a_k b_{n-k}| \leq \sum_{k=0}^n A(R^{-1} + \epsilon)^k B(R^{-1} + \epsilon)^{n-k} \leq AB(n+1)(R^{-1} + \epsilon)^n.$$

Thus, we have

$$|c_n|^{1/n} \leq (AB)^{1/n} (n+1)^{1/n} (R^{-1} + \epsilon).$$

Since $(AB)^{1/n} \rightarrow 1$ and $(n+1)^{1/n} \rightarrow 1$, we have that

$$\limsup_{n \rightarrow \infty} |c_n|^{1/n} \leq R^{-1} + \epsilon.$$

Since $\epsilon > 0$ is arbitrary, we have $\limsup_{n \rightarrow \infty} |c_n|^{1/n} \leq R^{-1}$, hence the radius of convergence of $C(s)$ is at least R .

Now, since we need to compare between two power series $|s| < R$, it is enough to show that the coefficient of s^k in $A(s)B(s)$ and $C(s)$ are same. But the coefficient of s^k in $A(s)B(s)$ can be found out easily by multiplying $A(s)B(s)$ term by term (alternatively taking the k -th derivative at 0) and is given by the formula (2.7) for any $k \geq 0$. ■

From above discussion, we note that the generating function of $X+Y$ is given by product of the generating function of X and Y . Continuing this argument, if $X_1, X_2, \dots, X_n$ be **iid** random variables and $S_n = X_1 + X_2 + \dots + X_n$, then the generating function of S_n is given by n^{th} power of the generating function of X_1 . In other words, if $P(s)$ is the generating function of X_1 , then the generating function $G(s)$ of S_n is given by

$$G(s) = (P(s))^n.$$

Example 2.27. Let $\{X_i : i \geq 1\}$ be **iid** random variables and $S_n = \sum_{i=1}^n X_i$ for $n \geq 1$.

- a) If $X_i \sim \text{Ber}(p)$, $(0 < p < 1)$, then $S_n \sim \text{Bin}(n, p)$. Thus, generating function of a Binomial distribution is given by $(q + ps)^n$. (See all Example 2.4).
- b) Let $X \sim \text{Bin}(m, p)$ and $Y \sim \text{Bin}(n, p)$ be independent. Then $X+Y \sim \text{Bin}(m+n, p)$ and its pgf is given by $(q + ps)^{m+n}$.
- c) If $X_i \sim \text{Geo}(p)$ then S_n follows a negative Binomial distribution with parameters n and p . Thus, the generating function of a negative Binomial distribution with parameters n and p , is given by $(p/(1 - qs))^n = p^n/(1 - qs)^n$ (See all Example 2.4).
- d) If X_1 and X_2 are independent, with $X_1 \sim \text{Poi}(\lambda_1)$ and $X_2 \sim \text{Poi}(\lambda_2)$, then $X_1 + X_2 \sim \text{Poi}(\lambda_1 + \lambda_2)$. Thus, the generating function of $\text{Poi}(\lambda_1 + \lambda_2)$ is given by the product of the generating functions of $\text{Poi}(\lambda_1)$ and $\text{Poi}(\lambda_2)$. ■

Example 2.28. Suppose that $a_0 = 0$, $a_1 = 1$ and for $k \geq 2$, a_k be defined by

$$a_k = (k-1)a_1 + (k-2)a_2 + \dots + a_{k-1}.$$

We define two more sequences : $b_k = k$ for each $k \geq 0$ and $\{c_k\} = \{a_k\} \star \{b_k\}$. Now, we note that, $c_0 = a_0b_0 = 0$ and $c_1 = a_0b_1 + a_1b_0 = 0$. For $k \geq 2$,

$$c_k = \sum_{j=0}^k a_j b_{k-j} = \sum_{j=1}^{k-1} a_j b_{k-j} = a_k$$

since $a_0 = b_0 = 0$. Let $A(s), B(s)$ and $C(s)$ be the generating functions of $\{a_k\}$, $\{b_k\}$ and $\{c_k\}$ respectively. Then, we have

$$C(s) = A(s)B(s)$$

and

$$B(s) = \sum_{k=0}^{\infty} k s^k = \sum_{k=0}^{\infty} (k+1) s^k - \sum_{k=0}^{\infty} s^k = \frac{1}{(1-s)^2} - \frac{1}{(1-s)} = \frac{s}{(1-s)^2}.$$

Now, using the facts that $a_0 = 0, a_1 = 1, a_k = c_k$ for $k \geq 2$ and $c_0 = c_1 = 0$, we have

$$\begin{aligned} A(s) &= \sum_{k=0}^{\infty} a_k s^k = s + \sum_{k=2}^{\infty} a_k s^k = s + \sum_{k=2}^{\infty} c_k s^k \\ &= s + \sum_{k=0}^{\infty} c_k s^k = s + C(s) = s + A(s)B(s). \end{aligned}$$

Thus,

$$A(s) = \frac{s}{1 - B(s)} = \frac{s(1-s)^2}{(1-3s+s^2)}.$$

■

2.4.1 Random Sum

We end this section with the generating function of random sums of random variables. Let $\{X_i : i \geq 1\}$ be a family of random variables (we assume no condition X_i 's at this moment) and N be another non-negative integer valued random variable as follows:

$$S_N = \sum_{j=1}^N X_j = \begin{cases} X_1 + X_2 + \cdots + X_j & \text{if } N = j \geq 1 \\ 0 & \text{if } N = 0 \end{cases}. \quad (2.9)$$

We are using the convention that a sum with the upper index strictly smaller than the lower index is set to 0. We will be using this throughout.

Proposition 2.29. *Let X_i 's be a sequence of iid non-negative integer valued random variables with $P(s)$ as the common pgf. Let N be another non-negative integer valued random variable with the pgf given by $G(s)$. Further, we assume that the family $\{X_i : i \geq 1\}$ and N are independent. Then the pgf of S_N , defined as in (2.9), is given by $G(P(s))$.*

Proof : We have

$$\begin{aligned} \mathbb{E}(s^{S_N}) &= \sum_{n=0}^{\infty} \mathbb{E}(s^{S_N} \mathbb{I}_{\{n\}}(N)) = \sum_{n=0}^{\infty} \mathbb{E}(s^{S_n} \mathbb{I}_{\{n\}}(N)) = \sum_{n=0}^{\infty} \mathbb{E}(s^{S_n}) \mathbb{E}(\mathbb{I}_{\{n\}}(N)) \\ &= \sum_{n=0}^{\infty} (P(s))^n \mathbb{P}(N = n) = G(P(s)) \end{aligned}$$

where we used the fact that $\{X_1, X_2, \dots, X_n\}$ and N are independent. ■

Differentiating the generating function and using Proposition 2.8, we obtain the following result.

Proposition 2.30. *Let $\{X_i : i \geq 1\}$ be iid non-negative integer valued random variables. Let N be another independent non-negative integer valued random variable. Then,*

$$\mathbb{E}(S_N) = \mathbb{E}(N)\mathbb{E}(X_1). \quad (2.10)$$

Note that if either of $\mathbb{E}(N)$ or $\mathbb{E}(X_1)$ is infinite, the left hand side is infinite.

Exercise 2.31. (Variance of a Random Sum) Let $S_N = X_1 + \dots + X_N$ be a random sum as defined in (2.9), with $N, X_1, X_2, \dots$ independent and the X_i iid. Prove that

$$\mathbb{V}\text{ar}(S_N) = \mathbb{E}(N) \mathbb{V}\text{ar}(X_1) + (\mathbb{E}(X_1))^2 \mathbb{V}\text{ar}(N). \quad \blacktriangleleft$$

2.5 Fixed point of Generating Function

We prove a proposition about the probability generating function which will be used later. Suppose that $P(s)$ is the pgf of a non-negative integer valued random variable X . We are interested in finding out possible solutions of the equation $P(s) = s$ in $[0, 1]$. Note that $P(1) = \sum_{k=0}^{\infty} \mathbb{P}(X = k) = 1$ so that 1 is a solution of the equation $P(s) = s$.

Let us first eliminate one trivial case: if $\mathbb{P}(X = 1) = 1$, then $G(s) = s$, then every $s \in [0, 1]$ is a solution to the equation $P(s) = s$. Therefore we assume that the random variable is not taking value 1 with probability 1, i.e., $\mathbb{P}(X = 1) < 1$.

Theorem 2.32. *For a non-negative random variable X with $\mathbb{P}(X = 1) < 1$, the equation $G(s) = s$ admits a solution in $[0, 1)$ if and only if $\mathbb{E}(X) > 1$.*

Proof : Let $p_k = \mathbb{P}(X = k)$ for $k \geq 0$. Consider the function $P : [0, 1] \rightarrow [0, 1]$. We know that the function $s \rightarrow P(s)$ is a convex function in the interval $[0, 1]$ (see Properties e)). Therefore, it can intersect any line at most twice in $[0, 1]$. Since $P(1) = 1$, the equation $P(s) = s$ can have at most one more solution in $[0, 1)$.

If $p_0 = 0$, we have $P(0) = p_0 = 0$ and hence 0 is the other solution. In this case, $p_0 + p_1 = p_1 < 1$, hence for some $k \geq 2, p_k > 0$. Thus, we have $\mathbb{E}(X) = \sum_{k=1}^{\infty} k p_k = \sum_{k=1}^{\infty} p_k + \sum_{k=2}^{\infty} (k-1) p_k > 1$ as the first sum equals 1 and the second sum consists

of non-negative terms with at least one strictly positive term. So, we may assume that $p_0 > 0$.

If $p_0 > 0$, then $P(0) = \mathbb{P}(X = 0) > 0$ and the curve $s \rightarrow P(s)$ starts at $(0, p_0)$ and finishes at $(1, 1)$. Since $(1, 1)$ is a point of intersection of the graphs of $y = P(s)$ and $y = s$ and P is convex function, there are only two possible cases : (a) the graph of $y = P(s)$ always lies above the line $y = s$ (see Figure 2.1) for $0 \leq s < 1$ and hence only solution at $(1, 1)$ or (b) the graph of $y = P(s)$ intersect the line $y = s$ at some point $\sigma < 1$ and two solutions (σ, σ) and $(1, 1)$ with $\sigma < 1$ (see Figure 2.1).

- a) The graph of P lies entirely above the line $y = s$ for $0 \leq s < 1$. In this case, $1 - P(s) \leq 1 - s$ for all $s \in [0, 1]$. Thus, by Mean value theorem, we get $(1 - P(s))/(1 - s) = P'(\tau) \leq 1$ where $\tau = \tau(s) \in (s, 1)$ for any $s < 1$. Letting $s \uparrow 1$, we note that $\tau \rightarrow 1$. Thus, from Abel's Lemma, we have $\mathbb{E}(X) = \lim_{\tau \uparrow 1} P'(\tau) \leq 1$.

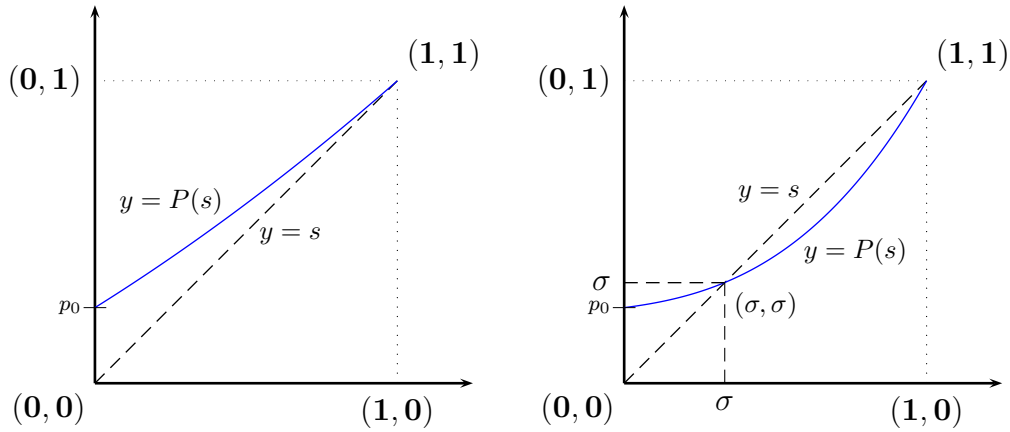


Figure 2.1: Two cases : $P(s) = s$ has no solution or exactly one in the interior of $(0, 1)$

- b) The graph of P intersects the $y = s$ at some point $\sigma < 1$ for some $0 < \sigma < 1$. First we note that P' must be strictly increasing on $[0, 1]$. Indeed, otherwise $P''(s) = 0$ for some $s \in (0, 1)$ by mean value theorem, but that would imply $p_k = 0$ for all $p_k \geq 2$ which in turn will imply $P(s) = p_0 + p_1 s$ for all $s \in [0, 1]$. But, then two straight lines $y = s$ and $y = P(s)$ will intersect at two points, viz., σ and 1, which is a contradiction.

Since we have $P(\sigma) = \sigma$ and $P(1) = 1$, by the mean value theorem, we must have

that $P'(\tau) = 1$ for some $\tau \in (\sigma, 1)$. Therefore, we have $\mathbb{E}(X) = \lim_{s \uparrow 1} P'(s) > P'(\tau)$. This completes the proof. ■

2.6 Supplementary Exercises

Exercise 2.33. Let P, P_1 and P_2 be pgfs.

- a) Show that $\lambda P_1 + (1 - \lambda)P_2$ is a pgf for any $0 \leq \lambda \leq 1$. Show that $P_1 P_2$ is a pgf.
- b) Is $\exp(P(s) - 1)$ a pgf? Fix $0 < \lambda < 1$. Is $P(\lambda s)/P(\lambda)$ necessarily a pgf? ◀

Exercise 2.34. Find the pgf for the following random variables and state their radius of convergence:

- a) X where $\mathbb{P}(X = n) = \frac{c}{n!}$ for $n \geq 0$. Find the constant c first.
- b) Y where $\mathbb{P}(Y = n) = \frac{c}{(n+1)(n+2)}$ for $n \geq 0$. Find the constant c first.
- c) Z where $\mathbb{P}(Z = n) = \frac{1}{2^{n+1}}$ for $n \geq 0$. ◀

Exercise 2.35. a) A random variable X whose pgf is given by $P(s) = \frac{1}{3-2s}$. Find the distribution of X .

- b) A random variable X whose pgf is given by $P(s) = \frac{1}{4-3s}$. Find $\mathbb{P}(X \geq n)$ for $n \geq 1$.
- c) Two dice are rolled. Let X be the sum of the outcomes. Find the pgf of X . Use this pgf to calculate the probability that the sum is divisible by 3.
- d) If the pgf of a random variable X is $P(s) = \frac{1}{2}e^s + \frac{1}{4}s^2 + \frac{1}{4}$, find $\mathbb{P}(X \geq 2)$. ◀

Exercise 2.36. a) (Combinatorial Sums) Use generating functions to prove the combinatorial identity: $\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$.

- b) Find the sequence whose generating function is $\frac{1}{(1-s)^4}$ by expressing it as a product of known generating functions and using convolution.
- c) Let a_n be the number of ways to tile a $2 \times n$ board with 1×2 dominoes for $n \geq 1$. Find a recurrence relation for a_n for $n \geq 3$. Set a_1, a_2 appropriately and, find its generating function. ◀

- Exercise 2.37** (Compound Poisson Process). a) Claims to an insurance company arrive as a Poisson process with rate λ . The size of each claim is an **iid** random variable Y with pgf $P_Y(s)$. Let S_n be the total claim amount up to time n . Find the pgf of S_n .
- b) A hen lays N eggs, where $N \sim \text{Poisson}(\lambda)$. Each egg hatches with probability p independently. Let K be the number of chicks. Find the pgf of K and identify its distribution.
- c) In the previous problem, find the expected number of chicks, $\mathbb{E}[K]$, using the random sum formula and verify by direct computation. ◀

- Exercise 2.38.** a) Suppose the pgf of a random variable satisfies the differential equation $P'(s) = \lambda P(s)$ with initial condition $P(0) = p_0 \in (0, 1)$. Solve for $P(s)$ and hence find the distribution.
- c) (Matching Problem) Suppose that n different letters are placed completely randomly (every arrangement has same probability) in n matching envelopes for $n \geq 2$. Let p_n be the probability that no letter is placed in the correct envelop. Show that $p_2 = 1/2$ and for $n \geq 3$,

$$p_n = \frac{1}{n}p_{n-2} + \left(1 - \frac{1}{n}\right)p_{n-1}.$$

Show that the generating function $A(s)$ satisfies the differential equation

$$(1 - s)A'(s) = sA(s) + s$$

with the boundary condition $A(0) = 0$. Solve for $A(s)$ and find an expression for p_n . Show that p_n converges to a limit. ◀

- Exercise 2.39.** a) Let X_n and X be random variables, taking non-negative integer values with P_n and P being the respective generating functions. Then, $\mathbb{P}(X_n = k) \rightarrow \mathbb{P}(X = k)$ for each $k \geq 1$ if and only if $P_n(s) \rightarrow P(s)$ for all $|s| < 1$.
- b) Suppose that $X_n \sim \text{Bin}(n, p_n)$ where $np_n \rightarrow \lambda > 0$ as $n \rightarrow \infty$. Show that the pgf of X_n converges to the pgf of X where $X \sim \text{Poi}(\lambda)$.
- c) Suppose that $X_n \sim \text{NB}(n, p_n)$ where $n(1 - p_n) \rightarrow \lambda > 0$ as $n \rightarrow \infty$. Show that the pgf of X_n converges to the pgf of X where $X \sim \text{Poi}(\lambda)$. ◀

Exercise 2.40 (Partial fraction expansion). Suppose $\{a_n : n \geq 0\}$ be a sequence with $A(s)$ being generating function.

a) Suppose the generating function can be written in a partial fraction expression:

$$A(s) = \frac{A_1}{1 - r_1 s} + \frac{A_2}{1 - r_2 s} + \cdots + \frac{A_m}{1 - r_m s}$$

where A_i 's are real numbers and r_i 's are distinct real numbers. Show that

$$A_i = \lim_{s \rightarrow r_i^{-1}} (1 - r_i s) A(s).$$

Conclude that

$$a_n = \sum_{i=1}^m A_i r_i^n.$$

Suppose that r_1 is such that $|r_1| = \max\{|r_i| : 1 \leq i \leq m\}$. Show that the radius of convergence of $A(s)$ is $|r_1|^{-1}$. Further, conclude that

$$a_n r_1^{-n} \rightarrow A_1$$

as $n \rightarrow \infty$.

b) Apply the above for Example 2.21 ($p \neq q$) to find the behaviour of $\mathbb{P}(T_{HT} > n)$.

c) (Repeated Linear Factors:) If $A(s)$ has a partial fraction expression which include repeated linear factors:

$$A(s) = \frac{B_1}{1 - rs} + \frac{B_2}{(1 - rs)^2} + \cdots + \frac{B_m}{(1 - rs)^m}$$

where B_i 's are real numbers and r is a real number. Show that

$$B_i = (-r)^{m-i} \lim_{s \rightarrow r^{-1}} \left(\frac{d^{m-i}}{ds^{m-i}} (1 - rs)^m A(s) \right) (0).$$

Conclude that

$$a_n = \sum_{i=1}^m B_i \binom{i+n-1}{n} r^n.$$

Show that the radius of convergence of $A(s)$ is $|r|$. Further, conclude that

$$a_n r^{-n} / \binom{m+n-1}{n} \rightarrow B_m$$

as $n \rightarrow \infty$. Use this in Example 2.21 ($p = q = 1/2$) to find the behaviour of $\mathbb{P}(T_{HT} > n)$.

- d) Decompose the following generating functions using partial fractions and find an expression for the n -th term:

$$i) \frac{1}{(1-2s)(1-3s)}; \quad ii) \frac{s}{(1-s)(1-2s)(1-3s)}; \quad iii) \frac{1+s}{(1-2s)^2}.$$

Exercise 2.41. a) Suppose $\{a_n : n \geq 0\}$ is a monotone sequence with

$$\lim_{s \uparrow 1} (1-s)A(s) = L.$$

Show that $\lim_{n \rightarrow \infty} a_n = L$.

- b) Suppose, for a sequence $\{a_n : n \geq 0\}$, we can express

$$A(s) = \frac{L}{1-s} + B(s)$$

where $B(s)$ admits a partial fraction expression as in Exercise 2.40 a) with $|r_1| < 1$ or c) with $|r| < 1$. Show that $\lim_{n \rightarrow \infty} a_n = L$.

- c) Let X be a non-negative integer valued random variable with $\mathbb{E}(X) = 1$ and $\text{Var}(X) = \sigma^2 < \infty$ having pgf $P(s)$. Show that

$$\lim_{s \uparrow 1} \frac{P(s) - s}{(1-s)^2} = \frac{\sigma^2}{2} \quad \text{and} \quad \lim_{s \uparrow 1} \left[\frac{1}{1-P(s)} - \frac{1}{1-s} \right] = \frac{\sigma^2}{2}. \quad \blacktriangleleft$$

Chapter 3

Branching Process

3.1 Introduction

Accurate models for the evolution of a population are notoriously difficult to handle, but there are simpler non-trivial models which are both tractable and mathematically interesting. The branching process is such a model. Suppose that a population evolves in generations, and let Z_n be the number of members of the n -th generation. Each member of the n -th generation gives birth to a family, possibly empty, of members of the $(n + 1)$ -th generation; the size of this family is a random variable. We shall make the following assumptions about these family sizes:

- a) the family sizes of the individuals of the branching process form a collection of independent random variables;
- b) all family sizes have the same probability generating function G .

The random evolution of the process is determined by these assumptions and the initial population size, specified here as $Z_0 = 1$. This is a general model without features specific to any single domain; it therefore provides a suitable framework for modeling diverse growth phenomena, including biological population expansion, nuclear fission processes, and infectious disease transmission.

We are interested in the following question: under what condition does the family dies with probability 1. Also, when the family survives with what probability does it survive?

In order to answer the above question, we consider the random variables $Z_0, Z_1, \dots$ of generation sizes, i.e., Z_n represents the number of individuals in the n -th generation. Generally we start with the single member, i.e., we set $Z_0 = 1$ with probability 1. Let us denote the number of offspring of the i -th member of the n -th generation by $\xi_{i;n}$. Then, we have, for $n \geq 0$,

$$Z_{n+1} = \sum_{i=1}^{Z_n} \xi_{i;n}. \quad (3.1)$$

Note we follow the usual convention the sum is 0 when the upper index is smaller than the lower index. This implies that if $Z_n = 0$, then $Z_{n+1} = 0$ for all $n \geq 1$.

By assumption (a), the random variables $\{\xi_{i;n} : i \geq 1, n \geq 1\}$ are **iid** having $\{p_k : k \geq 0\}$ as the common distribution, known as the *offspring distribution*. Now, for $n_1, n_2, \dots, n_k$, we can express the joint distribution of $\{Z_1, Z_2, \dots, Z_k\}$ as

$$\mathbb{P}(Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k) = \mathbb{P}(\xi_{1;0} = n_1, \sum_{j=1}^{n_1} \xi_{j;1} = n_2, \dots, \sum_{j=1}^{n_{k-1}} \xi_{j;k-1} = n_k).$$

When $n_k \geq 1$, we must also have $n_1, n_2, \dots, n_{k-1} \geq 1$. Then, using the **iid** nature of the random variables $\{\xi_{i;n}\}$, we can write,

$$\begin{aligned} & \mathbb{P}(Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k) \\ &= \sum_{\substack{n_{i;j} \geq 0, \sum_{i=1}^{n_j} n_{i;j} = n_{j+1} \\ 1 \leq j \leq k-1, 1 \leq i \leq n_j}} \mathbb{P}(\xi_{1;0} = n_1, \xi_{i;1} = n_{i;1}, 1 \leq i \leq n_1, \\ & \quad \xi_{i;2} = n_{i;2}, 1 \leq i \leq n_2, \dots, \xi_{i;k-1} = n_{i;k-1}, 1 \leq i \leq n_{k-1}) \\ &= \sum_{\substack{n_{i;j} \geq 0, \sum_{i=1}^{n_j} n_{i;j} = n_{j+1} \\ 1 \leq j \leq k-1, 1 \leq i \leq n_j}} p_{n_1} \prod_{i=1}^{n_1} p_{n_{i;1}} \prod_{i=1}^{n_2} p_{n_{i;2}} \cdots \prod_{i=1}^{n_{k-1}} p_{n_{i;k-1}}. \end{aligned} \quad (3.2)$$

When $n_k = 0$, we must have that, $n_1, n_2, \dots, n_{m-1} \geq 1, n_m = 0 = n_{m+1} = \dots = n_k$ for some $1 \leq m \leq k$. In such a case, we can write

$$\begin{aligned} & \mathbb{P}(Z_1 = n_1, Z_2 = n_2, \dots, Z_{m-1} = n_{m-1}, Z_m = n_m = 0 = Z_{m+1} = \dots = Z_k) \\ &= \sum_{\substack{n_{i;j} \geq 0, \sum_{i=1}^{n_j} n_{i;j} = n_{j+1} \\ 1 \leq j \leq m-1, 1 \leq i \leq n_j}} p_{n_1} \prod_{i=1}^{n_1} p_{n_{i;1}} \prod_{i=1}^{n_2} p_{n_{i;2}} \cdots \prod_{i=1}^{n_{m-2}} p_{n_{i;m-2}} \prod_{i=1}^{n_{m-1}} p_0. \end{aligned} \quad (3.3)$$

This shows that the joint distribution of $\{Z_1, Z_2, \dots, Z_k\}$ is determined by the offspring distribution $\{p_k : k \geq 0\}$. This may be used to compute probabilities for small values of k ; however as k grows this becomes rather inconvenient for any computation.

Let G be the common pgf, i.e., $G(s) = \sum_{k=0}^{\infty} p_k s^k$. Let us denote the expected size by μ , i.e.,

$$\mu = G'(1).$$

Let us denote $G_n(s) = \mathbb{E}(s^{Z_n})$ be the pgf of Z_n . From the above equation (3.1) and Proposition 2.29, we immediately have the following result.

Proposition 3.1. *For any $n \geq 0$, we have*

$$G_{n+1}(s) = G_n(G(s)). \quad (3.4)$$

The Proposition 3.1 can now be used inductively to obtain the following corollary:

Corollary 3.1.1. *For any $n \geq 0$, we have*

$$G_n(s) = \underbrace{G\left(G(\cdots (G(s)) \cdots)\right)}_{n\text{-fold composition}}$$

where the right hand side is the n -fold iterate of G .

We can re-write the above equation in a different way. Note that

$$G_{n+1}(s) = G\left(\underbrace{G(\cdots (G(s)) \cdots)}_{n\text{-fold composition}}\right) = G(G_n(s)). \quad (3.5)$$

The expression of G_n implies that the distribution of the whole branching process is uniquely determined by the pgf of the offspring distribution. In other words, if we have a sequence of pgfs $\{G_n : n \geq 1\}$ which can be written as above for a pgf G , then G_n must be the pgf of a branching process with G as the pgf of the offspring distribution.

Exercise 3.2. Consider a branching process starting with a single individual, $Z_0 = 1$.

- a) If the offspring distribution is given by $\mathbb{P}(\xi = 0) = 1/4$, $\mathbb{P}(\xi = 1) = 1/2$, $\mathbb{P}(\xi = 2) = 1/4$, compute $\mathbb{E}(Z_3)$ and $\text{Var}(Z_3)$. Also, find the probability that $\mathbb{P}(Z_3 = 0)$.
- b) If the offspring distribution is given by $\mathbb{P}(\xi = 0) = 1/6$, $\mathbb{P}(\xi = 1) = 1/2$, $\mathbb{P}(\xi = 2) = 1/6$, compute $\mathbb{E}(Z_2 Z_3)$ and $\text{Var}(Z_2 + Z_3)$. Also, find the probability $\mathbb{P}(Z_2 > 0, Z_3 = 0)$. ◀

Exercise 3.3. Suppose that $\{Z_n : n \geq 0\}$ is a branching process with $Z_0 = 1$. Fix any positive integer $r \geq 1$ and define $\{Y_n : n \geq 0\}$ as follows:

$$Y_n = Z_{nr} \text{ for } n \geq 0.$$

Check that $\{Y_n : n \geq 0\}$ is also a branching process. Find its probability generating function of the offspring distribution. ◀

Example 3.4. Suppose the offspring pgf is $G(s) = \frac{1}{2-s}$. Then, we have

$$G_2(s) = G(G(s)) = \frac{1}{2 - G(s)} = \frac{1}{2 - \frac{1}{2-s}} = \frac{2-s}{3-2s}.$$

We claim that $G_n(s) = (n - (n-1)s)/(n+1 - ns)$. For $n = 1, 2$, this is already verified. Assuming for n , we have

$$G_{n+1}(s) = G(G_n(s)) = \frac{1}{2 - G_n(s)} = \frac{1}{2 - \frac{n-(n-1)s}{n+1-ns}} = \frac{n+1-ns}{n+2-(n+1)s}$$

which verifies the claim. Observe that $G_n(0) \rightarrow 1$ as $n \rightarrow \infty$. ■

Exercise 3.5. For the branching process with offspring distribution having generating function $G(s) = q/(1 - ps)$ with $p \neq 1/2$, show that

$$G_n(s) = q \frac{p^n - q^n - (p^{n-1} - q^{n-1})ps}{p^{n+1} - q^{n+1} - (p^n - q^n)ps}.$$

Further conclude that

$$\lim_{n \rightarrow \infty} G_n(0) = \begin{cases} 1 & \text{if } p < q \\ \frac{q}{p} & \text{if } p > q. \end{cases} \quad \blacktriangleleft$$

We can now use Proposition 2.30 to compute the expected size of the generations. Indeed, we have

$$\mathbb{E}(Z_n) = \mathbb{E}(Z_{n-1})\mu = \mathbb{E}(Z_{n-2})\mu^2 = \cdots = \mathbb{E}(Z_0)\mu^n = \mu^n$$

since $Z_0 = 1$.

Exercise 3.6. Let $\{Z_n\}$ be a branching process with $Z_0 = 1$ and offspring mean μ .

- a) Assume that $G''(1) < \infty$. Show that $\mathbb{E}(Z_n Z_m) = \mu^{n-m} \mathbb{E}(Z_m^2)$ for $m \leq n$. Hence find the covariance and the correlation coefficient between Z_m and Z_n (in terms of μ) for $m \leq n$.
- b) Find $\mathbb{E}(Z_n | Z_m = k)$ for $m < n$ and $k \geq 0$. ◀

3.2 Extinction

Now we address the crucial question: when does the family becomes extinct? Before we proceed we will eliminate one trivial case. If we have the case that each individual produces exactly one child with probability 1, every generation will have exactly the same number of children. Clearly this is a trivial case and from now on, we will exclude it from our study.

It is obvious that if $Z_n = 0$, then the family is extinct on or before the n -th generation. Thus we are interested in the event $\{Z_n = 0 \text{ for some } n \geq 1\}$. It is clear that if $Z_n = 0$, then we have $Z_{n+1} = 0$. Therefore, the events $\{Z_n = 0\}$ is increasing. Thus, using Exercise 1.6, we obtain

$$\rho := \mathbb{P}(Z_n = 0 \text{ for some } n \geq 1) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} \{Z_n = 0\}\right) = \lim_{n \rightarrow \infty} \mathbb{P}(Z_n = 0). \quad (3.6)$$

Observe that $G_n(0) = \mathbb{P}(Z_n = 0)$. Thus,

$$\rho = \lim_{n \rightarrow \infty} G_n(0). \quad (3.7)$$

Example 3.7. For a branching process with offspring distribution given by a geometric distribution (see Example 3.3 and Exercise 3.5), it is clear that

$$\rho = \begin{cases} 1 & \text{if } p \leq 1/2 \\ \frac{q}{p} & \text{if } p > 1/2. \end{cases} \quad \blacksquare$$

Considering $G_{n+1}(0) = G(G_n(0))$ and taking limit as $n \rightarrow \infty$ and the fact that $G : [0, 1] \rightarrow [0, 1]$ is continuous, we have

$$\rho = \lim_{n \rightarrow \infty} G_{n+1}(0) = \lim_{n \rightarrow \infty} G(G_n(0)) = G\left(\lim_{n \rightarrow \infty} G_n(0)\right) = G(\rho).$$

Therefore, to find ρ , we need to solve the equation

$$s = G(s). \quad (3.8)$$

We need to separate the trivial case mentioned before, namely when $\mathbb{P}(\xi_{t;1} = 1) = 1$. In this case, every individual will have exactly one child and therefore every generation will have exactly the same number of children, viz., $Z_n = Z_0 = 1$ for all $n \geq 1$.

If $\mathbb{P}(\xi_{t;1} = 1) < 1$, from Theorem 2.32, we can now conclude that if $G'(1) = \mu \leq 1$, we have exactly one solution of (3.8), i.e., $\rho = 1$. In other words, if the expected family size is at most 1, the family will become extinct with probability 1.

In the case, $G'(1) = \mu > 1$, again by Theorem 2.32, we know that there is a solution of (3.8) in $[0, 1)$. So, there are two solutions of (3.8) and we need to identify which solution will provide us the value of ρ .

Suppose that τ is a solution of (3.8) in $[0, 1]$, i.e., $G(\tau) = \tau$. We claim that

$$\tau \geq G_n(0)$$

for any $n \geq 1$. Since $\tau \in [0, 1]$ and $G : [0, 1] \rightarrow [0, 1]$ is non-decreasing, we have

$$\tau = G(\tau) \geq G(0) = G_1(0).$$

Assume that the result holds for n , i.e., $\tau \geq G_n(0)$. Then we have $\tau = G(\tau) \geq G(G_n(0)) = G_{n+1}(0)$. This proves the claim.

Taking limit of both sides in the claim, we have

$$\tau \geq \lim_{n \rightarrow \infty} G_n(0) = \rho.$$

Thus, we have ρ as the smallest solution of (3.8). Hence, we obtain that if $\mu > 1$, ρ is given by the solution of (3.8) in $[0, 1)$. Hence, we have proved the following result.

Theorem 3.8. *For a branching process with offspring distribution which is not degenerate at 1, the extinction probability is 1 if and only if the expected size of the number of offspring is at most 1. If the expected number of offspring is strictly larger than 1, the extinction probability is given by the smallest solution of equation*

$$s = G(s).$$

Exercise 3.9. a) Find the extinction probability ρ for a branching process with offspring distribution:

i) $\mathbb{P}(\xi = 0) = 2/3, \mathbb{P}(\xi = 3) = 1/3$.

ii) $\mathbb{P}(\xi = 0) = 1/2, \mathbb{P}(\xi = 1) = \mathbb{P}(\xi = 3) = 1/4$.

b) Consider a branching process where each individual has a random number of offspring with a $\text{Poisson}(\lambda)$ distribution. Determine the critical value λ_c and find the extinction probability ρ for $\lambda > \lambda_c$.

c) A branching process starts with $Z_0 = N$ individuals, each evolving independently with the same offspring distribution. Show that the probability of ultimate extinction is ρ^N , where ρ is the extinction probability starting with one individual.

◀

Exercise 3.10. Suppose that the probability generating function of the offspring distribution is given by

$$G(s) = p_0 + p_1s + p_2s^2$$

with $p_2 > 0, p_k = 0$ for all $k \geq 3$ and $2p_2 + p_1 > 1$. Show that the extinction probability is given by

$$\rho = \frac{p_0}{p_2}. \quad \blacktriangleleft$$

Exercise 3.11. Suppose that the probability generating function of the offspring distribution is given by

$$G(s) = p_0 + p_1s + p_2s^2 + p_3s^3$$

with $p_3 > 0, p_k = 0$ for all $k \geq 4$ and $3p_3 + 2p_2 + p_1 > 1$. Show that the extinction probability is given by

$$\rho = \frac{\sqrt{(p_2 + p_3)^2 + 4p_0p_3} - (p_2 + p_3)}{2p_3}. \quad \blacktriangleleft$$

It is not always easy to find the branching process hidden in a problem. Let us consider the following example.

Example 3.12. Suppose a game of dice is being played. For any roll of the dice, we follow the rule: a) if we get 1, we lose the dice, b) if we roll 2, 3 or 4, we get back the dice, c) if we roll 5, we get back the dice along another extra fair dice, and d) finally if we roll 6, we get back the dice along two extra fair dice. We start with a single fair dice. The game continues if at any point of time we have at least one dice to roll. We are interested in finding the probability that we never run out dice.

We will enumerate the number of dice in hand in a particular way. Let us set $Z_0 = 1$ as we are starting with a single dice. Now, we roll that dice and see how many dice we are left with. We denote the number dice left after the roll as Z_1 . Let us enumerate these dice as $\xi_{1:1}, \xi_{2:1}, \dots, \xi_{Z_1:1}$. We will roll each of these dice at hand at this moment, i.e., $\xi_{1:1}, \xi_{2:1}, \dots, \xi_{Z_1:1}$ one by one without repeating the dice, even if we get it back or any extra dice obtained from the roll. After we have completed the rolls we will count the number of dice left at hand and we will denote them by Z_2 and again we will enumerate these dice as $\xi_{1:2}, \xi_{2:2}, \dots, \xi_{Z_2:2}$ and continue the same way. In general, we enumerate all the dice at the n -th enumeration as $\xi_{1:n}, \xi_{2:n}, \dots, \xi_{Z_n:n}$ and roll all of them one by one without repeating dice even if we get it back or any extra dice obtained from the roll.

After we have completed the rolls we will count the number of dice left at hand and we will denote them by Z_{n+1} . We need to compute

$$\mathbb{P}(Z_n > 0 \text{ for all } n \geq 1) = 1 - \mathbb{P}(Z_n = 0 \text{ for some } n \geq 1).$$

Now, with this formulation, we may now set it as branching process: for any dice, we will say that the number of dice that we get back, including itself is the number of children born to that dice. Therefore, the number of children of a dice roll is given by:

$$\xi = \begin{cases} 0 & \text{if roll results 1} \\ 1 & \text{if roll results in 2, 3 or 4} \\ 2 & \text{if roll results 5} \\ 3 & \text{if roll results 6.} \end{cases}$$

Thus, we have the offspring distribution given by: $\mathbb{P}(\xi = 0) = \mathbb{P}(\xi = 2) = \mathbb{P}(\xi = 3) = 1/6$ and $\mathbb{P}(\xi = 1) = 1/2$. Note, $\mathbb{P}(Z_n = 0 \text{ for some } n \geq 1)$ is the probability of extinction. Since $\mathbb{E}(\xi) = 1 \times \frac{3}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} > 1$, the probability of extinction is the smallest fixed point of $s = G(s)$ in $[0, 1]$, where $G(s)$ is the probability generating function of ξ , i.e.,

$$G(s) = \frac{1}{6} + \frac{3}{6}s + \frac{1}{6}s^2 + \frac{1}{6}s^3.$$

Hence we have that the equation $s = G(s)$ whose solutions are given by $1, \sqrt{2} - 1$ and $-\sqrt{2} - 1$. Therefore, the extinction probability is $\sqrt{2} - 1$ and hence the required probability is $1 - (\sqrt{2} - 1) = 2 - \sqrt{2}$.

Assume that in the above example, we need to compute the probability that you never run of dice given that first roll resulted in 5. In other words, we have $Z_1 = 2$. Hence, we need to find $\mathbb{P}(Z_n > 0 \text{ for all } n \geq 2 \mid Z_1 = 2)$. We can separate the children of both the individuals at generation 1 and start new branching process with each of them. Let us denote the family propagated by the first person as $\{Z_n^{(1)} : n \geq 2\}$ and the family propagated by the second person as $\{Z_n^{(2)} : n \geq 2\}$. Clearly, since all random variables are **iid**, each of the processes $\{Z_n^{(1)}\}$ and $\{Z_n^{(2)}\}$ are independent branching processes. Also the original branching process survives if either of them survives. Therefore, we have the required probability as

$$1 - (\sqrt{2} - 1)^2 = 1 - (2 + 1 - 2\sqrt{2}) = 2(\sqrt{2} - 1). \quad \blacksquare$$

Remark. The above argument holds in general for any generation. If we are given that the k -th generation of branching process comprises of m members. Then family will become extinct if and only if each of the m sub-family, propagated by the members of the k -th generation, will become extinct. Since the offspring distributions are **iid**, each of the sub-family is a branching process with same offspring distribution. Further they are independent. Therefore, the extinction probability of the family given that the k -th generation comprises of m members will be ρ^m where ρ is the probability of extinction of the family. We will return to this argument later. (see also Exercise 3.9 c)). $\blacktriangleright$

3.3 Extinction time

Definition 3.13. For a branching process, define the *extinction time*

$$T = \inf\{n \geq 1 : Z_n = 0\}. \quad (3.9)$$

It is clear that when the expected offspring size is at most 1, the extinction time is finite with probability 1, however when the expected offspring size is strictly larger than 1, the extinction time can be infinite with positive probability. In fact, we have

$$\mathbb{P}(T = +\infty) = 1 - \rho$$

where ρ is the probability of extinction.

We note that $T = n$ if and only if $Z_{n-1} > 0, Z_n = 0$. Thus, we have $\mathbb{P}(T = n) = \mathbb{P}(Z_{n-1} > 0, Z_n = 0) = \mathbb{P}(Z_{n-1} > 0) - \mathbb{P}(Z_n > 0)$. Therefore, $\mathbb{P}(T > n) = \mathbb{P}(Z_n > 0)$.

Exercise 3.14. Show that $\mathbb{P}(T = n) = qp^n$ when the pgf of the offspring distribution given by $G(s) = q + ps$. $\blacktriangleleft$

Exercise 3.15. For a branching process with offspring distribution given by a geometric distribution with parameter $0 < p < 1$, show that the extinction time has following distribution

$$\mathbb{P}(T = n) = \begin{cases} \frac{1}{n(n+1)} & \text{if } p = q, n \geq 1, n \in \mathbb{N} \\ \frac{p^{n-1}q^n(p-q)^2}{(p^n - q^n)(p^{n+1} - q^{n+1})} & \text{if } p \neq q, n \geq 1, n \in \mathbb{N} \\ 0 & \text{if } p \leq q, n = +\infty \\ 1 - \frac{q}{p} & \text{if } p > q, n = +\infty. \end{cases}$$

Show that $\mathbb{E}(T) < \infty$ if $p < q$ but $\mathbb{E}(T) = +\infty$ when $p = q = 1/2$. For the case $p < q$, compute $\mathbb{E}(T)$. $\blacktriangleleft$

We will separate our study into three distinct sub-cases, a) the subcritical case : $\mu = \mathbb{E}(X) < 1$, b) the critical case : $\mu = 1$ and c) the supercritical case : $\mu > 1$.

3.3.1 The subcritical case

In the subcritical case ($\mu = \mathbb{E}(X) < 1$) the family becomes extinct with probability 1. By Markov inequality, we have

$$\mathbb{P}(T > n) = \mathbb{P}(Z_n > 0) = \mathbb{P}(Z_n \geq 1) \leq \mathbb{E}(Z_n) = \mu^n. \quad (3.10)$$

So the extinction time tail decays at least exponentially fast. Therefore, T will have moments of all order. In fact $\mathbb{E}(\exp(\lambda T)) < \infty$ for any $\lambda < \mu$.

Theorem 3.16. *For a branching process with the offspring distribution having expectation $\mu < 1$, we have*

$$\mu^{-n} \mathbb{P}(T > n) \rightarrow C$$

as $n \rightarrow \infty$ where C is a non-negative constant, which depends only on the offspring distribution. If the variance of the offspring distribution σ^2 is finite, C is strictly positive.

Proof : Let ξ represent the offspring distribution with G being its pgf. We will express the function G using Taylor's series expansion near 1. For $s < 1$, we have

$$G(s) = G(1) - (1-s)G'(1) + \frac{1}{2}(1-s)^2 G''(s_*) \quad (3.11)$$

for some $s < s_* < 1$. Since $G''(s) \geq 0$ for any $s < 1$, the last term is always positive and hence, we obtain

$$G(s) \geq G(1) - (1-s)G'(1) = 1 - (1-s)\mu, \text{ i.e., } \mu(1-s) \geq 1 - G(s).$$

for any $s < 1$. Setting $s = G_n(0)$ to obtain

$$\mu(1 - G_n(0)) \geq 1 - G(G_n(0)) = 1 - G_{n+1}(0).$$

Multiplying both sides by μ^{-n-1} and observing that $\mathbb{P}(T > n) = \mathbb{P}(Z_n > 0) = 1 - P(Z_n = 0) = 1 - G_n(0)$, we get

$$\mu^{-n} \mathbb{P}(T > n) \geq \mu^{-n-1} \mathbb{P}(T > n+1).$$

Thus, the non-negative sequence $\{\mu^{-n} \mathbb{P}(T > n) : n \geq 0\}$ is decreasing, hence it must have a limit, say $C \geq 0$.

We know that $\mu^{-n}\mathbb{P}(T > n) \leq 1$, so the limit $C \leq 1$. However, there is no guarantee that the limit is positive in general. Now, we assume that the offspring has finite second moment. Now we observe that $M := G''(1) = \mathbb{E}(\xi(\xi - 1)) = \text{Var}(\xi) + \mathbb{E}(\xi) + (\mathbb{E}(\xi))^2 = \sigma^2 + \mu + \mu^2 < \infty$. Since G'' is a power series having non-negative terms, it must be the case that $G''(s) \leq G''(1) = M$ for all $s < 1$. Therefore, putting it in equation (3.11), we get that

$$G(s) \leq 1 - (1 - s)G'(1) + \frac{1}{2}(1 - s)^2M,$$

$$\text{i.e., } 1 - G(s) \geq \mu(1 - s) - \frac{1}{2}(1 - s)^2M = \mu(1 - s)\left(1 - \frac{1}{2\mu}M(1 - s)\right)$$

for any $s < 1$. Again putting $s = G_n(0)$ and multiplying by μ^{-n-1} , we have

$$\mu^{-n-1}\mathbb{P}(T > n + 1) \geq \mu^{-n}\mathbb{P}(T > n)\left[1 - \frac{1}{2\mu}M\mathbb{P}(T > n)\right] \geq \mu^{-n}\mathbb{P}(T > n)\left[1 - \frac{1}{2}M\mu^{n-1}\right]$$

where we have used the fact that $\mathbb{P}(T > n) \leq \mu^n$. Thus, we get

$$\frac{\mu^{-n-1}\mathbb{P}(T > n + 1)}{\mu^{-n}\mathbb{P}(T > n)} \geq \left[1 - \frac{1}{2}M\mu^{n-1}\right]$$

for any $n \geq 0$. Noting that $\mathbb{P}(T > 0) = 1$, we obtain that

$$\mu^{-n-1}\mathbb{P}(T > n + 1) = \prod_{k=0}^n \frac{\mu^{-k-1}\mathbb{P}(T > k + 1)}{\mu^{-k}\mathbb{P}(T > k)} \geq \prod_{k=0}^n \left[1 - \frac{1}{2}M\mu^{k-1}\right].$$

Thus, we have $\lim_{n \rightarrow \infty} \mu^{-n-1}\mathbb{P}(T > n + 1) \geq \prod_{k=0}^{\infty} \left[1 - \frac{1}{2}M\mu^{k-1}\right] > 0$ since the infinite product $\prod_{k=0}^{\infty} (1 - a_k)$ of the sequence $\{1 - a_k : k \geq 0\}$, (with $a_k < 1$) is positive if and only if $\sum_{k=0}^{\infty} a_k < \infty$. This completes the proof. ■

Exercise 3.17. For the branching processes where the expected number of offspring $\mu < 1$, show that the expected size of the total progeny, defined in (3.16), is finite. ◀

Exercise 3.18. For the branching processes where the offspring distribution given by a geometric distribution with $p < q$, compute μ and the constant C . ◀

3.3.2 The critical case

In the critical case ($\mu = \mathbb{E}(X) = 1$) the family also becomes extinct with probability 1. However, the tail behaviour is markedly different in the critical case.

Theorem 3.19. *For a branching process with the offspring distribution having expectation 1 and variance $0 < \sigma^2 < \infty$, we have*

$$n\mathbb{P}(T > n) \rightarrow \frac{2}{\sigma^2}$$

as $n \rightarrow \infty$ where Z_n represents the size of the n -th generation.

Proof : Let ξ represent the offspring distribution with G being its pgf. Note that $1 = \mathbb{E}(\xi) = G'(1)$ and $\sigma^2 = \mathbb{E}(\xi^2) - (\mathbb{E}(\xi))^2 = \mathbb{E}(\xi(\xi - 1)) + \mathbb{E}(\xi) - 1 = G''(1) + 1 - 1 = G''(1)$. By L'Hôpital's rule (see also Exercise 2.41 c))

$$\lim_{s \uparrow 1} \frac{1-s}{1-G(s)} = \frac{1}{G'(1)} = 1 \quad \text{and} \quad \lim_{s \uparrow 1} \frac{G(s) - s}{(1-s)^2} = \frac{\sigma^2}{2}.$$

Therefore, we have

$$\lim_{s \uparrow 1} \left[\frac{1}{1-G(s)} - \frac{1}{1-s} \right] = \lim_{s \uparrow 1} \left[\frac{G(s) - s}{(1-s)^2} \times \frac{1-s}{1-G(s)} \right] = \frac{\sigma^2}{2} \times 1 = \frac{\sigma^2}{2}.$$

Hence, for any sequence $s_n \rightarrow 1$, we must have

$$\lim_{n \rightarrow \infty} \left[\frac{1}{1-G(s_n)} - \frac{1}{1-s_n} \right] = \frac{\sigma^2}{2}. \quad (3.12)$$

Let G_n be the pgf of Z_n . Further, since the expected size of the offspring distribution is 1, the extinction probability is 1, i.e., $\lim_{n \rightarrow \infty} G_n(0) = 1$. Now we set $s_0 = 0$ and for $n \geq 1$, $s_n = G_n(0) = G(G_{n-1}(0)) = G(s_{n-1})$. Using this sequence $\{s_n\}$, we can rewrite the above equation in (3.12) as

$$\lim_{n \rightarrow \infty} \left[\frac{1}{1-s_{n+1}} - \frac{1}{1-s_n} \right] = \frac{\sigma^2}{2}.$$

Thus we have

$$\begin{aligned} \frac{G_n(0)}{n(1-G_n(0))} &= \frac{1}{n} \left[\frac{1}{1-G_n(0)} - 1 \right] = \frac{1}{n} \left[\frac{1}{1-s_n} - \frac{1}{1-s_0} \right] \\ &= \frac{1}{n} \sum_{k=1}^n \left[\frac{1}{1-s_k} - \frac{1}{1-s_{k-1}} \right] \rightarrow \frac{\sigma^2}{2} \end{aligned}$$

as $n \rightarrow \infty$, using the fact that Cesàro sum limit must be the same when the limit exists.

Finally, we have $\mathbb{P}(Z_n = 0) = G_n(0)$. Therefore, we have

$$n\mathbb{P}(T > n) = n\mathbb{P}(Z_n > 0) = n(1 - G_n(0)) = \frac{n(1 - G_n(0))}{G_n(0)} \times G_n(0) \rightarrow \frac{2}{\sigma^2}$$

since $G_n(0) \rightarrow 1$ as $n \rightarrow \infty$. ■

This shows that in the critical case, i.e., when the expected size of the progeny distribution is 1, the extinction time T has a tail probability $\mathbb{P}(T > n)$ behaving like $2/\sigma^2 n$ and hence the expected extinction time is $+\infty$.

3.3.3 The supercritical case

In the supercritical case ($\mu = \mathbb{E}(X) > 1$), the extinction time could be $+\infty$ with positive probability. However, one interesting question is that, how does the extinction time behave when it is conditioned that the family becomes extinct. We show that this conditional extinction time behaves like subcritical branching process and therefore has at least exponentially decaying tail.

Theorem 3.20. *Consider a branching process with the offspring distribution having expectation $\mu > 1$. The branching process conditioned to become extinct, has the same behaviour as that of a subcritical branching process.*

Proof : Suppose that the progeny distribution of the branching process is given by $\{p_k : k \geq 0\}$ with G as the pgf. Let ρ be the probability of extinction. We may consider $\rho > 0$, otherwise the extinction event has probability 0. We define the conjugate probability distribution as follows:

$$\hat{p}_k = p_k \rho^{k-1} \quad (3.13)$$

for $k \geq 0$. We have

$$\sum_{k=0}^{\infty} \hat{p}_k = \frac{1}{\rho} \sum_{k=0}^{\infty} p_k \rho^k = \frac{1}{\rho} G(\rho) = 1$$

since ρ is a fixed point of the equation $G(s) = s$. Therefore, the sequence $\{\hat{p}_k : k \geq 0\}$ is indeed a probability distribution. Next, we note that

$$\sum_{k=0}^{\infty} k \hat{p}_k = \sum_{k=0}^{\infty} k p_k \rho^{k-1} = G'(\rho).$$

Since we have assumed that $\mu = \sum_{k=1}^{\infty} k p_k > 1$, we must have that $p_k > 0$ for at least one $k \geq 2$. Therefore, the function $G'(s)$ is strictly increasing. We know that $G(\rho) = \rho$ and $G(1) = 1$. By mean value theorem, we get that $G'(\tau) = 1$ for some $\rho < \tau < 1$. Hence, we have $G'(\rho) < G'(\tau) = 1$, i.e., the expected value of the conjugate probability distribution is strictly smaller than 1.

We now show that the branching process $\{Z_n : n \geq 0\}$ conditioned to become extinct is same in distribution as of a branching process $\{\hat{Z}_n : n \geq 0\}$ having the offspring distribution given by the conjugate probability distribution $\{\hat{p}_k : k \geq 0\}$. Let A be the event that the branching process becomes extinct. Therefore, we need to show that: for

any $k \geq 1$ and $n_1, n_2, \dots, n_k \geq 0$

$$\begin{aligned}\mathbb{P}(Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k \mid A) &= \frac{1}{\rho} \mathbb{P}(\{Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k\} \cap A) \\ &= \mathbb{P}(\widehat{Z}_1 = n_1, \widehat{Z}_2 = n_2, \dots, \widehat{Z}_k = n_k)\end{aligned}$$

where we have used the fact that $\mathbb{P}(A) = \rho$.

We denote the number of offsprings of the i -th member of the n -th generation by $\xi_{i;n}$. We note that $\{Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k\}$ will depend only on the offsprings produced by the members in the generation at most $k-1$, i.e., $\{\xi_{i;n} : 0 \leq n \leq k-1\}$. In fact we can write,

$$\{Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k\} = \{\xi_{1;0} = n_1, \sum_{j=1}^{n_1} \xi_{j;1} = n_2, \dots, \sum_{j=1}^{n_{k-1}} \xi_{j;k-1} = n_k\}.$$

We now consider the event that there are $n_k \geq 1$ members in the k -th generation and the event A that the family becomes extinct. Consider the sub-family propagated by the i -th member (i.e., the sub-family with the given member, its children, grandchildren, and so on) of the k -th generation for $1 \leq i \leq n_k$. It is clear that the family will become extinct if and only if each of sub-families become extinct. Furthermore, the extinction of the sub-families depend only the random variables $\{\xi_{i;n} : n \geq k\}$. Therefore, we conclude that the event $\{Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k\}$ and A are independent. Let us denote by A_i the event that i -th sub-family becomes extinct. Since $\{\xi_{i;n} : n \geq 0, i \geq 1\}$ is a family of **iid** random variables, the events $\{A_i\}$ that the i -th family becomes extinct, are independent with $\mathbb{P}(A_i) = \rho$. Thus, we have

$$\begin{aligned}\mathbb{P}(\{Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k\} \cap A) &= \mathbb{P}(\{Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k\} \cap \cap_{i=1}^{n_k} A_i) \\ &= \mathbb{P}(Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k) \mathbb{P}(\cap_{i=1}^{n_k} A_i) \\ &= \mathbb{P}(Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k) \rho^{n_k}.\end{aligned}$$

When $n_k = 0$, the extinction has already happened. Then, we have $\{Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k = 0\} \cap A = \{Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k = 0\}$. Thus, we have

$$\mathbb{P}(\{Z_1 = n_1, Z_2 = n_2, \dots, Z_k = 0\} \cap A) = \mathbb{P}(Z_1 = n_1, Z_2 = n_2, \dots, Z_k = 0).$$

Combining both the above cases, we conclude that

$$\mathbb{P}(\{Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k\} \cap A) = \mathbb{P}(Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k) \rho^{n_k}. \quad (3.14)$$

We need to separate two cases: a) the family has not become extinct at time k , i.e., $n_1, n_2, \dots, n_k \geq 1$ and b) the family has become extinct on or before time k , i.e., for some $1 \leq m \leq k$, $n_1, n_2, \dots, n_{m-1} \geq 1, n_m = 0 = n_{m+1} = \dots = n_k$.

In case a), from above calculations in (3.14) and equation (3.2), we have

$$\begin{aligned}
& \mathbb{P}(Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k \mid A) \\
&= \mathbb{P}(Z_1 = n_1, Z_2 = n_2, \dots, Z_k = n_k) \rho^{n_k-1} \\
&= \rho^{n_k-1} \sum_{\substack{n_{i;j} \geq 0, \sum_{i=1}^{n_j} n_{i;j} = n_{j+1} \\ 1 \leq j \leq k-1, 1 \leq i \leq n_j}} p_{n_1} \prod_{i=1}^{n_1} p_{n_{i;1}} \prod_{i=1}^{n_2} p_{n_{i;2}} \cdots \prod_{i=1}^{n_{k-1}} p_{n_{i;k-1}} \\
&= \sum_{\substack{n_{i;j} \geq 0, \sum_{i=1}^{n_j} n_{i;j} = n_{j+1} \\ 1 \leq j \leq k-1, 1 \leq i \leq n_j}} (p_{n_1} \rho^{n_1-1}) \prod_{i=1}^{n_1} (p_{n_{i;1}} \rho^{n_{i;1}-1}) \\
&\quad \prod_{i=1}^{n_2} (p_{n_{i;2}} \rho^{n_{i;2}-1}) \cdots \prod_{i=1}^{n_{k-1}} (p_{n_{i;k-1}} \rho^{n_{i;k-1}-1}) \\
&= \sum_{\substack{n_{i;j} \geq 0, \sum_{i=1}^{n_j} n_{i;j} = n_{j+1} \\ 1 \leq j \leq k-1, 1 \leq i \leq n_j}} \hat{p}_{n_1} \prod_{i=1}^{n_1} \hat{p}_{n_{i;1}} \prod_{i=1}^{n_2} \hat{p}_{n_{i;2}} \cdots \prod_{i=1}^{n_{k-1}} \hat{p}_{n_{i;k-1}} \\
&= \mathbb{P}(\hat{Z}_1 = n_1, \hat{Z}_2 = n_2, \dots, \hat{Z}_k = n_k).
\end{aligned}$$

In case b), from above calculations in (3.14) and equation (3.3), we have

$$\begin{aligned}
& \mathbb{P}(Z_1 = n_1, Z_2 = n_2, \dots, Z_{m-1} = n_{m-1}, Z_m = Z_{m+1} = \dots = Z_k = 0 \mid A) \\
&= \frac{1}{\rho} \mathbb{P}(Z_1 = n_1, Z_2 = n_2, \dots, Z_{m-1} = n_{m-1}, Z_m = 0) \\
&= \frac{1}{\rho} \sum_{\substack{n_{i;j} \geq 0, \sum_{i=1}^{n_j} n_{i;j} = n_{j+1} \\ 1 \leq j \leq m-1, 1 \leq i \leq n_j}} p_{n_1} \prod_{i=1}^{n_1} p_{n_{i;1}} \cdots \prod_{i=1}^{n_{m-2}} p_{n_{i;m-2}} \prod_{i=1}^{n_{m-1}} p_0 \\
&= \sum_{\substack{n_{i;j} \geq 0, \sum_{i=1}^{n_j} n_{i;j} = n_{j+1} \\ 1 \leq j \leq m-1, 1 \leq i \leq n_j}} (p_{n_1} \rho^{n_1-1}) \prod_{i=1}^{n_1} (p_{n_{i;1}} \rho^{n_{i;1}-1}) \cdots \prod_{i=1}^{n_{m-2}} (p_{n_{i;m-2}} \rho^{n_{i;m-2}-1}) \prod_{i=1}^{n_{m-1}} (p_0 \rho^{-1}) \\
&= \mathbb{P}(\hat{Z}_1 = n_1, \hat{Z}_2 = n_2, \dots, \hat{Z}_{m-1} = n_{m-1}, \hat{Z}_m = 0) \\
&= \mathbb{P}(\hat{Z}_1 = n_1, \hat{Z}_2 = n_2, \dots, \hat{Z}_{m-1} = n_{m-1}, \hat{Z}_m = \hat{Z}_{m+1} = \dots = \hat{Z}_k = 0).
\end{aligned}$$

This completes the proof. ■

Exercise 3.21. For the branching processes with generating function of offspring distribution given by G , compute the expected extinction time conditioned on ultimate

extinction, when a) $G(s) = q/(1 - ps)$ for $p > q$, b) $G(s) = 1/4 + s/4 + s^2/2$, c) $G(s) = 1/3 + 2s^3/3$. ◀

3.4 Supplementary Exercises

Exercise 3.22. Let $G(s)$ be the pgf of a non-negative integer-valued random variable.

- a) Prove that the n -th iterate $G_n(s)$ is also a pgf for all $n \geq 0$.
- b) Compute $\mathbb{P}(Z_n = k)$ for a general n and k when $G(s) = \frac{1}{2-s}$.
- c) In the above, let Y be the total number generations which has size 1, i.e.,

$$Y = \sum_{n=0}^{\infty} \mathbb{I}_{\{1\}}(Z_n).$$

Show that $\mathbb{E}(Y) = \frac{1}{6}\pi^2$. ◀

Exercise 3.23. Suppose the generating function of the offspring distribution has the following form

$$G(s) = \frac{as + b}{1 - cs} \quad (3.15)$$

for some constants $a, b, c \geq 0$, with $a + b + c = 1$.

- a) Suppose that

$$A = \begin{bmatrix} a & b \\ -c & 1 \end{bmatrix} \quad \text{and} \quad A^n = \begin{bmatrix} a(n) & b(n) \\ c(n) & d(n) \end{bmatrix}$$

where A^n is the n -th power of the matrix A . Show that

$$G_n(s) = \frac{a(n)s + b(n)}{c(n)s + d(n)}.$$

- b) Show that, by choosing suitable parameters a, b, c , we can get the generating functions for geometric and Bernoulli cases. ◀

Exercise 3.24. Suppose that the probability generating function of the offspring distribution has the form

$$G(s) = 1 - \alpha(1 - s)^\beta$$

where $0 < \alpha, \beta < 1$. Show that the probability generating function of Z_n is given by

$$G_n(s) = 1 - \alpha^{1+\beta+\beta^2+\dots+\beta^{n-1}}(1-s)^{\beta^n}$$

for any $n \geq 0$. Show that

$$\lim_{n \rightarrow \infty} G_n(0) = 1 - \alpha^{\frac{1}{1-\beta}}. \quad \blacktriangleleft$$

Exercise 3.25. Suppose that the probability generating function of the offspring distribution has the form

$$G(s) = h^{-1}[P(h(s))]$$

where P is a probability generating function and h is a function such that G is a probability generating function. Show that the probability generating function of Z_n is given by

$$G_n(s) = h^{-1}[P_n(h(s))]$$

where P_n is n -fold composition of the generating function P .

For the special case $P(s) = s^m$ for some $m \geq 1$ and

$$h(s) = \frac{s}{\alpha - (\alpha - 1)s}$$

where $\alpha > 1$, show that

$$G_n(s) = \frac{s}{[\alpha^n - (\alpha^n - 1)s^m]^{1/m}}. \quad \blacktriangleleft$$

Exercise 3.26. For a branching process with the probability generating function of the offspring distribution given by G , define S_n as the total progeny until the n -th generation, i.e.,

$$S_n = 1 + Z_1 + Z_2 + \dots + Z_n.$$

Let H_n be the probability generating function of S_n . Show that, for all $n \geq 0$,

$$H_{n+1}(s) = sG(H_n(s)). \quad \blacktriangleleft$$

Exercise 3.27. (Branching with immigration). Suppose for a branching process, in each generation the population is augmented by a random number of immigrants who are indistinguishable from the other members of the population. Further assume the numbers of immigrants in different generations are independent of each other and of the past history of the branching process, with each such number having probability

generating function $H(s)$. Show that the probability generating function G_n of the size of the n -th generation (including the immigrants) is given by

$$G_{n+1}(s) = G(G_n(s))H(s).$$

Find $\mathbb{E}(Z_n)$ and $\text{Var}(Z_n)$ if the number of immigrants in each generation has mean ν and variance τ^2 . ◀

Exercise 3.28. (★) Assume that $p_1 < 1$. Show that for any $1 \leq k < \infty$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(Z_n = k) = 0.$$

For any $0 < s < 1$, show that

$$\lim_{n \rightarrow \infty} \mathbb{E}(Z_n s^{Z_n}) = 0.$$

Conclude that for any $0 < s < 1$, $\lim_{n \rightarrow \infty} G'_n(s) = 0$. ◀

Exercise 3.29. We set the *total progeny* of the branching process as

$$S = \sum_{n=0}^T Z_n \tag{3.16}$$

where T is the extinction time. Note, when the event $\{T = +\infty\}$ has positive probability, i.e., no extinction happens with positive probability, then $\{S = +\infty\}$ event has positive probability.

- a) Show that the generating function of S is given by the solution of the following equation

$$H(s) = sG(H(s))$$

for $0 < s < 1$, where the probability generating function of the offspring distribution is given by $G(s)$. Show that the above equation admits a unique solution when $0 \leq s < 1$.

- b) Show that the total progeny of the branching process has the generating function given by

$$H(s) = \frac{1}{2p} (1 - \sqrt{1 - 4pqs})$$

when the probability generating function of the offspring distribution is given by $G(s) = q + ps$. ◀

Exercise 3.30. a) Let $\phi_n(\theta) = \mathbb{E}(e^{-\theta Z_n})$ be the Laplace transform of Z_n . Derive a recursive functional equation for $\phi_n(\theta)$.

b) When G has the form given as in equation (3.15) with $G'(1) = 1$, show that

$$\lim_{n \rightarrow \infty} \sup_{s \in [0,1)} \left[\frac{1}{1 - G_n(s)} - \frac{1}{1 - s} \right] = \frac{\sigma^2}{2}.$$

c) In the above exercise, show that the distribution of Z_n/n , conditioned on $Z_n > 0$, converges to an exponential distribution. Find its rate parameter.

[**Seneta-Heyde Norming** : the above exercise holds for branching processes with $\mu = 1$ and $\sigma^2 < \infty$; however, this requires finer analysis.] ◀

Chapter 4

Random walk

4.1 Definition and Basic Example

A **random walk** (RW) is a mathematical object that describes a path consisting of a succession of random steps. A fundamental model is the walk on the integers $\mathbb{Z}$.

Definition 4.1 (Simple Symmetric Random Walk on $\mathbb{Z}$). Let $\epsilon_1, \epsilon_2, \dots$ be a sequence of **iid** random variables with $\mathbb{P}(\epsilon_k = +1) = \mathbb{P}(\epsilon_k = -1) = 1/2$. The sequence $\{S_n : n \geq 0\}$ defined by

$$S_0 = 0, \quad S_n = \sum_{k=1}^n \epsilon_k \quad \text{for } n \geq 1$$

is called the **simple symmetric random walk** (SSRW) on $\mathbb{Z}$.

This can be visualized as a particle starting at the origin, moving one step to the right for each Head of a fair coin toss, or one step to the left for each Tail.

Level Crossing and Recurrence

A natural question arises: if we draw a line some distance from the origin, how many times will the random walk cross it if allowed to continue indefinitely? A profound result is that the simple random walk on $\mathbb{Z}$ will almost surely cross every point an infinite number of times. This property is known as **recurrence**. We will study this in detail in Markov chain.

Now, consider a drunkard walking randomly in an infinite city arranged in a square grid (a two-dimensional lattice). At every intersection, he chooses one of the four possible

routes with equal probability. Formally, this is a random walk on $\mathbb{Z}^2$. Will the drunkard ever return to his starting point? It turns out that he will, and infinitely often. This is the two-dimensional analogue of the level-crossing problem.

However, this behavior changes with dimension. In three dimensions and above, a random walk may never return to its starting point. This is famously summarized by the quote, often attributed to George Pólya: *"A drunk man will eventually find his way home, but a drunk bird may be lost forever."*

To formalize this, we define the SSRW in d -dimensions ($d \geq 1$). Let $\{\epsilon_k : k \geq 1\}$ be i.i.d. random vectors such that $\mathbb{P}(\epsilon_k = \mathbf{e}_j) = \mathbb{P}(\epsilon_k = -\mathbf{e}_j) = 1/(2d)$ for $j = 1, 2, \dots, d$, where $\mathbf{e}_j$ is the unit vector in the j -th direction. The d -dimensional SSRW is defined as $\mathbf{S}_0 = \mathbf{0}$ and $\mathbf{S}_n := \sum_{k=1}^n \epsilon_k$ for $n \geq 1$.

The fundamental result concerning this behavior is **Pólya's Theorem**. Pólya 1921.

Theorem 4.2. *The simple symmetric random walk is **recurrent** in dimensions $d = 1$ and $d = 2$, and **transient** in dimensions $d \geq 3$.*

4.1.1 The Gambler's Ruin Problem

There are many other interesting formulations of random walks. A classical problem is the **Gambler's Ruin** Feller 1968.

Consider two gamblers, A and B, with initial capitals of Rs. a and Rs. b , respectively. They repeatedly play a fair coin-toss game where the loser pays Rs. 1 to the winner each round. The game ends when one player is bankrupt.

This scenario can be modeled as a random walk with an absorbing boundary. Let $S_0 = a$ represent A's initial capital. The change in A's capital at the k -th round is ϵ_k , with $\epsilon_k = +1$ if A wins and $\epsilon_k = -1$ if A loses. Then, A's capital after n rounds is $S_n = a + \sum_{k=1}^n \epsilon_k$. The game continues until the walk S_n hits either 0 (A is ruined) or $a+b$ (B is ruined). Analyzing the probability of each player's ruin is a classic application of random walk theory Ross 2019.

4.2 Simple symmetric random walk in one dimension

Section Contents

where $|A|$ is the number of elements in the set A . When $m = a = 0$, we will drop the super-script.

Note that the number of paths starting at (m, a) and ending at (n, b) remains the same if we translate the origin to (m, a) and look at the paths from $(0, 0)$ to $(n-m, b-a)$. Therefore, we have

$$N_{(n,b)}^{(m,a)} = N_{(n-m,b-a)}.$$

Clearly, this is 0 if $n - m + b - a$ is odd.

The expression for $N_{(n,b)}$ is easy to obtain. Each path is an arrangement of $+1$'s and -1 's. If there are p many $+1$'s and m many -1 's, then we have $p + m = n$ and $p - m = b$, i.e., $p = (n + b)/2$ and $m = (n - b)/2$. Since every such arrangement of p many $+1$'s and m many -1 's give rise to a path and vice versa, we have

$$N_{(n,b)} = \begin{cases} \binom{n}{(n+b)/2} & \text{if } n + b \text{ is even} \\ 0 & \text{otherwise.} \end{cases} \quad (4.1)$$

Also note that there are total 2^k many paths of length k . Each given path of length k is equally probable and hence has a probability of 2^{-k} . Thus, for any event A which depends on paths of length n , the probability is given by

$$\mathbb{P}(A) = \frac{N(A)}{2^k}$$

where $N(A)$ is the total number of paths of length k which satisfy the conditions of the event A .

4.2.2 Return Probabilities

Define the probability of return to origin (at time n), by

$$u_n := \mathbb{P}(S_n = 0). \quad (4.2)$$

From the above discussion, we have

$$u_n = \frac{N_{(n,0)}}{2^n} = \begin{cases} \binom{2k}{k} 2^{-2k} & \text{if } n = 2k \\ 0 & \text{otherwise.} \end{cases} \quad (4.3)$$

Notation: we set $u_0 = 1$. This is for convenience, so that we don't have to separate cases later.

Exercise 4.3. Show that for $n \geq 1$:

$$u_{2n} = (-1)^n \binom{-\frac{1}{2}}{n}$$

and hence show that the generating function $U(s) = \sum_{n=0}^{\infty} u_n s^n$ is given by

$$U(s) = \frac{1}{\sqrt{1-s^2}}.$$

(Compare with Example 2.2). ◀

Stirling's approximation:

$$n! \sim \sqrt{2\pi n} n^{n+\frac{1}{2}} e^{-n} \quad (4.4)$$

where we use the notation that for two sequences $\{a_n : n \geq 1\}$ and $\{b_n : n \geq 1\}$, $a_n \sim b_n$ if $a_n/b_n \rightarrow 1$ as $n \rightarrow \infty$.

Using Stirling's approximation, we note that

$$\begin{aligned} \sqrt{\pi n} u_{2n} &= \sqrt{\pi n} \times \frac{(2n)!}{n!n!} \frac{1}{2^{2n}} \\ &= \frac{(2n)!}{\sqrt{2\pi}(2n)^{2n+\frac{1}{2}}e^{-2n}} \times \left(\frac{\sqrt{2\pi}n^{n+\frac{1}{2}}e^{-n}}{n!} \right)^2 \times \frac{\sqrt{2\pi}(2n)^{2n+\frac{1}{2}}e^{-2n}\sqrt{\pi n}}{2\pi n^{2n+1}e^{-2n}} \times \frac{1}{2^{2n}} \\ &\rightarrow 1 \end{aligned}$$

as $n \rightarrow \infty$. Hence, we have

$$u_{2n} \sim \frac{1}{\sqrt{\pi n}}. \quad (4.5)$$

This immediately implies that

$$u_n \rightarrow 0 \text{ as } n \rightarrow \infty. \quad (4.6)$$

Further, by comparison test, we also have

$$\sum_{n=0}^{\infty} u_n = \infty. \quad (4.7)$$

4.2.3 First Return

We would like to compute the probability that the random walk will return to origin i.e., $\mathbb{P}(\cup_{n=1}^{\infty} \{S_n = 0\})$. Set $A_n = \{S_n = 0\}$ for $n \geq 1$. Clearly, the events A_n are not disjoint. In order to overcome this, we need to disjointify the events. Set

$$B_1 = A_1 \text{ and } B_k = A_k \cap (\cup_{i=1}^{k-1} A_i)^c \text{ for } k \geq 1.$$

Then, we have

$$\cup_{n=1}^{\infty} A_n = \cup_{n=1}^{\infty} B_n$$

and B_n 's are disjoint. Therefore, $\mathbb{P}(\cup_{n=1}^{\infty} A_n) = \mathbb{P}(\cup_{n=1}^{\infty} B_n) = \sum_{n=1}^{\infty} \mathbb{P}(B_n)$.

Now, we set $f_0 = 0$ and for $n \geq 1$:

$$f_n := \mathbb{P}(B_n) = \mathbb{P}(S_n = 0, S_j \neq 0 \text{ for } j = 1, 2, \dots, n-1). \quad (4.8)$$

Clearly, f_n represents the probability of the event that the first return to origin happens at time n . Again, if n is odd, $f_n = 0$. Hence, enough to show that $\sum_{n=0}^{\infty} f_{2n} = 1$. To this end, we will show that

Proposition 4.4. For $n \geq 1$:

$$f_{2n} = u_{2n-2} - u_{2n}. \quad (4.9)$$

Exercise 4.5. Show that for $n \geq 1$:

$$f_{2n} = \frac{1}{2n-1} u_{2n} = (-1)^{n-1} \binom{\frac{1}{2}}{n}$$

and hence show that the generating function $F(s) = \sum_{n=0}^{\infty} f_n s^n$ is given by

$$F(s) = 1 - \sqrt{1 - s^2}. \quad \blacktriangleleft$$

Now, using Proposition 4.4 and the equation (4.6), we have

$$\sum_{n=0}^{\infty} f_{2n} = \sum_{n=1}^{\infty} f_{2n} = \lim_{N \rightarrow \infty} \sum_{n=1}^N f_{2n} = \lim_{N \rightarrow \infty} (u_0 - u_{2N}) = \lim_{N \rightarrow \infty} (1 - u_{2N}) = 1.$$

Now, we can define:

$$\tau := \inf\{n \geq 1 : S_n = 0\}. \quad (4.10)$$

The random variable τ represents the time of first return to origin and $\mathbb{P}(\tau = n) = \mathbb{P}(S_n = 0, S_j \neq 0, j = 1, 2, \dots, n-1) = f_n$ for $n \geq 1$. Since $\sum_{n=0}^{\infty} f_{2n} = 1$, τ is proper random variable, i.e., $\mathbb{P}(\tau < \infty) = 1$. Using Proposition 4.4 and the expression for u_n in equation (4.3) and equation (4.5), we have

$$f_{2n} = u_{2n-2} - u_{2n} = \binom{2n-2}{n-1} 2^{-(2n-2)} - \binom{2n}{n} 2^{-2n} \quad (4.11)$$

$$= \binom{2n}{n} 2^{-2n} \left[\frac{4 \times n \times n}{(2n-1)(2n)} - 1 \right] = \frac{1}{2n-1} \binom{2n}{n} 2^{-2n} \sim \frac{1}{\sqrt{4\pi n^3}}. \quad (4.12)$$

Hence, we conclude that

$$\mathbb{E}(\tau) = \sum_{n=0}^{\infty} n f_n = \sum_{n=1}^{\infty} 2n f_{2n} = \infty. \quad (4.13)$$

Thus, the expected return time to origin is infinite, even though the random variable is finite almost surely. Note that we may also conclude the above using the generating function $F(s)$.

Exercise 4.6. Show that for any $\epsilon > 0$, $\mathbb{E}((\tau)^{1/2-\epsilon}) < \infty$ but $\mathbb{E}((\tau)^{1/2+\epsilon}) = \infty$. ◀

4.2.4 Renewal Equations

There is another relation between u_n and f_n which holds more generally. Any visit to the origin at time point $2n$ may be the first return, or else the first return occurs at time point $2k < 2n$, for $k = 1, 2, \dots, n-1$, and is followed by a renewed return $2n - 2k$ time units later. It is also clear that the events that the first return happens at time point $2k$ are disjoint for distinct values of k . Therefore, we have:

$$\begin{aligned} \mathbb{P}(S_{2n} = 0) &= \mathbb{P}(S_1 \neq 0, S_2 \neq 0, \dots, S_{2n-1} \neq 0, S_{2n} = 0) \\ &\quad + \sum_{k=1}^{n-1} \mathbb{P}(S_1 \neq 0, S_2 \neq 0, \dots, S_{2k-1} \neq 0, S_{2k} = 0, S_{2n} = 0) \\ &= f_{2n} + \sum_{k=1}^{n-1} \mathbb{P}(S_1 \neq 0, S_2 \neq 0, \dots, S_{2k-1} \neq 0, S_{2k} = 0, S_{2n} - S_{2k} = 0) \end{aligned}$$

For any $k = 1, 2, \dots, n-1$, we note that $\{S_1 \neq 0, S_2 \neq 0, \dots, S_{2k-1} \neq 0, S_{2k} = 0\} = \{\epsilon_1 \neq 0, \epsilon_1 + \epsilon_2 \neq 0, \dots, \epsilon_1 + \epsilon_2 + \dots + \epsilon_{2k-1} \neq 0, \epsilon_1 + \epsilon_2 + \dots + \epsilon_{2k-1} + \epsilon_{2k} = 0\}$ depends only on the random variables $\{\epsilon_1, \epsilon_2, \dots, \epsilon_{2k}\}$ while the event $\{S_{2n} - S_{2k} = 0\} = \{\epsilon_{2k+1} + \dots + \epsilon_{2n} = 0\}$ depends only on the random variables $\{\epsilon_{2k+1}, \epsilon_{2k+2}, \dots, \epsilon_{2n}\}$. Thus, the events are independent and hence we have:

$$\begin{aligned} &\mathbb{P}(S_1 \neq 0, S_2 \neq 0, \dots, S_{2k-1} \neq 0, S_{2k} = 0, S_{2n} - S_{2k} = 0) \\ &= \mathbb{P}(S_1 \neq 0, S_2 \neq 0, \dots, S_{2k-1} \neq 0, S_{2k} = 0) \mathbb{P}(S_{2n} - S_{2k} = 0) \\ &= \mathbb{P}(S_1 \neq 0, S_2 \neq 0, \dots, S_{2k-1} \neq 0, S_{2k} = 0) \mathbb{P}(\epsilon_{2k+1} + \dots + \epsilon_{2n} = 0) \\ &= \mathbb{P}(S_1 \neq 0, S_2 \neq 0, \dots, S_{2k-1} \neq 0, S_{2k} = 0) \mathbb{P}(\epsilon_1 + \dots + \epsilon_{2n-2k} = 0) \\ &= \mathbb{P}(S_1 \neq 0, S_2 \neq 0, \dots, S_{2k-1} \neq 0, S_{2k} = 0) \mathbb{P}(S_{2n-2k} = 0) \\ &= f_{2k} u_{2n-2k} \end{aligned}$$

where we have used the fact that ϵ_i 's are **iid** and hence the distribution of $\sum_{j=1}^{2n-2k} \epsilon_j$ is same as that of $\sum_{j=2k+1}^{2n} \epsilon_j$. Finally, using the fact that $f_0 = 0$ and $u_0 = 1$, we have the following proposition:

Proposition 4.7. *For every $n \geq 1$:*

$$u_{2n} = \sum_{k=0}^n f_{2k} u_{2n-2k}. \quad (4.14)$$

It should be noted that this proposition is more general and does not depend on the actual distribution of X_i 's as long as they are **iid**.

Exercise 4.8. Using Proposition 4.7, show that

$$F(s) = 1 - \frac{1}{U(s)}$$

and conclude that $\sum_{n=0}^{\infty} f_n = F(1) = 1$ if and only if $\sum_{n=0}^{\infty} u_n = U(1) = \infty$. ◀

4.3 Reflection Principle and Ballot Theorem

Consider that in an election, there are only two candidates, A and B . The candidate A scores p votes and candidate B scores q votes, where $p > q$. Thus, candidate A has won over B by $p - q$ votes in a overall total vote of $p + q$ votes. Assuming that *each voter has equal preference to both the candidates and is not influenced by others, the probability that the candidate A has lead over candidate B throughout is $(p - q)/(p + q)$* . This is known as the ballot theorem.

We can easily restate the above in terms of random walk. Let us represent the votes for candidate A by $+1$, while the votes for candidate B is represented by -1 . Thus, ϵ_k , for $k = 1, 2, \dots, p + q$ denote the vote of the k -th voter and $S_k = \sum_{i=1}^k \epsilon_i$ denotes the lead of candidate A over candidate after the k -th vote has been cast. Hence, Ballot Theorem states:

$$\mathbb{P}(S_1 > 0, S_2 > 0, \dots, S_{p+q} > 0 \mid S_{p+q} = p - q) = \frac{p - q}{p + q}.$$

From definition of conditional probability, the above probability is given by the ratio of two probabilities:

$$\mathbb{P}(S_1 > 0, \dots, S_{p+q} > 0 \mid S_{p+q} = p - q) = \frac{\mathbb{P}(S_1 > 0, S_2 > 0, \dots, S_{p+q} = p - q > 0)}{\mathbb{P}(S_{p+q} = p - q)}.$$

Each of these two probabilities can be computed using the usual method of path counting. The probability of any event is the ratio of number of paths satisfying the conditions of the event and the total number paths. Here total number of paths is 2^{p+q} . Since both the probabilities will have 2^{p+q} as the common denominator, the above ratio will become the ratio of the number of paths satisfying the conditions of events.

4.3.1 Ballot Theorem

Before we proceed, we establish a combinatorial lemma for counting a certain kind of paths. The lemma, though quite simple to state and prove, has a profound impact in probability theory and allows one to conclude many deep results in advanced probability theory.

Reflection Principle

Now, let $A = (m, a)$ and $B = (n, b)$ where $a, b, n > 0$ and $n > m \geq 0$. By reflection of A , we mean the point $A' = (m, -a)$. Then, we have following simple geometric result which is known as the reflection principle.

Lemma 4.9. (Reflection Principle) *The number of paths from A to B which touch or cross the x -axis is same as the number of paths from A' to B , i.e., N_{b+a}^{n-m} .*

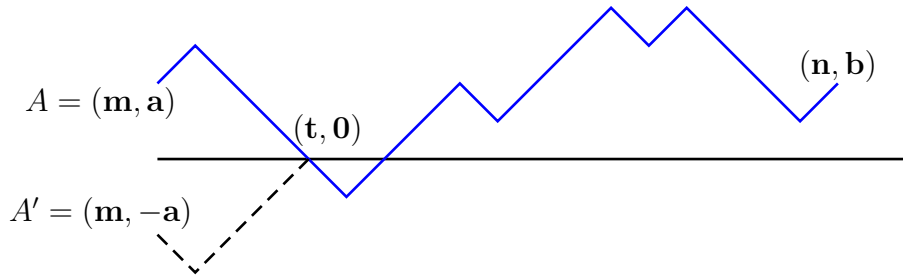


Figure 4.2: Reflection of a path from (m, a) to (n, b) which touches or crosses x -axis

Proof: Consider any path $\{(m, s_m = a), (m+1, s_{m+1}), \dots, (n, s_n = b)\}$ from $A = (m, a)$ to $B = (n, b)$ which touch or cross the x -axis. Let t be first time that the path hits the x -axis, i.e., $s_u > 0$ for $u = m, m+1, \dots, t-1$ and $s_t = 0$. Now, define a path $\{(m, -s_m), (m+1, -s_{m+1}), \dots, (t-1, -s_{t-1}), (t, s_t = 0), (t+1, s_{t+1}), \dots, (n, s_n = b)\}$. This is a path where the part of the path before hitting the x -axis has been reflected (see figure). Clearly this represents a path from A' to B . In a similar manner, it is easy

to check that any path from A' to B will give rise to a path from A to B which touch or cross x -axis. This completes the proof of the Lemma. $\blacksquare$

Returning to Ballot Theorem, we need to show that the number of paths starting from $(0, 0)$ and ending at $(p + q, p - q)$ and always remaining above the x -axis is $N_{(p+q, p-q)} \times (p - q)/(p + q)$. Clearly any path satisfying such a condition must have the property that $s_1 = 1$. Thus, the number of paths from $(1, 1)$ to $(p + q, p - q)$ which always remain above the x -axis, is obtained by subtracting the number of paths $(1, 1)$ to $(p + q, p - q)$ which touch or cross the x -axis from the total number of paths from $(1, 1)$ to $(p + q, p - q)$. So, by reflection principle, the total number of paths from $(1, 1)$ to $(p + q, p - q)$ which always stay above is given by $N_{(p+q-1, p-q-1)} - N_{(p+q-1, p-q+1)}$. Now, using the expression (4.1), we have:

$$\begin{aligned} N_{(p+q-1, p-q-1)} - N_{(p+q-1, p-q+1)} &= \binom{p+q-1}{p-1} - \binom{p+q-1}{p} \\ &= \frac{(p+q-1)!}{(p-1)!q!} - \frac{(p+q-1)!}{p!(q-1)!} = \frac{(p+q)!}{p!q!} \times \frac{1}{p+q} [p-q] \\ &= N_{(p+q, p-q)} \times \frac{p-q}{p+q}. \end{aligned}$$

Thus proving the result.

Remark. In the above argument, the fact that the votes are **iid** with $\mathbb{P}(X_i = +1) = \mathbb{P}(X_i = -1) = 1/2$ is not really necessary. As long as probability of a path which starts at $(0, 0)$ and ends at $(p + q, p - q)$ independent of the arrangement of $+1$'s (in turn -1 's), the Ballot Theorem holds. Besides the independent case, in many dependent cases, such as Pólya's urn scheme, the above is indeed true and the Ballot Theorem, therefore, holds. More generally, if the random variables are exchangeable, the above holds and hence the Ballot Theorem also holds. $\blacktriangleright$

4.3.2 Probability of No Return

Next we prove a result on the probability of no return.

Lemma 4.10. *The probability that no return to the origin occurs up to and including time $2n$ is the same as the probability that a return occurs at time $2n$, i.e., for $n \geq 1$:*

$$\mathbb{P}(S_1 \neq 0, S_2 \neq 0, \dots, S_{2n-1} \neq 0, S_{2n} = 0) = u_{2n}. \quad (4.15)$$

Using Lemma 4.10, Proposition 4.4 follows easily.

Proof of Proposition 4.4: Since at odd time points the walk cannot be at the origin, we can write the first return happens at time $2n$ as the difference of no return till time $2n - 2$ and no return till time $2n$, i.e.:

$$\begin{aligned} & \{S_1 \neq 0, S_2 \neq 0, \dots, S_{2n-2} \neq 0\} \\ &= \{S_1 \neq 0, S_2 \neq 0, \dots, S_{2n-2} \neq 0, S_{2n-1} \neq 0\} \\ &= \{S_1 \neq 0, S_2 \neq 0, \dots, S_{2n-2} \neq 0, S_{2n-1} \neq 0, S_{2n} \neq 0\} \\ &\quad \cup \{S_1 \neq 0, S_2 \neq 0, \dots, S_{2n-2} \neq 0, S_{2n-1} \neq 0, S_{2n} = 0\}. \end{aligned}$$

Therefore, we have:

$$\begin{aligned} f_{2n} &= \mathbb{P}\{S_1 \neq 0, S_2 \neq 0, \dots, S_{2n-2} \neq 0, S_{2n-1} \neq 0, S_{2n} = 0\} \\ &= \mathbb{P}\{S_1 \neq 0, S_2 \neq 0, \dots, S_{2n-2} \neq 0\} \\ &\quad - \mathbb{P}\{S_1 \neq 0, S_2 \neq 0, \dots, S_{2n-2} \neq 0, S_{2n-1} \neq 0, S_{2n} \neq 0\} \\ &= u_{2n-2} - u_{2n} \end{aligned}$$

using Lemma 4.10. This completes the proof. ■

When the event on the left occurs either all the S_k 's are positive, or all are negative. Again an easy correspondence of paths between the above two possible types of paths by reflecting them at the x -axis. Therefore, the probabilities of the above two events are same. Thus, we can restate Lemma 4.10 as follows:

Proposition 4.11. For $n \geq 1$:

$$\mathbb{P}(S_1 > 0, S_2 > 0, \dots, S_{2n-1} > 0, S_{2n} > 0) = \frac{1}{2}u_{2n}. \quad (4.16)$$

Proof: The possible values of S_{2n} are $2, 4, \dots$ (the terms above $2n$ will vanish and hence will not contribute). Therefore, we can write:

$$\begin{aligned} & \mathbb{P}(S_1 > 0, S_2 > 0, \dots, S_{2n} > 0) \\ &= \sum_{k=1}^{\infty} \mathbb{P}(S_1 > 0, S_2 > 0, \dots, S_{2n-1} > 0, S_{2n} = 2k) \\ &= \sum_{k=1}^{\infty} \frac{a_{2n}^{(2k)}}{2^{2n}} = \frac{1}{2^{2n}} \sum_{k=1}^{\infty} a_{2n}^{(2k)}. \end{aligned}$$

where $a_{2n}^{(2k)}$ is the total number paths $\pi = \{(0, 0), (1, s_1), \dots, (2n, s_{2n} = 2k)\}$ be a path such that $s_1 > 0, s_2 > 0, \dots, s_{2n-1} > 0, s_{2n} = 2k$. Clearly, $s_1 = 1$ and we may use the

same reflection argument as in the Ballot Theorem, to get:

$$a_{2n}^{(2k)} = N_{(2n-1, 2k-1)} - N_{(2n-1, 2k+1)}$$

for $k = 1, 2, \dots$. Hence, we have:

$$\begin{aligned} \sum_{k=1}^{\infty} a_{2n}^{(2k)} &= \sum_{k=1}^{\infty} (N_{(2n-1, 2k-1)} - N_{(2n-1, 2k+1)}) \\ &= N_{(2n-1, 1)} = \binom{2n-1}{n} = \frac{1}{2} \binom{2n}{n} = \frac{1}{2} N_{(2n, 0)}. \end{aligned}$$

Hence, the result follows from above equation. ■

Other interesting consequences arise from the above proposition.

Proposition 4.12. *For $n \geq 1$, we have:*

$$\mathbb{P}(S_1 \geq 0, S_2 \geq 0, \dots, S_{2n} \geq 0) = u_{2n}.$$

Proof: Let us consider the event $A = \{S_1 > 0, S_2 > 0, \dots, S_{2n} > 0\}$ and the event $B = \{S_1 \geq 0, S_2 \geq 0, \dots, S_{2n-1} \geq 0\}$. Let $\pi = \{(0, s_0 = 0), (1, s_1), \dots, (2n, s_{2n})\}$ be a path in the event A (see figure below). Then, $s_k > 0$ for $1 \leq k \leq 2n$. Since $s_1 > 0$, we must have $s_1 = 1$. Now shifting the origin to $(1, 1)$, we will obtain a path of length $2n - 1$, which lies above 0. More formally, define $s'_k = s_{k+1} - 1$ for $k = 0, 1, \dots, 2n - 1$ and define a path $\pi' = \{(0, s'_0 = 0), (1, s'_1), (2, s'_2), \dots, (2n - 2, s'_{2n-2}), (2n - 1, s'_{2n-1})\}$. Clearly this is a path of length $2n - 1$ such that $s'_k \geq 0$. Thus, this is a path in the event B . A converse argument can be easily made to construct a path from set B to a path in set A . Hence, we have $|A| = |B|$.

Therefore, we conclude that:

$$\mathbb{P}(S_1 \geq 0, \dots, S_{2n-1} \geq 0) = \frac{|B|}{2^{2n-1}} = \frac{|A|}{2^{2n-1}} = 2\mathbb{P}(S_1 > 0, \dots, S_{2n} > 0) = u_{2n}.$$

Finally, we conclude the result by noting S_{2n-1} must be odd, therefore $S_{2n-1} \geq 1$. Hence we must also have $S_{2n} \geq 0$. Thus:

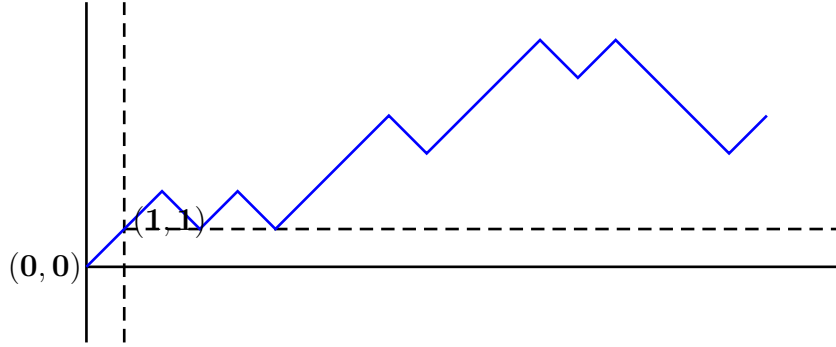
$$\mathbb{P}(S_1 \geq 0, \dots, S_{2n-1} \geq 0) = \mathbb{P}(S_1 \geq 0, \dots, S_{2n-1} \geq 0, S_{2n} \geq 0) = u_{2n}. \quad \blacksquare$$

Exercise 4.13. Show that:

$$\mathbb{P}(S_1 \geq -1, \dots, S_{2n-1} \geq -1) = 2u_{2n}. \quad \blacktriangleleft$$

Exercise 4.14. Show that:

$$\mathbb{P}(S_1 \geq 0, \dots, S_{2n-1} \geq 0, S_{2n} = 0) = 2f_{2n+2}. \quad \blacktriangleleft$$

Figure 4.3: Random walk paths which always stay above x -axis

4.3.3 Time of Last Return

For any $n \geq 1$, we define:

$$T_{2n} := \max\{k : 0 \leq k \leq n, S_{2k} = 0\}. \quad (4.17)$$

Since $S_0 = 0$, clearly 0 belongs in the set defined above and hence, the maxima is well defined. The random variable T_{2n} represents the last time when the walker returned to 0. Proposition 4.11 allows us to compute the distribution of T_{2n} .

Proposition 4.15. *For $k = 0, 1, \dots, n$, we have:*

$$\mathbb{P}(T_{2n} = 2k) = u_{2k}u_{2n-2k}. \quad (4.18)$$

Proof: If $k = 0$, result follows from Lemma 4.10. If $k = n$, the result follows from definition of u_{2n} . For any $k = 0, 1, \dots, n$, we have:

$$\begin{aligned} \mathbb{P}(T_{2n} = 2k) &= \mathbb{P}(S_{2k} = 0, S_{2k+1} \neq 0, \dots, S_{2n} \neq 0) \\ &= \mathbb{P}(S_{2k} = 0, S_{2k+1} - S_{2k} \neq 0, \dots, S_{2n} - S_{2k} \neq 0). \end{aligned}$$

Clearly, since $S_{2k} = \sum_{i=1}^{2k} \epsilon_i$, S_{2k} depends only on the random variables $\epsilon_1, \epsilon_2, \dots, \epsilon_{2k}$, while the random variables $S_{2k+1} - S_{2k}, \dots, S_{2n} - S_{2k}$ depends only on random variables $\epsilon_{2k+1}, \epsilon_{2k+2}, \dots, \epsilon_{2n}$, since $S_{2k+1} - S_{2k} = \epsilon_{2k+1}$, $S_{2k+2} - S_{2k} = \epsilon_{2k+1} + \epsilon_{2k+2}$, $\dots$, $S_{2n} - S_{2k} = \sum_{i=1}^{2n-2k} \epsilon_{2k+i}$. Therefore, the events $\{S_{2k} = 0\}$ and $\{S_{2k+1} - S_{2k} \neq 0, \dots, S_{2n} - S_{2k} \neq 0\}$ are independent. Further, setting $S_{2k+i} - S_{2k} = S'_i$ for $i = 1, 2, \dots, 2n - 2k$, since ϵ_i 's are **iid**, we note that:

$$(S'_1, S'_2, \dots, S'_{2n-2k}) \stackrel{d}{=} (S_1, S_2, \dots, S_{2n-2k}).$$

Using the above, we have:

$$\begin{aligned}
\mathbb{P}(T_{2n} = 2k) &= \mathbb{P}(S_{2k} = 0, S_{2k+1} - S_{2k} \neq 0, \dots, S_{2n} - S_{2k} \neq 0) \\
&= \mathbb{P}(S_{2k} = 0) \mathbb{P}(S_{2k+1} - S_{2k} \neq 0, \dots, S_{2n} - S_{2k} \neq 0) \\
&= \mathbb{P}(S_{2k} = 0) \mathbb{P}(S'_1 \neq 0, \dots, S'_{2n-2k} \neq 0) \\
&= \mathbb{P}(S_{2k} = 0) \mathbb{P}(S_1 \neq 0, \dots, S_{2n-2k} \neq 0) \\
&= u_{2k} u_{2n-2k}.
\end{aligned}$$

This completes the proof. ■

When $n \rightarrow \infty$ such that k and $n - k$ both tend to infinity, using the approximation of u_{2n} , given by equation (4.5), we have:

$$\mathbb{P}(T_{2n} = 2k) \sim \frac{1}{\pi} \frac{1}{\sqrt{k(n-k)}}. \quad (4.19)$$

4.3.4 Long Leads

We saw that, contrary to popular notions, it is quite likely that in a long coin-tossing game one of the players remains practically the whole time on the winning side, the other on the losing side. The next theorem elucidates the same phenomenon by an analysis of the fraction of the total time that the particle spends on the positive side. One feels intuitively that this fraction is most likely to be close to $1/2$ but the opposite is true: The possible values close to $1/2$ are least probable, whereas the extremes $k = 0$ and $k = n$ have the greatest probability.

Let us make this more precise. We say the particle spends the time interval $(i, i + 1)$ above the x-axis if at least one of the time points i and $i + 1$ the value is larger than 0. Otherwise, we say that the particle stays below the x-axis. Let us define:

$$L_{2n} = \sum_{i=1}^{2n} \mathbb{I}\{\text{the particle spends } (i-1, i) \text{ is above x-axis}\}.$$

Theorem 4.16. *The probability that in the time interval from 0 to $2n$ the particle spends $2k$ time units on the positive side (and hence $2n - 2k$ time units on the negative side) is given by:*

$$\mathbb{P}(L_{2n} = 2k) = u_{2k} u_{2n-2k}.$$

Proof: Consider paths of the fixed length $2n$ and denote by $b_{2n,2k}$, the probability that the particle stays exactly $2k$ time intervals lie above the x-axis. We have to prove

that:

$$b_{2n,2k} = \mathbb{P}(L_{2n} = 2k).$$

We will prove the result by induction. If $n = 1$, there are only two possible values of k , namely $k = 0, k = 1$. Clearly, we have:

$$b_{2,0} = \mathbb{P}(S_1 = -1) = \frac{1}{2} \text{ and } b_{2,2} = \mathbb{P}(S_1 = 1) = \frac{1}{2}$$

and:

$$u_0 = 1 \text{ and } u_2 = \mathbb{P}(S_2 = 0) = \mathbb{P}(S_1 = 1, S_2 = 0) + \mathbb{P}(S_1 = -1, S_2 = 0) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

This proves the base case.

Now, assume that the result is true for $n \leq n_0$. We want to prove the result for $n_0 + 1$. We will first prove the cases: $k = 0$ and $k = n_0 + 1$. Note that we have:

$$b_{2n_0+2,0} = \mathbb{P}(S_1 \leq 0, S_2 \leq 0, \dots, S_{2n_0+2} \leq 0) = \mathbb{P}(S_1 \geq 0, S_2 \geq 0, \dots, S_{2n_0+2} \geq 0) = u_{2n_0+2}$$

and:

$$b_{2n_0+2,2n_0+2} = \mathbb{P}(S_1 \geq 0, S_2 \geq 0, \dots, S_{2n_0+2} \geq 0) = u_{2n_0+2}$$

using Proposition 4.12. Using $u_0 = 1$, this proves the result for $k = 0$ and $k = n_0 + 1$.

Now, assume that $1 \leq k \leq n_0$. First, suppose that in the first step the particle goes above the x-axis. In such a case, it is easy to see that the particle must return to 0 within time $2k$, as otherwise it would have spent more than $2k$ time units above x-axis. So, we may suppose that particle returns to the origin for the first time in time $2t$ for $t = 1, 2, \dots, k$. Once the particle returns to the origin in time $2t$, it would have spent $2t$ time units above the x-axis. Therefore, in the remaining $2(n_0 + 1) - 2t$ time units, the particle must spend $2k - 2t$ time units above the x-axis. Therefore, we have the total $\frac{1}{2}f_{2t}2^{2t}$ many paths which will lie above the x-axis and returns for the first time to origin at time $2t$. For each such path, we will have $b_{2(n_0+1)-2t,2k-2t}2^{2(n_0+1)-2t}$ many paths which will spend $2k$ time units above the x-axis in $2(n_0 - t + 1)$ time units. Using induction hypothesis, this may be written as $u_{2k-2t}u_{2(n_0+1)-2k}2^{2(n_0+1)-2t}$ as $n_0 - t + 1 \leq n_0$. In this case, the total number of paths is given by:

$$\begin{aligned} \sum_{t=1}^k \frac{1}{2}f_{2t}2^{2t}u_{2k-2t}u_{2(n_0+1)-2k}2^{2(n_0+1)-2t} &= \frac{1}{2}2^{2(n_0+1)}u_{2(n_0+1)-2k} \sum_{t=1}^k f_{2t}u_{2k-2t} \\ &= \frac{1}{2}2^{2(n_0+1)}u_{2(n_0+1)-2k} \sum_{t=0}^k f_{2t}u_{2k-2t} = \frac{1}{2}2^{2(n_0+1)}u_{2(n_0+1)-2k}u_{2k} \end{aligned}$$

using $f_0 = 0$ and Proposition 4.7.

Next, suppose that in the first step the particle goes below the x-axis. In such a case, it is easy to see that the particle must return to 0 within time $2(n_0 + 1 - k)$, as otherwise it would have spent less than $2k$ time units above x-axis. So, we may suppose that particle returns to the origin for the first time in time $2t$ for $t = 1, 2, \dots, n_0 + 1 - k$. Once the particle returns to the origin in time $2t$, it would have spent 0 time units above the x-axis. Therefore, in the remaining $2(n_0 + 1) - 2t$ time units, the particle must spend $2k$ time units above the x-axis. Therefore, we have the total $\frac{1}{2}f_{2t}2^{2t}$ many paths which will lie below the x-axis and returns for the first time to origin at time $2t$. For each such path, we will have $b_{2(n_0+1)-2t, 2k}2^{2(n_0+1)-2t}$ many paths which will spend $2k$ time units above the x-axis in $2(n_0 - t + 1)$ time units. Using induction hypothesis, this may be written as $u_{2k}u_{2(n_0+1-k)-2t}2^{2(n_0+1)-2t}$ as $n_0 - t + 1 \leq n_0$. In this case, the total number of paths is given by:

$$\begin{aligned} \sum_{t=1}^{n_0+1-k} \frac{1}{2}f_{2t}2^{2t}u_{2k}u_{2(n_0+1-k)-2t}2^{2(n_0+1)-2t} &= \frac{1}{2}2^{2(n_0+1)}u_{2k} \sum_{t=1}^{n_0+1-k} f_{2t}u_{2(n_0+1-k)-2t} \\ &= \frac{1}{2}2^{2(n_0+1)}u_{2k} \sum_{t=0}^{n_0+1-k} f_{2t}u_{2(n_0+1-k)-2t} = \frac{1}{2}2^{2(n_0+1)}u_{2k}u_{2(n_0+1)-2k} \end{aligned}$$

again using $f_0 = 0$ and Proposition 4.7.

Combining both the cases, we have the total number of paths satisfying the condition that a total $2k$ time units are spent above the x-axis is given by $2^{2(n_0+1)}u_{2(n_0+1)-2k}u_{2k}$ in $2(n_0 + 1)$ time units. This proves the result. ■

Again, we may consider the case when $n \rightarrow \infty$ such that k and $n - k$ both tend to infinity. Using the approximation of u_{2n} , given by equation (4.5), we have:

$$\mathbb{P}(L_{2n} = 2k) \sim \frac{1}{\pi} \frac{1}{\sqrt{k(n-k)}}. \quad (4.20)$$

4.3.5 Distribution of Maxima

Perhaps the most interesting usage of the reflection principle is to find the distribution of the maxima of random walk. We define:

$$M_n := \max_{0 \leq k \leq n} S_k.$$

Since $S_0 = 0$, clearly $M_n \geq 0$.

Proposition 4.17. *For any $b \geq 0$:*

$$\mathbb{P}(M_n \geq b) = 2\mathbb{P}(S_n > b) + \mathbb{P}(S_n = b). \quad (4.21)$$

Proof: If $b = 0$, the left hand side is 1. Since ϵ_k 's are symmetric, S_n is also symmetric. Thus, $\mathbb{P}(S_n > 0) = \mathbb{P}(S_n < 0)$. So the right hand side is $\mathbb{P}(S_n > 0) + \mathbb{P}(S_n < 0) + \mathbb{P}(S_n = 0) = 1$.

For $b > 0$, we have:

$$\begin{aligned} \mathbb{P}(M_n \geq b) &= \mathbb{P}\left(\max_{0 \leq k \leq n} S_k \geq b\right) \\ &= \mathbb{P}\left(\max_{0 \leq k \leq n} S_k \geq b, S_n = b\right) + \mathbb{P}\left(\max_{0 \leq k \leq n} S_k \geq b, S_n > b\right) + \mathbb{P}\left(\max_{0 \leq k \leq n} S_k \geq b, S_n < b\right) \\ &= \mathbb{P}(S_n = b) + \mathbb{P}(S_n > b) + \mathbb{P}\left(\max_{0 \leq k \leq n} S_k \geq b, S_n < b\right) \end{aligned}$$

where the first two equalities follow as the events $\{S_n = b\}$ and $\{S_n > b\}$ is contained in the event $\{\max_{0 \leq k \leq n} S_k \geq b\}$. Therefore, it is enough to show that:

$$\mathbb{P}\left(\max_{0 \leq k \leq n} S_k \geq b, S_n < b\right) = \mathbb{P}(S_n > b).$$

Again, we carry out a reflection argument. For any path, starting at $(0,0)$ and finishing at (n, u) for $u < b$ and somewhere hitting or crossing the line $y = b$ must have a point at which it hits the line $y = b$ for the first time. Let $t < n$ be the first such point, i.e., $s_1 < b, s_2 < b, \dots, s_{t-1} < b, s_t = b$ for $t < n$. Now define a new path as: $(1, s_1), (2, s_2), \dots, (t-1, s_{t-1}), (t, s_t = b), (t+1, 2b-s_{t+1}), (t+2, 2b-s_{t+2}), \dots, (n, 2b-s_n)$. This is exactly the path reflected at the line $y = b$ after it hits the line $y = b$ (see Figure 4.4).

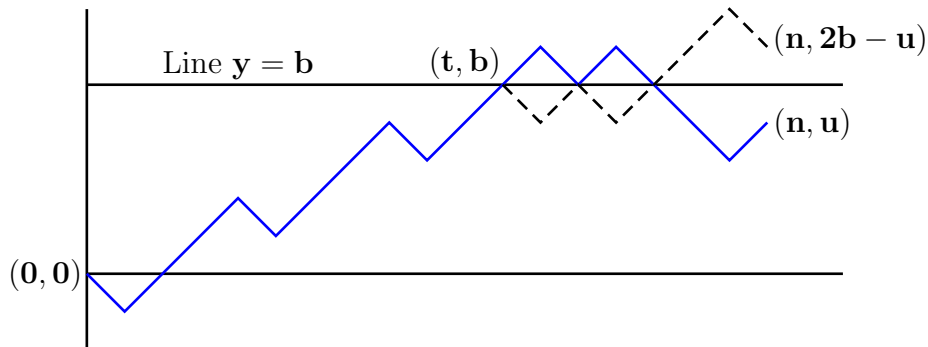


Figure 4.4: Random walk path reflected on maxima value

Since for the original path we had $s_n = u < b$, for the reflected path, we have $2b - s_n = 2b - u > b$. Conversely, for each path for which $s_n > b$, we must have a

$t < n$ such that $s_t = b$ for the first time. Again, define the reflected path after it hits the line $y = b$ as follows: $(1, s_1), (2, s_2), \dots, (t-1, s_{t-1}), (t, s_t = b), (t+1, 2b - s_{t+1}), (t+2, 2b - s_{t+2}), \dots, (n, 2b - s_n)$. This will then correspond to a path hits or crosses the line $y = b$ and finishes at a point lower than b . Thus, the number of paths from the origin which touch or cross the line $y = b$ but finishes at a point lower than b is the same as the number of paths from the origin which finishes above b . Thus, we have $\mathbb{P}(\max_{0 \leq k \leq n} S_k \geq b, S_n < b) = \mathbb{P}(S_n > b)$. This completes the proof. ■

4.4 Advanced Topics and Applications

Suggested Additional Content

1. **Arc-sine Laws:** Detailed treatment of the arc-sine law for last return times and time spent on the positive side.
2. **Limit Theorems:** Connection to Brownian motion via Donsker's theorem and functional central limit theorems.
3. **Random Walk in Random Environment:** Introduction to disordered systems and their effect on random walk behavior.
4. **Applications to Finance:** Random walk hypothesis in stock prices and option pricing models.
5. **Connection to Potential Theory:** Green's functions, harmonic functions, and their relation to random walks.
6. **Multi-dimensional Extensions:** Reflection principles in higher dimensions and their limitations.
7. **Random Walks on Graphs:** Extension to general graphs and connection to spectral graph theory.
8. **Queueing Theory Applications:** Random walks in queueing systems and waiting time distributions.

4.5 Supplementary Problems

Theoretical Problems

Exercise 4.18. Prove that for a simple symmetric random walk on $\mathbb{Z}$, the probability that the walk visits every integer infinitely often is 1.

Exercise 4.19. Show that the expected number of returns to the origin in n steps is $\sum_{k=0}^n u_k \sim \frac{2}{\pi}\sqrt{n}$ as $n \rightarrow \infty$.

Exercise 4.20. Using the reflection principle, derive the distribution of the first hitting time of level $a > 0$.

Exercise 4.21. Prove that for any $a, b > 0$, the probability that a random walk starting at 0 hits a before $-b$ is $b/(a + b)$.

Exercise 4.22. Show that the generating function for the maximum of a random walk up to time n satisfies a certain renewal-type equation.

Combinatorial Problems

Exercise 4.23. Count the number of paths of length $2n$ that never go below the x-axis (Catalan numbers).

Exercise 4.24. Using the reflection principle, prove the Chung-Feller theorem about the number of paths with given number of positive steps.

Exercise 4.25. Derive the number of paths that stay between two parallel lines using multiple reflections.

Advanced Problems

Exercise 4.26. Investigate the behavior of random walks with drift and their connection to ruin probabilities.

Exercise 4.27. Study the relationship between random walks and the heat equation via the discrete Laplacian.

Exercise 4.28. Explore the connection between random walks and electrical networks using Ohm's law and Kirchhoff's laws.

Exercise 4.29. Analyze the scaling limits of random walks and their convergence to Brownian motion.

Exercise 4.30. Investigate the effect of different step distributions on the recurrence/transience properties of random walks.

Chapter 5

Markov chain

5.1 Introduction

In general, a random process, more specifically a stochastic process, is a collection of random variables indexed by certain index set T . We will consider only discrete index set. More precisely, in our case, $T = \{0, 1, 2, \dots\}$, the set of all non-negative integers.

We have already seen examples where the random variables $X_0, X_1, X_2, \dots$ are independent. But more interesting and natural situation is when the evolution of the process is allowed to depend on the past values of the process. Here we will consider a particular kind of dependence in which the future evolution of the process depends on the past of the process only through the present value.

To make it more precise, let $\mathcal{S}$ be a countable set. Mostly, in examples, this is going to be some subset of $\mathbb{Z}^d$ where $d \geq 1$.

Definition 5.1. A random process is said to be a *Markov chain* with state space $\mathcal{S}$ if it satisfies the **Markov condition** i.e., for any $n \geq 1$ and states $i_0, i_1, i_2, \dots, i_{n-1}, i, j \in \mathcal{S}$

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i). \quad (5.1)$$

It can be also checked that Markov condition is equivalent to any of the following conditions:

- For any $n \geq 0, m \geq 1$ and states $i_0, i_1, i_2, \dots, i_{n-1}, i, j \in \mathcal{S}$

$$\mathbb{P}(X_{n+m} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = \mathbb{P}(X_{n+m} = j \mid X_n = i).$$

- For any $n \geq 0, k \geq 1$, with $0 \leq n_1 < n_2 < \dots < n_k \leq n$ and states $i_1, i_2, \dots, i_{k-1}$ and $i, j \in \mathcal{S}$

$$\mathbb{P}(X_{n+1} = j \mid X_{n_k} = i, X_{n_{k-1}} = i_{k-1}, \dots, X_{n_1} = i_1) = \mathbb{P}(X_{n+1} = j \mid X_{n_k} = i).$$

The distribution of X_0 will be called the *initial distribution* for the Markov chain. Let μ be a probability distribution on $\mathcal{S}$, i.e., μ is a vector $(\mu(i) : i \in \mathcal{S})$ where $\mu(i) \geq 0$ and $\sum_{i \in \mathcal{S}} \mu(i) = 1$. We will assume that

$$\mathbb{P}(X_0 = i) = \mu(i) \text{ for } i \in \mathcal{S}. \quad (5.2)$$

The evolution of the chain clearly depends on the initial distribution and the probabilities $\mathbb{P}(X_{n+1} = j \mid X_n = i)$. In general, this depends on i, j and n . We restrict our attention to the case where this probability is independent of n .

Definition 5.2. A Markov chain is called **homogeneous** (or time-homogeneous) if the conditional probabilities above is independent of n , i.e.,

$$\mathbb{P}(X_{n+1} = j \mid X_n = i) = \mathbb{P}(X_1 = j \mid X_0 = i)$$

for all $i, j \in \mathcal{S}$ and $n \geq 1$.

Henceforth, we consider only homogeneous Markov chains.

Definition 5.3. The **transition matrix** $\mathbf{P} = ((p_{ij}))$ is the matrix whose (i, j) -th entry is given by

$$p_{ij} = \mathbb{P}(X_1 = j \mid X_0 = i).$$

It is clear that p_{ij} 's, being conditional probabilities, must satisfy the conditions:

$$p_{ij} \geq 0 \text{ and } \sum_{j \in \mathcal{S}} p_{ij} = 1 \text{ for all } i \in \mathcal{S}.$$

Such matrices are called a *stochastic matrix*.

Note that the condition that a matrix is a stochastic matrix implies that

$$\mathbf{P}\mathbf{1}' = \mathbf{1}'$$

where $\mathbf{1}$ is the row vector having all entries 1. In other words, 1 is an *eigenvalue* of the matrix $\mathbf{P}$ with $\mathbf{1}'$ as the corresponding *eigenvector*.

Exercise 5.4. Let $\{X_n : n \geq 0\}$ be a Markov chain with state space $\mathcal{S}$ and $\mathcal{S}'$ be another set. Let $f : \mathcal{S} \rightarrow \mathcal{S}'$ be a one-to-one function. Show that $\{f(X_n) : n \geq 0\}$ is a Markov chain on $\mathcal{S}'$. Show by an example that when f is not one-to-one, $\{f(X_n) : n \geq 0\}$ need not be a Markov chain. ◀

Exercise 5.5. Let $\{X_n : n \geq 0\}$ and $\{Y_n : n \geq 0\}$ be Markov chains on the set of integers. Show that the sequence $\{Z_n := X_n + Y_n : n \geq 0\}$ is not necessarily a Markov chain. What happens if the processes are independent? ◀

Exercise 5.6. Show that, for a Markov chain, the future and the past are conditionally independent given the present. In other words, if $\{X_n : n \geq 0\}$ is a Markov chain, show that

$$\begin{aligned} & \mathbb{P}[X_0 = x_0, X_1 = x_1, \dots, X_{n-1} = x_{n-1}, X_{n+1} = x_{n+1} \mid X_n = x_n] \\ &= \mathbb{P}[X_0 = x_0, X_1 = x_1, \dots, X_{n-1} = x_{n-1} \mid X_n = x_n] \mathbb{P}[X_{n+1} = x_{n+1} \mid X_n = x_n] \end{aligned}$$

for $x_0, x_1, \dots, x_{n+1} \in \mathcal{S}$. ◀

Before we begin with examples, we prove a result which will be of great help in establishing the Markov property for a great number of examples.

Proposition 5.7. Let $\{\xi_t : t \geq 1\}$ be a sequence of **iid** random variables taking values in some arbitrary space $\mathcal{T}$. Assume that X_0 is independent of the sequence $\{\xi_t : t \geq 1\}$. Suppose $f : \mathcal{S} \times \mathcal{T} \rightarrow \mathcal{S}$ is a function. For $n \geq 0$, set

$$X_{n+1} = f(X_n, \xi_{n+1}). \quad (5.3)$$

Then $\{X_n : n \geq 0\}$ is a Markov chain.

Proof : First we will compute the joint distribution of the random variables $(X_0, X_1, \dots, X_n, X_{n+1})$. For any $k_0, k_1, \dots, k_{n+1} \in \mathcal{S}$, we have

$$\begin{aligned} & \mathbb{P}(X_{n+1} = k_{n+1}, X_n = k_n, X_{n-1} = k_{n-1}, \dots, X_1 = k_1, X_0 = k_0) \\ &= \mathbb{P}(X_{n+1} = f(X_n, \xi_{n+1}) = k_{n+1}, X_n = f(X_{n-1}, \xi_n) = k_n, \dots, \\ & \quad X_1 = f(X_0, \xi_1) = k_1, X_0 = k_0) \\ &= \mathbb{P}(f(k_n, \xi_{n+1}) = k_{n+1}, f(k_{n-1}, \xi_n) = k_n, \dots, f(k_0, \xi_1) = k_1, X_0 = k_0) \\ &= \prod_{t=0}^n \mathbb{P}(f(k_t, \xi_{t+1}) = k_{t+1}) \mathbb{P}(X_0 = k_0). \end{aligned}$$

In the last equality, we have used the fact that the random variables $\{\xi_t : t \geq 1\}$ are independent and hence the transformed random variables $\{f(k_t, \xi_t) : t \geq 1\}$ (for any $k_t \in \mathcal{S}$) are also independent as they are the functions of ξ_t and a non-random constant $k_t \in \mathcal{S}$.

Using the above calculations, for $i, j, i_0, i_1, \dots, i_{n-1} \in \mathcal{S}$, we have

$$\begin{aligned}
& \mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_1 = i_1, X_0 = i_0) \\
&= \frac{\mathbb{P}(X_{n+1} = j, X_n = i, X_{n-1} = i_{n-1}, \dots, X_1 = i_1, X_0 = i_0)}{\mathbb{P}(X_n = i, X_{n-1} = i_{n-1}, \dots, X_1 = i_1, X_0 = i_0)} \\
&= \frac{\mathbb{P}(f(i, \xi_{n+1}) = j) \mathbb{P}(f(i_{n-1}, \xi_n) = i) \prod_{t=0}^{n-2} \mathbb{P}(f(i_t, \xi_{t+1}) = i_{t+1}) \mathbb{P}(X_0 = i_0)}{\mathbb{P}(f(i_{n-1}, \xi_n) = i) \prod_{t=0}^{n-2} \mathbb{P}(f(i_t, \xi_{t+1}) = i_{t+1}) \mathbb{P}(X_0 = i_0)} \\
&= \mathbb{P}(f(i, \xi_{n+1}) = j).
\end{aligned} \tag{5.4}$$

For the other part, we observe that

$$\mathbb{P}(X_{n+1} = j \mid X_n = i) = \frac{\mathbb{P}(X_{n+1} = j, X_n = i)}{\mathbb{P}(X_n = i)}.$$

We can write the above marginal distributions by summing over other co-ordinates as follows:

$$\begin{aligned}
& \mathbb{P}(X_{n+1} = j, X_n = i) \\
&= \sum_{k_0, k_1, \dots, k_{n-1} \in \mathcal{S}} \mathbb{P}(X_{n+1} = j, X_n = i, X_{n-1} = k_{n-1}, \dots, X_1 = k_1, X_0 = k_0) \\
&= \sum_{k_0, k_1, \dots, k_{n-1} \in \mathcal{S}} \mathbb{P}(f(i, \xi_{n+1}) = j) \mathbb{P}(f(k_{n-1}, \xi_n) = i) \prod_{t=0}^{n-2} \mathbb{P}(f(k_t, \xi_{t+1}) = k_{t+1}) \mathbb{P}(X_0 = k_0) \\
&= \mathbb{P}(f(i, \xi_{n+1}) = j) \sum_{k_0, k_1, \dots, k_{n-1} \in \mathcal{S}} \mathbb{P}(f(k_{n-1}, \xi_n) = i) \prod_{t=0}^{n-2} \mathbb{P}(f(k_t, \xi_{t+1}) = k_{t+1}) \mathbb{P}(X_0 = k_0).
\end{aligned}$$

Using the same arguments, we have

$$\mathbb{P}(X_n = i) = \sum_{k_0, k_1, \dots, k_{n-1} \in \mathcal{S}} \mathbb{P}(f(k_{n-1}, \xi_n) = i) \prod_{t=0}^{n-2} \mathbb{P}(f(k_t, \xi_{t+1}) = k_{t+1}) \mathbb{P}(X_0 = k_0).$$

Therefore, we obtain that $\mathbb{P}(X_{n+1} = j \mid X_n = i) = \mathbb{P}(f(i, \xi_{n+1}) = j)$. This completes the proof. $\blacksquare$

With the above representation, we can easily identify the transition matrix $\mathbf{P} = ((p_{ij}))$. Indeed, from equation (5.4), we have

$$p_{ij} = \mathbb{P}(f(i, \xi_1) = j). \tag{5.5}$$

Note: we have set no conditions on the sequence of random variables $\{\xi_i : i \geq 1\}$. Even these could be random vectors, as long as they are independent and identically distributed.

5.2 Examples of Markov Chains

Now we will look at examples of Markov chains.

Example 5.8. A sequence of **iid** random variables will, of course, constitute a Markov chain. Let $\{\xi_i : i \geq 0\}$ represent **iid** random variables with $\mathbb{P}\{\xi_i = i\} = \alpha_i, \alpha_i \geq 0$ and $\sum_{i=-\infty}^{+\infty} \alpha_i = 1$. Clearly, we have

$$\begin{aligned} \mathbb{P}(\xi_{n+1} = j \mid \xi_n = i, \xi_{n-1} = i_{n-1}, \dots, \xi_1 = i_1, \xi_0 = i_0) \\ = \mathbb{P}(\xi_{n+1} = j) = \alpha_j = \mathbb{P}(\xi_{n+1} = j \mid \xi_n = i). \end{aligned}$$

Thus, the process $\{\xi_n : n \geq 0\}$ is a homogeneous Markov chain, having state space $\mathcal{S} = \mathbb{Z}$, with the transition matrix $\mathbf{P} = ((p_{ij}))$ where $p_{ij} = \alpha_j$ for $i, j \in \mathcal{S}$, i.e.,

$$\mathbf{P} = \begin{array}{c} \begin{array}{cccccc} \cdots & -1 & 0 & 1 & 2 & \cdots \end{array} \\ \begin{array}{c} \vdots \\ -1 \\ 0 \\ 1 \\ \vdots \\ i \\ \vdots \end{array} \begin{bmatrix} \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \cdots & \alpha_{-1} & \alpha_0 & \alpha_1 & \alpha_2 & \cdots \\ \cdots & \alpha_{-1} & \alpha_0 & \alpha_1 & \alpha_2 & \cdots \\ \cdots & \alpha_{-1} & \alpha_0 & \alpha_1 & \alpha_2 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \cdots & \alpha_{-1} & \alpha_0 & \alpha_1 & \alpha_2 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \end{array}$$

Each row being identical expresses the fact that the random variable X_{n+1} is independent of X_n . ■

Example 5.9. (Two state chain) We have two coins : Coin I and Coin II, with probabilities of success being α and β respectively. We start with some coin and toss it : if the toss results in a success, we continue with the same coin at the next time point. If the toss results in a failure then we switch to the other coin at the next time point. Let X_n be the index of the coin that we are using at time n , i.e., $X_n = 1$ if we are using

Coin I at time n , while $X_n = 2$ if we are using Coin II at time n . Thus, the state space $\mathcal{S}$ here is $\mathcal{S} = \{1, 2\}$. Clearly, for $i_0, i_1, \dots, i_{n-1}, i \in \{1, 2\}$.

$$\mathbb{P}(X_{n+1} = 1 \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_1 = i_1, X_0 = i_0) = \begin{cases} \alpha & \text{if } i = 1 \\ 1 - \beta & \text{if } i = 2. \end{cases}$$

It is clearly seen that : for $i \in \{1, 2\}$,

$$\mathbb{P}(X_{n+1} = 1 \mid X_n = i) = \begin{cases} \alpha & \text{if } i = 1 \\ 1 - \beta & \text{if } i = 2. \end{cases}$$

It can be similarly seen for the other case, i.e., $X_{n+1} = 2$. Hence, the above process is a Markov chain with transition matrix $\mathbf{P} = ((p_{ij}))$ where

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{bmatrix} \alpha & 1 - \alpha \\ 1 - \beta & \beta \end{bmatrix} \end{matrix}.$$

We will take generic values of the probability of successes and denote $\mathbf{P}$ by

$$\begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \end{matrix}$$

with the usual condition that $p_{11} + p_{12} = 1 = p_{21} + p_{22}$.

This is also useful to cast this example in mould of Proposition 5.7. Here, we can take $\{\xi_i : i \geq 1\}$ as a vector, $\xi_n = (\xi_n^{(1)}, \xi_n^{(2)})$ where $\xi_n^{(j)}$ represents the success or failure (1 or 0) for the j -th coin ($j = 1$ or 2) at the n -th toss. Then, ξ_n takes values in $\{0, 1\}^2$ and we can define the function $f : \mathcal{S} \times \{0, 1\}^2 = \{1, 2\} \times \{0, 1\}^2$ as follows:

$$f(i, (\xi_1, \xi_2)) = \begin{cases} 1 & \text{if } i = 1, \xi_1 = 1 \\ 2 & \text{if } i = 1, \xi_1 = 0 \\ 2 & \text{if } i = 2, \xi_2 = 1 \\ 1 & \text{if } i = 2, \xi_2 = 0. \end{cases}$$

An intuitive way of thinking about the above is the following: assume that at every time point we toss both the coins regardless of which state the Markov chain is. The results

of the tosses at time n , are given by random vector $\xi_{n+1} = (\xi_{n+1}^{(1)}, \xi_{n+1}^{(2)})$. Depending on the state of the chain, we select which component of the vector we need to use. In other words, if we are at state 1, we need to use the first coin, and therefore use the first component $\xi_{n+1}^{(1)}$ which gives the result of toss of the first coin and vice versa. It is easy to check $f(X_n, \xi_{n+1})$ has the same distribution as X_{n+1} described above. ■

5.2.1 Random walks

We will now describe some examples of Markov chain which are broadly classified as random walks.

Example 5.10. Let $\{\xi_i \geq 1\}$ be any sequence of **iid** random vectors taking values in $\mathbb{Z}^d$. Let $\mathbf{x}_0 \in \mathbb{Z}^d$. Now, define $S_0 = \mathbf{x}_0$ and $S_n = S_0 + \sum_{i=1}^n \xi_i$ for $n \geq 1$. We will call the sequence of random variables $\{S_n : n \geq 0\}$ a random walk.

Let us define $f : \mathbb{Z}^d \times \mathbb{Z}^d \rightarrow \mathbb{Z}^d$ by

$$f(\mathbf{i}, \mathbf{j}) = \mathbf{i} + \mathbf{j}.$$

Clearly, we have $S_{n+1} = f(S_n, \xi_{n+1})$ for all $n \geq 0$. Therefore, using Proposition 5.7, we conclude that $\{S_n : n \geq 0\}$ is a Markov chain with state space $\mathbb{Z}^d$ and the transition matrix is given by $\mathbf{P} = ((p_{ij}))$ where $p_{ij} = \mathbb{P}(\xi_1 = j - i)$. ■

Example 5.11. Simple random walk in one dimension : We return to the random walk example on the integers where the random walker (or a particle) moves one step to the right or one step to the left with equal probability. Here, we consider more general case : if the walker is at x at time n , at time $n + 1$ it jumps to $x + 1$ with probability p and to $x - 1$ with probability $q = 1 - p$. The evolution of the walk from time n on depends entirely on the position at time n and not on positions prior to time n .

Formally, let $\{\xi_n : n \geq 1\}$ be a sequence of **iid** random variables with $\mathbb{P}(\xi_n = 1) = p = 1 - \mathbb{P}(\xi_n = -1)$. The random variable ξ_n can be thought as the value of the jump size of the walker at time n . Define, for any $x_0 \in \mathbb{Z}$,

$$S_0 = x_0 \text{ and } S_n = x_0 + \sum_{i=1}^n \xi_i \text{ for } n \geq 1.$$

We have already proved the Markov property of the process $\{S_n : n \geq 0\}$ in Example 5.10. Note that here ξ_n takes values in $\{-1, 1\}$. Therefore, from a state i , the chain can

only move to $f(i, \xi_n) = i + \xi_n \in \{i - 1, i + 1\}$. The transition probabilities are easy to compute. Indeed, we have

$$p_{ij} = \mathbb{P}(i + \xi_1 = j) = \begin{cases} p & \text{if } j = i + 1 \\ q & \text{if } j = i - 1 \\ 0 & \text{otherwise.} \end{cases}$$

The transition matrix is given by

$$\mathbf{P} = \begin{matrix} & \cdots & -1 & 0 & 1 & 2 & \cdots & i-1 & i & i+1 & \cdots \\ \begin{matrix} \vdots \\ 0 \\ 1 \\ \vdots \\ i \\ \vdots \end{matrix} & \begin{bmatrix} \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \cdots & q & 0 & p & 0 & \cdots & 0 & 0 & 0 & \cdots \\ \cdots & 0 & q & 0 & p & \cdots & 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \cdots & 0 & 0 & 0 & 0 & \cdots & q & 0 & p & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \end{matrix}.$$

In later section, we include a state transition diagram to visualize the transitions from different states. ■

Example 5.12. Simple symmetric random walk in d -dimension : The classical symmetric random walk in d -dimensions admits the following Markov chain formulation.

We start with the two dimensional case. Think of a (infinite) square grid. At any time point of time, the walker has four neighbouring vertices - one in each of direction of top, left, bottom and right. The walker chooses one of the four neighbouring vertices with equal probability and moves to it. The same process is repeated at each time point. More mathematically, we can identify the state space as $\{(i_1, i_2) : i_1, i_2 \in \mathbb{Z}\}$. Clearly from any point (i_1, i_2) the walker can move to each of the vertices $(i_1, i_2 + 1)$ (top), $(i_1 - 1, i_2)$ (left), $(i_1, i_2 - 1)$ (bottom) and $(i_1 + 1, i_2)$ (right) with $1/4$ probability each, i.e.,

$$p_{(i_1, i_2), (j_1, j_2)} = \begin{cases} \frac{1}{4} & \text{if } (j_1, j_2) \in \{(i_1, i_2 + 1), (i_1 - 1, i_2), (i_1, i_2 - 1), (i_1 + 1, i_2)\} \\ 0 & \text{otherwise.} \end{cases}$$

We will show state transitions through a state transition diagram later.

b) For three dimensions, we consider the (infinite) three dimensional lattices points $\{(i_1, i_2, i_3) : i_1, i_2, i_3 \in \mathbb{Z}\}$. At any time point of time, the walker has six neighbouring vertices - one in each of direction of top, left, bottom, right, up and down and the walker moves to one these six neighbours with equal probability.

c) For general d -dimensions, the state space is identified with the set of all integral lattice points of $\mathbb{Z}^d$, i.e., a state is an d -tuple $\mathbf{i} = (i_1, i_2, \dots, i_d), i_s \in \mathbb{Z}$ for $s = 1, 2, \dots, d$. In such a case, every vertex has $2d$ many neighbours, and the walker moves to one these neighbours with equal probability. The transition probability matrix $\mathbf{P} = ((p_{\mathbf{i}\mathbf{j}}))$ where

$$p_{\mathbf{i}\mathbf{j}} = \begin{cases} \frac{1}{2d} & \text{if } \sum_{s=1}^d |i_s - j_s| = 1 \text{ where } \mathbf{i} = (i_1, \dots, i_d), \mathbf{j} = (j_1, \dots, j_d) \\ 0 & \text{otherwise.} \end{cases} \quad (5.6)$$

In the one-dimensional simple symmetric random walk case, we have the **iid** random variables $\{\xi_i : i \geq 1\}$ taking values in $\{-1, 1\} \subseteq \mathbb{Z}$ with $\mathbb{P}(\xi_i = 1) = 1/2 = 1 - \mathbb{P}(\xi_i = -1)$ for any $i \geq 1$. In higher dimensional Random walk, we have

$$\mathbb{P}(\xi_n = \mathbf{k}) = \begin{cases} \frac{1}{2d} & \text{if } \sum_{s=1}^d |k_s| = 1 \text{ where } \mathbf{k} = (k_1, \dots, k_d) \\ 0 & \text{otherwise} \end{cases}$$

i.e., ξ_n 's gives equal probability to each of the $2d$ -unit vectors $\{\mathbf{e}_1, -\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_d, -\mathbf{e}_d\}$ where $\mathbf{e}_s = (\underbrace{0, 0, \dots, 0}_{s-1}, 1, \underbrace{0, 0, \dots, 0}_{d-s})$. ■

Example 5.13. Random walk on the circle : Let us assume that the possible states be $\mathcal{S} = \{0, 1, 2, \dots, N-1\}$, but they are ordered cyclically so that $N-1$ has the neighbour 0 and $N-2$ and 0 has neighbours 2 and $N-1$ (one should take it as a modulo (N) operation). In order to have a more visual picture, think of a N -gon, whose vertices are numbered 0, 1 and so on sequentially and a particle is doing a random walk on the set of vertices. If, as before, the particle passes either to the right or to the left neighbour, the position of the particle at time n , will constitute a Markov chain rows

with transition matrix $\mathbf{P}$ given by

$$\mathbf{P} = \begin{array}{c} \begin{array}{cccccccc} & 0 & 1 & 2 & 3 & \cdots & N-3 & N-2 & N-1 \\ \begin{array}{c} 0 \\ 1 \\ 2 \\ \vdots \\ N-2 \\ N-1 \end{array} & \begin{bmatrix} 0 & p & 0 & 0 & \cdots & 0 & 0 & q \\ q & 0 & p & 0 & \cdots & 0 & 0 & 0 \\ 0 & q & 0 & p & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & q & 0 & p \\ p & 0 & 0 & 0 & \cdots & 0 & q & 0 \end{bmatrix} \end{array} \end{array}.$$

A more general situation will be where the particle is allowed to have jumps of size more than 1. Let $\alpha_0, \alpha_1, \dots, \alpha_{N-1}$ be, respectively, the probability of staying fixed or moving $1, 2, \dots, N-1$ units to the right (where k units to the right is the same as $N-k$ units to the left). Then $\mathbf{P}$ is the transition matrix

$$\mathbf{P} = \begin{array}{c} \begin{array}{cccccc} & 0 & 1 & 2 & \cdots & N-2 & N-1 \\ \begin{array}{c} 0 \\ 1 \\ 2 \\ \vdots \\ N-1 \end{array} & \begin{bmatrix} \alpha_0 & \alpha_1 & \alpha_2 & \cdots & \alpha_{N-2} & \alpha_{N-1} \\ \alpha_{N-1} & \alpha_0 & \alpha_1 & \cdots & \alpha_{N-3} & \alpha_{N-2} \\ \alpha_{N-2} & \alpha_{N-1} & \alpha_0 & \cdots & \alpha_{N-4} & \alpha_{N-3} \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots \\ \alpha_1 & \alpha_2 & \alpha_3 & \cdots & \alpha_{N-1} & \alpha_0 \end{bmatrix} \end{array} \end{array}.$$

One important observation that will be used later is that the transition matrix is circular, hence its column sum is also 1. Such a matrix is called a *doubly stochastic matrix*. This gives us: $\mathbf{1P} = \mathbf{1}$ where $\mathbf{1}$ is the N -dimensional row vector $\mathbf{1} = (1, 1, \dots, 1)$. ■

In general, for random walks, the transition probabilities do not change when states are translated, i.e.,

$$p_{i+k, j+k} = \mathbb{P}(\xi_1 = j - i) = p_{i, j} \text{ for all } k \in \mathcal{S}.$$

This implies, in some sense, using the same coin for every state. However, we may consider that different coins, with varying probability of success, are used at different states. Such models are classified under a generic model of *birth and death* chains.

Birth and death chains

The name birth and death chain comes from the transitions that specify that the transitions can happen to a state to only its nearest neighbours including itself, i.e., $p_{ij} = 0$ if $|i - j| > 1$. The transition of i to $i + 1$ can be interpreted as a birth, while the transition from i to $i - 1$ is interpreted as a death. Clearly, different choices of the parameters will yield the models described above. Various models used in physical literature fall into this category, whose examples are provided below.

Example 5.14. Finite state birth and death chains : These models are defined on a finite state space $\mathcal{S} = \{0, 1, 2, \dots, N - 1, N\}$. The transition matrix $\mathbf{P}$ for the finite state space is given as follows: let $p_i > 0, q_i > 0, r_i \geq 0$ for $i = 1, 2, \dots, N - 1$ with $p_i + q_i + r_i = 1$ and $p_0 \geq 0, r_0 \geq 0$ with $p_0 + r_0 = 1$ and $q_N \geq 0, r_N \geq 0$ with $q_N + r_N = 1$. The transition matrix is given by

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & \cdots & N-2 & N-1 & N \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ \vdots \\ N-1 \\ N \end{matrix} & \begin{bmatrix} r_0 & p_0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ q_1 & r_1 & p_1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & q_2 & r_2 & p_2 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & q_{N-1} & r_{N-1} & p_{N-1} \\ 0 & 0 & 0 & 0 & \cdots & 0 & q_N & r_N \end{bmatrix} \end{matrix}.$$

We note that when $r_0 = 1$ (i.e., $p_0 = 0$), we have $p_{00} = 1$. This implies that once the chain enters the state 0, it stays forever after at 0, i.e., if $X_n = 0$, then $X_{n+k} = 0$ for all $k \geq 1$. Note, the same thing happens if $r_N = 1$, i.e., if the chain hits the state N , then it stays there. We will show that if $r_0 = 1$ and $r_N < 1$, then chain will get fixed at 0, while if $r_0 < 1, r_N = 1$, the chain will get fixed at N . If both $r_0 = r_N = 1$, then the chain will get fixed at either 0 or N .

Example 5.15. Infinite state birth and death chains : These models are defined on the infinite state space $\mathcal{S} = \{0, 1, 2, \dots\}$. The transition matrix $\mathbf{P}$ for the infinite state space is given as follows: let $p_i > 0, q_i > 0, r_i \geq 0$ for $i \geq 1$ with $p_i + q_i + r_i = 1$

and $p_0 \geq 0, r_0 \geq 0$ with $p_0 + r_0 = 1$ and the transition matrix is given by

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & \cdots & i-1 & i & i+1 & \cdots \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ \vdots \\ i \\ \vdots \end{matrix} & \begin{bmatrix} r_0 & p_0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots \\ q_1 & r_1 & p_1 & 0 & \cdots & 0 & 0 & 0 & \cdots \\ 0 & q_2 & r_2 & p_2 & \cdots & 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots \\ 0 & 0 & 0 & 0 & \cdots & q_i & r_i & p_i & \cdots \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots \end{bmatrix} \end{matrix}.$$

Example 5.16. Random walk with absorbing boundaries – gambler’s ruin :

A random walk on a finite number of states with two end points being sticky is of particular interest to us. This is known as the gambler’s ruin problem. Think of two gamblers, Gambler I and Gambler II, who has sequence of bets of Rs. 1, each bet being played independently, until one of them is bankrupt. Each bet, Gambler I can win with probability p and Gambler II with probability $q = 1 - p$. If Gambler I wins the bet, his fortune increases by Rs. 1, while the fortune of Gambler II reduces by Rs. 1. Let us assume that starting fortune of Gambler I is Rs. a while the starting fortune of Gambler II is Rs. b . Set $N = a + b$, the total fortune. In order to formulate this as a Markov chain, set $\mathcal{S} = \{0, 1, \dots, N\}$ and the fortune of Gambler I at time n by S_n . We note that if $S_n = 0$ for some $n \geq 1$, then the Gambler I has lost all his money and is bankrupt and the game ends at that point. Then, we can set $S_{n+1} = 0$. Hence, we must have $p_{00} = 1$. Similarly, if $S_n = N$ for some $n \geq 1$, then the Gambler II is bankrupt. Then, we can set $S_{n+1} = N$ and $p_{NN} = 1$. If $S_n = i$, for $1 \leq i \leq N - 1$, we have $S_{n+1} = i + 1$ with probability p and $S_{n+1} = i - 1$ with probability $q = 1 - p$. Thus, the transition matrix $\mathbf{P}$ is given by

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & \cdots & N-2 & N-1 & N \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ \vdots \\ N-1 \\ N \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ q & 0 & p & 0 & \cdots & 0 & 0 & 0 \\ 0 & q & 0 & p & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & q & 0 & p \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 1 \end{bmatrix} \end{matrix}.$$

In order to show that this is a Markov chain, we consider the **iid** sequence of random variables $\{\xi_n : n \geq 1\}$ with $\mathbb{P}(\xi_n = 1) = p = 1 - q = 1 - \mathbb{P}(\xi_n = -1)$. Now, consider the function $f : \{0, 1, \dots, N\} \times \{1, -1\} \rightarrow \{0, 1, \dots, N\}$ as follows:

$$f(i, \xi) = i + \xi \mathbb{I}_{\{0, N\}^c}(i).$$

We will prove later that the game cannot continue forever and also compute the probability that the Gambler I or Gambler II is bankrupt, i.e., the Markov chain will eventually be fixed in on the states 0 or N .

Playing against infinitely rich adversary : An interesting variant of the above model is the random walk on the non-negative integers with only one absorbing state. Here, the state space is $\mathcal{S} = \{0, 1, 2, \dots\}$, with $p_{00} = 1$ and for $i \geq 1$, $p_{i,i+1} = p$ and $p_{i,i-1} = q = 1 - p$. This can be interpreted as the gambler's ruin problem with one player having infinite fortune. The transition matrix $\mathbf{P}$ is given by

$$\mathbf{P} = \begin{array}{c} \begin{array}{cccccccc} 0 & 1 & 2 & 3 & \dots & i-1 & i & i+1 & \dots \end{array} \\ \begin{array}{c} 0 \\ 1 \\ \vdots \\ i \\ \vdots \end{array} \begin{bmatrix} 1 & 0 & 0 & 0 & \dots & 0 & 0 & 0 & \dots \\ q & 0 & p & 0 & \dots & 0 & 0 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots & \dots \\ 0 & 0 & 0 & 0 & \dots & q & 0 & p & \dots \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots & \dots \end{bmatrix} \end{array}.$$

We may modify the function f defined above by dropping the $i = N$ case. This will prove that this process is a Markov chain. It is clear that the player can become bankrupt with positive probability. The interesting question here is that when is this probability 1? ■

Example 5.17. Random walk with reflecting barriers : An interesting variant of the preceding example is represented by the chain with possible states $\mathcal{S} = \{0, 1, 2, \dots, N\}$ and transition probability matrix $\mathbf{P}$ given by

$$\mathbf{P} = \begin{array}{c} \begin{array}{ccccccccc} 0 & 1 & 2 & 3 & \dots & N-2 & N-1 & N \end{array} \\ \begin{array}{c} 0 \\ 1 \\ \vdots \\ N-1 \\ N \end{array} \begin{bmatrix} 0 & 1 & 0 & 0 & \dots & 0 & 0 & 0 \\ q & 0 & p & 0 & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & q & 0 & p \\ 0 & 0 & 0 & 0 & \dots & 0 & 1 & 0 \end{bmatrix} \end{array}.$$

This chain may be interpreted in gambling language by considering two players playing for unit stakes with the agreement that every time a player is bankrupt, his adversary gives him Rs. 1 in order that the game can be continued. In the particle system terminology, this can be interpreted as: the particle gets reflected at 0 and N , otherwise it does a random walk.

An infinite state space version of this may also be considered, which can be interpreted that the particle gets reflected only at 0. The transition matrix is given by

$$\mathbf{P} = \begin{array}{c} \begin{array}{cccccccccc} & 0 & 1 & 2 & 3 & \cdots & i-1 & i & i+1 & \cdots \\ \begin{array}{c} 0 \\ 1 \\ \vdots \\ i \\ \vdots \end{array} & \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots \\ q & 0 & p & 0 & \cdots & 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots \\ 0 & 0 & 0 & 0 & \cdots & q & 0 & p & \cdots \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots \end{bmatrix} \end{array} \end{array}.$$

Another version of the model is also considered, which is referred as the *retaining* or *elastic* barrier. Here, it is assumed that when the particle hits the barrier it may spend some time at the barrier before being reflected back. This is achieved by setting $p_{00} = q$ and $p_{01} = p$. The properties of both the models are quite similar (but not exactly same though). In this case, the transition probability matrix is given by

$$\mathbf{P} = \begin{array}{c} \begin{array}{cccccccccc} & 0 & 1 & 2 & 3 & \cdots & i-1 & i & i+1 & \cdots \\ \begin{array}{c} 0 \\ 1 \\ \vdots \\ i \\ \vdots \end{array} & \begin{bmatrix} q & p & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots \\ q & 0 & p & 0 & \cdots & 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots \\ 0 & 0 & 0 & 0 & \cdots & q & 0 & p & \cdots \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots \end{bmatrix} \end{array} \end{array}.$$

Example 5.18. The Ehrenfest model of diffusion : Ehrenfest described a conceptual urn experiment where N molecules are distributed in two urns A and B. At each trial a molecule is chosen at random and moved from its urn to the other. The state of the system is determined by the number of molecules in the urn A. Suppose that at a certain moment there are exactly i molecules in the urn A. At the next trial the system passes into $i-1$ or $i+1$ according to whether a molecule in urn A or urn B is chosen; the corresponding probabilities are i/N and $(N-i)/N = 1 - i/N$.

Clearly the above is an example of a model of a finite state space birth and death chain. We consider a chain with the $N+1$ states, i.e., $\mathcal{S} = \{0, 1, \dots, N\}$ which represents the number of molecules in the urn A. Here, we set $r_i = 0, p_i = 1 - i/N$ and $q_i = i/N$ so that the transition matrix $\mathbf{P}$ is given by :

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & \cdots & N-2 & N-1 & N \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ \vdots \\ N-1 \\ N \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ \frac{1}{N} & 0 & \frac{N-1}{N} & 0 & \cdots & 0 & 0 & 0 \\ 0 & \frac{2}{N} & 0 & \frac{N-2}{N} & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \\ 0 & 0 & 0 & 0 & \cdots & \frac{N-1}{N} & 0 & \frac{1}{N} \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 & 0 \end{bmatrix} \end{matrix}.$$

A modified version of the above model has also been studied which is known as the *modified Ehrenfest chain*. In this model, at each discrete time, independently of the past, we select an atom at random and a urn at random. We then put the chosen atom in the chosen urn. Therefore, if the chosen atom is from the chosen urn, no change in state happens, whereas if the chosen atom is from the urn which has not been chosen the state changes by 1. Thus, this again a model of a finite state space birth and death chain. Here, the transition probabilities are given by

$$p_{i,i-1} = \frac{i}{2N}, \quad p_{i,i} = \frac{1}{2} \quad \text{and} \quad p_{i,i+1} = \frac{N-i}{2N}.$$

This chain can also be interpreted as diffusion with a central force, that is, a random walk in which the probability of a step to the right varies with the position. From the state j , the particle is more likely to move to a state which is to the right or to the left according as $j < N/2$ or $j > N/2$. This will imply that the particle has a tendency to move toward the central state $N/2$, which corresponds to an attractive elastic force increasing in direct proportion to the distance. ■

Example 5.19. The Bernoulli-Laplace model of diffusion : A model similar to the Ehrenfest model was proposed by D. Bernoulli as a probabilistic analogue for the flow of two incompressible liquids between two containers. This time we have a total of $2N$ particles among which N are black and N white. Since these particles are supposed to represent incompressible liquids the densities must not change, and so the number N of particles in each urn remains constant. We say that the system is in state k , where

$k = 0, 1, 2, \dots, N$ if the first urn contains k white particles. This implies that it contains $N - k$ black particles while the second urn contains $N - k$ white and k black particles. At each trial one particle is chosen from each urn and these two particles are interchanged. The transition probabilities are then given by

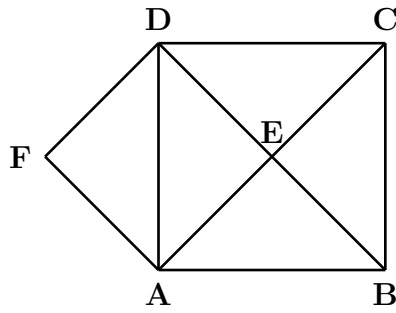
$$p_{j,j-1} = \frac{j^2}{N^2}, \quad p_{j,j} = \frac{2j(N-j)}{N^2} \quad \text{and} \quad p_{j,j+1} = \frac{(N-j)^2}{N^2}. \quad \blacksquare$$

Example 5.20. Random walk on graphs : The above example of random walks on a d -dimensional Euclidean lattice can be thought in general (connected) graphs. Let us consider a graph $G = (V, E)$ where V is the vertex set and E is the edge set. We impose the condition that there are no multiple edges or self loops in the graphs, i.e., there can be at most one edge between two vertices and there is no edge from a vertex to itself. For any vertex u , let $N(u)$ be set of vertices which are neighbours of the vertex u (the set of all vertices which share an edge with u). We assume that for each $u \in V$, the set $N(u)$ is finite. The cardinality of the set $N(u)$ is called the degree of the vertex and is denoted by $\deg(u)$.

Suppose a walker is performing a random walk on the set V in the following manner: at any given time point, the walker is at the vertex u . At the next time point, the walker moves to a one vertex chosen at random from the set $N(u)$. The position of the walker forms a Markov chain, with the state space being the set of vertices V of the graph. We leave the construction of the function f (see Proposition 5.7) to the reader. The transition matrix $\mathbf{P} = (p_{uv})$ is given by

$$p_{uv} = \frac{1}{\deg(u)} \mathbb{I}_{N(u)}(v) = \begin{cases} \frac{1}{\deg(u)} & \text{if } v \in N(u) \\ 0 & \text{otherwise.} \end{cases}$$

We illustrate the above with an example: suppose that graph $G = (V, E)$ is given by the following figure where the vertex set $V = \{A, B, C, D, E, F\}$ and the edge-set $E = \{(A, B), (A, D), (A, E), (A, F), (B, C), (B, E), (C, D), (C, E), (D, F)\}$. Observe that $N(A) = \{B, D, E, F\}$, $N(B) = \{A, C, E\}$ etc. The random walk on the graph is a Markov chain on $\mathcal{S} = V$ with the transition matrix $\mathbf{P}$:



$$\mathbf{P} = \begin{matrix} & \begin{matrix} A & B & C & D & E & F \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \\ E \\ F \end{matrix} & \begin{bmatrix} 0 & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 \\ 0 & \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 \\ \frac{1}{4} & 0 & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix} \end{matrix}.$$

It is clear that for the d -dimensional Euclidean lattice, every vertex has degree $2d$. Therefore, the random walk on the d -dimensional Euclidean lattice is actually the simple symmetric random walk described in Examples 5.11 and 5.12.

A more generalized version can be thought where each pair of vertices u and v in a graph will have non-negative weight $w_{u,v}$ which is symmetric, i.e., $w_{u,v} = w_{v,u}$. The absence of the edge between two vertices may be thought as $w_{u,v} = w_{v,u} = 0$. Assume also that $w_{u,u} = 0$ for all vertices $u \in V$. Further, we assume that the total weight emanating is finite for each vertex $u \in V$, i.e., $w_{u,\cdot} := \sum_{z \in V} w_{u,z} < \infty$ for all $u \in V$. Now, we may consider the random walker taking the step from a vertex u to another vertex v with probabilities proportional to the corresponding weight of the edge. Hence, for $u, v \in V$, we have

$$p_{uv} = \frac{w_{uv}}{w_{u,\cdot}}.$$

If we take $w_{u,v} = 1$ if the vertices share an edge and otherwise 0, this model reduces to the previous version. A modified version of this model has been used in Google's Page Rank algorithm. In the page rank model, each web site is thought as a vertex of the so called internet graph. If there is a link on of the pages which points to the other page (multiple links are ignored or counted as a single link), then they are said to have an edge between them. The random walker will traverse between edges for a long time and pages are given ranks according to the frequency of the long term visit counts. This is related to the stationary distribution of the Markov Chain which we will discuss in later.

In the modified model one thinks of a dampening effect of continuing with the random walk. Let $\mathcal{A} = \{v_1, v_2, \dots, v_k\}$ be a set of fixed vertices. At each time point, it is postulated that a coin with success probability α is tossed. On success, the web surfer

uses the random walk model by choosing a vertex uniformly from the neighbour. However, on failure, the surfer chooses one the vertices in the set $\mathcal{A}$ uniformly and visits that page. More precisely, we set

$$p_{uv}^{(\mathcal{A})} = \frac{1}{|\mathcal{A}|} \mathbb{I}_{\mathcal{A}}(v) = \begin{cases} \frac{1}{|\mathcal{A}|} & \text{if } v \in \mathcal{A} \\ 0 & \text{otherwise.} \end{cases}$$

The modified transition probability is given by the mixture of two transition probabilities:

$$p_{uv}^{(\text{modified})} = \alpha p_{uv}^{(\mathcal{A})} + (1 - \alpha) p_{uv}. \quad \blacksquare$$

Exercise 5.21. Suppose that a chess piece (King, Queen, Rook, Bishop or Knight), placed at a square on a chessboard, moves to one of the squares it can legally move picked at random. Formulate the movement of the piece as a random walk on a suitably defined graph. Find its transition matrix. $\blacksquare$

5.2.2 Queueing model

Example 5.22. We now describe a discrete queueing model as a Markov chain. Customers arrive for service and take their place in a waiting line. During each period of time a single customer is served, provided that at least one customer is present. If no customer awaits service then during this period no service is performed. To fix ideas, we can imagine a taxi stand at which a taxis arrives at fixed time intervals to give service. If no one is present the cab immediately departs. During a service period new customers may arrive. We suppose the actual number of arrivals in the n -th period is a random variable ξ_n whose distribution function is independent of the period and is given by $\mathbb{P}(\xi_n = k) = \alpha_k$, for $k = 0, 1, \dots$. We also assume the random variables ξ_n are independent. The state of the system at the start of each period is defined to be the number of customers waiting in line for service.

We set the function $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ as follows :

$$f(i, \xi) = \max(i - 1, 0) + \xi = (i - 1)^+ + \xi.$$

It is clear that we have $X_{n+1} = f(X_n, \xi_{n+1})$ and it is a Markov chain with state space

$\mathcal{S} = \{0, 1, 2, \dots\}$. Thus, transition probabilities are given by

$$p_{ij} = \mathbb{P}(f(i, \xi) = j) = \begin{cases} \alpha_j & \text{if } i = 0, j \geq 0 \\ \alpha_{j-(i-1)} & \text{if } i \geq 1, j \geq i-1 \\ 0 & \text{otherwise.} \end{cases} \quad (5.7)$$

Thus the transition probability matrix $\mathbf{P}$ is given by

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & \dots \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ \vdots \end{matrix} & \begin{bmatrix} \alpha_0 & \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 & \dots \\ \alpha_0 & \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 & \dots \\ 0 & \alpha_0 & \alpha_1 & \alpha_2 & \alpha_3 & \dots \\ 0 & 0 & \alpha_0 & \alpha_1 & \alpha_2 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \dots \end{bmatrix} \end{matrix}.$$

Note here that the chain can jump to the right by any distance however it may jump to the left by only one step, i.e., $p_{ij} = 0$ for $j < i - 1$. Such chains are called *left non-skipping*. Similarly, if $p_{ij} = 0$ for $j > i + 1$, the chain is called *right non-skipping*.

One more observation that is important is that the transition probabilities behave in a regular fashion. For $i, j \geq 1$, we have that $p_{i,i+k} = p_{j,j+k} = p_{1,1+k}$ for all $k \geq -1$. This is a space stationarity, i.e., if the particle is at $i \geq 1$, it is unable to differentiate between states and behave the same way as if it was at state 1. We will come back to this at a later stage. ■

5.2.3 Renewal event chains

Example 5.23. Renewal events and residual waiting times : The chain with state space $\mathcal{S} = \{0, 1, 2, \dots\}$ and transition probability matrix $\mathbf{P}$ given by

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & \dots \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ \vdots \end{matrix} & \begin{bmatrix} f_1 & f_2 & f_3 & f_4 & f_5 & \dots \\ 1 & 0 & 0 & 0 & 0 & \dots \\ 0 & 1 & 0 & 0 & 0 & \dots \\ 0 & 0 & 1 & 0 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \dots \end{bmatrix} \end{matrix}.$$

The probabilities $\{f_k : k \geq 1\}$ are arbitrary except that they must add to unity. To visualize the process suppose that it starts from the initial state 0. If the first step leads to k the system is bound to pass successively through $k-1, k-2, k-3$ and so on and at the k^{th} step the system returns to 0, whence the process starts from scratch. The successive returns to 0 thus represent a recurrent event with the distribution $\{f_k : k \geq 1\}$ for the recurrence times. The state of the system at any time represents the residual renewal time of the system. ■

Example 5.24. Another Markov chain connected with recurrent events is described as follows: let the state space be $\mathcal{S} = \{0, 1, 2, \dots\}$. Consider the following transition matrix $\mathbf{P}$ on $\mathcal{S}$

$$\mathbf{P} = \begin{array}{c} \begin{array}{cccccc} & 0 & 1 & 2 & 3 & 4 & \dots \\ \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \\ \vdots \end{array} & \begin{bmatrix} q_0 & p_0 & 0 & 0 & 0 & \dots \\ q_1 & 0 & p_1 & 0 & 0 & \dots \\ q_2 & 0 & 0 & p_2 & 0 & \dots \\ q_3 & 0 & 0 & 0 & p_3 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \dots \end{bmatrix} \end{array} \end{array}.$$

where $p_k + q_k = 1$ for all $k \geq 0$. We may interpret the state i as representing the *age* of the system. When the system reaches age k the ageing process continues with probability p_k , but with probability q_k it dies and the age is reset to 0. With probability p_0 the system starts afresh with age zero. The successive passages through the state 0 again represent a recurrent event. The case when $p_0 = 0$, the process gets stuck at 0. This clearly happens with positive probability, but the interesting question is when will happen with probability 1?

Success Run length : As a special case, consider a sequence of Bernoulli trials. We say that the n^{th} trial the system is in the state k if the last failure occurred at the trial number $n - k$ for $k = 0, 1, \dots$. In other words, the index k equals the length of the uninterrupted block of successes ending at the n -th trial. The transition probabilities are those of the preceding example with $p_k = p$ and $q_k = q$ for all k . ■

5.2.4 Genetics

Example 5.25. Wright Fisher model in population genetics The following idealized genetics model was introduced by S. Wright to investigate the fluctuation of gene

frequency under the influence of mutation and selection. We begin by describing a so-called simple haploid model of random reproduction, disregarding mutation pressures and selective forces. We assume that we are dealing with a fixed population size of N genes composed of type-a and type-A individuals. The makeup of the next generation is determined by N independent binomial trials as follows: If the parent population consists of j a-genes and $N - j$ A-genes then each trial results in a or A with probabilities

$$p_j = \frac{j}{N} \quad \text{and} \quad q_j = \frac{N - j}{N}$$

respectively. Repeated selections are done with replacement. By this procedure we generate a Markov chain $\{X_n : n \geq 0\}$ where X_n is the number of a-genes in the n^{th} generation among a constant population size of N elements. The state space contains the $N + 1$ values $\mathcal{S} = \{0, 1, 2, \dots, N\}$. The transition probability matrix is computed according to the binomial distribution as

$$p_{jk} = \binom{N}{k} p_j^k q_j^{N-k} \quad (5.8)$$

for $j, k \in \mathcal{S}$. Observe that $p_{00} = p_{N,N} = 1$. If $X_n = 0$, the population consists of only a-genes, while if $X_n = N$, the population consists of only A-genes. These are called pure states. We will show that under this model, population will get fixed in one of these two states.

A more realistic model takes account of mutation pressures. We assume that prior to the formation of the new generation each gene has the possibility to mutate, that is, to change into a gene of the other kind. Specifically, we assume that for each gene the mutation $a \rightarrow A$ occurs with probability α_1 and $A \rightarrow a$ occurs with probability α_2 . Again we assume that the composition of the next generation is determined by N independent binomial trials. The relevant value of p_j and q_j when the parent population consists of j a-genes are now taken to be

$$p_j = \frac{j}{N}(1 - \alpha_1) + \frac{N - j}{N}\alpha_2 \quad \text{and} \quad q_j = \frac{j}{N}\alpha_1 + \frac{N - j}{N}(1 - \alpha_2) \quad (5.9)$$

The idea is as follows: we assume that the mutation pressures are operating in the beginning, after which a new gene is determined by selecting at random from the population. Now the probability of selecting an a-gene after the mutation forces have acted is just $1/(N)$ times the number of a-genes present: hence the average probability (averaged with respect to the possible mutations) is simply $1/N$ times the average number of a-genes

after mutation. But this average number is clearly $\frac{j}{N}(1 - \alpha_1) + \frac{N-j}{N}\alpha_2$ and the transition probabilities are given by (5.8). If $\alpha_1\alpha_2 > 0$, one should observe that $p_{00} < 1, p_{N,N} < 1$. We will show that later that under the condition $\alpha_1\alpha_2 > 0$, the population does not get fixed at the pure states, rather will approach a steady state distribution.

We can also introduce the concept of a selection force operating in favour of, say, a-genes. Suppose we wish to impose a selective advantage for a-genes over A-genes so that the relative number of offspring have expectations proportional to $1 + s$ and 1, respectively, where s is small and positive. We replace $p_j = j/N$ and $q_j = (N - j)/N$ by

$$p_j = \frac{(1 + s)j}{N + sj} \quad \text{and} \quad q_j = 1 - p_j \quad (5.10)$$

and the transition probabilities are given by (5.8). This is easy to see that the ratio of expected population size of the a-genes to A-genes will get multiplied by a factor $(1 + s)$ in each generation. Observe that, in this case also, $p_{0,0} = p_{N,N} = 1$ and again the population will eventually be fixed in one of these two pure states 0, N . ■

Example 5.26. Markov chain in cell genetics : A biological problem which may be described roughly as follows. Each cell of a certain organism contains N particles, some of which are of type A, the others of type B. The cell is said to be in state j , for $j = 0, 1, 2, \dots, N$, if it contains exactly j particles of type A. Daughter cells are formed by cell division, but prior to the division each particle replicates itself; the daughter cell inherits N particles chosen at random from the $2j$ particles of type A and $N - 2j$ particles of type B present in the parental cell. The probability that a daughter cell is in state k is then given by the hypergeometric probability

$$p_{jk} = \frac{\binom{2j}{k} \binom{2N-2j}{N-k}}{\binom{2N}{N}}$$

for $j, k \in \mathcal{S}$ with the convention that $\binom{a}{b} = 0$ if $a < b$. We should note here also that $p_{00} = p_{N,N} = 1$. ■

5.2.5 Miscellaneous examples

Example 5.27. Branching process : We have already studied branching process (see Chapter 3). We can frame branching process also as a Markov chain $\{Z_n : n \geq 0\}$ on the set of non-negative integers. We take $\{(\xi_t^{(k)} : t \geq 1) : k \geq 1\}$ as the independent

sequence. Then, we have $f : \{0, 1, \dots\} \times \{\{\xi_t : t \geq 1\}\} \rightarrow \{0, 1, \dots\}$ by

$$f(i, \{\xi_t : t \geq 1\}) = \begin{cases} 0 & \text{if } i = 0 \\ \sum_{t=1}^i \xi_t & \text{otherwise.} \end{cases}$$

Then, $Z_{n+1} = f(Z_n, \{\xi_t^{(n+1)} : t \geq 1\})$. Thus, $\{Z_n : n \geq 0\}$ is a Markov chain. ■

Example 5.28. Let us consider a generic example of a Markov chain. Consider the state space $\mathcal{S} = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and a transition matrix $\mathbf{P}$ given by

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \end{matrix} & \begin{bmatrix} \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{2}{5} & 0 & 0 & \frac{1}{5} & 0 & \frac{2}{5} \\ \frac{1}{3} & 0 & 0 & \frac{2}{3} & 0 & 0 & 0 & 0 \\ 0 & \frac{2}{3} & \frac{1}{6} & 0 & 0 & \frac{1}{6} & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{9} & \frac{2}{9} & 0 & 0 & 0 & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & \frac{1}{3} & 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} \end{bmatrix} \end{matrix}.$$

It is easy to see that we can construct a process which is a Markov chain on this state space with $\mathbf{P}$ as the transition probability matrix. We will use this example to carry out several computations to illustrate different properties of a finite state Markov chain. ■

Before we proceed, we provide an example of a process which is not a Markov chain.

Example 5.29. Urn model : An urn contains b black and r red balls. A ball is drawn at random. It is replaced and, moreover, c balls of the color drawn are added to the urn. A new random drawing is made from the urn (now containing $r + b + c$ balls), and this procedure is repeated. For $n = 0, 1, 2, \dots$, let X_n denote the colour (R for read and B for black) of ball the ball drawn at the $(n + 1)$ -th draw. Now, we note that

$$\begin{aligned} \mathbb{P}(X_2 = R, X_1 = R, X_0 = R) &= \frac{r}{r+b} \times \frac{r+c}{r+b+c} \times \frac{r+2c}{r+b+2c} \\ \mathbb{P}(X_2 = R, X_1 = R, X_0 = B) &= \frac{b}{r+b} \times \frac{r}{r+b+c} \times \frac{r+c}{r+b+2c} \\ \mathbb{P}(X_1 = R, X_0 = R) &= \frac{r}{r+b} \times \frac{r+c}{r+b+c} \\ \mathbb{P}(X_1 = R, X_0 = B) &= \frac{b}{r+b} \times \frac{r}{r+b+c}. \end{aligned}$$

Therefore, we have

$$\begin{aligned}
 \mathbb{P}(X_1 = R) &= \mathbb{P}(X_1 = R, X_0 = R) + \mathbb{P}(X_1 = R, X_0 = B) \\
 &= \frac{r}{r+b} \times \frac{r+c}{r+b+c} + \frac{b}{r+b} \times \frac{r}{r+b+c} \\
 &= \frac{r}{r+b}; \\
 \mathbb{P}(X_2 = R, X_1 = R) &= \mathbb{P}(X_2 = R, X_1 = R, X_0 = R) \\
 &\quad + \mathbb{P}(X_2 = R, X_1 = R, X_0 = B) \\
 &= \frac{r}{r+b} \times \frac{r+c}{r+b+c} \times \frac{r+2c}{r+b+2c} + \frac{b}{r+b} \times \frac{r}{r+b+c} \times \frac{r+c}{r+b+2c} \\
 &= \frac{r}{r+b} \times \frac{r+c}{r+b+c}.
 \end{aligned}$$

Thus, we have

$$\begin{aligned}
 \mathbb{P}(X_2 = R | X_1 = R) &= \frac{\mathbb{P}(X_2 = R, X_1 = R)}{\mathbb{P}(X_1 = R)} \\
 &= \frac{r}{r+b} \times \frac{r+c}{r+b+c} \times \frac{r+b}{r} = \frac{r+c}{r+b+c}
 \end{aligned}$$

and

$$\begin{aligned}
 \mathbb{P}(X_2 = R | X_1 = R, X_0 = R) &= \frac{\mathbb{P}(X_2 = R, X_1 = R, X_0 = R)}{\mathbb{P}(X_1 = R, X_0 = R)} \\
 &= \frac{r}{r+b} \times \frac{r+c}{r+b+c} \times \frac{r+2c}{r+b+2c} \times \frac{r+b}{r} \times \frac{r+b+c}{r+c} \\
 &= \frac{r+2c}{r+b+2c}.
 \end{aligned}$$

Thus, clearly $\{X_n : n \geq 1\}$ is not a Markov chain, unless $c = 0$. In case $c = 0$, we have the X_n 's are independent and identically distributed. ■

Exercise 5.30. Let $\{\xi_n : n \geq 0\}$ be a sequence of independent identically distributed random variables taking values in $\{-1, 1\}$ with $\mathbb{P}(\xi_n = 1) = p = 1 - \mathbb{P}(\xi_n = -1)$. Let $X_n = \xi_n \xi_{n+1}$, for all $n \geq 0$. Is $\{X_n : n \geq 0\}$ a Markov chain? ◀

Exercise 5.31. Let $\{\xi_n : n \geq 1\}$ be a sequence of independent identically distributed integer-valued random variables. Let $S_0 = 0$ and for $n \geq 1$, $S_n = \sum_{i=1}^n \xi_i$. Also define $X_0 = 0$, $X_1 = \xi_1$ and for $n \geq 2$, $X_n = \xi_n + \xi_{n-1}$ and $Y_0 = 0$ and for $n \geq 1$, $Y_n = \sum_{k=1}^n S_k$. Which of the following constitute Markov chains : (a) $\{X_n : n \geq 0\}$, (b) $\{Y_n : n \geq 0\}$ and (c) the sequence of pairs $\{(S_n, Y_n) : n \geq 0\}$? ◀

Exercise 5.32. A die is rolled repeatedly. The results be denoted by $\{\xi_n : n \geq 1\}$. Which of the following are Markov chains? For those that are, find the transition matrix.

1. The largest number $\{M_n : n \geq 0\}$ shown up to the $(n + 1)$ -th roll, i.e., $M_n = \max\{\xi_1, \dots, \xi_{n+1}\}$.
2. The number N_n of sixes in n rolls, i.e., $N_n = \sum_{k=1}^n \mathbb{I}(\epsilon_{k+1} = 6)$.
3. The number of throws since the last six, i.e., $L_n = n - \max\{k \leq n : \epsilon_{k+1} = 6\}$, with the convention that if the set $\{k \leq n : \epsilon_{k+1} = 6\}$ is empty, the maximum is taken to be 0.
4. The number of throws until the next six, $R_n = \min\{k > n : \epsilon_{k+1} = 6\} - n$. ◀

Exercise 5.33. Suppose that $\{\xi_n : n \geq 1\}$ denote the scores obtained at the n -th throw of a fair dice. Set $S_n = (\epsilon_{n+1} + \epsilon_{n+2}) \bmod 6$ for $n \geq 0$. Show that $\{S_n : n \geq 1\}$ is a Markov Chain. Find the transition probability matrix for this chain. ◀

Exercise 5.34. Assume that a gambler, playing against an opponent with infinite capital, has the following strategy: he bets Rs. 1 unless he has won last two bets, in which case he bets Rs 2. Suppose that his initial capital was Rs. $a > 0$ and the probability that he wins the n^{th} bet is $0 < p < 1$, independent of all other things. Let $\{S_n : n \geq 0\}$ be the fortune of the gambler at time n . Show that $\{S_n : n \geq 0\}$, $\{Y_n := (S_n, S_{n+1}) : n \geq 0\}$ are not Markov chains. Show that $\{Z_n := (S_n, S_{n+1}, S_{n+2}) : n \geq 0\}$ is a Markov chain on $\mathcal{S} \times \mathcal{S} \times \mathcal{S}$ and find its transition matrix. ◀

Exercise 5.35. Let $\{S_n : n \geq 0\}$ be the simple symmetric ($p = 1/2$) random walk in one dimension with $S_0 = 0$.

- Show that $\{|S_n| : n \geq 0\}$ a Markov chain on $\{0, 1, 2, \dots\}$. Find its transition matrix.
- Let $M_n = \max\{S_k : 0 \leq k \leq n\}$, and show that $Y_n = M_n - S_n$ defines a Markov chain. Find the transition probabilities of this chain.

What happens if the random walk is simple but asymmetric? What happens if the starting position is not the origin, i.e., $S_0 \neq 0$. ◀

Exercise 5.36. The snake chain : Let $\{X_n : n \geq 0\}$ be a Markov chain on the state space $\mathcal{S}$ and transition matrix $\mathbf{P}$. Set, for $k \geq 1$ and $n \geq 0$,

$$Y_n := (X_n, X_{n+1}, \dots, X_{n+k}). \quad (5.11)$$

Show that the process $\{Y_n : n \geq 0\}$, taking values $\mathcal{S}^{k+1}$, is a Markov chain. What is its transition matrix. The Markov chain $\{Y_n : n \geq 0\}$ is often referred as the snake chain of length $k + 1$ associated with $\{X_n : n \geq 0\}$.

Consider the chain $\{X_n : n \geq 0\}$, where X_n 's represents the result of independent tosses of a coin with probability of success being p . For $k = 2$, we sample space consists of all possible patterns of length 3, i.e., $\{(H, H, H), (H, H, T), (H, T, H), (H, T, T), (T, H, H), (T, H, T), (T, T, H), (T, T, T)\}$ (we will drop the separating comma in future). Find the transition matrix this snake chain. ◀

Exercise 5.37. Let $\{X_n^{(j)} : n \geq 0\}$ be independent sequences of Markov chains for $1 \leq j \leq k$ on the state space $\mathcal{S}_j$ and transition matrix $\mathbf{P}_j$. Set, for $n \geq 0$,

$$Y_n := (X_n^{(1)}, X_n^{(2)}, \dots, X_n^{(k)}). \quad (5.12)$$

Show that the process $\{Y_n : n \geq 0\}$, taking values $\prod_{j=1}^k \mathcal{S}_j$, is a Markov chain. What is its transition matrix. ◀

Exercise 5.38. Suppose that $\{X_n : n \geq 0\}$ and $\{Y_n : n \geq 0\}$ be independent Markov chains on the state space $\mathcal{S}_j$ and transition matrices $\mathbf{P}_j$ and $\mathbf{Q}$. Let $\{\xi_n : n \geq 1\}$ be a sequence of **iid** random variables with $\mathbb{P}(\xi_n = 1) = \alpha = 1 - \mathbb{P}(\xi_n = 0)$. We also assume that $\{\xi_n : n \geq 1\}$ are independent of the sequences $\{X_n : n \geq 0\}$ and $\{Y_n : n \geq 0\}$. Assume that $Z_0 \in \mathcal{S}$ according to some initial distribution and for $n \geq 1$, define

$$Z_n := \begin{cases} X_n & \text{if } \xi_n = 1; \\ Y_n & \text{if } \xi_n = 0. \end{cases} \quad (5.13)$$

Show that the process $\{Z_n : n \geq 0\}$ is a Markov chain on $\mathcal{S}$. What is its transition matrix.

If we take $\mathbf{Q}$ as the identity matrix on $\mathcal{S}$ (how does the Markov chain associated with it behave?), this chain is often referred as the *lazy* Markov chain. ◀

5.3 State Transition Diagrams

It is very useful to visualize the state transitions through a diagram, often referred as the *state transition diagram*. State transition diagrams are essentially a directed graph where the vertex set is the state space of the chain. We draw directed edges from a state i to a state j if $p_{ij} > 0$. This also implies that if $p_{ii} > 0$, there will be a loop at the vertex i . However, there are no multi-edges in the graph.

Example 5.39. a) For the Example 5.9 with state space $\mathcal{S} = \{1, 2\}$ with $p_{11} + p_{12} = 1 = p_{21} + p_{22}$, the state transition diagram can be visualized as follows:

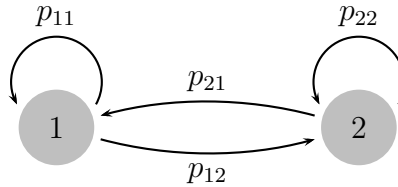


Figure 5.1: State transition diagram for two state chain

b) For the simple random walk in one dimension in Example 5.11, the state transition diagram is represented by



Figure 5.2: State transition diagram for one dimensional simple random walk

c) For the simple symmetric random walk in two dimensions (see Example 5.11), the state transition diagram is represented by

d) For the gambler's ruin problem in Example 5.16, the state transition diagram is represented by

5.4 Supplementary Exercises

Exercise 5.40. Let $\{X_n : n \geq 0\}$ be a sequence of independent random variables. Show that $\{X_n\}$ is a Markov chain. Find its transition probabilities. How does the

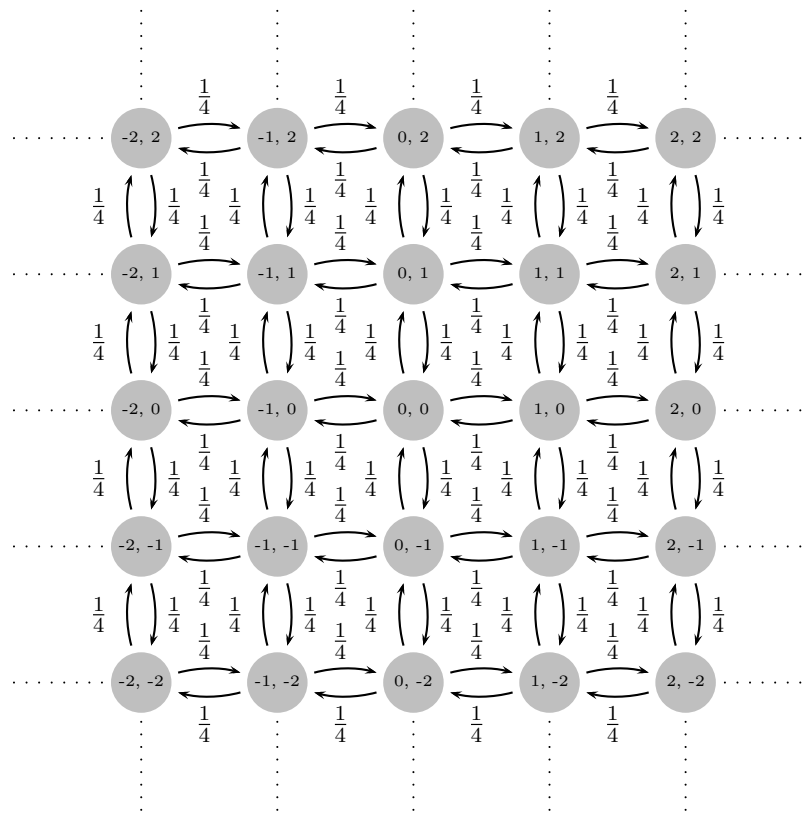


Figure 5.3: State transition diagram for two dimensional simple symmetric random walk

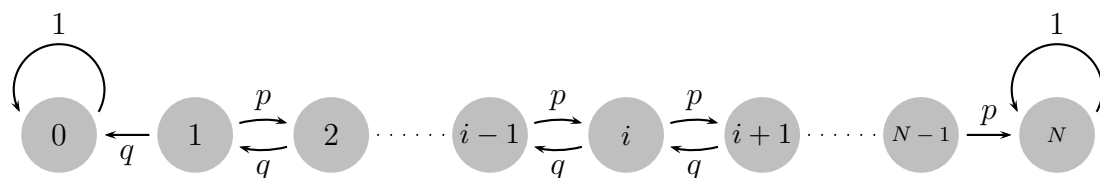


Figure 5.4: State transition diagram for gambler's ruin problem

initial distribution relate to the overall process?

Exercise 5.41. Consider the two-state Markov chain from Example 5.9 with transition matrix:

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} \alpha & 1 - \alpha \\ 1 - \beta & \beta \end{bmatrix} \end{matrix}.$$

Assume the initial distribution is $\mathbb{P}(X_0 = 0) = p, \mathbb{P}(X_0 = 1) = 1 - p$.

1. Compute $\mathbb{P}(X_1 = 0)$ and $\mathbb{P}(X_1 = 1)$.
2. Compute $\mathbb{P}(X_2 = 0)$ and $\mathbb{P}(X_2 = 1)$.
3. Find $\mathbb{P}(X_2 = 0 \mid X_0 = 0)$.

Exercise 5.42. Consider a simple random walk on the integers $\{0, 1, 2, 3\}$ with absorbing boundaries at 0 and 3. That is, $p_{00} = 1$, $p_{33} = 1$, and for $i = 1, 2$, $p_{i,i+1} = p$, $p_{i,i-1} = q = 1 - p$.

1. Write down the transition matrix $\mathbb{P}$.
2. If the chain starts at state 1, what are the possible states it can be in after 2 steps? Compute the probabilities of each.
3. Is state 2 accessible from state 1? Is state 1 accessible from state 2?

Exercise 5.43. Consider a random walk on a circle with 5 states $\{0, 1, 2, 3, 4\}$ (with addition modulo 5). At each step, the walker moves to the right with probability p or to the left with probability $q = 1 - p$.

1. Write down the transition matrix.
2. Show that the chain is doubly stochastic (each column also sums to 1).
3. If $p = 1$, describe the behavior of the chain.

Exercise 5.44. Identify which of the following Markov chains are birth-death chains (i.e., $p_{ij} = 0$ if $|i - j| > 1$). For those that are, write them in the standard birth-death form.

1. The two-state chain from Example 5.9.
2. The simple random walk on integers (Example 5.11).
3. The random walk on a circle with $N \geq 4$ (Example 5.13).
4. The Ehrenfest model (Example 5.18).

Exercise 5.45. For each of the following Markov chains, describe a function f and a sequence of independent random variables $\{\xi_n\}$ such that $X_{n+1} = f(X_n, \xi_{n+1})$, as in Proposition 5.7.

1. The two-state chain.
2. Simple symmetric random walk on $\mathbb{Z}$.
3. The random walk on the circle with 4 states.

Exercise 5.46. Explain why each of the following processes is **not** a Markov chain.

1. $X_n = \xi_n + \xi_{n-1}$, where $\{\xi_n\}$ are i.i.d.
2. The process $\{M_n\}$ where $M_n = \max\{\xi_1, \xi_2, \dots, \xi_n\}$ and $\{\xi_n\}$ are i.i.d. (Hint: Consider what information M_n gives about past values.)
3. The Pólya urn process described in Example 5.29 when $c \neq 0$.

Exercise 5.47. Prove that for a Markov chain $\{X_n\}$, the future $\{X_m : m > n\}$ and the past $\{X_m : m < n\}$ are conditionally independent given the present X_n . That is, for any events A in the σ -algebra generated by the future and B in the σ -algebra generated by the past,

$$\mathbb{P}(A \cap B \mid X_n) = \mathbb{P}(A \mid X_n)\mathbb{P}(B \mid X_n).$$

(This is a more formal version of Exercise 5.6.)

Exercise 5.48. Give an example of a Markov chain that is **not** time-homogeneous. (Hint: Consider a process where the transition probabilities depend explicitly on the time n .)

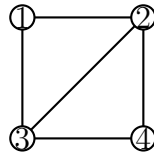
Exercise 5.49. Consider the gambler's ruin chain on $\{0, 1, 2, 3, 4\}$ with $p = 0.6$, $q = 0.4$. The chain starts at state 2.

1. Write the transition matrix.
2. List all possible paths of length 3 that start and end at state 2, and compute their probabilities.
3. What is the probability of being absorbed at state 0 within 2 steps? (Note: This means being at state 0 at time 2, not necessarily for the first time.)

Exercise 5.50. Consider the discrete queue model from Example 5.22. Assume that the number of arrivals per period ξ_n follows a Poisson distribution with mean λ , so that $\alpha_k = e^{-\lambda} \lambda^k / k!$.

1. Write the transition probabilities p_{0j} and p_{ij} for $i \geq 1$ explicitly in terms of λ .
2. Compute p_{0j} for $j = 0, 1, 2, 3$ when $\lambda = 1$.
3. If the queue starts empty ($X_0 = 0$), what is the probability that there is exactly one customer in the queue after one period?

Exercise 5.51. Consider a random walk on the following graph:



1. Write the transition matrix.
2. If the walker starts at vertex 1, what is the probability distribution after 2 steps?
3. Is this chain doubly stochastic? Explain.

Exercise 5.52. Consider a branching process $\{Z_n\}$ as defined in Example 5.27. Suppose the offspring distribution is $\mathbb{P}(\xi = 0) = 0.2, \mathbb{P}(\xi = 1) = 0.5, \mathbb{P}(\xi = 2) = 0.3$.

1. Compute the transition probabilities p_{0j}, p_{1j}, p_{2j} for $j = 0, 1, 2, 3, 4$.
2. If the population starts with a single individual ($Z_0 = 1$), what is the probability that the population is extinct ($Z_1 = 0$) after one generation?

3. What is the probability that the population has exactly 2 individuals after one generation?

Exercise 5.53. Let $\{X_n\}$ and $\{Y_n\}$ be independent Markov chains on state spaces $\mathcal{S}_X$ and $\mathcal{S}_Y$ with transition matrices $\mathbf{P}$ and $\mathbf{Q}$ respectively.

1. Show that $\{(X_n, Y_n)\}$ is a Markov chain on $\mathcal{S}_X \times \mathcal{S}_Y$ and find its transition probabilities.
2. Show that $\{Z_n = X_n + Y_n\}$ is not necessarily a Markov chain. Provide a counterexample.
3. Under what condition on $\mathbf{P}$ and $\mathbf{Q}$ would $\{Z_n\}$ be a Markov chain?

Exercise 5.54. Describe a procedure to simulate a Markov chain with a given transition matrix $\mathbf{P}$ and initial distribution μ . Your description should be detailed enough to implement on a computer. (Hint: Think about how to use Proposition 5.7 practically.)

Exercise 5.55. Construct a simple example of a Markov chain on three states that exhibits "mean-reverting" behavior. That is, the chain has a tendency to return to a central state. Write its transition matrix and explain why it has this property.

Exercise 5.56. Consider a Markov chain on the states $\{0, 1, 2\}$ with transition matrix:

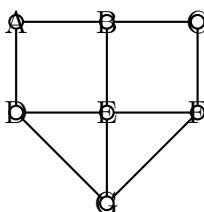
$$\mathbf{P} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \end{matrix}.$$

1. Draw the state transition diagram using pstricks.
2. If the chain starts at state 0, what are the possible states at time $n = 1, 2, 3, 4, 5$?
3. Show that $\mathbb{P}(X_n = 0) = 1$ if n is a multiple of 3, and 0 otherwise.
4. This chain is called *periodic*. Can you find another example of a periodic chain?

Exercise 5.57. Consider the random walk with reflecting barriers from Example 5.17 on the state space $\{0, 1, 2, 3\}$.

1. Write the transition matrix for this chain.
2. Draw the state transition diagram using pstricks.
3. If the chain starts at state 1, what is the probability it returns to state 1 in exactly 2 steps?
4. Is state 0 accessible from state 3? Is state 3 accessible from state 0?

Exercise 5.58. Consider a random walk on the following more complex graph:



1. Write the transition matrix for this random walk.
2. If the walker starts at vertex E, what is the probability it reaches vertex C within 2 steps?
3. Calculate the stationary distribution for this chain (if you have covered this topic).

Chapter 6

Higher Order Transitions

6.1 Joint distribution of Markov chain

The joint distribution for a Markov chain is fairly straightforward to compute. Indeed, for any $i_0, i_1, \dots, i_n \in \mathcal{S}$, the probability

$$\begin{aligned} & \mathbb{P}[X_n = i_n, X_{n-1} = i_{n-1}, \dots, X_1 = i_1 \mid X_0 = i_0] \\ &= \mathbb{P}[X_n = i_n \mid X_{n-1} = i_{n-1}, \dots, X_1 = i_1, X_0 = i_0] \\ & \quad \times \mathbb{P}[X_{n-1} = i_{n-1}, \dots, X_1 = i_1 \mid X_0 = i_0] \\ &= \mathbb{P}[X_n = i_n \mid X_{n-1} = i_{n-1}] \mathbb{P}[X_{n-1} = i_{n-1}, \dots, X_1 = i_1 \mid X_0 = i_0] \\ &= p_{i_{n-1}i_n} \mathbb{P}[X_{n-1} = i_{n-1}, \dots, X_1 = i_1 \mid X_0 = i_0]. \end{aligned}$$

Using the above inductively, we obtain,

$$\mathbb{P}[X_n = i_n, X_{n-1} = i_{n-1}, \dots, X_1 = i_1 \mid X_0 = i_0] = \prod_{t=1}^n p_{i_{t-1}i_t}. \quad (6.1)$$

Suppose that $C \subseteq \underbrace{\mathcal{S} \times \mathcal{S} \times \dots \times \mathcal{S}}_{m \text{ times}}$ and $D \subseteq \underbrace{\mathcal{S} \times \mathcal{S} \times \dots \times \mathcal{S}}_{(n+1) \text{ times}}$. Then, for a fixed state i , using equation (6.1), we have

$$\begin{aligned} & \mathbb{P}[(X_{n+1}, X_{n+2}, \dots, X_{n+m}) \in C, X_n = i, (X_0, X_1, \dots, X_n) \in D \mid X_0 = i_0] \\ &= \sum_{\substack{(i_0, i_1, \dots, i_n) \in D, i_n = i \\ (i_{n+1}, i_{n+2}, \dots, i_{n+m}) \in C}} \mathbb{P}[X_t = i_t \text{ for } t = 0, 1, \dots, n+m \mid X_0 = i_0] \\ &= \sum_{\substack{(i_0, i_1, \dots, i_n) \in D, i_n = i \\ (i_{n+1}, i_{n+2}, \dots, i_{n+m}) \in C}} \prod_{t=1}^{n+m} p_{i_{t-1}i_t} = \sum_{\substack{(i_{n+1}, \dots, i_{n+m}) \in C, \\ i_n = i}} \prod_{t=n+1}^{n+m} p_{i_{t-1}i_t} \sum_{\substack{(i_0, i_1, \dots, i_n) \in D, \\ i_n = i}} \prod_{t=1}^n p_{i_{t-1}i_t}. \end{aligned}$$

Also we have, using equation (6.1) again,

$$\mathbb{P}\left[(X_1, X_2, \dots, X_m) \in C \mid X_0 = i\right] = \sum_{\substack{(i_{n+1}, \dots, i_{n+m}) \in C, \\ i_n = i}} \prod_{t=n+1}^{n+m} p_{i_{t-1}i_t}$$

and

$$\mathbb{P}\left[(X_0, X_1, \dots, X_n) \in D, X_n = i \mid X_0 = i_0\right] = \sum_{\substack{(i_0, i_1, \dots, i_n) \in D, \\ i_n = i}} \prod_{t=1}^n p_{i_{t-1}i_t}.$$

Therefore, we have

$$\begin{aligned} & \mathbb{P}\left[(X_{n+1}, X_{n+2}, \dots, X_{n+m}) \in C, X_n = i, (X_0, X_1, \dots, X_n) \in D \mid X_0 = i_0\right] \\ &= \mathbb{P}\left[(X_1, X_2, \dots, X_m) \in C \mid X_0 = i\right] \mathbb{P}\left[(X_0, X_1, \dots, X_n) \in D, X_n = i \mid X_0 = i_0\right]. \end{aligned} \quad (6.2)$$

Thus we have proved the following proposition.

Proposition 6.1. *For a Markov chain, the future process $\{X_{n+k} : k \geq 1\}$ is independent of the past process $(X_0, \dots, X_n)$ if we know the precise present state, i.e., precise value of X_n . Further, the future process behaves like an independent Markov chain starting at the present state.*

We should observe in the above that knowledge of the precise state of the present is very important. In other words, if we condition the present being in set consisting of at least two elements, the Markov condition may not hold.

Exercise 6.2. Find an example of a Markov chain $\{X_n : n \geq 0\}$ on the state space $\mathcal{S}$ and $x_0, x_2 \in \mathcal{S}$ and $A_1 \subseteq \mathcal{S}$ (having at least two elements) such that

$$\mathbb{P}(X_2 = x_2 \mid X_1 \in A_1, X_0 = x_0) \neq \mathbb{P}(X_2 = x_2 \mid X_1 \in A_1). \quad \blacktriangleleft$$

However, in certain Markov chains, we may be able to group of some states into an aggregated set and the represented aggregated set at different time points will form a Markov chain on the state space of class of aggregated sets. Let $\mathcal{S}$ be the state space and $\{A_\lambda : \lambda \in \Lambda\}$ be a partition of the state space. Suppose $\{X_n : n \geq 0\}$ is a Markov chain on $\mathcal{S}$ and transition matrix $\mathbf{P}$. Let us define a new sequence of random variables on Λ as follows

$$Y_n = \lambda \text{ if } X_n \in A_\lambda.$$

Proposition 6.3. Suppose that for any $\beta \in \Lambda$, $\sum_{j \in A_\beta} p_{ij}$ is independent of $i \in A_\alpha$ for any $\alpha \in \Lambda$. In other words, we have

$$\sum_{j \in A_\beta} p_{i_1, j} = \sum_{j \in A_\beta} p_{i_2, j}$$

for any $i_1, i_2 \in A_\alpha$ for any $\alpha, \beta \in \Lambda$. Then the process $\{Y_n : n \geq 0\}$ a Markov chain with transition probabilities given by : for $\alpha, \beta \in \Lambda$

$$\hat{p}_{\alpha, \beta} = \sum_{j \in A_\beta} p_{ij}.$$

We will refer the chain Y_n as the *aggregated* chain of X_n . This are often useful for reducing computational effort in some chains.

Proof : For $n \geq 1$, let $\lambda_{n+1}, \lambda_n, \lambda_{n-1}, \dots, \lambda_0 \in \Lambda$, we have

$$\begin{aligned} & \mathbb{P}(Y_{n+1} = \lambda_{n+1}, Y_n = \lambda_n, \dots, Y_0 = \lambda_0) \\ &= \mathbb{P}(X_{n+1} \in A_{\lambda_{n+1}}, X_n \in A_{\lambda_n}, \dots, X_0 \in A_{\lambda_0}) \\ &= \sum_{i \in A_{\lambda_n}} \mathbb{P}(X_{n+1} \in A_{\lambda_{n+1}}, X_n = i, \dots, X_0 \in A_{\lambda_0}). \end{aligned}$$

Now, for any sets $B_{n+1}, B_n, \dots, B_0$, using the fact the future and past are independent at the presence of the current state (see Proposition 6.1), we have

$$\begin{aligned} & \mathbb{P}(X_{n+1} \in B_{n+1}, X_n = i, \dots, X_0 \in B_0) \\ &= \mathbb{P}(X_1 \in B_{n+1} \mid X_0 = i) \mathbb{P}(X_n = i, \dots, X_0 \in B_0) \\ &= \sum_{j \in B_n} p_{ij} \times \mathbb{P}(X_n = i, \dots, X_0 \in B_0). \end{aligned} \tag{6.3}$$

Therefore, specializing to the case, $B_{n+1} = A_{\lambda_{n+1}}$ and using the fact that $\sum_{j \in A_\beta} p_{ij}$ is independent of $i \in A_\alpha$ for all $\alpha, \beta \in \Lambda$, we have

$$\begin{aligned} & \mathbb{P}(Y_{n+1} = \lambda_{n+1}, Y_n = \lambda_n, \dots, Y_0 = \lambda_0) \\ &= \hat{p}_{\lambda_n, \lambda_{n+1}} \sum_{i \in A_{\lambda_n}} \mathbb{P}(X_n = i, \dots, X_0 \in A_{\lambda_0}) \\ &= \hat{p}_{\lambda_n, \lambda_{n+1}} \mathbb{P}(Y_n = \lambda_n, \dots, Y_0 = \lambda_0). \end{aligned}$$

Therefore, we conclude

$$\mathbb{P}(Y_{n+1} = \lambda_{n+1} \mid Y_n = \lambda_n, \dots, Y_0 = \lambda_0) = \hat{p}_{\lambda_n, \lambda_{n+1}}.$$

Further, we have

$$\begin{aligned}
 \mathbb{P}(Y_{n+1} = \lambda_{n+1}, Y_n = \lambda_n) &= \mathbb{P}(X_{n+1} \in A_{\lambda_{n+1}}, X_n \in A_{\lambda_n}) \\
 &= \sum_{i \in A_{\lambda_n}} \mathbb{P}(X_{n+1} \in A_{\lambda_{n+1}}, X_n = i, X_{n-1} \in \mathcal{S}, \dots, X_0 \in \mathcal{S}) \\
 &= \hat{p}_{\lambda_n, \lambda_{n+1}} \sum_{i \in A_{\lambda_n}} \mathbb{P}(X_n = i, X_{n-1} \in \mathcal{S}, \dots, X_0 \in \mathcal{S}) \\
 &= \hat{p}_{\lambda_n, \lambda_{n+1}} \mathbb{P}(X_n \in A_{\lambda_n})
 \end{aligned}$$

where we have used (6.3) in the third step. Therefore, we obtain that $\mathbb{P}(Y_{n+1} = \lambda_{n+1} \mid Y_n = \lambda_n) = \hat{p}_{\lambda_n, \lambda_{n+1}}$, which completes the proof. ■

Exercise 6.4. Consider the Snake chain of length 3 (i.e., $k = 2$) (see Exercise 5.36) associated with independent tosses of a coin with probability p of success. Set $A_0 = \{HHT, HTT, THT, TTT\}$, $A_1 = \{HTH, TTH\}$, $A_2 = \{THH\}$ and $A_3 = \{HHH\}$. Show that the aggregated process is a Markov chain on $\{0, 1, 2, 3\}$. Find its transition matrix. ◀

Exercise 6.5. We return to the Exercise 5.21. Suppose that a Rook, placed at a square on a chessboard, moves to one of the squares it can legally move, picked at random. Assuming the usual algebraic notation for chessboard, consider $A_0 = \{A1\}$, $A_1 = \{A2, A3, A4, A5, A6, A7, B1, C1, D1, E1, F1, G1\}$, $A_2 = \{H2, H3, H4, H5, H6, H7, B8, C8, D8, E8, F8, G8\}$, $A_3 = \{A8, H1\}$, $A_4 = \{H8\}$. Let A_5 be the remaining squares of the chessboard. Show that the aggregated process is a Markov chain on $\{0, 1, 2, 3, 4, 5\}$. Find its transition matrix. ◀

6.2 Higher order transition probabilities

We denote by $p_{ij}^{(n)}$ the probability of a transition from the state i to the state j in exactly n -steps. In other words, we define,

$$p_{ij}^{(n)} := \mathbb{P}(X_n = j \mid X_0 = i). \quad (6.4)$$

For higher order transition probabilities, using the joint distribution formula in (6.1), we have

$$p_{ij}^{(2)} = \mathbb{P}(X_2 = j \mid X_0 = i) = \sum_{k \in \mathcal{S}} \mathbb{P}[X_2 = j, X_1 = k \mid X_0 = i] = \sum_{k \in \mathcal{S}} p_{ik} p_{kj} = (\mathbf{P}^2)_{ij}$$

where $(\mathbf{P}^2)_{ij}$ is the (i, j) -th entry of the matrix $\mathbf{P}^2$. Thus, we have $\mathbf{P}^2 = ((p_{ij}^{(2)}))$. Using induction, it is easy to show that $\mathbf{P}^n = ((p_{ij}^{(n)}))$ for every $n \geq 1$. It is also easy to check that the entries of $\mathbf{P}^n$ are non-negative and it is a stochastic matrix since $\mathbf{P}^n \mathbf{1}' = \mathbf{P}^{n-1} \mathbf{1}' = \dots = \mathbf{P} \mathbf{1}' = \mathbf{1}'$.

Exercise 6.6. Show that if $\mathbf{P}$ is a doubly stochastic matrix, then $\mathbf{P}^n$ is also a doubly stochastic matrix for any $n \geq 1$. ◀

6.2.1 Chapman Kolmogorov Equations

The above identification gives us the following identity: for any $m, n \geq 1$,

$$p_{ij}^{(m+n)} = \sum_{k \in S} p_{ik}^{(m)} p_{kj}^{(n)}. \quad (6.5)$$

This is known as the Chapman Kolmogorov equation. It reflects the simple fact that the first m steps lead from i to some intermediate state k , and that the probability of a subsequent passage from k to j does not depend on the manner in which k was reached.

For notational sake, we set $p_{ij}^{(0)} = \delta_{ij}$ where

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise.} \end{cases}$$

This will mean that $\mathbf{P}^0 = \mathbf{I}$, the identity matrix. With this identification, we observe that equation (6.5) holds for any $m, n \geq 0$.

Exercise 6.7. If $\{X_n : n \geq 0\}$ is a Markov chain with the transition probability matrix $\mathbf{P}$, then for any $k \geq 1$, $\{X_{nk} : n \geq 0\}$ is also a Markov chain with transition probability matrix $\mathbf{P}^k$. ◀

Exercise 6.8. Show that not all stochastic matrices can be achieved as $\mathbf{P}^2$. For 2×2 , matrices show that a stochastic matrix $\mathbf{M}$ is a square of a stochastic matrix if and only if the trace of the matrix $\mathbf{M}$ is at least 1. ◀

Explicit expressions for the higher-order transition probabilities are usually hard to come by, but fortunately they are only of minor interest. In some cases, we are able to compute this explicitly.

Example 6.9. Independent trials (see Example 5.8): As an important, if trivial, exception we note the special case of independent trials. This case arises when all rows of $\mathbf{P}$ are identical with a given probability distribution, and it is clear without calculations that this implies $\mathbf{P}^n = \mathbf{P}$ for all $n \geq 2$. ■

Example 6.10. Markov chain with two states (see Example 5.9): For the 2-state Markov chain, it is possible to set up a recurrence relation, which can be solved to find the exact formula for $p_{ij}^{(n)}$. Let us set $\lambda_2 = (p_{11} + p_{22} - 1)$. Observe that λ_2 is the second eigenvalue of $\mathbf{P}$, the first eigenvalue being 1. We assume that $|\lambda_2| < 1$, i.e., $0 < p_{11} < 1, 0 < p_{22} < 1$. We note that, for any $n \geq 0$, $p_{11}^{(n)} + p_{12}^{(n)} = 1$ as $\mathbf{P}^n$ is also a stochastic matrix. Using Chapman-Kolmogorov equation, for any $n \geq 0$, we have

$$\begin{aligned} p_{11}^{(n)} &= p_{11}^{(n-1)}p_{11} + p_{12}^{(n-1)}p_{21} = p_{11}^{(n-1)}p_{11} + (1 - p_{11}^{(n-1)})p_{21} \\ &= p_{21} + (p_{11} - p_{21})p_{11}^{(n-1)} = p_{21} + \lambda_2 p_{11}^{(n-1)}. \end{aligned}$$

Iterating the above, and using the fact that $p_{11}^{(0)} = 1$, we have

$$\begin{aligned} p_{11}^{(n)} &= p_{21} + p_{21}\lambda_2 + \lambda_2^2 p_{11}^{(n-2)} = \cdots = \sum_{k=0}^{n-1} p_{21}\lambda_2^k + \lambda_2^n \\ &= \frac{p_{21}(1 - \lambda_2^n)}{1 - \lambda_2} + \lambda_2^n = \frac{p_{21}}{p_{21} + p_{12}} + \left[1 - \frac{p_{21}}{p_{21} + p_{12}}\right]\lambda_2^n. \end{aligned}$$

Thus, we note that $p_{11}^{(n)} \rightarrow \frac{p_{21}}{p_{21} + p_{12}} := \pi(1)$ as $n \rightarrow \infty$. Further, it is easy to see that

$$|p_{11}^{(n)} - \pi(1)| \leq |\lambda_2|^n$$

for all $n \geq 1$. Hence, the convergence is exponentially fast.

Interchanging the roles of the state 1 and state 2, we may obtain $|p_{22}^{(n)} - \pi(2)| \leq |\lambda_2|^n$ where $\pi(2) := \frac{p_{12}}{p_{21} + p_{12}}$. Now, observing that $\pi(1) + \pi(2) = 1$ and the fact that $p_{21}^{(n)} + p_{22}^{(n)} = 1$, we get that

$$|p_{21}^{(n)} - \pi(1)| = |p_{22}^{(n)} - \pi(2)| \leq |\lambda_2|^n \text{ and } |p_{12}^{(n)} - \pi(2)| = |p_{11}^{(n)} - \pi(1)| \leq |\lambda_2|^n.$$

Thus, we observe that the limit is independent of the initial state.

We also make another observation : let $\pi = (\pi(1), \pi(2))$. Then, it is straightforward to show that

$$\pi = \pi\mathbf{P}.$$

We will return to this observation at a later stage. ■

Example 6.11. Random walk in one dimension (see Example 5.11): It is easy to compute $p_{ij}^{(n)}$ for the random walk on the integers. Starting at i , we must have $(n + j - i)/2$ steps to the right and $(n - j + i)/2$ steps to the left. Therefore, we have

$$p_{ij}^{(n)} = \begin{cases} \binom{n}{(n+j-i)/2} p^{(n+j-i)/2} (1-p)^{(n-j+i)/2} & \text{if } n + j - i \text{ is even} \\ 0 & \text{otherwise.} \end{cases} \quad (6.6)$$

Observe that $p_{ij}^{(n)} \rightarrow 0$ as $n \rightarrow \infty$ for all $i, j \in \mathcal{S}$. ■

Example 6.12. Success runs (see Example 5.24): The higher transition probabilities are easy to compute. If $j = i + n$, then we must take n -steps to the right from i to reach j , which happens with probability p^n . For $j \leq n - 1$, out of n steps last j steps must be to the positive and the $n - j - 1$ -th step must be a step to 0, with no restriction of movements on the first $n - j - 2$ steps. In other case, the event is not possible. Thus, we have

$$p_{ij}^{(n)} = \begin{cases} qp^j & \text{for } j = 0, 1, \dots, n - 1 \\ p^n & \text{for } j = i + n \\ 0 & \text{otherwise.} \end{cases}$$

In this case, it is clear that $p_{ij}^{(n)}$ converges to qp^j for every $i \in \mathcal{S}$. ■

Exercise 6.13. Let $\{\epsilon_n : n \geq 1\}$ represent the results (H or T) of independent coin tosses with probability of getting head (H) being $0 < p < 1$. Define $\{X_n : n \geq 0\}$ on the state space $\mathcal{S} = \{HH, HT, TH, TT\}$ by setting

$$X_n = \begin{cases} HH & \text{if } \epsilon_{n+1} = H, \epsilon_{n+2} = H; \\ HT & \text{if } \epsilon_{n+1} = H, \epsilon_{n+2} = T; \\ TH & \text{if } \epsilon_{n+1} = T, \epsilon_{n+2} = H; \\ TT & \text{if } \epsilon_{n+1} = T, \epsilon_{n+2} = T \end{cases}$$

for $n \geq 0$ (Also look at Exercise 5.36). Compute $\lim_{n \rightarrow \infty} p_{ij}^{(n)}$ for states $i, j \in \mathcal{S}$. ◀

Exercise 6.14. Set $Z_0 = 0$ and $Z_n = \max\{X_1, X_2, \dots, X_n\}$ for $n \geq 1$ where X_i are iid random variables taking values in $\{0, 1, 2, \dots\}$. Show that $\{Z_n : n \geq 1\}$ is a Markov chain. Find $p_{ij}^{(n)}$. ◀

Chapter 7

Recurrence and transience

7.1 Recurrence and transience

We return to the question that is of fundamental importance : will the chain return to the state, where it started from; if so, how often? Define, for $i, j \in \mathcal{S}$ and $n \geq 1$,

$$f_{ij}^{(n)} := \mathbb{P}(X_n = j, X_t \neq j, t = 1, 2, \dots, n-1 \mid X_0 = i). \quad (7.1)$$

For notation, we set $f_{ij}^{(0)} = 0$. Clearly we have,

$$\mathbb{P}(X_n = j \text{ for some } n \geq 1 \mid X_0 = i) = \sum_{n=1}^{\infty} f_{ij}^{(n)}. \quad (7.2)$$

For $i, j \in \mathcal{S}$, we set

$$f_{ij}^* = \sum_{n=1}^{\infty} f_{ij}^{(n)}. \quad (7.3)$$

Definition 7.1. We say a state $i \in \mathcal{S}$ is called recurrent (or persistent) if $f_{ii}^* = 1$ and it is called transient if $f_{ii}^* < 1$.

This concept may be described in terms (first) return times. We will discuss the return time more generally in Chapter 11. Fix a state $i \in \mathcal{S}$, and let τ_i be the first time that the chain will return to the state i (see also Definition 11.7):

$$\tau_{\{i\}} = \tau_i = \inf\{n \geq 1 : X_n = i\}. \quad (7.4)$$

It is clear from the definition that

$$\mathbb{P}(\tau_j = n \mid X_0 = i) = f_{ij}^{(n)} \quad (7.5)$$

for all $n \geq 1$. Further, we have

$$\mathbb{P}(\tau_j < \infty \mid X_0 = i) = \sum_{n=1}^{\infty} \mathbb{P}(\tau_j = n \mid X_0 = i) = \sum_{n=1}^{\infty} f_{ij}^{(n)} = f_{ij}^*. \quad (7.6)$$

We may summarize the above as follows: if the state i is recurrent, the random variable τ_i is finite with probability 1 (i.e., it is a proper random variable); if the state i is transient, it is the case that τ_i finite with probability $f_{ii}^* < 1$. So, we can interpret this as an improper random variable with $\tau_i = +\infty$ with probability $1 - f_{ii}^* > 0$.

If the state is recurrent, then τ_i is a proper random variable and hence, we may compute the expectation of τ_i . For a recurrent state i , let us denote

$$\mu_i = \mathbb{E}(\tau_i \mid X_0 = i) = \sum_{n=1}^{\infty} n \mathbb{P}(\tau_i = n \mid X_0 = i) = \sum_{n=1}^{\infty} n f_{ii}^{(n)}. \quad (7.7)$$

Definition 7.2. A recurrent state is called positive recurrent if $\mu_i < \infty$. A recurrent state is called null recurrent if $\mu_i = \infty$.

The literal meaning of *recurrent* is *occurring often or repeatedly*. The next proposition justifies the meaning.

Proposition 7.3. For a recurrent state i , the number of returns to the state i must be infinite with probability 1 when the chain starts from the state i , but for a transient state i , number of returns must be finite (no matter where the chain starts), i.e., if i recurrent,

$$\mathbb{P}(N(i) = \infty \mid X_0 = i) = 1$$

but if i transient, for any $j \in \mathcal{S}$,

$$\mathbb{P}(N(i) < \infty \mid X_0 = j) = 1.$$

We will prove a general result in the next Proposition, which implies the previous proposition. Furthermore, it will show that the number of returns to a transient state, starting from the state is actually a geometric random variable.

Proposition 7.4. For states $i, j \in \mathcal{S}$, let $N(j)$ be the number of visits to the state j . Then, for $k \geq 1$,

$$\mathbb{P}(N(j) \geq k \mid X_0 = i) = f_{ij}^* (f_{jj}^*)^{k-1}.$$

Note that to prove Proposition 7.3, we set $i = j$ to see that $\mathbb{P}(N(i) \geq k \mid X_0 = i) = 1$ for any $k \geq 1$ when i is recurrent and therefore, $\mathbb{P}(N(i) = \infty \mid X_0 = i) = 1$. If i is transient, we see that $\mathbb{P}(N(i) \geq k \mid X_0 = i) \rightarrow 0$ as $k \rightarrow \infty$ since $f_{ii}^* < 1$. Thus, $\mathbb{P}(N(i) < \infty \mid X_0 = j) = 1$. In fact, here we have the exact distribution of the number of returns. Note, for any $k \geq 0$, we have

$$\begin{aligned} \mathbb{P}(N(i) = k \mid X_0 = j) &= \mathbb{P}(N(i) \geq k \mid X_0 = j) - \mathbb{P}(N(i) \geq k+1 \mid X_0 = j) \\ &= f_{ji}^* (1 - f_{ii}^*) (f_{jj}^*)^{k-1}. \end{aligned}$$

Therefore, by taking $i = j$, we see that $\mathbb{P}(N(i) = k \mid X_0 = i) = (1 - f_{ii}^*) (f_{ii}^*)^k$. Thus, $N(i)$ follows a geometric distribution with parameter $1 - f_{ii}^*$ given that the Markov chain starts at i .

Proof : We note that, from definition of

$$\mathbb{P}(N(j) \geq 1 \mid X_0 = i) = \mathbb{P}(X_n = j \text{ for some } n \geq 1 \mid X_0 = i) = \sum_{n=1}^{\infty} f_{ij}^{(n)} = f_{ij}^*.$$

We prove the result for $k = 2$, the others being similar. If $N(j) \geq 2$, then we will find $1 \leq n_1 < n_2$ at which point the chain will visit the state j for the first time and for the second time respectively. Therefore, we can write

$$\{N(j) \geq 2\} = \bigcup_{1 \leq n_1 < n_2} \{X_{n_1} = X_{n_2} = j, X_t \neq j, t = 1, 2, \dots, n_1 - 1, n_1 + 1, \dots, n_2 - 1\}$$

For any fixed (n_1, n_2) , using Proposition 6.1, we observe that

$$\begin{aligned} &\mathbb{P}(X_{n_1} = X_{n_2} = j, X_t \neq j, t = 1, 2, \dots, n_1 - 1, n_1 + 1, \dots, n_2 - 1 \mid X_0 = i) \\ &= \mathbb{P}(X_{n_2 - n_1} = j, X_t \neq j, t = 1, 2, \dots, n_2 - n_1 - 1 \mid X_0 = j) \\ &\quad \times \mathbb{P}(X_{n_1} = j, X_t \neq j, t = 1, 2, \dots, n_1 - 1 \mid X_0 = i) \\ &= f_{jj}^{(n_2 - n_1)} f_{ij}^{(n_1)}. \end{aligned}$$

For distinct pairs of (n_1, n_2) , the above events are disjoint. Therefore, we have

$$\begin{aligned} &\mathbb{P}(N(j) \geq 2 \mid X_0 = i) \\ &= \sum_{1 \leq n_1 < n_2}^{\infty} \mathbb{P}(X_{n_1} = X_{n_2} = j, X_t \neq j, t = 1, 2, \dots, n_1 - 1, n_1 + 1, \dots, n_2 - 1 \mid X_0 = i) \\ &= \sum_{1 \leq n_1 < n_2} f_{ij}^{(n_1)} f_{jj}^{(n_2 - n_1)} = \sum_{n_1=1}^{\infty} f_{ij}^{(n_1)} \sum_{n_2=n_1+1}^{\infty} f_{jj}^{(n_2 - n_1)} = f_{ij}^* f_{jj}^*. \end{aligned}$$

This proves the result. ■

Proposition 7.3 implies that the Markov chain cannot stay in a finite collection of transient states. We will use this later to conclude about the finite state Markov chains.

7.1.1 Computation of ever returning

In some cases, it is possible to explicitly compute $f_{ii}^{(n)}$ and hence f_{ii}^* . We will later try to find alternative methods for computing it.

Example 7.5. In Example 5.9, it is easy to see that

$$f_{11}^{(n)} = \begin{cases} p_{11} & \text{if } n = 1 \\ p_{12}(p_{22})^{n-2}p_{21} & \text{if } n \geq 2. \end{cases}$$

It is easy to see that,

$$f_{11}^* = p_{11} + \sum_{n=2}^{\infty} p_{12}(p_{22})^{n-2}p_{21} = p_{11} + \frac{p_{12}p_{21}}{1 - p_{22}} = p_{11} + p_{12} = 1$$

and

$$\begin{aligned} \mu_1 &= p_{11} + \sum_{n=2}^{\infty} n p_{12}(p_{22})^{n-2}p_{21} = p_{11} + p_{12}p_{21} \sum_{n=0}^{\infty} [(n+1)(p_{22})^n + (p_{22})^n] \\ &= p_{11} + p_{12}p_{21} \left[\sum_{n=1}^{\infty} n(p_{22})^{n-1} + \sum_{n=0}^{\infty} (p_{22})^n \right] \\ &= p_{11} + \frac{p_{12}p_{21}}{(1 - p_{22})^2} + \frac{p_{12}p_{21}}{1 - p_{22}} = 1 + \frac{p_{12}}{p_{21}} = \frac{(p_{12} + p_{21})}{p_{21}}. \end{aligned}$$

Note that, $\mu_1 = 1/\pi(1) = 1/\lim_{n \rightarrow \infty} p_{11}^{(n)}$ (see Example 5.9). ■

Example 7.6. In the simple symmetric 1-dimensional random walk (see Example 5.11), we know that

$$f_{00}^{(n)} = \begin{cases} \frac{1}{2^{k-1}} \binom{2k}{k} 2^{-2k} & \text{if } n = 2k, k \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

In this case, it is the case that $f_{00}^* = 1$ but

$$\mu_0 = \sum_{n=1}^{\infty} n f_{00}^{(n)} = \sum_{k=1}^{\infty} 2k f_{00}^{(2k)} = \infty.$$

We should also note here that $\lim_{n \rightarrow \infty} p_{00}^{(n)} = 0 = 1/\infty$. ■

Example 7.7. In success run length Markov Chain in Example 5.24, with $p_k = p$ (and hence $q_k = 1 - p_k = 1 - p = q$), we note that

$$f_{00}^{(n)} = \mathbb{P}(X_n = 0, X_k \neq 0 \text{ for } 1 \leq k < n) = qp^{n-1}.$$

This is a geometric distribution and hence, $f_{00}^* = \sum_{n=1}^{\infty} f_{00}^{(n)} = 1$ and $\mu_0 = \sum_{n=1}^{\infty} n f_{00}^{(n)} = 1/p$. Again, observe that $\mu_0 = 1/\lim_{n \rightarrow \infty} p_{00}^{(n)}$. ■

7.2 Basic Renewal Equation

In most cases, it is very difficult to compute $f_{ii}^{(n)}$. Fortunately, it is not important to have an exact expression of $f_{ii}^{(n)}$. To this end, we develop the basic renewal relation.

Proposition 7.8. For $i, j \in \mathcal{S}$ and $n \geq 1$, we have

$$p_{ij}^{(n)} = \sum_{t=0}^n f_{ij}^{(t)} p_{jj}^{(n-t)}. \quad (7.8)$$

Note that if $i = j$ and $n = 0$, this relation does not hold as $p_{ii}^{(0)} = 1$ while the right hand side of the above equation will give 0. However, if $i \neq j$, the relation holds for $n = 0$.

Proof : Our aim is to suitably break the event $\{X_n = j\}$ into disjoint parts to obtain the above relation. Let us consider the first return time of the Markov chain to the state j . Since $X_n = j$, we have that the first return time to j is at most n . Therefore, we consider the all possible values $1, 2, \dots, n$ of the first return to the state j . We consider the events that the first return to j happens at t , i.e., $\{X_s \neq j \text{ for } 1 \leq s < t, X_t = j\}$ for $t = 1, 2, \dots, n$. Hence, we have

$$\{X_n = j\} = \cup_{t=1}^n \{X_n = j\} \cap \{X_s \neq j \text{ for } 1 \leq s < t, X_t = j\}$$

Note that these events are disjoint for distinct t 's. When $t = n$, we have

$$\mathbb{P}(X_n = j, X_s \neq j \text{ for } 1 \leq s < n \mid X_0 = i) = f_{ij}^{(n)}.$$

For $t < n$, we have

$$\begin{aligned} & \mathbb{P}(X_n = j, X_s \neq j \text{ for } 1 \leq s < t, X_t = j \mid X_0 = i) \\ &= \mathbb{P}(X_{n-t} = j \mid X_0 = j) \mathbb{P}(X_s \neq j \text{ for } 1 \leq s < t, X_t = j \mid X_0 = i) = p_{jj}^{(n-t)} f_{ij}^{(t)} \end{aligned}$$

where we have used Proposition 6.1 in the second step. Thus, we have

$$p_{ij}^{(n)} = f_{ij}^{(n)} + \sum_{t=1}^{n-1} p_{jj}^{(n-t)} f_{ij}^{(t)}.$$

Finally the result follows from the observation that $p_{ii}^{(0)} = 1$ and $f_{ii}^{(0)} = 0$. ■

Exercise 7.9. For any $i, j \in \mathcal{S}$, show that

$$\sup_{n \geq 1} p_{ij}^{(n)} \leq f_{ij}^* \leq \sum_{n=1}^{\infty} p_{ij}^{(n)}.$$
 ■

The equation (7.8) yields relations between generating functions of $\{p_{ij}^{(n)} : n \geq 0\}$ and $\{f_{ij}^{(n)} : n \geq 0\}$. Let $P_{ij}(s)$ and $F_{ij}(s)$ be the generating functions of the sequences $\{p_{ij}^{(n)} : n \geq 0\}$ and $\{f_{ij}^{(n)} : n \geq 0\}$ respectively.

Proposition 7.10. For states $i, j \in \mathcal{S}$, we have

$$P_{ij}(s) = \begin{cases} 1 + P_{ii}(s)F_{ii}(s) & \text{if } i = j; \\ F_{ij}(s)P_{jj}(s) & \text{if } i \neq j. \end{cases} \quad (7.9)$$

Proof : Case $i = j$. Set $\{g_n\} = \{p_{ii}^{(n)}\} \star \{f_{ii}^{(n)}\}$. Thus, letting $G(s) = \sum_{n=0}^{\infty} g_n s^n$, we have that $G(s) = P_{ii}(s)F_{ii}(s)$. We observe that $g_n = p_{ii}^{(n)}$ for all $n \geq 1$ and $p_{ii}^{(0)} = 1$ while $g_0 = 0$. Therefore, we have

$$\begin{aligned} P_{ii}(s) &= \sum_{n=0}^{\infty} p_{ii}^{(n)} s^n = 1 + \sum_{n=1}^{\infty} p_{ii}^{(n)} s^n = 1 + \sum_{n=1}^{\infty} g_n s^n \\ &= 1 + \sum_{n=0}^{\infty} g_n s^n = 1 + G(s) = 1 + P_{ii}(s)F_{ii}(s). \end{aligned}$$

Case $i \neq j$. Set $\{h_n\} = \{p_{ij}^{(n)}\} \star \{f_{jj}^{(n)}\}$. Thus, letting $H(s) = \sum_{n=0}^{\infty} h_n s^n$, we have $H(s) = P_{ij}(s)F_{jj}(s)$. We observe that $h_n = p_{ij}^{(n)}$ for all $n \geq 0$. Therefore, we have

$$P_{ij}(s) = \sum_{n=0}^{\infty} p_{ij}^{(n)} s^n = \sum_{n=0}^{\infty} h_n s^n = H(s) = P_{ij}(s)F_{jj}(s).$$

This completes the proof. ■

The immediate consequence of Proposition 7.10 is the following result:

Theorem 7.11. A state i is recurrent if and only if $\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty$.

Proof : From equation (7.9), we have

$$F_{ii}(s) = 1 - \frac{1}{P_{ii}(s)}.$$

Letting $s \uparrow 1$ and using Abel's summation formula, we conclude that $f_{ii}^* = \sum_{n=0}^{\infty} f_{ii}^{(n)} = 1 - 1/\sum_{n=0}^{\infty} p_{ii}^{(n)}$. This proves the result. ■

Corollary 7.11.1. *If j is transient, then $f_{jj}^* = 1 - 1/\sum_{n=0}^{\infty} p_{jj}^{(n)}$. Further, for any $i \in \mathcal{S}, i \neq j$, $\sum_{n=0}^{\infty} p_{ij}^{(n)} = f_{ij}^* \sum_{n=0}^{\infty} p_{jj}^{(n)} < \infty$.*

Proof : The first part is trivial. For the second part, for $i \neq j$, we have $P_{ij}(s) = F_{ij}(s)P_{jj}(s)$. Letting $s \uparrow 1$, using Abel's summation formula, we conclude the result. ■

Corollary 7.11.2. *If j is transient, for any $i \in \mathcal{S}$,*

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = 0.$$

Proof : Since $\sum_{n=0}^{\infty} p_{ij}^{(n)} < \infty$, the result is obvious. ■

Assuming that $\lim_{n \rightarrow \infty} p_{jj}^{(n)}$ exists, we can find $\lim_{n \rightarrow \infty} p_{ij}^{(n)}$. First we prove an analytic lemma.

Lemma 7.12. *If $\{a_n\}$ and $\{b_n\}$ are two sequences such that $b_n \rightarrow b$ and $\sum_{n=0}^{\infty} |a_n| < \infty$, then, as $n \rightarrow \infty$,*

$$\sum_{k=0}^n a_k b_{n-k} \rightarrow b \sum_{n=0}^{\infty} a_n.$$

Proof : We have,

$$\sum_{k=0}^n a_k b_{n-k} = b \sum_{k=0}^n a_k + \sum_{k=0}^n a_k (b_{n-k} - b).$$

The first term, of course, will converge to the desired limit $b \sum_{n=0}^{\infty} a_k$. In order to show that the second term converges to 0, fix $\epsilon > 0$ and choose N large so that $|b_n - b| < \epsilon$ for $n \geq N$ and $\sum_{n=N}^{\infty} |a_n| < \epsilon$. Also, let $M_b = \sup\{|b_n - b| : n \geq 1\}$. Then, for $n \geq 2N$, we have

$$\begin{aligned} \left| \sum_{k=0}^n a_k (b_{n-k} - b) \right| &\leq \left| \sum_{k=N}^n a_k (b_{n-k} - b) \right| + \left| \sum_{k=1}^{N-1} a_k (b_{n-k} - b) \right| \\ &\leq M_b \sum_{k=N}^n |a_k| + \sum_{k=1}^{N-1} |a_k| |b_{n-k} - b| \\ &\leq M_b \epsilon + \epsilon \sum_{k=1}^{N-1} |a_k| \leq M_b \epsilon + \epsilon \sum_{k=1}^{\infty} |a_k| \end{aligned}$$

as $|b_{n-k} - b| < \epsilon$ for $1 \leq k \leq N - 1$ as $n - k \geq n - N \geq N$ for $n \geq 2N$. This proves the Lemma. $\blacksquare$

If we assume that if $\lim_{n \rightarrow \infty} p_{jj}^{(n)}$ exists, using the basic recurrence equation (7.8) and the above Lemma, we have

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \lim_{n \rightarrow \infty} p_{jj}^{(n)} \sum_{n=0}^{\infty} f_{ij}^{(n)} = f_{ij}^* \lim_{n \rightarrow \infty} p_{jj}^{(n)}. \quad (7.10)$$

Remark. We have already observed that the state i is recurrent if and only if $\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty$. We also have a criteria for null recurrent states, which we establish at a later stage.

A state is i is *null recurrent* if and only if

$$\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty \quad \text{and} \quad \lim_{n \rightarrow \infty} p_{ii}^{(n)} = 0. \quad (7.11)$$

This will imply for any null recurrent state i ,

$$\lim_{n \rightarrow \infty} p_{ji}^{(n)} = 0 \quad (7.12)$$

for any state $j \in \mathcal{S}$. $\blacktriangleright$

Remark. At a later stage, we will show for a certain type of positive recurrent states i the limit $p_{ii}^{(n)}$ converges to $\pi(i) > 0$. In such a case we have that $\lim_{n \rightarrow \infty} p_{ji}^{(n)} = f_{ji}^* \pi(i)$. Furthermore, we will prove that when both i, j are recurrent, then f_{ji}^* must be either 0 or 1 (see Proposition 8.11 and the discussion preceding it). This implies that the limit is either 0 or $\pi(i)$ and is independent of the state j . $\blacktriangleright$

We now show how the Theorem 7.11 can be used to conclude about transience and recurrence of the states.

Example 7.13. Returning to the example of the two state Markov chain (see Example 5.9), we conclude that the state 1 is recurrent as $p_{11}^{(n)} \rightarrow \pi(1) > 0$ (see Example 6.10) and hence $\sum_{n=0}^{\infty} p_{11}^{(n)} = \infty$. Further, the limit is positive, hence the state is not a null recurrent state, i.e., it must be a positive recurrent state. Note, we have done an explicit calculation to show that 1 is actually a positive recurrent state (see Example 7.5). Similar argument can also be carried out to show $p_{22}^{(n)} \rightarrow \pi(2) > 0$, implying that the state 2 is also positive recurrent. $\blacksquare$

Example 7.14. We return to the chain of success run length in Example 5.24. We note that $p_{00}^{(n)} = q$ for $n \geq 1$ (see Example 6.12), hence $\lim_{n \rightarrow \infty} p_{00}^{(n)} = q > 0$. Hence, using

the same arguments as above, 0 is a positive recurrent state. Note, in Example 7.7, we gave direct proof that 0 is a positive recurrent state. ■

Example 7.15. For the one dimensional simple random walk (not necessarily symmetric), (see Example 5.11), we have already seen that

$$p_{00}^{(n)} = p_{ii}^{(n)} = \begin{cases} \binom{2k}{k} (pq)^k & \text{if } n = 2k \\ 0 & \text{otherwise.} \end{cases}$$

Using Stirling's approximation, we conclude that

$$p_{00}^{(2n)} \sim \frac{(4pq)^n}{\sqrt{\pi n}}.$$

Then, if $p \neq 1/2$, $4pq < 1$. Hence $\sum_{n=0}^{\infty} p_{00}^{(n)} < \infty$. Thus, the state 0 is transient.

In fact, we can compute this sum exactly. Note that, the generating function, $P_{00}(s) = \sum_{n=0}^{\infty} \binom{2n}{n} (pq)^n s^{2n} = \sum_{n=0}^{\infty} \binom{2n}{n} (pq s^2)^n = (1 - 4pq s^2)^{-1/2}$. Hence, we have, $\sum_{n=0}^{\infty} p_{00}^{(n)} = (1 - 4pq)^{-1/2}$, so that $f_{00}^* = 1 - (1 - 4pq)^{1/2} = 1 - |p - q|$.

If $p = 1/2$, we have $\sum_{n=0}^{\infty} p_{00}^{(n)} = \infty$. Therefore, for the simple symmetric random walk, the state 0 is recurrent. We further observe that $\lim_{n \rightarrow \infty} p_{00}^{(n)} = 0$. Therefore, we can conclude that 0 is null recurrent. Note in Example 7.6, we have shown that the state 0 is null recurrent. ■

Example 7.16. For the two dimensional simple symmetric random walk (see Example 5.12), we note that $p_{\mathbf{0},\mathbf{0}}^{(n)} = 0$ if n is odd. For even n , we have

$$p_{\mathbf{0},\mathbf{0}}^{(2n)} = \sum_{k=0}^n \frac{(2n)! 4^{-2n}}{k! k! (n-k)! (n-k)!} = \binom{2n}{n} 4^{-2n} \sum_{k=0}^n \binom{n}{k} \binom{n}{n-k} = \binom{2n}{n}^2 4^{-2n}.$$

Again, using Stirling's approximation, we have that

$$p_{\mathbf{0},\mathbf{0}}^{(2n)} \sim \frac{1}{\pi n}.$$

Therefore, we conclude that $\sum_{n=0}^{\infty} p_{\mathbf{0},\mathbf{0}}^{(n)} = \infty$ but $p_{\mathbf{0},\mathbf{0}}^{(n)} \rightarrow 0$ as $n \rightarrow \infty$. Therefore, the state $\mathbf{0}$ is a null recurrent state. ■

Example 7.17. In three dimensional random walk, we again have $p_{\mathbf{0},\mathbf{0}}^{(n)} = 0$ if n is odd. For even n , we have

$$p_{\mathbf{0},\mathbf{0}}^{(2n)} = \sum_{j,k \geq 0, j+k \leq n} \frac{(2n)!}{j! j! k! k! (n-j-k)! (n-j-k)!} 6^{-2n}.$$

Multiplying numerator and denominator by $(n!)^2$ and factoring out a term gives 2^{-2n} , we have

$$p_{\mathbf{0},\mathbf{0}}^{(2n)} = \binom{2n}{n} 2^{-2n} \sum_{j,k \geq 0, j+k \leq n} \left[\frac{(n)!}{j!k!(n-j-k)!} \right]^2 3^{-2n} \leq C_n \binom{2n}{n} 2^{-2n} 3^{-n}$$

where

$$C_n = \max_{j,k \geq 0, j+k \leq n} \left[\frac{(n)!}{j!k!(n-j-k)!} \right].$$

Here, we have used the fact that

$$\sum_{j,k \geq 0, j+k \leq n} \frac{(n)!}{j!k!(n-j-k)!} 3^{-n} = 1.$$

It is easy to see that for large values of n , the maxima is attained when $j = k \sim n/3$. Using this and Stirling's approximation, we have that

$$p_{\mathbf{0},\mathbf{0}}^{(n)} \leq C n^{-3/2}$$

for a suitably chosen positive constant C . Therefore, we conclude that $\sum_{n=0}^{\infty} p_{\mathbf{0},\mathbf{0}}^{(n)} < \infty$. Hence the state $\mathbf{0}$ (similarly any other state), is transient. This explains the comment of Pólya. ■

7.3 Classification of states

We have so far discussed how a given state can be classified as a recurrent or transient state. Now we will see how we can describe the behaviour of one state if we know about the behaviour of another state.

Definition 7.18. We say that a state j can be *reached* from a state i if there exists some $n \geq 0$ such that $p_{ij}^{(n)} > 0$. We denote this by $i \rightsquigarrow j$. We say that the states i and j *communicate* if j can be reached from i and i can be reached from j , i.e., for some $n_1, n_2 \geq 0$, $p_{ij}^{(n_1)} > 0$ and $p_{ji}^{(n_2)} > 0$. We denote this by $i \longleftrightarrow j$.

If two states i and j do not communicate, then either

$$p_{ji}^{(n)} = 0 \text{ for all } n > 0 \quad \text{or} \quad p_{ij}^{(n)} = 0 \text{ for all } n > 0 \quad \text{or} \quad \text{both relations hold.}$$

Exercise 7.19. Show that the following statements are equivalent:

- a) $i \rightsquigarrow j$.

b) There exists $n \geq 1$ and states $i_1, i_2, \dots, i_{n-1} \in \mathcal{S}$ such that

$$p_{i,i_1} p_{i_1,i_2} \cdots p_{i_{n-1},j} > 0.$$

c) There exists $n \geq 1$ and distinct states $i_1, i_2, \dots, i_{n-1} \in \mathcal{S}$ such that

$$p_{i,i_1} p_{i_1,i_2} \cdots p_{i_{n-1},j} > 0.$$

d) $\mathbb{P}(X_n = j \text{ for some } n \geq 1 \mid X_0 = i) > 0.$ ◀

Exercise 7.20. For states $i, j \in \mathcal{S}$ with $i \rightsquigarrow j$ in a Markov chain having N states show that there exists $n \leq N - 1$ such that $p_{ij}^{(n)} > 0.$ ◀

Example 7.21. In the random walk example, (see Example 5.11), any state can be reached from any other state. Note that if $i > j$, $p_{ij}^{(i-j)} \geq q^{i-j} > 0$ and $i < j$, then $p_{ij}^{(j-i)} > p^{j-i} > 0$. Similar argument holds for other random walk examples. So, every pair of states communicate.

In Gambler's ruin problem (see Example 5.16), from $i = 1, 2, \dots, N - 1$, all states can be reached. However, from the states 0, the only state that can be reached is 0. Similarly, from the state N , the only state that can be reached is N . ■

The concept of communication is an equivalence relation on the state space $\mathcal{S}$.

- **Reflexive:** $i \rightsquigarrow i$ is a consequence of the definition of $p_{ii}^{(0)} = 1$.
- **Symmetry:** If $i \rightsquigarrow j$, then $j \rightsquigarrow i$, from the definition of communication.
- **Transitive:** If $i \rightsquigarrow j$ and $j \rightsquigarrow k$, then $i \rightsquigarrow k$. Since $i \rightsquigarrow j$, there exist integers n_1 and n_2 , such that $p_{ij}^{(n_1)} > 0$ and $p_{ji}^{(n_2)} > 0$. Similarly, since $j \rightsquigarrow k$, there exist integers n_3 and n_4 such that $p_{jk}^{(n_3)} > 0$ and $p_{kj}^{(n_4)} > 0$. Hence, we have

$$p_{ik}^{(n_1+n_3)} = \sum_{l \in \mathcal{S}} p_{il}^{(n_1)} p_{lk}^{(n_3)} \geq p_{ij}^{(n_1)} p_{jk}^{(n_3)} > 0;$$

$$p_{ki}^{(n_2+n_4)} = \sum_{l \in \mathcal{S}} p_{kl}^{(n_4)} p_{li}^{(n_2)} \geq p_{kj}^{(n_4)} p_{ji}^{(n_2)} > 0.$$

Any equivalence relation partition the state space into disjoint equivalence classes. The states in an equivalence class are those which *communicate* with each other, hence from now on, we will refer to these equivalence classes as *communicating classes*. Understanding the communicating class will be of utmost importance.

Example 7.22. In the example of two state Markov chain (see Example 5.9), there is a single communicating class if $p_{12} > 0$ and $p_{21} > 0$. If either of them is 0, then we will have two communicating classes, namely, $\{1\}$ and $\{2\}$. In random walk examples (see Examples 5.11, 5.12, 5.13) as well as the random walk on any *connected* Graph (see Example 5.20), there is a single communicating class, namely the whole state space.

For the Markov chain the queueing example (see Example 5.22), if $\alpha_0 > 0$ and $\alpha_0 + \alpha_1 < 1$, then the whole state space is the only communicating class. In finite birth and death chains (see Example 5.14, both $r_0 = r_N = 1$, there are three communicating classes, namely $\{0\}$, $\{N\}$ and $\{1, 2, \dots, N-1\}$. If $r_0 = 1$ but $r_N < 1$, then we have two communicating classes $\{0\}$ and $\{1, 2, \dots, N-1, N\}$. Similarly for the other case, if $r_0 < 1$ but $r_N = 1$, then we have $\{0, 1, \dots, N-1\}$ and $\{N\}$ as the communicating classes. If we have $r_0, r_N < 1$, there is only one communicating class, namely the whole state space. In the infinite state birth and death chains, if $r_0 = 1$, there are two communicating classes $\{0\}$ and $\{1, 2, \dots\}$ while if $r_0 < 1$, again the whole state space is the only communicating class. In the Wright Fisher model as well as Wright Fisher model with selection (see Example 5.25) and in Example 5.26, we have three communicating classes namely $\{0\}$, $\{N\}$ and $\{1, 2, \dots, N-1\}$. However, in Wright Fisher model with migration, the whole state space is the only communicating class.

In Example 5.28, it is clear by inspection that $1 \rightsquigarrow 4 \rightsquigarrow 1$, $3 \rightsquigarrow 6 \rightsquigarrow 8 \rightsquigarrow 3$ and $2 \rightsquigarrow 5 \rightsquigarrow 2$. Also, $5 \rightsquigarrow 3, 2 \rightsquigarrow 6$ but $3 \not\rightsquigarrow 5$ and $6 \not\rightsquigarrow 2$. Also, $7 \rightsquigarrow 1$ but $1 \not\rightsquigarrow 7$. Therefore, the communicating classes in this chain are $\{1, 4\}$, $\{3, 6, 8\}$, $\{2, 5\}$ and $\{7\}$. ■

We note that if two states $i \neq j$ communicate, then we can find $m, n > 0$ such that $p_{ij}^{(m)}, p_{ji}^{(n)} > 0$. Using the Chapman Kolmogorov equations, for any $k \geq 0$, we obtain that

$$p_{ii}^{(m+n+k)} \geq p_{ij}^{(m)} p_{jj}^{(k)} p_{ji}^{(n)} = C(i, j) p_{jj}^{(k)} \text{ and } p_{jj}^{(m+n+k)} \geq p_{ji}^{(n)} p_{ii}^{(k)} p_{ij}^{(m)} = C(i, j) p_{ii}^{(k)} \quad (7.13)$$

where $C(i, j) = p_{ij}^{(m)} p_{ji}^{(n)} > 0$. Indeed, the left hand side of the first inequality, stands for all possible ways of returning to i from i in $m + n + k$ steps, while the right hand represents only a particular way of returning to i from i - namely, after first m steps the chain goes to j to i , from there returning to j after another k steps, and finally returning to i from j after another n steps. Similar arguments hold for the other inequality.

This proves that, if i and j communicate, $\sum_{k=0}^{\infty} p_{ii}^{(k)}$ and $\sum_{k=0}^{\infty} p_{jj}^{(k)}$ behave in the same way, i.e., either both converge or both diverge. This yields a very important proposition.

Proposition 7.23. *All states in a communicating class behave the same way. More precisely, suppose that $i \leftrightarrow j$.*

- a) *If i is recurrent, then j is recurrent.*
- b) *If i is transient, then j is transient.*

The Proposition 7.23 shows recurrence / transience is a class property. Therefore, from now on, a communicating class containing at least one (and hence all) recurrent / transient state will be referred as a *recurrent / transient* (communicating) class.

Proof : a) We make use of Theorem 7.11. If i is recurrent, then $\sum_{k=0}^{\infty} p_{ii}^{(k)} = \infty$. Therefore, by above discussion, $\sum_{k=0}^{\infty} p_{jj}^{(k)} = \infty$. This implies that j is recurrent.

b) Note if i transient but j is recurrent, since i and j communicate, by part a) will imply i also recurrent, which is a contradiction. ■

Remark. Part b) can also be proved in the same way, i.e., if $\sum_{n=0}^{\infty} p_{ii}^{(n)} < \infty$, then $\sum_{n=0}^{\infty} p_{jj}^{(n)} < \infty$. ►

Remark. Besides these results, we also have the following results. Suppose i and j communicate

- a) *If i is null recurrent, then j is null recurrent.*
- b) *If i is positive recurrent, then j is positive recurrent.*

To prove these results, we will require the result that $\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty, \lim_{n \rightarrow \infty} p_{ii}^{(n)} = 0$ if and only if i is null recurrent. Assuming the result, we note that if i is null recurrent, we already know from a) of Proposition 7.23 that j must be recurrent, and hence $\sum_{k=0}^{\infty} p_{jj}^{(k)} = \infty$. Further, we have $\lim_{n \rightarrow \infty} p_{ii}^{(n)} = 0$. From the inequalities in (7.13), we now have $\lim_{n \rightarrow \infty} p_{jj}^{(n)} = 0$. Thus, we conclude that j is also null recurrent.

Finally, suppose that i is positive recurrent and j is null recurrent. Then, by the first part, since j is null recurrent and $i \leftrightarrow j$, we have that i must be null recurrent which is a contradiction. ►

It may be possible, starting in one communicating class, to enter some other communicating class with positive probability; if so happens, however, it is clearly impossible to return to the initial class, or else the two classes would together form a single class. This can be seen in the Gambler's ruin problem where from the class $\{1, 2, \dots, N-1\}$, it is possible to enter either of the classes $\{0\}$ or $\{N\}$, however once the chain is in either

of the two above classes, they cannot get out of the classes $\{0\}$ or $\{N\}$. This motivates us to define the concept of **closed** sets.

Definition 7.24. A set of states $\mathcal{C} \subseteq \mathcal{S}$, is called *closed* if the chain is unable to exit the set $\mathcal{C}$ once it enters the set, i.e., for all $i \in \mathcal{C}$,

$$\sum_{j \notin \mathcal{C}} p_{ij} = 0. \quad (7.14)$$

Equivalently, we must have $\sum_{j \in \mathcal{C}} p_{ij} = 1$ for all $i \in \mathcal{C}$.

- (a) When a closed set is a singleton, the state will be called an *absorbing* state.
- (b) A Markov chain is irreducible if there exists only one communicating class, namely the whole state space.

If a set $\mathcal{C}$ is closed, then for any $i \in \mathcal{C}$ and $j \notin \mathcal{C}$, we have

$$p_{ij}^{(2)} = \sum_{k \in \mathcal{S}} p_{ik} p_{kj} = \sum_{k \in \mathcal{C}} p_{ik} p_{kj} + \sum_{k \notin \mathcal{C}} p_{ik} p_{kj} = 0$$

since in the first sum, for any $k \in \mathcal{C}$, we have $p_{kj} = 0$, while in the second sum, for any $k \notin \mathcal{C}$, we have $p_{ik} = 0$. Inductively, it is easy to show that if a set $\mathcal{C}$ is closed then $p_{ij}^{(n)} = 0$ whenever $i \in \mathcal{C}$ and $j \notin \mathcal{C}$ for all $n \geq 0$.

Using the above arguments, we have the obvious result.

Theorem 7.25. *If in the matrix $\mathbf{P}$ all rows and all columns corresponding to states outside the closed set $\mathcal{C}$ are deleted, there remain stochastic matrix for which the fundamental relations - namely the Chapman Kolmogorov equations, hold.*

This means that we have a Markov chain defined on $\mathcal{C}$, and this sub-chain can be studied independently of all other states, which matches the original Markov chain when it starts in the closed set $\mathcal{C}$.

Observe that the whole state space is always closed by definition. Also, union of closed sets is also closed. Though the closed set is not directly associated with communicating classes, as we will later see, it is most useful for communicating classes. It is easy to observe that any proper subset of a communicating class is not closed. Indeed, suppose $\mathcal{C}$ is a communicating class and $\mathcal{D} \subsetneq \mathcal{C}$, then there exists $i \in \mathcal{D}$ and $j \in \mathcal{C} \setminus \mathcal{D}$ such that $p_{ij}^{(n)} > 0$ for some $n \geq 1$, which by above arguments will be a contradiction. Thus, $\mathcal{D}$ is not closed.

Example 7.26. In random walk examples (see Examples 5.11, 5.12, 5.13) as well as the random walk on any *connected* Graph (see Example 5.20), the whole state space is the only communicating class (see Example 7.22). So no proper subset of the state space is closed. Therefore, the Markov chain is irreducible in these cases. In the example of queueing model (see Example 5.22), when $\alpha_0 > 0, \alpha_0 + \alpha_1 < 1$, the only closed set is the whole state space. Hence the chain is irreducible. In the case of length of success runs (see Example 5.24) with $0 < p_i < 1$ for all $i \geq 0$, the only communicating class is the whole state space and hence the chain is irreducible.

In the example of two state Markov chain (see Example 5.9), there is a single communicating class if $p_{12} > 0$ and $p_{21} > 0$, hence $\{1, 2\}$ is the only closed set and the chain is irreducible. If $p_{11} = 1, p_{21} > 0$, note then 1 is absorbing state while $\{2\}$ is not a closed set. Similarly, If $p_{22} = 1, p_{12} > 0$, note then 2 is absorbing state while $\{1\}$ is not a closed set. If $p_{11} = p_{22} = 1$, then both 1 and 2 are absorbing states.

In the finite state birth and death chain (see Example 5.14, also see Example 7.22), if $r_0 = r_N = 1$, then both 0 as well as N are absorbing states and no other proper subset of state space other than $\{0, N\}$, including the other communicating class $\{1, 2, \dots, N-1\}$ is a closed set. If $r_0 = 1, r_N < 1$, the state 0 is absorbing, and all other subsets are not closed. Similarly, $r_0 < 1, r_N = 1$, the state N is absorbing, and other subsets are not closed. If $r_0, r_N < 1$, then the whole state space is the only communicating class and hence the only closed set. In the infinite state birth and death chain (see Example 5.15, also see Example 7.22), if $r_0 = 1$, 0 is an absorbing state and no other proper subset of the state space $\mathcal{S}$ is closed. If $r_0 < 1$, then the whole state space is the only closed set and hence the chain is irreducible.

In the Wright Fisher model as well as Wright Fisher model with selection (see Example 5.25) and in Example 5.26, both the states 0 and N are absorbing and no other proper subset other than $\{0, N\}$ of the state space $\mathcal{S}$ is closed. However, for the Wright Fisher model with mutation under condition of $\alpha_1 > 0, \alpha_2 > 0$, the whole state space is the only closed set and hence the chain is irreducible.

In Example 5.28, we have observed that $\{1, 4\}, \{3, 6, 8\}, \{2, 5\}$ and $\{7\}$ are the communicating classes. This is easily seen for $\{1, 4\}$ as $p_{11} + p_{14} = 1 = p_{41} + p_{44}$. Therefore, $\{1, 4\}$ is closed. Similarly, we can check that $\{3, 6, 8\}$ is also closed. However, both the subsets $\{2, 5\}$ and $\{7\}$ are not closed since $2 \rightsquigarrow 6$ and $7 \rightsquigarrow 1$. Therefore, the only closed sets are $\{1, 4\}, \{3, 6, 8\}$, their union and the whole state space. ■

Next we prove the simple Proposition which will help us classifying the states into recurrent and transient states. We will be able to characterize the states for a finite state space.

Proposition 7.27. *A recurrent communicating class must be closed.*

Proof : Suppose to the contrary that a recurrent communicating class $\mathcal{C}$ is not closed. Therefore, we will have $i \in \mathcal{C}$ and $j \notin \mathcal{C}$ such that $p_{ij} > 0$. Then, we must have $p_{ji}^{(n)} = 0$ for all $n \geq 1$, otherwise $j \rightsquigarrow i$. But that would imply that $i \longleftrightarrow j$ and hence $j \in \mathcal{C}$. So, we have

$$\begin{aligned} 1 - f_{ii}^* &= \mathbb{P}(X_n \neq i \text{ for all } n \geq 1 \mid X_0 = i) \geq \mathbb{P}(X_1 = j, X_n \neq i \text{ for all } n \geq 2 \mid X_0 = i) \\ &= \mathbb{P}(X_1 = j \mid X_0 = i) \times \mathbb{P}(X_n \neq i \text{ for all } n \geq 1 \mid X_0 = j) = p_{ij} \times 1 = p_{ij} > 0. \end{aligned}$$

But this contradicts the fact that i is recurrent. Hence $\mathcal{C}$ must be closed. ■

The Proposition 7.27 implies that if we have a communicating class, which is not closed, it must not be recurrent. Therefore it must consist of transient states.

Corollary 7.27.1. *Any communicating class which is not closed must be transient.*

Furthermore, the Proposition 7.27 also implies that if a state j can be reached from a recurrent state i , then j must also be recurrent. Indeed, if we consider that the communicating class $\mathcal{C}$ containing the state i , then j must belong to $\mathcal{C}$; otherwise $\mathcal{C}$ is not closed. So, $j \in \mathcal{C}$ and hence j is recurrent.

Corollary 7.27.2. *Only recurrent states can be reached from a recurrent state.*

The communicating (equivalence) classes partition the state space into disjoint sets. In view of Proposition 7.27, we have the following decomposition theorem.

Theorem 7.28. Decomposition theorem. *The state space $\mathcal{S}$ can be partitioned uniquely as*

$$\mathcal{S} = \mathcal{T} \cup \mathcal{C}_1 \cup \mathcal{C}_2 \cup \dots$$

where $\mathcal{T}$ is the set of transient states, and the $\mathcal{C}_i$ are closed communicating classes of recurrent states.

This essentially means that the transition matrix can be partitioned into blocks where the blocks correspond to the classes $\mathcal{C}_i$ and $\mathcal{T}$. More precisely, we can write

$$\mathbf{P} = \begin{array}{c} \mathcal{C}_1 \\ \mathcal{C}_2 \\ \vdots \\ \mathcal{T} \end{array} \begin{array}{ccccc} & \mathcal{C}_1 & \mathcal{C}_2 & \cdots & \mathcal{T} \\ \left[\begin{array}{cccc} \mathbf{P}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{P}_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & \vdots \\ \mathbf{B}_1 & \mathbf{B}_2 & \cdots & \mathbf{Q} \end{array} \right] \end{array}$$

where each $\mathbf{P}_i$ is the transition probability matrix associated with the states in the closed recurrent class $\mathcal{C}_i$ for $i \geq 1$, $\mathbf{Q}$ is the block matrix obtained from the transition matrix when we keep only the transient states in $\mathcal{T}$ and $\mathbf{B}_i$ is the block matrix obtained from the transition matrix when we keep the rows corresponding to the transient states and columns corresponding to the recurrent class $\mathcal{C}_i$ for $i \geq 1$.

It should be noted that $\mathcal{T}$ may be empty, for example in simple symmetric random walks in dimensions one or two. On the other hand, there may be no recurrent states (for infinite state space case). For example in simple symmetric random walks in dimensions three or higher.

For the Gambler's ruin problem, using Corollary 7.27.1, it is easy to see that the equivalence class $\{1, 2, \dots, N-1\}$ is not closed as $1 \rightsquigarrow 0$. Therefore, all the states $1, 2, \dots, N-1$ must be transient. Note that it is trivial to compute f_{00}^* and f_{NN}^* since $f_{00}^{(1)} = p_{00} = 1$ and $f_{NN}^{(1)} = p_{NN} = 1$. We will now prove that the states 0 and N are both recurrent via a more generic method.

We now prove a simple yet very useful result for finite state space Markov chains. This will allow us to identify the recurrent classes in finite state space Markov chain.

Theorem 7.29. *In a finite state space Markov chain, there must be at least one recurrent state.*

Proof : Let the state space be denoted by $\mathcal{S} = \{1, 2, \dots, N\}$. Suppose, if possible, that all states are transient. Now, fix any state $i \in \mathcal{S}$. Since every state $j \in \mathcal{S}$ is transient we have $p_{ij}^{(n)} \rightarrow 0$ as $n \rightarrow \infty$ (see Corollary 7.11.2). But, we have $1 = \sum_{j \in \mathcal{S}} p_{ij}^{(n)} \rightarrow 0$ as $n \rightarrow \infty$. This is a contradiction. Therefore, we must have at least one recurrent state. ■

We have shown that if the communicating class is not closed, it must be transient.

Note, we show that if the closed communicating class is finite, it must be a recurrent class.

Proposition 7.30. *Any finite communicating class which is closed must be recurrent.*

Proof : By Theorem 7.25, we may consider the Markov chain corresponding to the closed communicating class. By Theorem 7.29, this sub-chain will contain at least one recurrent state. However, the behaviour of the original chain starting inside the closed set is exactly same as the sub-chain. Therefore, any state which is recurrent in the sub-chain will be recurrent in the original chain. Since this is an communicating class, all states must be recurrent. ■

Note that the above result does not hold when the chain has infinite state space. The one-dimensional simple but asymmetric random walk or the simple symmetric random walk in three or higher dimensions is irreducible but transient.

Remark. We can strengthen the Proposition 7.30 to say that every finite closed communicating class must be positive recurrent.

Consider the Markov chain corresponding to the finite closed communicating class, say $\mathcal{C}$. Since this is finite closed communicating class, each state must be recurrent. Now, if any state, say $i \in \mathcal{C}$, is null recurrent, then all the states must be null recurrent. We have noted in equation (7.12), that $\lim_{n \rightarrow \infty} p_{ij}^{(n)} = 0$ for any j which is null recurrent.

We have that $\sum_{j \in \mathcal{C}} p_{ij}^{(n)} = 1$ for every $n \geq 0$. Now, letting $n \rightarrow \infty$, we have,

$$1 = \lim_{n \rightarrow \infty} \sum_{j \in \mathcal{C}} p_{ij}^{(n)} = \sum_{j \in \mathcal{C}} \lim_{n \rightarrow \infty} p_{ij}^{(n)} = 0.$$

This is a contradiction. Hence, no state in $\mathcal{C}$ can be null recurrent, i.e., every state must be positive recurrent. ►

Remark. This also proves that in a finite state Markov chain, there can be no null recurrent states, i.e., every recurrent state must be positive recurrent. ►

We have now a complete characterization of transient and recurrent states in a finite state Markov chain. We summarize the results in the following theorem.

Theorem 7.31. *The states belonging to a communicating class that is not closed must be transient. Any state belonging to a finite communicating class that is closed must be recurrent (in fact positive recurrent). Consequently, for a finite state Markov chain, all closed communicating classes must be (positive) recurrent.*

Let us now summarize the recurrence / transience of the (finite state) examples that we have discussed so far.

Example 7.32. For a finite state birth and death chain, if $r_0 = r_N = 1$, the states 0 and N are recurrent and rest of the states are transient. The same holds in the Wright Fisher model as well as Wright Fisher model with selection (see Example 5.25) and in Example 5.26. In the finite state birth and death chain, if $r_0 = 1, r_N < 1$, then 0 is the only recurrent state and rest of the states are transient; if $r_0 < 1, r_N = 1$, then N is the only recurrent state and rest of the states are transient, while if $r_0, r_N < 1$ (which includes the Ehrenfest model of diffusion in Example 5.18 or the Bernoulli-Laplace model of diffusion in Example 5.19), all states are recurrent (in fact positive recurrent). The same holds for random walk on finite connected graphs, random walk on the circle or Wright Fisher model with mutation under condition of $\alpha_1 > 0, \alpha_2 > 0$. In Example 5.28, the states 1, 3, 4, 6, 8 are (positive) recurrent, but the states 2, 5, 7 are transient. ■

For infinite state models, such as the infinite state birth and death chains with $r_0 = 1$ or the branching process, having offspring distribution with $\alpha_0 > 0, \alpha_0 + \alpha_1 < 1$, we have 0 as the only recurrent state and the rest of the states are transient. In irreducible chains, such as the simple random walks on one-dimension, or simple symmetric random walk in two or higher dimensions, or in success run length Markov Chain in Example 5.24, with $p_k = p$ (and hence $q_k = 1 - p_k = 1 - p = q$), we have shown the recurrence or transience via direct computation. But for other irreducible chain, we have so far no concrete way deciding whether they are transient or recurrent. In the next chapter, we will develop some methods which will resolve some of these questions.

7.4 Periodicity of states

Before we continue, we describe another property of a state.

Definition 7.33. A state i is said to be of period $\lambda(i) > 1$ if $p_{ii}^{(n)} = 0$ unless n is a multiple of $\lambda(i)$, i.e., $n = k\lambda(i)$ for some $k \geq 1$ and $\lambda(i)$ is the largest integer with this property. The state is said to aperiodic if no such integer $\lambda(i) > 1$ exists. A aperiodic positive recurrent state is called an **ergodic** state.

Let us denote

$$I(i) = \{n \geq 1 : p_{ii}^{(n)} > 0\}. \quad (7.15)$$

Formally speaking, the period of the state i is defined as

$$\lambda(i) = \gcd(I(i)) \quad (7.16)$$

where $\gcd(A)$ is the greatest common divisor of the elements of A . If the set A is empty, then we take $\gcd(A) = +\infty$. When the set A is non-empty, 1 is always a divisor, therefore the set of divisors is non-empty. Further, no number bigger than any element of the set can be a divisor of the set A . Thus, the set of divisors is non-empty and finite and hence the gcd (maximum of the set of divisors) is uniquely defined.

In general, it is easy to find the period of a state from the transition probability matrix. Note that for the two state Markov chain (see Example 5.9), the states are aperiodic if at least one row has two positive entries. Similarly, in Example 5.24, the state 0 is aperiodic if $p_{00} = q_0 > 0$. But for the simple random walk we observe that $p_{00}^{(n)} = 0$ unless $n = 2m$ for some $m \geq 1$. Therefore, it has period at least 2. But $p_{00}^{(2)} > 0$ and hence the period can be at most 2. So the period is 2. Similar logic applies for simple random walks in higher dimensions. For the Gambler's ruin problem, we note that all the states $1, 2, \dots, N-1$ have period 2, however the states 0 or N have period 1. For the random walk on a graph (see Example 5.20), $G = (V, E)$ where $V = \{A, B, C, D, E, F\}$ and the edge set $E = \{(A, B), (A, D), (A, E), (A, F), (B, C), (B, E), (C, D), (C, E), (D, F)\}$, we have $p_{AA}^{(2)} \geq p_{AB} \times p_{BA} = 1/4 \times 1/4 > 0$ and $p_{AA}^{(3)} \geq p_{AB} \times p_{BE} \times p_{EA} = 1/4 \times 1/3 \times 1/4 > 0$. So, the period must be 1, as the largest number that divides both 2 and 3 is 1.

Exercise 7.34. For the Markov chains on $\{1, 2, 3, 4\}$ with transition probability matrix given by

$$a) \mathbf{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \end{bmatrix} \end{matrix}; \quad b) \mathbf{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{bmatrix} \end{matrix}; \quad c) \mathbf{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix};$$

find the period of each state. ◀

If $m, n \in I(i)$, then using Chapman Kolmogorov equations,

$$p_{ii}^{(m+n)} = \sum_{j \in \mathcal{S}} p_{ij}^{(m)} p_{ji}^{(n)} \geq p_{ii}^{(m)} p_{ii}^{(n)} > 0.$$

Therefore, $m + n \in I(i)$. So $I(i)$ is closed under addition.

Proposition 7.35. *If i is an aperiodic state, then there exists $N \geq 1$ such that $p_{ii}^{(n)} > 0$ for all $n \geq N$.*

We first illustrate this with an example. Suppose $p_{ii}^{(4)} > 0, p_{ii}^{(7)} > 0$, i.e., $4, 7 \in I(i)$. Using the above property $p_{ii}^{(8)} > 0$. So, $7, 8 \in I(i)$ and we have $1 = 8 - 7$. Now, we claim that if $n \geq 42 = 7 \times 6$ then $p_{ii}^{(n)} > 0$. Indeed, for any $n \geq 42$, we should be able to write it as $n = K7 + r$ where $r = 0, 1, 2, 3, 4, 5$ or 6 and $K \geq 6$. So, we can write $n = K7 + r(8 - 7) = (K - r)7 + r8$. Since $K \geq 6$, we have $K - r \geq 0$. Thus, by the property that $I(i)$ is closed under addition, we obtain that $(K - r)7 \in I(i)$ and $r8 \in I(i)$ and hence $n = (K - r)7 + r8 \in I(i)$. Note here that 42 is not the least such integer, we can do much better. Indeed a little calculation would show that if $n \geq 18$, $p_{ii}^{(n)} > 0$.

Proof : Our first goal is to prove that $I(i)$ contains two consecutive integers. Let $n_0, n_0 + k \in I(i)$. If $k = 1$, we are done. If $k > 1$, there is an $n_1 \in I(i)$ so that k is not a divisor of n_1 since the greatest common divisor of $I(i)$ is 1. Then $n_1 = mk + r$ with $0 < r < k$. Since $I(i)$ is closed under addition, $(m + 1)(n_0 + k) > (m + 1)n_0 + n_1$ are both in $I(i)$. Their difference is $k(m + 1) - n_1 = k - r < k$.

Repeating the last argument (at most k times), we eventually arrive at a pair of consecutive integers $m_0, m_0 + 1 \in I(i)$. It is now easy to show that the result holds for $N = m_0(m_0 - 1)$. Let $n \geq N$ and write $n = Lm_0 + r$ with $0 \leq r < m_0$ and $L \geq m_0 - 1$. Then $n = Lm_0 + r(m + 1 - m) = (L - r)m + r(m + 1) \in I(i)$. This completes the proof. ■

Exercise 7.36. Consider an irreducible aperiodic Markov chain.

a) For any pair of states $i, j \in \mathcal{S}$, show that there exists $M = M(i, j)$ such that $p_{ij}^{(n)} > 0$ for all $n \geq M$.

(b) If the chain has N states, show that there exists $M = M(N)$ such that $p_{ij}^{(n)} > 0$ for all states i, j and $n \geq M$.

(c) Show further that $M(4) \geq 18$. In general, $M(N) \geq 2(N - 1)^2$. ◀

Another property of communicating classes is the preservation of periodicity of the states.

Proposition 7.37. *All states in a communicating class have the same period. More precisely, suppose $i \leftrightarrow j$. If i has period λ , then j has period λ .*

Proof : Let us consider $I(i) = \{n \geq 1 : p_{ii}^{(n)} > 0\}$ and $I(j) = \{n \geq 1 : p_{jj}^{(n)} > 0\}$. Assume that $\lambda_i = \gcd(I(i))$ and $\lambda_j = \gcd(I(j))$.

Choose n_1 and n_2 such that $p_{ij}^{(n_1)} > 0$ and $p_{ji}^{(n_2)} > 0$. Since $p_{jj}^{(n_1+n_2)} \geq p_{ji}^{(n_2)} p_{ij}^{(n_1)} > 0$, we have that $(n_1 + n_2) \in I(j)$. Thus, $\lambda_j \mid (n_1 + n_2)$. Now, for any $n \in I(i)$, i.e., $p_{ii}^{(n)} > 0$, we have $p_{jj}^{(n_1+n_2+n)} \geq p_{ji}^{(n_2)} p_{ii}^{(n)} p_{ij}^{(n_1)} > 0$ by the Chapman Kolmogorov equations. Thus, $(n_1 + n_2 + n) \in I(j)$. Hence $\lambda_j \mid (n_1 + n_2 + n)$. Since $\lambda_j \mid (n_1 + n_2)$, we have $\lambda_j \mid n$ for every $n \in I(i)$, i.e., λ_j is a divisor of the $I(i)$. Since λ_i is the gcd of $I(i)$, we have that $\lambda_i \geq \lambda_j$.

Reversing the roles, $\lambda_j \geq \lambda_i$. Thus, $\lambda_i = \lambda_j$. ■

Exercise 7.38. Consider an irreducible Markov chain on N states with λ as the period of the chain. (Since the chain is irreducible, all states have the same period). Show that

a) Consider the set $J = \{n \geq 1 : p_{ii}^{(n)} > 0 \text{ for some } i \in \mathcal{S}\} = \cup_{i \in \mathcal{S}} I(i)$. Show that

$$\lambda = \gcd(J).$$

b) Set $J_k = J \cap \{1, 2, \dots, k\}$. Show that

$$\lambda = \gcd(J_N).$$

Show by an example that we may have $\lambda < \gcd(J_{N-1})$.

c) Fix any state i and consider $I_k(i) = I(i) \cap \{1, 2, \dots, k\}$. Show that

$$\lambda = \gcd(I_{2N-1}(i)).$$

Show by an example that we may have $\lambda < \gcd(I_{2N-2}(i))$. ◀

Chapter 8

First step analysis

We now introduce one of the most used techniques in Markov chain theory to compute various quantities through a system of linear equations. This is generally achieved by looking at the first step of the chain and using the Markov property, hence the name *first step analysis*.

8.1 Entering a closed set

The number of visits to a transient state, from Proposition 7.3, is finite with probability 1. Therefore, for a finite set of transient states, the total number of visits to the transient states, if any, can be at most finite with probability 1. Note, once the chain visits a recurrent state, it must be confined to the communicating recurrent class to which the state belongs. Therefore, for any Markov chain having finitely many transient states, can remain in the set of transient states for only a finite amount of time and then must enter one of the recurrent communicating classes.

To make it more formal, suppose that $\mathcal{C}$ is a closed class. We wish to compute the probability of entering a closed set starting from a transient state. For a closed set $\mathcal{C}$ and $i \in \mathcal{S}$, we set

$$\alpha_i^{(\mathcal{C})} = \mathbb{P}(X_n \in \mathcal{C} \text{ for some } n \geq 0 \mid X_0 = i). \quad (8.1)$$

When the set $\mathcal{C}$ is unambiguously defined, we will drop the superscript. Also, if the closed set consists of a single (absorbing) state $\{i_0\}$, we will denote it by $\alpha_i^{(i_0)}$. Note that we only need to concern ourselves with the case when i is transient. If i is recurrent and belongs to the recurrent class $\mathcal{D}$, say, then either $\mathcal{C} = \mathcal{D}$ or $\mathcal{C} \cap \mathcal{D} = \emptyset$. In the first case,

by definition, $\alpha_i^{(\mathcal{C})} = 1$ while in the second case, $\alpha_i^{(\mathcal{C})} = 0$ as $\mathcal{D}$ must be closed and hence the chain can never go out of $\mathcal{D}$.

We can express the above event as a union of disjoint events. So, we may write

$$\begin{aligned}\alpha_i^{(\mathcal{C})} &= \sum_{n=1}^{\infty} \mathbb{P}(X_n = i_n, X_k \notin \mathcal{C} \text{ for } k = 1, 2, \dots, n-1 \mid X_0 = i) \\ &= \sum_{n=1}^{\infty} \sum_{i_1, i_2, \dots, i_{n-1} \in \mathcal{S} \setminus \mathcal{C}, i_n \in \mathcal{C}} p_{i, i_1} p_{i_1, i_2} \cdots p_{i_{n-2}, i_{n-1}} p_{i_{n-1}, i_n}.\end{aligned}\quad (8.2)$$

Observe that if $\mathcal{C} = \{i_0\}$, i.e., i_0 is an absorbing state, we can see that

$$\alpha_i^{(\mathcal{C})} = \sum_{n=1}^{\infty} \mathbb{P}(X_n = i_0, X_k \neq i_0 \text{ for } k = 1, 2, \dots, n-1 \mid X_0 = i) = \sum_{n=1}^{\infty} f_{i, i_0}^{(n)} = f_{i, i_0}^*.$$

However, the above does not provide a suitable method for computation of $\alpha_i^{(\mathcal{C})}$. Denote by $\alpha_i^{(\mathcal{C})}(n)$ the probability that the chain visits the closed set $\mathcal{C}$ on or before the n -th step, i.e.,

$$\alpha_i^{(\mathcal{C})}(n) = \mathbb{P}(X_n \in \mathcal{C} \mid X_0 = i) = \sum_{j \in \mathcal{C}} \mathbb{P}(X_n = j \mid X_0 = i) = \sum_{j \in \mathcal{C}} p_{ij}^{(n)}. \quad (8.3)$$

We observe that, from the definition of closed set, $\{X_n \in \mathcal{C}\}$ is a sequence of increasing events. Thus, we have

$$\alpha_i^{(\mathcal{C})}(n) = \mathbb{P}(X_n \in \mathcal{C} \mid X_0 = i) \uparrow \mathbb{P}(\cup_{n=0}^{\infty} \{X_n \in \mathcal{C}\} \mid X_0 = i) = \alpha_i^{(\mathcal{C})}.$$

For a chain having finite number of transient states, suppose that the state space is partitioned into $\mathcal{S} = \mathcal{T} \cup \cup_{r \in \Lambda} \mathcal{C}_r$ where $\mathcal{T}$ is the set of transient states and $\mathcal{C}_r$ are the (closed) recurrent communicating classes for $r \in \Lambda$. The number of recurrent classes may be finite or infinite, i.e., Λ may be finite or infinite. Then, for any transient state i , we must have

$$1 = \sum_{j \in \mathcal{S}} p_{ij}^{(n)} = \sum_{j \in \mathcal{T}} p_{ij}^{(n)} + \sum_{r \in \Lambda} \sum_{j \in \mathcal{C}_r} p_{ij}^{(n)} = \sum_{j \in \mathcal{T}} p_{ij}^{(n)} + \sum_{r \in \Lambda} \alpha_i^{(\mathcal{C}_r)}(n).$$

Note that $p_{ij}^{(n)} \rightarrow 0$ for any transient state j (see Corollary 7.11.2). Using the fact that $\mathcal{T}$ is finite, and using MCT, if required (when Λ is infinite, we note $\alpha_i^{(\mathcal{C}_r)}(n) \uparrow \alpha_i^{(\mathcal{C}_r)}$), we obtain

$$1 = \sum_{r \in \Lambda} \alpha_i^{(\mathcal{C}_r)}. \quad (8.4)$$

Therefore, we conclude that, if there is only one recurrent class, it will enter that class with probability 1. In case there are more than one recurrent classes, the chain will enter into one of these recurrent classes and will stay inside the class afterwards. Thus, for the finite birth and death chains with $r_0 = r_N = 1$ (which includes the Gambler's ruin problem, see Example 5.16), we must have that the chain will end up either at 0 or at N . The same conclusion holds in the Wright Fisher model as well as Wright Fisher model with selection (see Example 5.25) and in Example 5.26.

If the set transient states is infinite, it may be the case that the Markov chain remains in the set of transient states with positive probability, even if there is only one recurrent class. An example of this is seen in the branching process, where 0 is the absorbing state and all states $1, 2, \dots$ are transient. But under conditions that $\alpha_1 < 1$ and $\sum_{k=1}^{\infty} k\alpha_k > 1$ (see Theorem 3.8), the extinction probability, which here is the probability of entering the closed set $\{0\}$, is strictly smaller than 1. We will now develop a method for computing the probabilities for entering different classes.

We show that the probabilities $\{\alpha_i^{(c)} : i \in \mathcal{T}\}$ satisfy the system of inhomogeneous equations: for $i \in \mathcal{T}$

$$y_i = \sum_{k \in \mathcal{C}} p_{ik} + \sum_{j \in \mathcal{T}} p_{ij} y_j = p_{i,\mathcal{C}} + \sum_{j \in \mathcal{T}} p_{ij} y_j \quad (8.5)$$

where $p_{i,\mathcal{C}} = \sum_{k \in \mathcal{C}} p_{ik}$. The system of equations (8.5) may admit of several independent solutions, but we show that among them there exists, a minimal solution defined in the obvious manner, which gives the required probability.

Theorem 8.1. *The probabilities $\{\alpha_i^{(c)} : i \in \mathcal{T}\}$, of entering in the closed set $\mathcal{C}$ are given by the minimal solution of (8.5) taking values in $[0, 1]$.*

Before we proceed, we write the set of equations (8.5) in an algebraic notation. We have already noted that the transition matrix can be partitioned into block matrices where each block corresponds to either one of the recurrent class or the set of transient states (see Theorem 7.28 and discussion following it). Let us now consider only the parts of the transition matrix where rows corresponding to the transient states $\mathcal{T}$, and columns corresponding to the closed set $\mathcal{C}$ and the transient states $\mathcal{T}$ are kept. More formally, we consider the matrix

$$\begin{array}{cc} & \begin{array}{cc} \mathcal{C} & \mathcal{T} \end{array} \\ \begin{array}{c} \mathcal{T} \end{array} & \begin{bmatrix} \mathbf{B} & \mathbf{Q} \end{bmatrix} \end{array}$$

We now construct a vector $\mathbf{b} = \mathbf{B}\mathbf{1}_C$ where $\mathbf{1}_C$ the column vector consisting of all 1's. Therefore, the i -th entry of the vector $\mathbf{b}$ is given by $\sum_{j \in C} p_{ij} = p_{i,C}$. Let $\mathbf{y}$ be the row vector $(\alpha_i^{(C)} : i \in \mathcal{T})$. Hence, we can write, the system of equations in (8.5) as

$$\mathbf{y}' = \mathbf{b} + \mathbf{Q}\mathbf{y}' \text{ i.e., } (\mathbf{I} - \mathbf{Q})\mathbf{y}' = \mathbf{b}. \quad (8.6)$$

Proposition 8.2. *When the number of transient states is finite, the equations in (8.6) admit a unique solution.*

Proof : When the number of transient states is finite, we want to show that $\mathbf{I} - \mathbf{Q}$ is invertible. This will imply that the equations in (8.6) or equivalently in (8.5) admit an unique solution, in case there are finitely many transient states. The unique solution is given by

$$\mathbf{y}' = (\mathbf{I} - \mathbf{Q})^{-1}\mathbf{b}.$$

Since this is the only solution we have that, for $i \in \mathcal{T}$

$$\alpha_i^{(C)} = ((\mathbf{I} - \mathbf{Q})^{-1}\mathbf{b})_i$$

where the right hand side is the i -th co-ordinate of the vector $(\mathbf{I} - \mathbf{Q})^{-1}\mathbf{b}$.

We now show that $\mathbf{I} - \mathbf{Q}$ is invertible. We can consider this matrix as an element of $\mathbb{R}^{|\mathcal{T}|^2}$ where $|\mathcal{T}|$ is the number of states in $\mathcal{T}$. Note that the convergence in $\mathbb{R}^{|\mathcal{T}|^2}$ is equivalent to convergence of each of the entries in $\mathbb{R}$.

Let us define the matrix $\mathbf{T}_n = \mathbf{I} + \mathbf{Q} + \mathbf{Q}^2 + \cdots + \mathbf{Q}^n$. Note that, (i, j) -th entry of the matrix $\mathbf{Q}^n$ is given by $p_{ij}^{(n)}$. Thus, the (i, j) -th entry of the matrix $\mathbf{T}_n$ is given by $\sum_{t=0}^n p_{ij}^{(t)} \rightarrow \sum_{t=0}^{\infty} p_{ij}^{(t)} < \infty$ for $i, j \in \mathcal{T}$. Thus, we see that $\mathbf{T}_n \rightarrow \mathbf{T}$ whose (i, j) -th entry is given by $\sum_{t=0}^{\infty} p_{ij}^{(t)}$. Now, we observe that

$$\begin{aligned} \mathbf{T}_n(\mathbf{I} - \mathbf{Q}) &= (\mathbf{I} - \mathbf{Q})\mathbf{T}_n \\ &= (\mathbf{I} - \mathbf{Q})(\mathbf{I} + \mathbf{Q} + \mathbf{Q}^2 + \cdots + \mathbf{Q}^n) \\ &= (\mathbf{I} + \mathbf{Q} + \mathbf{Q}^2 + \cdots + \mathbf{Q}^n) - (\mathbf{Q} + \mathbf{Q}^2 + \cdots + \mathbf{Q}^n + \mathbf{Q}^{n+1}) \\ &= \mathbf{I} - \mathbf{Q}^{n+1}. \end{aligned}$$

Since the matrix multiplication is a linear operation in its entries and here finitely many terms are involved, it is continuous. Therefore, we have $\mathbf{T}_n(\mathbf{I} - \mathbf{Q}) \rightarrow \mathbf{T}(\mathbf{I} - \mathbf{Q})$ and $(\mathbf{I} - \mathbf{Q})\mathbf{T}_n \rightarrow (\mathbf{I} - \mathbf{Q})\mathbf{T}$. On the other hand, $\mathbf{Q}^{n+1} \rightarrow \mathbf{0}$ since $\sqrt{\sum_{i,j \in \mathcal{T}} (p_{ij}^{(n+1)})^2} \leq$

$\sum_{i,j \in \mathcal{T}} p_{ij}^{(n+1)} \rightarrow 0$ as the n -th term of a convergent series tends to 0 and here finitely many terms are involved. This proves that

$$(\mathbf{I} - \mathbf{Q})\mathbf{T} = \mathbf{T}(\mathbf{I} - \mathbf{Q}) = \mathbf{I}. \quad (8.7)$$

Therefore, $\mathbf{I} - \mathbf{Q}$ is invertible. ■

The matrix $(\mathbf{I} - \mathbf{Q})^{-1}$ is known as the *fundamental matrix* for an absorbing chain. For an infinite state chain, the above system of equations in (8.5) (or equivalently (8.6)) may have more than one solution, as we will see shortly.

Proof : Since we are dealing with the single closed class, we will drop the superscript $\mathcal{C}$ in the proof. With $\alpha_i(n)$ as defined in equation (8.3), we note that $\alpha_i(0) = 0$ for all $i \in \mathcal{T}$ and for $n \geq 1$,

$$\alpha_i(n+1) = p_{i,\mathcal{C}} + \sum_{j \in \mathcal{T}} p_{ij} \alpha_j(n). \quad (8.8)$$

Indeed, we see that the above relation holds for $n = 0$ for $i \in \mathcal{T}$. Assuming that the equation (8.8) holds for all $i \in \mathcal{T}$ for some $n \geq 0$, we have

$$\begin{aligned} \alpha_i(n+1) &= \mathbb{P}(X_{n+1} \in \mathcal{C} \mid X_0 = i) = \sum_{j \in \mathcal{S}} \mathbb{P}(X_1 = j, X_{n+1} \in \mathcal{C} \mid X_0 = i) \\ &= \sum_{j \in \mathcal{C}} \mathbb{P}(X_1 = j, X_{n+1} \in \mathcal{C} \mid X_0 = i) + \sum_{j \in \mathcal{T}} \mathbb{P}(X_1 = j, X_{n+1} \in \mathcal{C} \mid X_0 = i) \end{aligned}$$

Note that we have removed the sums on any closed set which do not intersect $\mathcal{C}$ since, if X_1 enters such a set then X_{n+1} will remain in the same set and will never enter $\mathcal{C}$. For $j \in \mathcal{C}$, we must have $X_{n+1} \in \mathcal{C}$ if $X_1 = j$. Thus, for $j \in \mathcal{C}$ we have that $\mathbb{P}(X_1 = j, X_{n+1} \in \mathcal{C} \mid X_0 = i) = \mathbb{P}(X_1 = j \mid X_0 = i) = p_{ij}$. For any $j \in \mathcal{T}$, using Proposition 6.1, we have

$$\mathbb{P}(X_1 = j, X_{n+1} \in \mathcal{C} \mid X_0 = i) = \mathbb{P}(X_1 = j \mid X_0 = i) \mathbb{P}(X_n \in \mathcal{C} \mid X_0 = j) = p_{ij} \alpha_j^{(n)}.$$

Combining the above relations, we have

$$\alpha_i(n+1) = \sum_{j \in \mathcal{C}} p_{ij} + \sum_{j \in \mathcal{T}} p_{ij} \alpha_j(n) = p_{i,\mathcal{C}} + \sum_{j \in \mathcal{T}} p_{ij} \alpha_j(n).$$

Now, we take limit as $n \rightarrow \infty$ in (8.8). The left hand side obviously converge to α_i . For the right hand side, the sum $\sum_{j \in \mathcal{T}} p_{ij} \alpha_j(n) \rightarrow \sum_{j \in \mathcal{T}} p_{ij} \alpha_j$ by DCT as $p_{ij} \alpha_j(n) \rightarrow p_{ij} \alpha_j$ and $p_{ij} \alpha_j(n) \leq p_{ij}$ with $\sum_{j \in \mathcal{T}} p_{ij} \leq 1$. Thus, we observe that $\{\alpha_i : i \in \mathcal{T}\}$ satisfy the system of equations in (8.5).

If y_i is any non-negative solution of (8.5) in $[0, 1]$, then $y_i \geq \alpha_j(0) = 0$. Then, we claim that $y_i \geq \alpha_j(n)$ for all $n \geq 0$. Assume this for n . Then, for $i \in \mathcal{T}$,

$$y_i = p_{i,c} + \sum_{j \in \mathcal{T}} p_{ij} y_j \geq p_{i,c} + \sum_{j \in \mathcal{T}} p_{ij} \alpha_j(n) = \alpha_j(n+1).$$

Thus, $y_i \geq \alpha_i$ for $i \in \mathcal{T}$. This completes the proof of the Theorem. ■

Example 8.3. We return to the Example 5.28. Let $\mathcal{C}_1 = \{1, 4\}$ and $\mathcal{C}_2 = \{3, 6, 8\}$ and $\mathcal{T} = \{2, 5, 7\}$. Now, we want to find $\alpha_i^{(\mathcal{C}_1)}$ and $\alpha_i^{(\mathcal{C}_2)}$ for $i \in \mathcal{T}$. We have the systems of equations from (8.5):

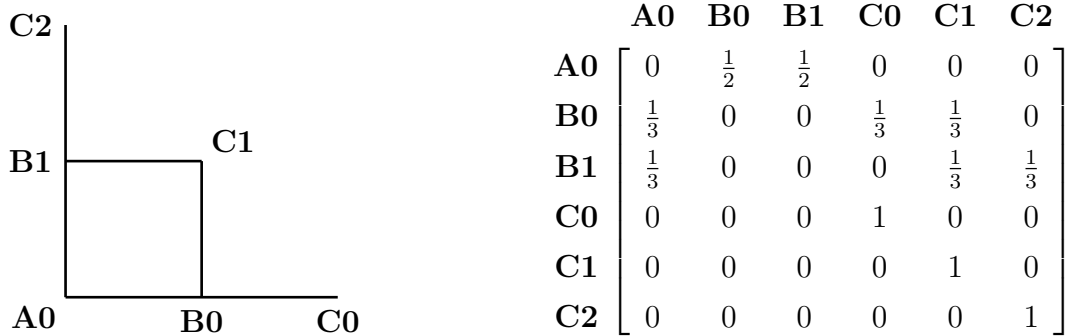
$$\alpha_2^{(\mathcal{C}_1)} = \frac{1}{2}\alpha_5^{(\mathcal{C}_1)}, \alpha_5^{(\mathcal{C}_1)} = \frac{2}{3}\alpha_2^{(\mathcal{C}_1)} \text{ and } \alpha_7^{(\mathcal{C}_1)} = \frac{1}{3} + \frac{1}{9}\alpha_2^{(\mathcal{C}_1)} + \frac{1}{6}\alpha_5^{(\mathcal{C}_1)} + \frac{1}{6}\alpha_7^{(\mathcal{C}_1)}.$$

Solving, we have $\alpha_2^{(\mathcal{C}_1)} = \alpha_7^{(\mathcal{C}_1)} = 0$ and $\alpha_5^{(\mathcal{C}_1)} = 2/5$.

Since the state space is finite, the chain must end in a recurrent class (see equation (8.4)), we must have $\alpha_i^{(\mathcal{C}_1)} + \alpha_i^{(\mathcal{C}_2)} = 1$ for $i = 2, 5$ or 7 . Hence, $\alpha_2^{(\mathcal{C}_2)} = 1, \alpha_5^{(\mathcal{C}_2)} = 1$ and $\alpha_7^{(\mathcal{C}_2)} = 3/5$. ■

Exercise 8.4. Consider the random walk on the graphs given below (with some modifications).

a) Consider



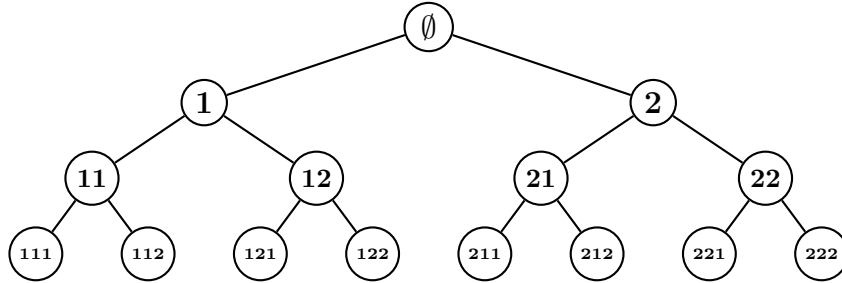
with the changes that the vertices **C0**, **C1** and **C2** are absorbing states. Find the probability $\alpha_{\mathbf{A}}^{(\mathbf{C0})}$ and $\alpha_{\mathbf{A}}^{(\mathbf{C1})}$.

b) Consider

	A0	B0	B1	C0	C1	C2	D0	D1	D2	D3
A0	0	$\frac{1}{2}$	$\frac{1}{2}$	0	0	0	0	0	0	0
B0	$\frac{1}{3}$	0	0	$\frac{1}{3}$	$\frac{1}{3}$	0	0	0	0	0
B1	$\frac{1}{3}$	0	0	0	$\frac{1}{3}$	$\frac{1}{3}$	0	0	0	0
C0	0	$\frac{1}{3}$	0	0	0	0	$\frac{1}{3}$	$\frac{1}{3}$	0	0
C1	0	$\frac{1}{4}$	$\frac{1}{4}$	0	0	0	0	$\frac{1}{4}$	$\frac{1}{4}$	0
C2	0	0	$\frac{1}{3}$	0	0	0	0	0	$\frac{1}{3}$	$\frac{1}{3}$
D0	0	0	0	0	0	0	1	0	0	0
D1	0	0	0	0	0	0	0	1	0	0
D2	0	0	0	0	0	0	0	0	1	0
D3	0	0	0	0	0	0	0	0	0	1

with the changes that the vertices **D0**, **D1**, **D2** and **D3** are absorbing states. Find the probability $\alpha_{\mathbf{A}}^{(\mathbf{D0})}$ and $\alpha_{\mathbf{A}}^{(\mathbf{D1})}$. ◀

Exercise 8.5. Consider a modified random walk on the rooted regular binary tree till depth 3 (and hence finite) shown below:



Suppose that the leaf nodes (the vertices **111**, **112**, **121**, **122**, **211**, **212**, **221** and **222**) are absorbing states. Show that the probability of absorption at any of these states are the same, when the walker starts from the root $\emptyset$ and hence equals $1/8$. What is the probability of absorption at each of these states when walker starts from **1**. Generalize the above for the rooted regular binary tree till depth n . ◀

Example 8.6. (Birth and death chains with finitely many states) : We consider a finite state space birth and death chain with two absorbing states 0 and N , i.e., $r_0 = r_N = 1$. Note that if we have $r_0 = 1, r_N < 1$ or $r_0 < 1, r_N = 1$, then the chain will get absorbed in the only closed class with probability 1.

Let $\mathcal{C}_1 = \{0\}$, $\mathcal{C}_2 = \{N\}$ and $\mathcal{T} = \{1, 2, \dots, N-1\}$. We wish to compute $\alpha_i^{(\mathcal{C}_1)}$ for $i \in \mathcal{T}$. We will drop the superscript in the computation below. We have following set of equations from (8.5):

$$\begin{aligned} \alpha_1 &= q_1 + r_1\alpha_1 + p_1\alpha_2 \\ \alpha_i &= q_i\alpha_{i-1} + r_i\alpha_i + p_i\alpha_{i+1} \quad \text{for } i = 2, 3, \dots, N-2 \\ \alpha_{N-1} &= r_{N-1}\alpha_{N-1} + q_{N-1}\alpha_{N-2}. \end{aligned}$$

Setting $\alpha_0 = 1$ and $\alpha_N = 0$, and noting that $p_i + r_i + q_i = 1$, we see that the above equations can be written as

$$p_i(\alpha_{i+1} - \alpha_i) = q_i(\alpha_i - \alpha_{i-1})$$

for $i = 1, 2, \dots, N-1$. Therefore, we have, for $i = 1, 2, \dots, N-1$,

$$\begin{aligned} \alpha_{i+1} - \alpha_i &= \frac{q_i}{p_i}(\alpha_i - \alpha_{i-1}) = \frac{q_i q_{i-1}}{p_i p_{i-1}}(\alpha_{i-1} - \alpha_{i-2}) = \dots \\ &= \frac{q_i q_{i-1} \dots q_1}{p_i p_{i-1} \dots p_1}(\alpha_1 - \alpha_0). \end{aligned}$$

We set

$$a_i = \begin{cases} \prod_{j=1}^i \frac{q_j}{p_j} & \text{if } i \geq 1 \\ 1 & \text{if } i = 0. \end{cases} \quad (8.9)$$

Therefore, we have $\alpha_{i+1} - \alpha_i = a_i(\alpha_1 - 1)$ for $i \geq 0$. Summing these over $i = 0, 1, 2, \dots, N-1$, we have,

$$-1 = \alpha_N - \alpha_0 = \sum_{i=0}^{N-1} (\alpha_{i+1} - \alpha_i) = (\alpha_1 - \alpha_0) \sum_{i=0}^{N-1} a_i = (\alpha_1 - 1) \sum_{i=0}^{N-1} a_i$$

so that

$$\alpha_1 = 1 - \frac{1}{\sum_{i=0}^{N-1} a_i}.$$

Thus, for $i = 2, 3, \dots, N-1$, we have

$$\alpha_i - 1 = \sum_{j=0}^{i-1} (\alpha_{j+1} - \alpha_j) = (\alpha_1 - 1) \sum_{j=0}^{i-1} a_j = -\frac{\sum_{j=0}^{i-1} a_j}{\sum_{i=0}^{N-1} a_i}.$$

Therefore, we have, for $i = 0, 1, \dots, N$

$$\alpha_i^{(\mathcal{C}_1)} = f_{i0}^* = 1 - \frac{\sum_{j=0}^{i-1} a_j}{\sum_{i=0}^{N-1} a_i}. \quad (8.10)$$

For the closed set $\mathcal{C}_2 = \{N\}$, again we note that for all $i \in \mathcal{T}$, $\alpha_i^{(\mathcal{C}_1)} + \alpha_i^{(\mathcal{C}_2)} = 1$. Hence, for $i = 1, 2, \dots, N-1$, we have

$$\alpha_i^{(\mathcal{C}_2)} = f_{iN}^* = \frac{\sum_{j=0}^{i-1} a_j}{\sum_{i=0}^{N-1} a_i}. \quad (8.11)$$

Also observe that, the expression matches also for $i = 0$ as well as $i = N$.

For the gambler's ruin problem in Example 5.16, we have $p_k = p$ and $q_k = q$ for all $k = 1, 2, \dots, N-1$. Therefore, for $i \geq 0$, we have,

$$a_i = \prod_{j=1}^i \frac{q}{p} = \frac{p^i}{q^i} = \begin{cases} 1 & \text{if } p = q \\ \left(\frac{q}{p}\right)^i & \text{if } p \neq q. \end{cases} \quad (8.12)$$

Thus, the probability of ruin of Gambler I starting with capital i , i.e., for $i = 1, 2, \dots, N-1$, absorption to state 0 starting at i is given by

$$\alpha_i^{(\mathcal{C}_1)} = f_{i0}^* = \begin{cases} \frac{N-i}{N} & \text{if } p = q \\ \frac{(q/p)^N - (q/p)^i}{(q/p)^N - 1} & \text{if } p \neq q \end{cases} \quad (8.13)$$

and the probability of ruin of Gambler II is given by

$$\alpha_i^{(\mathcal{C}_2)} = f_{iN}^* = \begin{cases} \frac{i}{N} & \text{if } p = q \\ \frac{(q/p)^i - (q/p)^N}{(q/p)^N - 1} & \text{if } p \neq q. \end{cases} \quad (8.14)$$

■

Example 8.7. (Birth and death chains with infinitely many states) For the birth and death chain on the set of all non-negative integers (see Example 5.15) with 0 as the absorbing state, i.e., $r_0 = 1$, the system of equations in (8.5) yields the following equations:

$$\begin{aligned} y_1 &= q_1 + r_1 y_1 + p_1 y_2 \\ y_i &= q_i y_{i-1} + r_i y_i + p_i y_{i+1} \end{aligned} \quad \text{for } i \geq 2.$$

Again we set $y_0 = 1$ and noting that $p_i + r_i + q_i = 1$ for all $i \geq 1$, we see that the above equations can be written as

$$p_i(y_{i+1} - y_i) = q_i(y_i - y_{i-1})$$

for all $i \geq 1$. Hence, as before, we have $y_{i+1} - y_i = (y_1 - 1)a_i$, for all $i \geq 1$, where a_i is as defined in equation (8.9). Hence, for $i \geq 2$, we have

$$y_i = 1 - (1 - y_1) \sum_{j=0}^{i-1} a_j = 1 - (1 - y_1)A_i \quad (8.15)$$

where $A_i = \sum_{j=0}^{i-1} a_j$. Note, for $i \geq 1$, $A_i > 0$ and $A_i \uparrow A_\infty =: \sum_{j=0}^{\infty} a_j$ as $i \rightarrow \infty$.

We make a few observations first: any choice of y_1 will provide a solution $\{y_i : i \geq 1\}$ of the system of equations in (8.5) by putting the value of y_1 in equation (8.15). Furthermore, these are all possible of solutions of equations in (8.5). Thus, every solution of (8.5) will correspond to a value of y_1 . Further each y_i is a strictly increasing function of y_1 . So, the minimal solution of (8.5) will correspond to the minimal value of $y_1 \in [0, 1]$ such that $0 \leq y_i \leq 1$ for all $i \geq 1$. Now we need to separate between two cases.

- Case $A_\infty = \infty$: if $y_1 < 1$, then letting $i \rightarrow +\infty$ in equation (8.15), we have $y_i \rightarrow 1 - (1 - y_1)A_\infty = -\infty$. Hence, for all sufficiently large i , we must have $y_i < 0$. Thus, such a value of y_1 is not feasible as the corresponding solution is not feasible as it violates the condition that $y_i \in [0, 1]$ for all $i \geq 1$. If $y_1 = 1$, we have $y_i = 1$ for $i \geq 2$. Thus, the only feasible value of y_1 is 1. Therefore we have $\alpha_i^{(\{0\})} = 1$ for all $i \geq 1$. This means that the absorption to 0 from any state $i \geq 1$ is certain.
- Case $A_\infty < \infty$: if $y_1 < 1 - 1/A_\infty$ i.e., $(1 - y_1)A_\infty > 1$, then letting $i \rightarrow +\infty$ in equation (8.15)

$$y_i \rightarrow 1 - (1 - y_1)A_\infty < 0.$$

Therefore, as above, such values of y_1 are not feasible.

Next consider $y_1 \geq 1 - 1/A_\infty$ i.e., $(1 - y_1)A_\infty \leq 1$. First observe that we have $y_i = 1 - (1 - y_1)A_i \leq 1$ as $y_1 \leq 1$ and $A_i > 0$. Further since $A_i \uparrow A_\infty$, and hence $y_i \downarrow \lim_{i \rightarrow \infty} y_i = 1 - (1 - y_1)A_\infty \geq 0$. Thus, all values of $y_1 \in [1 - 1/A_\infty, 1]$ are feasible. Hence, the minimality condition will imply that the absorption probabilities are obtained by taking $y_1 = 1 - 1/A_\infty$. Therefore, for all $i \geq 1$, from equation (8.15), we have

$$\alpha_i = f_{i0}^* = 1 - \frac{A_i}{A_\infty} = 1 - \frac{\sum_{j=0}^{i-1} a_j}{\sum_{j=0}^{\infty} a_j}.$$

For the special case of random walk on the non-negative integers with 0 as the only absorbing state (see the case of playing against infinitely rich adversary in Example

5.16), we have $p_k = p$ and $q_k = q$ for all $k \geq 1$. Thus, $a_j = q^j/p^j$ for $j \geq 0$ (see equation (8.12) also). Thus, we have

$$A_\infty = \sum_{j=0}^{\infty} a_j = \sum_{j=0}^{\infty} \left(\frac{q}{p}\right)^j = \begin{cases} \infty & \text{if } q \geq p, \text{ i.e., } p \leq 1/2 \\ \frac{p}{p-q} < \infty & \text{if } q < p, \text{ i.e., } p > 1/2. \end{cases}$$

Thus, the absorption probability at 0 is given by

$$\alpha_i = f_{i0}^* = \begin{cases} 1 & \text{if } p \leq 1/2, \\ \left(\frac{q}{p}\right)^i & \text{if } p > 1/2. \end{cases} \quad \blacksquare$$

Exercise 8.8. For the following birth and death chains, with $r_0 = 1$, decide whether they are absorbed at 0 with probability 1.

1. For $i \geq 1$,

$$p_i = p, r_i = r \text{ and } q_i = q$$

where $p + q + r = 1$.

- 2.

$$\begin{aligned} p_i = p, r_i = r \text{ and } q_i = q & \quad \text{if } i \text{ is odd, } i \geq 1 \\ p_i = q, r_i = r \text{ and } q_i = p & \quad \text{if } i \text{ is even, } i \geq 1 \end{aligned}$$

where $p + q + r = 1$.

3. For $i \geq 1$,

$$p_i = \frac{i}{2(i+1)}, r_i = \frac{1}{i+1} \text{ and } q_i = \frac{i}{2(i+1)}.$$

4. For $i \geq 1$,

$$p_i = \frac{i+2}{2(i+1)}, r_i = 0 \text{ and } q_i = \frac{i}{2(i+1)}.$$

5. For $i \geq 1$,

$$p_i = \frac{1}{2}, r_i = \frac{1}{2(i+1)} \text{ and } q_i = \frac{i}{2(i+1)}.$$

6. For $i \geq 1$,

$$p_i = \frac{i+1}{2i+1}, r_i = 0 \text{ and } q_i = \frac{i}{2i+1}.$$

7.

$$\begin{aligned} p_i &= \frac{i-1}{2i}, r_i = 0 \text{ and } q_i = \frac{i+1}{2i} & \text{if } i \geq 2 \\ p_i &= \frac{1}{2}, r_i = 0 \text{ and } q_i = \frac{1}{2} & \text{if } i = 1. \end{aligned} \quad \blacktriangleleft$$

Exercise 8.9. Consider a modified random walk on the infinite rooted regular k -ary trees (for $k \geq 2$), i.e., every node has k -many child nodes, with the modification that the root node is the only absorbing state. Compute the probability of absorption at the root node starting from any node. $\blacktriangleleft$

Example 8.10. We return to the Example 5.24. Here the state space $\mathcal{S} = \{0, 1, \dots\}$. Observe that, if $0 < p_k < 1$ for all $k \geq 1$ but $q_0 = 1$, i.e., $p_0 = 0$, then 0 is the only absorbing state. Now, the system of equations in (8.5) is given by

$$y_i = q_i + p_{i+1}y_{i+1} = q_i + (1 - q_i)y_{i+1} \quad \text{for } i \geq 1.$$

Therefore, we have $1 - y_i = (1 - q_i)(1 - y_{i+1})$. Thus, for $i \geq 2$, we have

$$1 - y_{i+1} = \frac{1}{q_i}(1 - y_i) = \frac{1}{q_i q_{i-1}}(1 - y_{i-1}) = \dots = \frac{1}{q_i q_{i-1} \dots q_1}(1 - y_1).$$

Let us set

$$A_i = \begin{cases} \frac{1}{\prod_{j=1}^{i-1} q_j} & \text{if } i \geq 1 \\ 1 & \text{if } i = 0. \end{cases} \quad (8.16)$$

Hence, for $i \geq 2$, we have

$$y_i = 1 - (1 - y_1)A_i. \quad (8.17)$$

Rest of the analysis is same as in the previous example. We note here that $A_i \uparrow A_\infty$ where $A_\infty = 1/\prod_{j=1}^\infty q_j$. Observe that $A_\infty = \infty$ if and only if $\prod_{j=1}^\infty q_j = 0$. Therefore, for $i \geq 1$, we have

$$\alpha_i = f_{i0}^* = \begin{cases} 1 & \text{if } \prod_{j=1}^\infty q_j = 0 \\ 1 - \frac{A_i}{A_\infty} = 1 - \frac{\prod_{j=1}^{i-1} q_j}{\prod_{j=1}^\infty q_j} = 1 - \prod_{j=i}^\infty q_j & \text{if } \prod_{j=1}^\infty q_j > 0. \end{cases}$$

In the special case of success runs, we have $p_k = p$ and $q_k = q = 1 - p$ where $0 < p < 1$. Thus, $A_i = 1/q^{i-1} \rightarrow \infty$ as $i \rightarrow \infty$. Therefore, $\alpha_i = f_{i0}^* = 1$, i.e., streak of success runs, however long, must end with probability 1. $\blacksquare$

8.2 Probability of visiting a state

If a state i is recurrent and another state j is transient, we have $f_{ij}^* = 0$ since recurrent states can only lead to recurrent states (see Corollary 7.27.2). For recurrent states $i \neq j$, f_{ij}^* must be either 0 or 1. Indeed, for two recurrent states i and j , consider their communicating classes $\mathcal{C}(i)$ and $\mathcal{C}(j)$ with $i \in \mathcal{C}(i)$ and $j \in \mathcal{C}(j)$. Then either $\mathcal{C}(i)$ and $\mathcal{C}(j)$ are identical or they have no intersection. If $\mathcal{C}(i) \cap \mathcal{C}(j) = \emptyset$, then $i \not\leftrightarrow j$. Hence, $f_{ij}^* = 0$.

Proposition 8.11. *If $i \neq j \in \mathcal{C}$ where $\mathcal{C}$ is a communicating recurrent class, then $f_{ij}^* = 1$.*

Proof : Suppose that $f_{ij}^* < 1$. Hence, $\mathbb{P}(X_n \neq j \text{ for all } n \geq 1 \mid X_0 = i) = 1 - f_{ij}^* > 0$. Since i is reachable from j , for some $m \geq 1$, we must have $p_{ji}^{(m)} > 0$. Since j is recurrent, the state j must be visited infinitely many times, in particular, must be visited after m starting from j . Therefore, $\mathbb{P}(X_n \neq j \text{ for some } n > m \mid X_0 = j) = 0$. On the other hand,

$$\begin{aligned} \mathbb{P}(X_n \neq j \text{ for } n > m \mid X_0 = j) &\geq \mathbb{P}(X_n \neq j \text{ for } n > m, X_m = i \mid X_0 = j) \\ &= \mathbb{P}(X_n \neq j \text{ for } n \geq 1 \mid X_0 = i) \mathbb{P}(X_m = i \mid X_0 = j) = (1 - f_{ij}^*) p_{ji}^{(m)} > 0. \end{aligned}$$

This yields a contradiction. Therefore, we must have $f_{ij}^* = 1$ ■

If i is transient and j is recurrent, we consider the (recurrent and closed) communicating class $\mathcal{C}$ which contains j . We claim that

$$f_{i,j}^* = \alpha_i^{(\mathcal{C})}.$$

Intuitively this is obvious since, once the chain visits the closed class $\mathcal{C}$, the chain must remain inside $\mathcal{C}$, and from any state in the recurrent class $\mathcal{C}$, the chain must visit j with probability 1. Formally, the event $\{X_n = j \text{ for some } n \geq 1\}$ is a subset of $\{X_n \in \mathcal{C} \text{ for some } n \geq 1\}$. Therefore, we have $f_{ij}^* \leq \alpha_i^{(\mathcal{C})}$. On the other hand, we have if $\{X_n = j \text{ for some } n \geq 1\}$ is a strict subset of $\{X_n \in \mathcal{C} \text{ for some } n \geq 1\}$, then we must have that the chain will visit $\mathcal{C}$ at some time point but it will never visit the state j then on. When the chain visits $\mathcal{C}$ for the first time, it will obviously land at a state $k \neq j$ in $\mathcal{C}$. Therefore

we have

$$\begin{aligned}
& \alpha_i^{(C)} - f_{ij}^* \\
& \leq \mathbb{P}\left(\bigcup_{n=1}^{\infty} \bigcup_{k \in C \setminus \{j\}} \{X_s \notin C \text{ for } 1 \leq s < n, X_n = k, X_{n+t} \neq j \text{ for all } t \geq 1\} \mid X_0 = i\right) \\
& \leq \sum_{n=1}^{\infty} \sum_{k \in C \setminus \{j\}} \mathbb{P}(X_s \notin C \text{ for } 1 \leq s < n, X_n = k, X_{n+t} \neq j \text{ for all } t \geq 1 \mid X_0 = i) \\
& \leq \sum_{n=1}^{\infty} \sum_{k \in C \setminus \{j\}} \mathbb{P}(X_s \notin C \text{ for } 1 \leq s < n, X_n = k \mid X_0 = i) \mathbb{P}(X_t \neq j \text{ for all } t \geq 1 \mid X_0 = k) \\
& \leq \sum_{n=1}^{\infty} \sum_{k \in C \setminus \{j\}} (1 - f_{kj}^*) \mathbb{P}(X_s \notin C \text{ for } 1 \leq s < n, X_n = k \mid X_0 = i) = 0
\end{aligned}$$

where we have used Proposition 6.1 in the third step.

However, when both states are transient, we do not have any suitable method for calculating the probabilities. We now adapt the methods of the previous section to compute this. Let us consider a transient state i_0 . We will update the transition matrix $\mathbf{P}$ by making a given state i_0 as absorbing, i.e., we define $\tilde{\mathbf{P}} = ((\tilde{p}_{ij}))$ as follows:

$$\tilde{p}_{i,j} = \begin{cases} \delta_{i_0,j} & \text{if } i = i_0 \\ p_{i,j} & \text{if } i \neq i_0. \end{cases} \quad (8.18)$$

In the updated chain, the state i_0 is an absorbing state and other transient states remain transient. However, observe that, using (8.2), we have

$$\begin{aligned}
\tilde{\alpha}_i^{(i_0)} &= \sum_{n=1}^{\infty} \sum_{i_1, i_2, \dots, i_{n-1} \neq i_0} \tilde{p}_{i,i_1} \tilde{p}_{i_1,i_2} \cdots \tilde{p}_{i_{n-2},i_{n-1}} \tilde{p}_{i_{n-1},i_0} \\
&= \sum_{n=1}^{\infty} \sum_{i_1, i_2, \dots, i_{n-1} \neq i_0} p_{i,i_1} p_{i_1,i_2} \cdots p_{i_{n-2},i_{n-1}} p_{i_{n-1},i_0} \\
&= \sum_{n=1}^{\infty} \mathbb{P}(X_t \neq i_0 \text{ for } 1 \leq t < n, X_n = i_0 \mid X_0 = i) = f_{i,i_0}^*. \quad (8.19)
\end{aligned}$$

Observe that in this argument we have not actually used the fact that i is a transient state. Therefore, it holds in general.

Therefore, we have f_{i,i_0}^* as the minimal solutions of the equations

$$y_i = p_{i,i_0} + \sum_{j \in T \setminus \{i_0\}} p_{ij} y_j.$$

As earlier, if the state space is finite, we have a unique solution which provides us the required values.

Now, to compute the return probabilities (f_{ii}^*) , we may use a generic equation. Observe that, from any state i , the return may happen at the first step with probability p_{ii} or in the first step the chain may move to a state j and from j , the state will eventually visit i , which happens with probability f_{ji}^* . Therefore, we have

$$f_{ii}^* = p_{ii} + \sum_{j \neq i} p_{ij} f_{ji}^*. \quad (8.20)$$

Note that, if i is transient, we can remove all recurrent states j as from j we can never reach the transient state i .

Example 8.12. Consider the Markov Chain on $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$ with the transition matrix $\mathbf{P}$ given by

$$\mathbf{P} = \begin{array}{c} \begin{array}{cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} & \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{2}{5} & \frac{3}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{6} & \frac{1}{3} & 0 & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & 0 & \frac{1}{8} & 0 \end{bmatrix} \end{array} \end{array}.$$

There are only two communicating classes, namely $\mathcal{C}_1 = \{1, 2\}$ and $\mathcal{C}_2 = \{3, 4, 5, 6\}$. Also note that $\mathcal{C}_1$ is closed and $\mathcal{C}_2$ is not closed. Therefore, the states 1, 2 are recurrent and the states 3, 4, 5, 6 are all transient.

We now use the above methods to compute f_{33}^* . From equation (8.20), we have

$$f_{33}^* = p_{33} + \sum_{i \neq 3} p_{3i} f_{i3}^* = \frac{1}{6} + \sum_{i=4}^6 p_{3i} f_{i3}^* = \frac{1}{6} + \frac{1}{3} f_{43}^* + \frac{1}{6} f_{63}^*$$

using the fact that 3 is transient and 1 and 2 are recurrent and hence $f_{i3}^* = 0$ for $i = 1, 2$. Therefore, we need to compute $\{f_{i3}^* : i \geq 4\}$. For that we have the system of linear equations

$$\begin{aligned} f_{43}^* &= \frac{1}{8} + \frac{1}{8} f_{43}^* + \frac{1}{8} f_{53}^* + \frac{1}{8} f_{63}^*; \\ f_{53}^* &= \frac{1}{6} + \frac{1}{6} f_{43}^* + \frac{1}{6} f_{53}^* + \frac{1}{6} f_{63}^*; \\ f_{63}^* &= \frac{1}{8} + \frac{1}{8} f_{53}^*. \end{aligned}$$

Solving, we have $f_{43}^* = \frac{9}{44}$, $f_{53}^* = \frac{3}{11}$ and $f_{63}^* = \frac{7}{44}$ and hence $f_{33}^* = \frac{23}{88}$. We can use similar method to compute other return probabilities. ■

8.3 Recurrence of irreducible chains

We have already observed that for finite state spaces any closed communicating class must be recurrent (see Proposition 7.30). However, this can't be generalized to the infinite state space, as can be in the case of simple but asymmetric random walk (see Example 7.15). Therefore it is of great interest to find whether an infinite irreducible chain is recurrent or transient. We now apply the results of the previous sections in deciding whether the irreducible chain on an infinite state space is recurrent or transient.

Theorem 8.13. *An irreducible Markov chain with state space $\{0, 1, 2, \dots\}$, the state 0 is recurrent (hence the whole chain is recurrent), if and only if the system of linear equations: for $i \geq 1$*

$$x_i = \sum_{k=1}^{\infty} p_{ik} x_k \quad (8.21)$$

admits of no solution with $0 \leq x_i \leq 1$ except $x_i = 0$ for all i .

Proof : We show that the result can be obtained from Theorem 8.1 after making suitable modifications. Since the chain is irreducible, if $f_{i0}^* < 1$ for some $i \geq 1$, then 0 must be transient (see Proposition 8.11). On the other hand, if $f_{i0}^* = 1$ for all $i \geq 1$, we have from equation (8.20) that

$$f_{00}^* = p_{00} + \sum_{i=1}^{\infty} p_{0i} f_{i0}^* = p_{00} + \sum_{i=1}^{\infty} p_{0i} = 1.$$

Therefore the chain is recurrent if and only if $f_{i0}^* = 1$ for all $i \geq 1$.

We prove the result by modifying the transition matrix by making the state 0 an absorbing state. Following the same method as in (8.18), we define $\tilde{\mathbf{P}} = ((\tilde{p}_{ij}))$ as follows:

$$\tilde{p}_{i,j} = \begin{cases} \delta_{0,j} & \text{if } i = 0 \\ p_{i,j} & \text{if } i \geq 1. \end{cases}$$

Therefore, in the updated chain we have that 0 is the only absorbing state and rest of the states $i \geq 1$ are transient state. Hence, from Theorem 8.1, we conclude that $\tilde{\alpha}_i^{(0)}$ is

given by minimal solution of the system of equations

$$y_i = p_{i0} + \sum_{j=1}^{\infty} p_{ij} y_j \quad (8.22)$$

for $i \geq 1$. Furthermore, we should also observe that $\tilde{\alpha}_i^{(0)} = f_{i,0}^*$ for any $i \geq 1$ (see equation (8.19)).

Let $\{y_i : i \geq 1\}$ be a solution of the system of equation in (8.22). Then, we claim $\{1 - y_i : i \geq 1\}$ is a solution of the system of equations in (8.21) and vice versa. Indeed, we have

$$1 - y_i = \sum_{j=0}^{\infty} p_{ij} - p_{i0} - \sum_{j=1}^{\infty} p_{ij} y_j = \sum_{j=1}^{\infty} p_{ij} (1 - y_j).$$

The converse also follows similarly.

We can now conclude that if the system of equations in (8.21) has only the trivial solution, then the system of equations in (8.22) has only solution as $y_i = 1$ for all $i \geq 1$. Therefore, we have $f_{i,0}^* = 1$ for all $i \geq 1$. If the system of equations in (8.21) has a non-trivial solution then the system of equations in (8.22) has a solution such that $y_i < 1$ for some $i \geq 1$. Since $f_{i,0}^*$ is the minimal solution of (8.22), we have that $f_{i,0}^* < 1$. This proves the result. ■

Since the system of equations in (8.21), multiplying both sides of the equation by a constant still yields a solution of (8.21), we have the following corollary.

Corollary 8.13.1. *An irreducible Markov chain with state space $\{0, 1, 2, \dots\}$ is recurrent if and only if the system of linear equations in (8.21) admit no non-negative bounded solution other than the trivial solution.*

Example 8.14. (Birth and death chains with infinitely many states) : We now consider an irreducible birth and death chain with infinitely many states. Note that, we need that $r_0 < 1$ for the irreducibility. We have observed that the system of equations in (8.21) can be solved from the system of equations in (8.22), and we have already solved the system of equations in (8.22) in Example 8.7. Hence, using those calculations, we may conclude that the chain is recurrent if and only if

$$\sum_{n=0}^{\infty} \prod_{j=1}^n \frac{q_j}{p_j} = \infty.$$

For the special case of reflecting barriers or retaining barriers on the set of non-negative integers (see Example 5.17 and also Example 8.7) the chain is recurrent if and only if $q \geq p$. ■

Example 8.15. We now return to the Example 5.24. We now consider the irreducible chain, i.e., $p_0 > 0$. However, we have considered the version of the chain where 0 is absorbing state in Example 8.10). Using the same calculations, we can now conclude that the chain is recurrent if and only if

$$\prod_{j=1}^{\infty} p_j = 0.$$

In the success run length Markov chain in Example 5.24, we have $p_k = p \in (0, 1)$. Therefore, the chain is recurrent. ■

Sometimes we may not be able to solve the system of linear equations explicitly. However, in some cases, we may be able to exhibit the existence of a non-trivial solution of the system of equations in (8.21) which is enough to show transience of the model.

Example 8.16. We return the discrete queuing model described in Example 5.22. Let us assume that the number of customers arrive with a probability distribution $\{\alpha_k : k \geq 0\}$ which satisfy the condition that $\alpha_1 < 1, \sum_{k=0}^{\infty} k\alpha_k > 1$. Let us consider the generating function $P(s) = \sum_{k=0}^{\infty} \alpha_k s^k$ of the arrival distribution. Since we have the expected number of arrivals is larger than 1, i.e., $\sum_{k=0}^{\infty} k\alpha_k > 1$ we know that $P(s) = s$ has a unique positive solution σ in $[0, 1)$ (see Theorem 2.32). We verify that $\{x_i = 1 - \sigma^i : i \geq 1\}$ is a solution of system of linear equations in (8.21). Note that for $i \geq 1$, the transition probabilities are given by (see equation (5.7)) :

$$p_{ij} = \begin{cases} \alpha_{j-(i-1)} & \text{if } j \geq i-1 \\ 0 & \text{otherwise.} \end{cases}$$

Therefore, for $i \geq 1$,

$$\begin{aligned} \sum_{j=1}^{\infty} p_{ij}(1 - \sigma^j) &= \sum_{j=i-1}^{\infty} \alpha_{j-(i-1)}(1 - \sigma^j) = \sum_{j=i-1}^{\infty} \alpha_{j-(i-1)} - \sum_{j=i-1}^{\infty} \alpha_{j-(i-1)} \sigma^{j-(i-1)+(i-1)} \\ &= \sum_{j=0}^{\infty} \alpha_j - \sigma^{i-1} \sum_{j=0}^{\infty} \alpha_j \sigma^j = 1 - \sigma^{i-1} P(\sigma) = 1 - \sigma^i. \end{aligned}$$

Thus, we have established a bounded non-trivial solution of (8.21) and hence the chain is transient. ■

Note however we do not have a full resolution of all the cases, namely, what happens if $\sum_{k=0}^{\infty} k\alpha_k \leq 1$. We will answer this at a later stage using martingale.

However, since we do not have the general solutions of (8.21), we are unable to claim that this is the maximal solution. For that we will use a more direct computation.

8.4 Visiting a state before another

A often asked question in Markov chain is whether the chain visits a given state before visiting another set of states. Let i be a given state and $\mathcal{C}$ be a set of states (of course not containing i) and we wish to find the probability that starting from state j , whether the chain visits the state i before visiting any of the states in $\mathcal{C}$, i.e.,

$$\begin{aligned}\alpha_{j;\mathcal{C}}^{(i)} &= \mathbb{P}(\text{the chain visits } i \text{ before visiting } \mathcal{C} \mid X_0 = j) = \mathbb{P}(T_i < T_{\mathcal{C}}) \\ &= \mathbb{P}(X_n = i \text{ for some } n \geq 1, X_k \notin \mathcal{C} \text{ for } 1 \leq k < n \mid X_0 = j)\end{aligned}\quad (8.23)$$

where T_i and $T_{\mathcal{C}}$ are respectively the hitting times of the state i and the set $\mathcal{C}$.

Now, if the state i is an absorbing state and $\mathcal{C}$ is closed, the problem is equivalent to visiting the absorbing state i but it would mostly be the case that neither the state i is absorbing nor the set $\mathcal{C}$ is closed. However, we may consider a derived transition matrix where the state i is absorbing and the set of states $\mathcal{C}$ is closed and other other transition probabilities are unchanged. To fix ideas, let us define $\tilde{\mathbf{P}} = ((\tilde{p}_{ij}))$ as follows:

$$\tilde{p}_{i,j} = \begin{cases} \delta_{k,j} & \text{if } k = i \text{ or } k \in \mathcal{C} \\ p_{k,j} & \text{otherwise.} \end{cases}$$

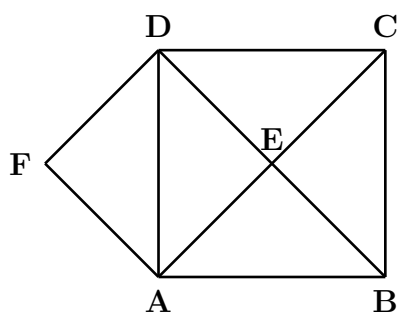
The probability of ever visiting the absorbing state i in this modified chain $\tilde{X}_n$ is the probability that the chain visits i before visiting $\mathcal{C}$. Intuitively, this is obvious as the transition probabilities from i are required only after the chain has reached i . Since each of the state in $\mathcal{C}$ has been transformed into an absorbing state in the modified chain, if the chain $\tilde{X}_n$ has reached the state i , it must not have visited any of the states in $\mathcal{C}$ as in that case it would have get stuck in that state. Thus, the equations in (8.5) have the form

$$y_j = p_{ji} + \sum_{k \in \mathcal{S} \setminus \mathcal{C}, k \neq i} p_{jk} y_j \quad (8.24)$$

and we wish to find the minimal solution of the the above equations. Again, if the number of states is finite, as before we have a single solution and hence it would provide the values.

We can use the above computation of the probability of ever visiting a state (when the state is transient) or even when the

Example 8.17. We return to the specific example of random walk on graph, given in Example 5.20:



$$\mathbf{P} = \begin{matrix} & \begin{matrix} A & B & C & D & E & F \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \\ E \\ F \end{matrix} & \begin{bmatrix} 0 & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 \\ 0 & \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 \\ \frac{1}{4} & 0 & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix} \end{matrix}.$$

Suppose that we are interested in computing the probability of reaching the vertex D before vertex C starting from vertex A . Therefore, we consider the transformed transition matrix:

$$\tilde{\mathbf{P}} = \begin{matrix} & \begin{matrix} A & B & C & D & E & F \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \\ E \\ F \end{matrix} & \begin{bmatrix} 0 & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix} \end{matrix}.$$

Hence we have the following equations:

$$\begin{aligned} y_A &= \frac{1}{4} + \frac{1}{4}y_B + \frac{1}{4}y_E + \frac{1}{4}y_F, & y_B &= \frac{1}{3}y_A + \frac{1}{3}y_E, \\ y_E &= \frac{1}{4} + \frac{1}{4}y_A + \frac{1}{4}y_B, & y_F &= \frac{1}{2} + \frac{1}{4}y_A. \end{aligned}$$

Note here that we have finitely many states, and hence the solution is unique. Solving the above system of equations, we get $y_A = \alpha_{A:\{C\}}^{(D)} = \frac{82}{129}$, $y_B = \alpha_{B:\{C\}}^{(D)} = \frac{49}{129}$, $y_E = \alpha_{E:\{C\}}^{(D)} = \frac{65}{129}$ and $y_F = \alpha_{F:\{C\}}^{(D)} = \frac{85}{129}$. ■

Example 8.18. We return to Exercise 5.21 again (see also Exercise 6.5). Suppose a rook is moving at random to a square where it can move legally and we are interested in computing the probability of the rook landing at the bottom left square ($A1$) before

landing the top right square ($H8$) if it starts from square on the first rank and the fifth file ($A5$). Now, we can compute this as in the previous equation and we will to solve a huge system (62 equations in 62 variables) to compute it. However, there is a way to minimize these calculations by aggregating the states as in Exercise 6.5.

As in Exercise 6.5, we partition the state space as $A_0 = \{A1\}$, $A_1 = \{A2, A3, A4, A5, A6, A7, B1, C1, D1, E1, F1, G1\}$, $A_2 = \{H2, H3, H4, H5, H6, H7, B8, C8, D8, E8, F8, G8\}$, $A_3 = \{A8, H1\}$, $A_4 = \{H8\}$. Let A_5 be the remaining squares of the chessboard. Intuitively, it should be understood that A_1 consists of those squares from where the rook may move to $A1$ in one move, but can't move to $H8$ in one move. Similarly, A_2 is the set of squares from where the rook may move to $H8$ in one move but but can't move to $A1$ in one move. The set A_3 represents those square from where the rook may move to both $A1$ and $H8$ in the next move and A_5 is the set of squares from where the rook will take 2 move to reach $A1$ as well as $H8$.

Let us denote the position of the rook after n -th move by X_n and the aggregated chain by Y_n . First observe that, for any $j \notin \{A1, H8\}$,

$$\alpha_{j:\{H8\}}^{(A1)} = \beta_{\kappa:\{4\}}^{(1)}$$

where $\beta_{\kappa:\{4\}}^{(0)}$ represents the probability of reaching the state 0(= $\{A1\}$) before the state 4(= $\{H8\}$) in the aggregated chain and κ represents where set in which the square j belong. In our case, $j = A5$, so that $\kappa = 1$. This is intuitively obvious as the piece (rook) will behave in the same fashion from each of the squares in the sets given above. Therefore the exact state (the square of the chess board) of the piece will be unimportant, only the set in which the square belongs will be required. (Justify this using formal arguments). Now, we can easily solve it:

$$\begin{aligned} y_1 &= \frac{1}{14} + \frac{5}{14}y_1 + \frac{1}{14}y_2 + \frac{1}{14}y_3 + \frac{6}{14}y_5 \\ y_2 &= \frac{1}{14}y_1 + \frac{5}{14}y_2 + \frac{1}{14}y_3 + \frac{6}{14}y_5 \\ y_3 &= \frac{1}{14} + \frac{6}{14}y_1 + \frac{6}{14}y_2 \\ y_5 &= \frac{2}{14}y_1 + \frac{2}{14}y_2 + \frac{10}{14}y_5. \end{aligned}$$

Solving we get $y_1 = \frac{11}{20}$, $y_2 = \frac{9}{20}$, $y_3 = y_5 = 1/2$. Note that the symmetry of the squares can be seen from the fact that $y_3 = y_5 = 1/2$ and that $y_1 + y_2 = 1$. It is indeed this reasoned intuitively (can as well be made formal) that every path from any square in A_3 or A_5 which gets absorbed in $A1$ will correspond to a path from the square obtained

by reflecting the squares by the diagonal. Also, from any square in $A1$ will correspond to a square in $A2$ by reflecting on the diagonal. The path from a square in $A1$ which gets absorbed in $A1$ can similarly be associated with a path which gets absorbed in $H8$ from the reflected square. ■

Exercise 8.19. Suppose a coin is being flipped until either of the patterns HTH or HHH appear. Find the probability that HTH appears before HHH . ◀

Exercise 8.20. You are playing a game which consists of tossing a coin until either one of the patterns of success and failures, one chosen by you and other by your opponent, appear. Assuming that the coin tosses are fair, show that for any pattern of length 3 chosen by your opponent, you can choose a pattern of length 3 so that the game is favourable to you.

Conclude that the game is not transitive. In other words, show that there exist three patterns p_1, p_2, p_3 such that the pattern p_1 is favourable to p_2 and p_2 is favourable to p_3 , but p_3 is favourable to p_1 .

[Hint : Representing H by 1 and tail by 0, for a pattern $\epsilon_1\epsilon_2\epsilon_3$, chosen by your opponent, choose the pattern $(1 - \epsilon_2)\epsilon_1, \epsilon_2]$ ◀

This section is to be written.

Chapter 9

Stationary distribution

9.1 Stationary probability distribution

We start with a definition.

Definition 9.1. A probability distribution π on $\mathcal{S}$ is called a stationary or invariant probability distribution if

$$\pi = \pi \mathbf{P} \quad \text{i.e.,} \quad \pi(j) = \sum_{i \in \mathcal{S}} \pi(i) p_{ij} \quad (9.1)$$

for every $j \in \mathcal{S}$.

The equation $\pi = \pi \mathbf{P}$ says that π is a left eigenvector of the matrix $\mathbf{P}$ with (right) eigenvalue 1. In addition to that, π must be a probability. We have already noted that the matrix $\mathbf{P}$ has the eigenvalue 1 always. These equations are known as *balance equations*.

The physical significance of balance equations becomes apparent if we imagine a large number of processes going on simultaneously. To be specific, consider N particles performing independently the same type of random behaviour on a set of states and move between the states with certain transition probabilities. Suppose that the distribution of particles at different states have reached equilibrium, i.e., the number of particles over different time points do not change. The number of particles at state j , will thus be $N_j = N \mathbb{P}(\text{equilibrium probability at state } j)$ whereas the number of particles that the state j receives from all states is given $\sum_{i \in \mathcal{S}} N_i p_{ji} = \sum_{i \in \mathcal{S}} N \mathbb{P}(\text{equilibrium probability$

at state i) p_{ji} . Therefore, in order to achieve balance, we must have

$$N\mathbb{P}(\text{equilibrium probability at state } j) = \sum_{i \in \mathcal{S}} N\mathbb{P}(\text{equilibrium probability at state } i)p_{ji}.$$

Dividing by N and denoting the equilibrium probabilities by $\pi(\cdot)$, we achieve the balance equations. Conversely, when the balance equations hold, we would say that the system is in equilibrium. The distribution π is therefore also called *equilibrium distribution*. It should be understood that when the system is in equilibrium, there are transitions happening in the system in such a way that they maintain balance between all states that the meta manifestations do not change over time.

Iterating the equation in (9.1), we obtain that for every $n \geq 1$,

$$\pi(j) = \sum_{k \in \mathcal{S}} \pi(k)p_{kj} = \sum_{k \in \mathcal{S}} \sum_{l \in \mathcal{S}} \pi(l)p_{lk}p_{kj} = \sum_{l \in \mathcal{S}} \pi(l) \sum_{k \in \mathcal{S}} p_{lk}p_{kj} = \sum_{l \in \mathcal{S}} \pi(l)p_{lj}^{(2)}.$$

It is easy to check by induction that, for every $n \geq 1$,

$$\sum_{j \in \mathcal{S}} \pi(j)p_{ji}^{(n)} = \pi(i). \quad (9.2)$$

Suppose π is a stationary distribution of a Markov chain with transition matrix $\mathbf{P}$. Further assume that the initial distribution is given by π , i.e., $\mathbb{P}(X_0 = i) = \pi(i)$ for all $i \in \mathcal{S}$. Using (9.2), we have, for all $n \geq 0$

$$\mathbb{P}(X_n = i) = \sum_{j \in \mathcal{S}} \mathbb{P}(X_0 = j)\mathbb{P}(X_n = i|X_0 = j) = \sum_{j \in \mathcal{S}} \pi(j)p_{ji}^{(n)} = \pi(i).$$

It should be noted that not only the marginal distributions are identical but any joint distributions are also the same. Indeed, for any $k \geq 1$, we have

$$\mathbb{P}(X_0 = i_0, X_1 = i_1, \dots, X_k = i_k) = \pi(i_0)p_{i_0,i_1}p_{i_1,i_2} \cdots p_{i_{k-1},i_k}$$

and for any $n \geq 1$,

$$\begin{aligned} \mathbb{P}(X_n = i_0, X_{n+1} = i_1, \dots, X_{n+k} = i_k) &= \mathbb{P}(X_n = i_0)p_{i_0,i_1}p_{i_1,i_2} \cdots p_{i_{k-1},i_k} \\ &= \pi(i_0)p_{i_0,i_1}p_{i_1,i_2} \cdots p_{i_{k-1},i_k}. \end{aligned}$$

Therefore, we can conclude that the processes $\{X_k : k \geq 0\}$ and $\{X_{n+k} : k \geq 0\}$ have identical distribution. Therefore, the process $\{X_k : k \geq 0\}$ is a stationary process. This justifies the name of stationary distribution.

Before we proceed, we prove a quick proposition to show that transient states will provide no mass to the stationary distribution.

Proposition 9.2. *If a chain possesses an invariant probability distribution π then $\pi(j) = 0$ for each transient state j .*

Proof : If j is a transient state, then $p_{ij}^{(n)} \rightarrow 0$ (see Corollary 7.11.2) for any $i \in \mathcal{S}$ as $n \rightarrow \infty$. Thus, letting $n \rightarrow \infty$ in equation (9.2) and using DCT, we conclude that $\pi(j) = 0$. ■

Remark. We will later show that for a null recurrent state $j \in \mathcal{S}$, $\pi(j) = 0$.

Writing the equations in this form is not always the best thing to do. Instead of eliminating equations, we introduce more, expecting to be able to choose, amongst them, a set of independent equations which can be more easily solved. The convenience is often suggested by the topological structure of the chain, as we will see in examples.

Let us introduce the notion of *probability flow* from a set A of states to its complement under a distribution π :

$$F(A, A^c) = \sum_{i \in A} \sum_{j \in A^c} \pi(i) p_{ij}.$$

Think of $\pi(i) p_{ij}$ as a *current* flowing from i to j . So $F(A, A^c)$ is the total current from A to A^c . We have

Proposition 9.3. *π is a stationary distribution if and only if $\pi \mathbf{1} = \mathbf{1}$ and*

$$F(A, A^c) = F(A^c, A)$$

for all $A \subseteq S$.

Proof : If the latter condition holds, then choose $A = \{i\}$. Then see that

$$\sum_{j \neq i} \pi(i) p_{ij} = F(\{i\}, \{i\}^c) = F(\{i\}^c, \{i\}) = \sum_{j \neq i} \pi(j) p_{ji}.$$

Summing over $\pi(i) p_{ii}$ to both sides, we have

$$\pi(i) = \pi(i) \left[\sum_{j \neq i} p_{ij} + p_{ii} \right] = \sum_{j \neq i} \pi(j) p_{ji} + \pi(i) p_{ii} = \sum_j \pi(j) p_{ji};$$

hence we obtain the balance equations.

Conversely, suppose that the balance equations hold. Fix a set $A \subseteq S$ and we can write, using $\sum_j p_{ij} = 1$,

$$\begin{aligned} \sum_{j \in A} \pi(i) p_{ij} + \sum_{j \in A^c} \pi(i) p_{ij} &= \pi(i) \sum_j p_{ij} \\ &= \pi(i) = \sum_j \pi(j) p_{ji} = \sum_{j \in A} \pi(j) p_{ji} + \sum_{j \in A^c} \pi(j) p_{ji}. \end{aligned}$$

This is true for all $i \in \mathcal{S}$. Summing over $i \in A$, we see that

$$\begin{aligned} \sum_{i \in A} \sum_{j \in A} \pi(i) p_{ij} + \sum_{i \in A} \sum_{j \in A^c} \pi(i) p_{ij} &= \sum_{i \in A} \left[\sum_{j \in A} \pi(i) p_{ij} + \sum_{j \in A^c} \pi(i) p_{ij} \right] \\ &= \sum_{i \in A} \left[\sum_{j \in A} \pi(j) p_{ji} + \sum_{j \in A^c} \pi(j) p_{ji} \right] = \sum_{i \in A} \sum_{j \in A} \pi(j) p_{ji} + \sum_{i \in A} \sum_{j \in A^c} \pi(j) p_{ji}. \end{aligned}$$

The first term of the left hand side and the right hand side cancel. Cancelling these terms, we have

$$F(A, A^c) = \sum_{i \in A} \sum_{j \in A^c} \pi(i) p_{ij} = \sum_{i \in A} \sum_{j \in A^c} \pi(j) p_{ji} = \sum_{j \in A^c} \sum_{i \in A} \pi(j) p_{ji} = F(A^c, A).$$

This completes the proof. ■

The extra degree of freedom provided by the last result, gives us flexibility. One can ask how to choose a minimal complete set of linearly independent equations, but we shall not do this here.

Example 9.4. Consider the 2-state Markov chain (see Example 5.9) with $p_{12}, p_{21} \in (0, 1)$. Consider $\pi = (\pi(1), \pi(2))$ and $A = \{1\}$. We observe that

$$\pi(1)p_{12} = \pi(2)p_{21}, \text{ i.e., } \frac{\pi(1)}{p_{21}} = \frac{\pi(2)}{p_{12}}.$$

Now, using the fact that $\pi(1) + \pi(2) = 1$, we have $\pi(1) = p_{21}/(p_{21} + p_{12})$ and $\pi(2) = p_{12}/(p_{21} + p_{12})$. ■

Example 9.5. Consider the Markov chain on $\{0, 1, 2, \dots\}$ with random walks with elastic barriers (see Example 5.17). The balance equations are:

$$\pi(0) = q\pi(1) \text{ and } \pi(i) = p\pi(i-1) + q\pi(i+1), \text{ for } i \geq 1.$$

But we can make them simpler by choosing $A = \{0, 1, \dots, i-1\}$. If $i \geq 1$, we have

$$F(A, A^c) = \pi(i-1)p = \pi(i)q = F(A^c, A).$$

These are much simpler than the previous ones. In fact, we can solve them immediately. For $i \geq 1$,

$$\pi(i) = (p/q)^i \pi(0).$$

Does this provide a stationary distribution? Yes, provided that $\sum_{i=0}^{\infty} \pi(i) = 1$. This is possible if and only if $p < q$, i.e., $p < 1/2$. ■

9.1.1 Reversible chains

Many laws of physics imply that the evolution of a process would be identical in nature if somehow the process is run with time reversed. Though in general it may not be expected that such a condition should be true, but it is often the case that when the process is in equilibrium, the evolution of the process with time reversed is identical in nature to that of the process running forward in time.

To set the ideas, let us assume that $\{X_n : 0 \leq n \leq N\}$ be a Markov chain with transition matrix $\mathbf{P}$ and a stationary distribution π . Let us define the time-reversed chain $\{Y_n = X_{N-n} : 0 \leq n \leq N\}$. We should note that in general the reversed time process is not a Markov chain, however when the Markov chain is stationary, i.e., the initial distribution is also given by π .

Proposition 9.6. *The reversed chain $\{Y_n : 0 \leq n \leq N\}$ is a Markov chain with transition probabilities given by*

$$q_{ij} = \mathbb{P}(Y_{n+1} = j \mid Y_n = i) = \frac{\pi(j)}{\pi(i)} p_{ji}. \quad (9.3)$$

Proof : For $i_0, i_1, \dots, i_n = i$ and j , we have

$$\begin{aligned} & \mathbb{P}(Y_{n+1} = j \mid Y_n = i_n, Y_{n-1} = i_{n-1}, \dots, Y_0 = i_0) \\ &= \frac{\mathbb{P}(Y_{n+1} = j, Y_n = i_n, Y_{n-1} = i_{n-1}, \dots, Y_0 = i_0)}{\mathbb{P}(Y_n = i_n, Y_{n-1} = i_{n-1}, \dots, Y_0 = i_0)} \\ &= \frac{\mathbb{P}(X_{N-n-1} = j, X_{N-n} = i_n, X_{N-n+1} = i_{n-1}, \dots, X_N = i_0)}{\mathbb{P}(X_{N-n} = i_n, X_{N-n+1} = i_{n-1}, \dots, X_N = i_0)} \\ &= \frac{\pi(j) p_{j, i_n} p_{i_n, i_{n-1}} \cdots p_{i_1, i_0}}{\pi(i_n) p_{i_n, i_{n-1}} \cdots p_{i_1, i_0}} \\ &= \frac{\pi(j)}{\pi(i_n)} p_{j, i_n} \\ &= \frac{\pi(j)}{\pi(i)} p_{ji}. \end{aligned}$$

This proves the Markov property of the family $\{Y_n : 0 \leq n \leq N\}$. ■

Definition 9.7. Let $X = \{X_n : n \geq 0\}$ be an irreducible Markov chain such that X_n has the stationary distribution π . The chain is called reversible in π if the transition matrices of X and its time-reversal are the same. In other words, for all $i, j \in \mathcal{S}$,

$$\pi(i) p_{ij} = \pi(j) p_{ji}. \quad (9.4)$$

Reversible chains: More generally we say that a transition matrix $\mathbf{P}$ and a distribution π are in *detailed balance* if

$$\pi(i)p_{ij} = \pi(j)p_{ji}$$

for all $i, j \in \mathcal{S}$. An irreducible chain X having a stationary distribution π is called *reversible in equilibrium* if its transition matrix $\mathbf{P}$ is in detailed balance with π .

Theorem 9.8. *Let $\mathbf{P}$ be the transition matrix of a chain X , and suppose that there exists a distribution π such that*

$$\pi(i)p_{ij} = \pi(j)p_{ji}.$$

for all $i, j \in \mathcal{S}$. Then π is a stationary distribution of the chain. Furthermore, if X is irreducible, then X is reversible in equilibrium.

Proof : Suppose that π satisfies the conditions of the theorem. Then we have, for $i, j \in \mathcal{S}$,

$$\pi(i) = \sum_{j \in \mathcal{S}} \pi(j)p_{ji} = \sum_{j \in \mathcal{S}} \pi(j)p_{ji}.$$

Thus, π is a stationary distribution. The reversibility in equilibrium of the transition matrix with π is obvious from definition. ■

Exercise 9.9. Consider an irreducible chain having finitely many states. Suppose that for a fixed state $i_0 \in \mathcal{S}$, we have $p_{ii_0} > 0$ for all $i \in \mathcal{S}$. Show that if

$$p_{i_0 i} p_{ij} p_{ji_0} = p_{i_0 j} p_{ji} p_{ii_0} \text{ for all } i, j \in \mathcal{S},$$

then there is a stationary distribution π and the chain is reversible in π .

[Hint: take $\pi(i) = C p_{i_0 i} / p_{ii_0}$ for a suitable choice of C] ◀

Example 9.10. Random Walk on Graphs with weights : We return to the example of random walk on a graph with weights, see Example 5.20. Now, for any vertex u , consider $w(u, \cdot) = \sum_{v \in V} w(u, v)$ where V is the vertex set of the graph and $w(u, v)$ is the edge weight of the edge between u and v . Further, let us define $w(\cdot, \cdot) = \sum_{v \in V} w(v, \cdot)$. We assume that $w(\cdot, \cdot) < \infty$. Observe that $\{\pi(v) : v \in V\}$ is a probability where

$$\pi(v) = \frac{w(v, \cdot)}{w(\cdot, \cdot)}. \quad (9.5)$$

We observe, for $u, v \in V$

$$\pi(v)p_{v,u} = \frac{w(v, \cdot)}{w(\cdot, \cdot)} \times \frac{w(v, u)}{w(v, \cdot)} = \frac{w(v, u)}{w(\cdot, \cdot)} = \frac{w(u, v)}{w(\cdot, \cdot)} = \frac{w(u, \cdot)}{w(\cdot, \cdot)} \times \frac{w(u, v)}{w(u, \cdot)} = \pi(u)p_{u,v}.$$

Thus, π satisfy the detailed balance condition.

When the weights are taken to be 1 if an edge is present and 0 if it is absent, then $w(u, \cdot)$ will represent the degree of the vertex u . In such a case, for finite graphs, i.e., $|V| < \infty$, a stationary distribution is given by

$$\pi(u) = \frac{\deg(u)}{\sum_{w \in V} \deg(w)}.$$

We will later show under minimal conditions this is the unique distribution. ■

Example 9.11. Birth and death chain : Now we consider an irreducible birth and death chain, i.e., $p_i > 0, q_i > 0$ for all $i \geq 0$ if the state-space is infinite. For the infinite case, we have the following equations

$$\begin{aligned} \pi(0)p_0 &= \pi(0)p_{01} = \pi(1)p_{10} = \pi(1)q_1 \\ \pi(i+1)q_{i+1} &= \pi(i+1)p_{i+1,1} = \pi(i)p_{i,i+1} = \pi(i)p_i \quad \text{for } i \geq 1. \end{aligned}$$

Hence, by induction, we conclude that, for all $i \geq 0$

$$\pi(i+1) = \pi(i) \frac{p_i}{q_{i+1}} = \pi(0) \frac{p_0 p_1 \cdots p_i}{q_1 q_2 \cdots q_{i+1}} = \pi(0) a_{i+1}$$

where

$$a_0 = 1 \text{ and } a_i = \frac{p_0 p_1 \cdots p_{i-1}}{q_1 q_2 \cdots q_i} \text{ for } i \geq 1.$$

Since $\sum_{i=0}^{\infty} \pi(i) = 1$, we must have, $\sum_{i=0}^{\infty} a_i < \infty$, which implies

$$\sum_{i=1}^{\infty} \frac{p_0 p_1 \cdots p_{i-1}}{q_1 q_2 \cdots q_i} < \infty \tag{9.6}$$

so that we have, for $i \geq 0$

$$\pi(i) = \frac{a_i}{\sum_{i=0}^{\infty} a_i}.$$

Thus, π satisfies the detailed balance condition.

One can consider the reflected random walk on the infinite half line, i.e., $p_0 = 1$ and $q_i = q$ and $p_i = p$ for all $i \geq 1$. Then the condition (9.6) turns out as $\sum_{i=1}^{\infty} p^{i-1} q^{-i} = q^{-1} \sum_{i=1}^{\infty} (pq^{-1})^{i-1} < \infty$. Clearly, this is true if and only if $p < q$. We have already seen

that the chain is recurrent if and only if $p \leq q$, i.e., $p \leq 1/2$ (see Example 5.17). Thus, the chain admits a stationary distribution which satisfy the detailed balance equations.

For the finite case, we have the same equations for $i = 0, 1, 2, \dots, N-1$. Again, we have $\pi(i) = \pi(0)a_i$ for $i = 1, 2, \dots, N$ where a_i is as defined above. Since we have finitely many states, this must be the case that $\sum_{i=1}^N a_i < \infty$. Thus, the unique stationary distribution is given by

$$\pi(i) = \frac{a_i}{\sum_{i=0}^N a_i} \quad (9.7)$$

for $i = 0, 1, \dots, N$. ■

Example 9.12. The Ehrenfest model of diffusion (see Example 5.18) : For this, setting $p_j = 1 - j/N$ and $r_j = 0$, we note that the equation (9.7) reduces to

$$\pi(i) = \binom{N}{i} 2^{-N} \quad (9.8)$$

for $i = 0, 1, \dots, N$. The chain has period 2, the states have finite mean recurrence times, and the binomial distribution with $p = 1/2$ is invariant.

This result can be interpreted as follows: Whatever the initial number of molecules in the first container, after a long time the probability of finding k molecules in it is nearly the same as if the a molecules had been distributed at random, each molecule having probability $1/2$ to be in the first container. This is a typical example of how our result gains physical significance. ■

Example 9.13. The Bernoulli-Laplace model of diffusion (see Example 5.19) : For the Bernoulli Laplace model, setting $p_j = (N-j)^2/N^2$, $q_j = j^2/N^2$ and $r_j = 2j(N-j)/N^2$, the equation (9.7) reduces to

$$\pi(i) = \frac{\binom{N}{i}^2}{\binom{2N}{N}} \quad (9.9)$$

for $i = 0, 1, \dots, N$. It is a hypergeometric distribution. This means that in the state of equilibrium the distribution of colors in each container is the same as if the N particles in it had been chosen at random from a collection of N black and N white particles. ■

9.2 Construction of a stationary distribution

We first define a related concept called the *stationary measure*.

Definition 9.14. Any non-negative solution $\mu = (\mu(i) : i \in \mathcal{S})$ of the balance equation is called a stationary measure. In other words, suppose that $\mu(i) \geq 0$ for $i \in \mathcal{S}$ and $\mu(i) = \sum_{j \in \mathcal{S}} \mu(j)p_{ji}$, then μ is a stationary measure.

Note that we are not insisting on the fact that μ is a probability i.e., $\sum_{i \in \mathcal{S}} \mu(i) = 1$. It is straightforward to check that if μ is a stationary measure, then any non-negative multiplier of μ , i.e., $c\mu = (c\mu(i) : i \in \mathcal{S})$ is also a stationary measure for any $c \geq 0$. Note that two trivial stationary measures exist for any chain, namely the zero measure, i.e., $\mu = (\mu(i) = 0 : i \in \mathcal{S})$ and the infinite measure $\mu = (\mu(i) = +\infty : i \in \mathcal{S})$. Further, any stationary probability distribution is automatically a stationary measure. On the other hand, if we have the finiteness of the total measure, i.e., $0 < \mu(\mathcal{S}) = \sum_{i \in \mathcal{S}} \mu(i) < \infty$, then if we divide μ by the total sum $\mu(\mathcal{S})$, we obtain a stationary measure. Therefore, our aim will be to construct stationary measures and obtain conditions under which the finiteness of the total measure can be obtained.

Suppose that μ is a stationary measure, i.e., $\mu = \mu\mathbf{P}$. Then, we have

$$\mu = \mu\mathbf{P} = (\mu\mathbf{P})\mathbf{P} = \mu\mathbf{P}^2 = \cdots = \mu\mathbf{P}^n$$

for any $n \geq 1$.

Before we discuss the construction of a stationary measure, we prove a small proposition which shows the non-triviality.

Proposition 9.15. Suppose that μ is a stationary measure. Suppose that there is at least one state i for which $0 < \mu(i) < \infty$. Then, for all j such that i and j communicate ($i \rightsquigarrow j$), we must have $0 < \mu(j) < \infty$.

Proof : Since $i \rightsquigarrow j$, there exists $n \geq 1$, such that $p_{ij}^{(n)} > 0$. Hence, we have

$$\mu(j) = \sum_{k \in \mathcal{S}} \mu(k)p_{kj}^{(n)} \geq \mu(i)p_{ij}^{(n)} > 0$$

as both $\mu(i)$ and $p_{ij}^{(n)} > 0$.

We also have $j \rightsquigarrow i$. Thus, we may choose $m \geq 1$ such that $p_{ji}^{(m)} > 0$. Hence, we have

$$\mu(i) = \sum_{k \in \mathcal{S}} \mu(k)p_{ki}^{(m)} \geq \mu(j)p_{ji}^{(m)};$$

and hence $\mu(j) \leq \mu(i)/p_{ji}^{(m)} < \infty$. This completes the proof. ■

We will now construct a stationary measure. Later we use this stationary measure to show the uniqueness property of the irreducible positive recurrent chains. Fix a recurrent state $a \in \mathcal{S}$. We refer the state a as the *ground state*. Suppose τ_a is the first return time to a (see equation 7.4), i.e., $\tau_a = \inf\{n \geq 1 : X_n = a\}$. Now, for any $i \in \mathcal{S}$, we define the following quantity

$$m^{[a]}(i) := \mathbb{E}_a\left(\sum_{n=0}^{\tau_a-1} \mathbb{I}(X_n = i)\right) = \mathbb{E}\left(\sum_{n=0}^{\tau_a-1} \mathbb{I}(X_n = i) \mid X_0 = a\right). \quad (9.10)$$

Since the state a is recurrent, $\tau_a < \infty$ with probability one. By definition of τ_a , we have $X_{\tau_a} = X_0 = a$ and we can re-write

$$m^{[a]}(i) = \mathbb{E}\left(\sum_{n=1}^{\tau_a} \mathbb{I}(X_n = i) \mid X_0 = a\right).$$

Before, we proceed we note some of the properties of $m^{[a]}$.

- Since $X_n \neq a$ for $1 \leq n < \tau_a$, we have $m^{[a]}(a) = 1$ and $X_{\tau_a} = 1$, we have $m^{[a]}(a) = \mathbb{E}\left(\sum_{n=1}^{\tau_a} \mathbb{I}(X_n = a) \mid X_0 = a\right) = \mathbb{E}(1 \mid X_0 = a) = 1$.
- We can express the above expectation as sum of probabilities. For any $i \in \mathcal{S}$, we have

$$m^{[a]}(i) = \mathbb{E}\left(\sum_{n=1}^{\tau_a} \mathbb{I}(X_n = i) \mid X_0 = a\right) = \mathbb{E}\left(\sum_{k=1}^{\infty} \sum_{n=1}^k \mathbb{I}(X_n = i) \mathbb{I}(\tau_a = k) \mid X_0 = a\right).$$

Since the terms inside the sums are non-negative, we may interchange the order of summation. Thus, we will that sum may be rewritten as

$$\begin{aligned} \sum_{k=1}^{\infty} \sum_{n=1}^k \mathbb{I}(X_n = i) \mathbb{I}(\tau_a = k) &= \sum_{n=1}^{\infty} \sum_{k=n}^{\infty} \mathbb{I}(X_n = i) \mathbb{I}(\tau_a = k) \\ &= \sum_{n=1}^{\infty} \mathbb{I}(X_n = i) \sum_{k=n}^{\infty} \mathbb{I}(\tau_a = k) = \sum_{n=1}^{\infty} \mathbb{I}(X_n = i) \mathbb{I}(\tau_a \geq n). \end{aligned}$$

Putting together, we have

$$m^{[a]}(i) = \sum_{n=1}^{\infty} \mathbb{P}(X_n = i, \tau_a \geq n \mid X_0 = a) = \sum_{n=1}^{\infty} l_n^{[a]}(i) \quad (9.11)$$

where, for any $i \in \mathcal{S}$ and $n \geq 1$,

$$l_n^{[a]}(i) = \mathbb{P}(X_n = i, \tau_a \geq n \mid X_0 = a). \quad (9.12)$$

- Also note that

$$\begin{aligned}
m^{[a]}(\mathcal{S}) &:= \sum_{i \in \mathcal{S}} m^{[a]}(i) = \mathbb{E} \left(\sum_{i \in \mathcal{S}} \sum_{n=0}^{\tau_a-1} \mathbb{I}(X_n = i) \mid X_0 = a \right) \\
&= \mathbb{E} \left(\sum_{n=0}^{\tau_a-1} \sum_{i \in \mathcal{S}} \mathbb{I}(X_n = i) \mid X_0 = a \right) = \mathbb{E} \left(\sum_{n=0}^{\tau_a-1} 1 \mid X_0 = a \right) \\
&= \mathbb{E}(\tau_a \mid X_0 = a) = \mu_a.
\end{aligned} \tag{9.13}$$

Therefore, we observe that $m^{[a]}(\mathcal{S}) < \infty$ if a is a positive recurrent state; however, when a is null recurrent state, $m^{[a]}(\mathcal{S}) = \infty$.

Proposition 9.16. *For any recurrent state a , $m^{[a]}$, defined in (9.10), satisfies the balance equations:*

$$m^{[a]} = m^{[a]}\mathbf{P}, \text{ i.e., } m^{[a]}(i) = \sum_{j \in \mathcal{S}} m^{[a]}(j)p_{ji}, \tag{9.14}$$

for all $i \in \mathcal{S}$. Moreover, for any state $b \in \mathcal{S}$ such that $a \rightsquigarrow b$ (a leads to b), we have

$$0 < m^{[a]}(b) < \infty.$$

Proof : We start the Markov chain with $X_0 = a$, where a is the fixed recurrent ground state whose existence was assumed.

Noting $\tau_a \geq 1$ with probability 1 and $m^{[a]}(a) = 1$, we have

$$l_1^{[a]}(i) = \mathbb{P}(X_1 = i, \tau_a \geq 1 \mid X_0 = a) = \mathbb{P}(X_1 = i \mid X_0 = a) = p_{a,i} = p_{a,i}m^{[a]}(a). \tag{9.15}$$

For $n \geq 2$, we may proceed by introducing the last but one step. We note that

$$l_n^{[a]}(i) = \mathbb{P}(X_n = i, \tau_a \geq n \mid X_0 = a) = \sum_{j \in \mathcal{S}, j \neq a} \mathbb{P}(X_n = i, X_{n-1} = j, \tau_a \geq n \mid X_0 = a).$$

We observe that, for any $j \neq a$, we have $\{X_n = i, X_{n-1} = j, \tau_a \geq n\} = \{X_n = i, X_{n-1} = j, \tau_a \geq n-1\}$ so that $\mathbb{P}(X_n = i, X_{n-1} = j, \tau_a \geq n \mid X_0 = a) = \mathbb{P}(X_n = i, X_{n-1} = j, \tau_a \geq n-1 \mid X_0 = a)$. Hence,

$$\begin{aligned}
&\mathbb{P}(X_n = i, X_{n-1} = j, \tau_a \geq n-1 \mid X_0 = a) \\
&= \mathbb{P}(X_n = i \mid X_{n-1} = j) \mathbb{P}(X_{n-1} = j, \tau_a \geq n-1 \mid X_0 = a) \\
&= p_{ji} \mathbb{P}(X_{n-1} = j, \tau_a \geq n-1 \mid X_0 = a) \\
&= p_{ji} l_{n-1}^{[a]}(j).
\end{aligned}$$

Thus, we have

$$l_n^{[a]}(i) = \sum_{j \in \mathcal{S}, j \neq a} p_{ji} l_{n-1}^{[a]}(j). \quad (9.16)$$

Hence, summing over all $n \geq 2$, we have

$$\sum_{n=2}^{\infty} l_n^{[a]}(i) = \sum_{n=2}^{\infty} \sum_{j \in \mathcal{S}, j \neq a} p_{ji} l_{n-1}^{[a]}(j) = \sum_{j \in \mathcal{S}, j \neq a} p_{ji} \sum_{n=2}^{\infty} l_{n-1}^{[a]}(j) = \sum_{j \in \mathcal{S}, j \neq a} p_{ji} m^{[a]}(j).$$

Now, combining with the term for $n = 1$ given by equation (9.15), we have

$$m^{[a]}(i) = p_{a,i} m^{[a]}(a) + \sum_{j \in \mathcal{S}, j \neq a} p_{ji} m^{[a]}(j) = \sum_{j \in \mathcal{S}} p_{ji} m^{[a]}(j).$$

The second part of the result follows from the fact that a is recurrent and $a \rightsquigarrow b$. Therefore, $b \rightsquigarrow a$ (see Proposition 7.27). This proves the result. ■

Remark. If μ satisfies the balance equation, multiplying both sides of the balance equation by a positive constant c , we observe that $c\mu$ is also a solution of the balance equation. We will soon show these are all the solutions of the balance equation under some restrictions. ►

The last result was shown under the assumption that a is a recurrent state. We are now going to assume more, namely, that a is positive recurrent. We now have:

Corollary 9.16.1. *Suppose that the Markov chain possesses a positive recurrent state a . Define*

$$\pi^{[a]}(i) := \frac{1}{\mathbb{E}(\tau_a \mid X_0 = a)} m^{[a]}(i) = \frac{1}{\mathbb{E}(\tau_a \mid X_0 = a)} \mathbb{E}\left(\sum_{n=0}^{\tau_a-1} \mathbb{I}(X_n = i) \mid X_0 = a\right). \quad (9.17)$$

This $\pi^{[a]}$ is a stationary distribution.

Proof : Since $0 < \mathbb{E}(\tau_a \mid X_0 = a) < \infty$, dividing both sides of (9.10) by $\mathbb{E}(\tau_a \mid X_0 = a)$, we see that balance equation holds for $\pi^{[a]}$. Now, that $\pi^{[a]}$ is a probability, follows from equation (9.13) and the definition of $\pi^{[a]}$. This completes the proof. ■

9.3 Uniqueness of the stationary distribution

A natural question is that how many stationary distributions exist. We stress that a stationary distribution may not be unique.

Example 9.17. Let $\mathcal{S} = \{0, 1, 2\}$ and $p_{00} = p_{22} = 1$ and $p_{1i} = 1/3$ for $i = 0, 1, 2$. Clearly, the states $\{0\}$ and $\{2\}$ are positive recurrent and 1 is transient. It is easy to check that for any $\alpha \in [0, 1]$, $\pi_\alpha = (\alpha, 0, 1 - \alpha)$ is a stationary distribution. ■

In view of the above example, we should ask if we impose more restrictions can be have more specific result. Indeed the following result answers the question.

Thorem 9.18. *Consider irreducible recurrent Markov chain. Then there exists a positive root μ of the balance equation $\mu = \mu\mathbf{P}$, which is unique up to a multiplicative constant. The chain is positive recurrent if $\sum_{i \in \mathcal{S}} \mu(i) < \infty$ and null recurrent if $\sum_{i \in \mathcal{S}} \mu(i) = \infty$.*

Proof : We have already proved the existence of a solution of the balance equation in Proposition 9.16.

Now, suppose that $\{X_n : n \geq 0\}$ be a irreducible recurrent Markov chain with transition matrix $\mathbf{P}$ and μ is a solution of the balance equation which has at least one positive finite entry. By Proposition 9.15, all entries must be positive and finite.

Now, let us fix a ground state $a \in \mathcal{S}$. For $i, j \in \mathcal{S}$, define a new chain $\{Y_n : n \geq 0\}$ with transition probability given by

$$q_{ij} = \frac{\mu(j)}{\mu(i)} p_{ji}.$$

Clearly $q_{ij} \geq 0$. Also, we have

$$\sum_{j \in \mathcal{S}} q_{ij} = \sum_{j \in \mathcal{S}} \frac{\mu(j)}{\mu(i)} p_{ji} = \frac{1}{\mu(i)} \sum_{j \in \mathcal{S}} \mu(j) p_{ji} = 1$$

as μ is a solution of the balance equation.

Now, we claim that

$$q_{ij}^{(n)} = \frac{\mu(j)}{\mu(i)} p_{ji}^{(n)}. \quad (9.18)$$

We have verified this for $n = 1$. Assume that it is true for $n = m$. Then we have

$$q_{ij}^{(m+1)} = \sum_{k \in \mathcal{S}} q_{ik} q_{kj}^{(m)} = \sum_{k \in \mathcal{S}} \frac{\mu(k)}{\mu(i)} p_{ki} \times \frac{\mu(j)}{\mu(k)} p_{jk}^{(m)} = \frac{\mu(j)}{\mu(i)} \sum_{k \in \mathcal{S}} p_{jk}^{(m)} p_{ki} = \frac{\mu(j)}{\mu(i)} p_{ji}^{(m+1)}.$$

This proves the claim.

Further, we have $q_{ii}^{(n)} = p_{ii}^{(n)}$. Since the original chain is recurrent, we have $\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty$. Hence, the new chain is also recurrent. Since the original chain was irreducible

we must have $p_{ij}^{(n)} > 0$ and $p_{ji}^{(m)} > 0$ for some $n, m \geq 1$. This in turn implies that $q_{ji}^{(n)} = \frac{\mu(i)}{\mu(j)} p_{ij}^{(n)} > 0$ and $q_{ij}^{(m)} = \frac{\mu(j)}{\mu(i)} p_{ji}^{(m)} > 0$. Thus, the new chain is also irreducible.

Now, we claim that, for any $n \geq 1$,

$$\begin{aligned} & \mathbb{P}(Y_n = a, Y_t \neq a \text{ for } 1 \leq t < n \mid Y_0 = i) \\ &= \frac{\mu(a)}{\mu(i)} \mathbb{P}(X_n = i, X_t \neq a \text{ for } 1 \leq t < n \mid X_0 = a) = \frac{\mu(a)}{\mu(i)} l_n^{[a]}(i). \end{aligned} \quad (9.19)$$

It is easy to check that $\mathbb{P}(X_n = i, X_t \neq a \text{ for } 1 \leq t < n \mid X_0 = a) = \mathbb{P}(X_n = i, T_a \geq n \mid X_0 = a) = l_n^{[a]}(i)$ where $T_a = \min\{t > 0 : X_t = a\}$ and $l_n^{[a]}(i)$ is as defined in equation (??).

If $n = 1$, the left hand side is given by q_{ia} and the right hand side is given by $\frac{\mu(a)}{\mu(i)} \mathbb{P}(X_n = i, T_a \geq 1 \mid X_0 = a) = \frac{\mu(a)}{\mu(i)} p_{ia}$. Hence, we have the equality from the equation (9.18). Assume that the equality holds for $n = m$. Now, we have, from equation (9.16),

$$l_{m+1}^{[a]}(i) = \sum_{k \in \mathcal{S}, k \neq a} l_m^{[a]}(k) p_{ki}$$

and hence

$$\begin{aligned} \frac{\mu(a)}{\mu(i)} l_{m+1}^{[a]}(i) &= \sum_{k \in \mathcal{S}, k \neq a} \frac{\mu(a)}{\mu(i)} l_m^{[a]}(k) p_{ki} = \sum_{k \in \mathcal{S}, k \neq a} \frac{\mu(a)}{\mu(k)} l_m^{[a]}(k) \times \frac{\mu(k)}{\mu(i)} p_{ki} \\ &= \sum_{k \in \mathcal{S}, k \neq a} \mathbb{P}(Y_m = a, Y_t \neq a \text{ for } 1 \leq t < m \mid Y_0 = x) q_{ik} \\ &= \mathbb{P}(Y_{m+1} = a, Y_t \neq a \text{ for } 1 \leq t < m+1 \mid Y_0 = i). \end{aligned}$$

This proves the claim.

Now, summing over $n \geq 1$ and noting that the modified chain is also irreducible and recurrent, we have that $\sum_{n=1}^{\infty} \mathbb{P}(Y_n = a, Y_t \neq a \text{ for } 1 \leq t < n \mid Y_0 = i) = \mathbb{P}(Y_n = a \text{ for some } n \geq 1 \mid Y_0 = i) = 1$. Therefore, we have,

$$1 = \sum_{n=1}^{\infty} \frac{\mu(a)}{\mu(i)} l_n^{[a]}(i) = \frac{\mu(a)}{\mu(i)} m^{[a]}(i).$$

Thus, for any $i \in \mathcal{S}$, we have $\mu(i) = \mu(a) m^{[a]}(i)$. This holds for any state i . Thus, we have, for any two states i_1, i_2

$$\frac{\mu(i_1)}{\mu(i_2)} = \frac{\mu(a) m^{[a]}(i_1)}{\mu(a) m^{[a]}(i_2)} = \frac{m^{[a]}(i_1)}{m^{[a]}(i_2)}.$$

This proves the first part of the result.

For the second part, let μ be a solution of the balance equation. Fix a ground state a and observe that $m^{[a]}(i) = c\mu(i)$ for some $c > 0$. From equation (??), we have $\mathbb{E}(T_a \mid X_0 = a) = \sum_{i \in \mathcal{S}} c\mu(i)$. Thus, $\mathbb{E}(T_a \mid X_0 = a) < \infty$ if and only if $\sum_{i \in \mathcal{S}} \mu(i) < \infty$. Therefore a is positive recurrent if and only if $\sum_{i \in \mathcal{S}} \mu(i) < \infty$. Since the chain is irreducible, we have the result. ■

If an irreducible recurrent chain has a stationary distribution, then, by Theorem 9.18, the chain must be positive recurrent.

For a Markov chain with at least one positive recurrent state, we already know from Corollary 9.16.1 that there is a stationary distribution. Now, consider an irreducible positive recurrent Markov chain. It must have at least one stationary distribution. Further, by irreducibility property all solutions of the balance equation must be constant multiple of each other. Therefore, if we have two stationary distributions, μ and ν , say, then we must have

$$\mu(i) = c\nu(i) \text{ for all } i \in \mathcal{S}$$

for some $c > 0$. But both are stationary distributions. So, summing over $i \in \mathcal{S}$, we must have

$$1 = \sum_{i \in \mathcal{S}} \mu(i) = c \sum_{i \in \mathcal{S}} \nu(i) = c.$$

Thus, the stationary distribution is unique. Therefore, we have proved the following result:

Theorem 9.19. *Consider a recurrent irreducible Markov chain. If the chain admits a stationary distribution, then the chain is positive recurrent. In this case, it is the unique stationary distribution of the chain.*

Now, the above result has very important corollaries. For a positive recurrent irreducible Markov chain, there is a unique stationary distribution. Thus, for any two states $a, b \in \mathcal{S}$, the stationary distributions, $\pi^{[a]}$ and $\pi^{[b]}$, must be equal. Hence, we have the following Corollary.

Corollary 9.19.1. *Consider a positive recurrent irreducible Markov chain and two states $a, b \in \mathcal{S}$. Then*

$$\pi^{[a]} = \pi^{[b]}.$$

In particular, we have the cycle formula:

$$\frac{\mathbb{E}(\sum_{n=0}^{T_a-1} \mathbb{I}(X_n = i) \mid X_0 = a)}{\mathbb{E}(T_a \mid X_0 = a)} = \frac{\mathbb{E}(\sum_{n=0}^{T_b-1} \mathbb{I}(X_n = i) \mid X_0 = b)}{\mathbb{E}(T_b \mid X_0 = b)} = \pi(i). \quad (9.20)$$

Now, considering $i = a$, in the cycle formula, we have the following important Corollary.

Corollary 9.19.2. *Consider a positive recurrent irreducible Markov chain with stationary distribution π . Then, for all $a \in \mathcal{S}$,*

$$\pi(a) = \frac{1}{\mathbb{E}(T_a \mid X_0 = a)}. \quad (9.21)$$

We return to examples of random Walks on graphs with weights (Example 9.10), Birth and death chain (Example 9.11), the Ehrenfest model of diffusion (Example 9.12) and the Bernoulli-Laplace model of diffusion (Example 9.13).

Example 9.20. Random Walk on Graphs with weights : We assume that the graphs is connected when we consider the edge set as the set of all edges for which the edge weights are positive. This will imply that the graph is irreducible. This will imply that the Markov chain is irreducible. Therefore whenever the stationary distribution exists, it must be unique. Hence, whenever $w(\cdot, \cdot) < \infty$, we have a unique stationary distribution.

Now, we assume that $w(\cdot, \cdot) = \infty$. Let $\mu = (w(v, \cdot) : v \in V)$. Now, observe that

$$\sum_{v \in V} \mu(v) p_{v,u} = \sum_{v \in V} \frac{w(v, u)}{w(v, \cdot)} \times w(v, \cdot) = \sum_{v \in V} w(v, u) = \sum_{v \in V} w(u, v) = w(u, \cdot) = \mu(u).$$

Thus, μ is a stationary measure and we note that $\sum_{u \in V} \mu(u) = \sum_{u \in V} w(u, \cdot) = w(\cdot, \cdot) = \infty$, by Theorem 9.18, the random walk on the graph with weights is either null recurrent or transient. It should be noted that both possibilities of null recurrence or transience are possible which can be seen by looking at simple symmetric random walk in one or two dimensions and in higher dimensions. ■

Example 9.21. Birth and death chain : When $p_i > 0, q_i > 0$ for all $i \geq 0$ then Markov chain is irreducible. Thus, for the finite state space case, the stationary distribution given by equation 9.7) must be unique. The Ehrenfest model of diffusion (Example 9.12) and the Bernoulli-Laplace model of diffusion are examples of finite state birth and death chains. Hence the stationary distribution obtain in equations (9.8) and (9.9) respectively must be unique.

In the infinite state space, we have

$$a_0 = 1 \text{ and } a_i = \frac{p_0 p_1 \cdots p_{i-1}}{q_1 q_2 \cdots q_i} \text{ for } i \geq 1.$$

For any $\alpha > 0$ and $i \geq 0$, set

$$\mu(i) = \alpha a_i.$$

We should observe that $\{\mu(i) : i \geq 0\}$ is a solution of the balance equation. Therefore, $\sum_{i=0}^{\infty} \mu(i) = \alpha \sum_{i=0}^{\infty} a_i$. If $\sum_{i=0}^{\infty} a_i < \infty$, it must be a positive recurrent chain. However if $\sum_{i=0}^{\infty} a_i = \infty$, the chain must be null recurrent. Therefore, the chain is positive recurrent if and only if the equation (9.6) holds.

One can consider the reflected random walk on the infinite half line, i.e., $p_0 = 1$ and $q_i = q$ and $p_i = p$ for all $i \geq 1$. Then the condition (9.6) turns out as $\sum_{i=1}^{\infty} p^{i-1} q^{-i} = q^{-1} \sum_{i=1}^{\infty} (pq^{-1})^{i-1} < \infty$. Clearly, this is true if and only if $p < q$. We have already seen that the chain is recurrent if and only if $p \leq q$, i.e., $p \leq 1/2$ (see Example 5.17). Thus, the chain is null recurrent if $p = q = 1/2$, positive recurrent if $p < q$ and transient if $p > q$. ■

Example 9.22. Doubly stochastic matrices : For a doubly stochastic matrix, (see, for example, Example 5.13) we have already observed $\mathbf{1P} = \mathbf{1}$, where $\mathbf{1} = (1, 1, \dots, 1)$. Now, assume that the chain is irreducible. Then, if the number of states is infinite, we note that $\sum_{i \in \mathcal{S}} \mathbf{1}(i) = \sum_{i \in \mathcal{S}} 1 = \infty$. Thus, if the chain is recurrent, it must be null recurrent. This is, for example, the case in simple symmetric random walk. If the number of states is finite, say N , then note that the chain must be recurrent (since it has finite state space and is irreducible) and further $\pi = (1/N, 1/N, \dots, 1/N)$ is a stationary distribution. Therefore, it is the unique stationary distribution.

The above two corollaries provide us also other information. For example, for an irreducible Markov chain with finite state space having a doubly stochastic transition matrix, we must have

$$\mu_a = \mathbb{E}(T_a \mid X_0 = a) = \frac{1}{\pi(a)} = N$$

where N is the number of states. Also, for any $b \in \mathcal{S}$, we have

$$\begin{aligned} & \mathbb{E}(\text{No. of returns to } b \text{ between two visits to } a) \\ &= \mathbb{E}\left(\sum_{n=0}^{T_a-1} \mathbb{I}(X_n = b) \mid X_0 = a\right) = \frac{\pi(b)}{\pi(a)} = 1. \end{aligned} \quad \blacksquare$$

Remark. It should also be noted that for the asymmetric random walk also the transition matrix is doubly stochastic. So, for infinite state space irreducible Markov chains, doubly stochastic matrices will correspond to either transient or null recurrent chains. ►

Example 9.23. Returning to the Example 5.24 about recurrent events, the equations for an invariant probability distribution are

$$\begin{aligned}\pi(0) &= \pi(0)q_0 + \pi(1)q_1 + \pi(2)q_2 + \dots \\ \pi(k) &= \pi(k-1)p_{k-1} \quad \text{for } k \geq 1.\end{aligned}$$

It is easily seen that $\pi(k) = \pi(0)p_0p_1 \cdots p_{k-1}$ for $k \geq 1$. Further, we have, $\pi(k)q_k = \pi(k)(1-p_k) = \pi(k) - \pi(k)p_k = \pi(k) - \pi(k+1)$ for $k \geq 0$. Thus, the sum $\sum_{k=0}^N \pi(k)q_k = u_0 - u_{N+1} \rightarrow u_0$ as $N \rightarrow \infty$ if $u_N \rightarrow 0$ as $N \rightarrow \infty$. Thus, the first condition will be automatically satisfied if $u_n \rightarrow 0$. Since $\pi(k)$ will be a probability, we will have

$$\pi(0)\left(1 + \sum_{k=1}^{\infty} p_0p_1 \cdots p_{k-1}\right) = 1$$

which gives us the condition that

$$\sum_{k=1}^{\infty} p_0p_1 \cdots p_{k-1} < \infty. \quad (9.22)$$

In this case, the stationary distribution is given by, for $n \geq 0$,

$$\pi(n) = \frac{p_0p_1 \cdots p_{n-1}}{1 + \sum_{k=1}^{\infty} p_0p_1 \cdots p_{k-1}}.$$

If $p_k = p$ for all k (see success run length part in Example 5.24), the condition (9.22) is satisfied for $p < 1$. Hence, the stationary distribution is given by : $\pi(k) = qp^k$ for all $k \geq 0$ (see also Example 6.12).

Example 9.24. Recurrent Events : Returning to Example 5.23, we obtain the equations for the stationary distribution as

$$\pi(k) = \pi(k+1) + f_{k+1}\pi(0) \quad \text{for } k \geq 0.$$

Now, set $r_n = \sum_{k=n+1}^{\infty} f_k$. Clearly, $\pi(k) \rightarrow 0$ as $k \rightarrow \infty$, since π is a probability distribution and hence $\sum_{k=0}^{\infty} \pi(k) = 1$. Thus, we have

$$\pi(n) - \pi(N+1) = \sum_{k=n}^N (\pi(k) - \pi(k+1)) = \sum_{k=n}^N \pi(0)f_{k+1} = \pi(0) \sum_{k=n}^N f_{k+1}.$$

Now, letting $N \rightarrow \infty$, we conclude that $\pi(n) = \pi(0) \sum_{k=n}^{\infty} f_{k+1} = \pi(0)r_n$. Since $\pi(k)$ will be a probability, we will have

$$\pi(0)\left(1 + \sum_{k=1}^{\infty} r_k\right) = \pi(0)\mu$$

where $\mu = 1 + \sum_{k=1}^{\infty} r_k = \sum_{k=1}^{\infty} k f_k$ which gives us the condition that

$$\mu = \sum_{k=1}^{\infty} k f_k < \infty. \quad (9.23)$$

In this case, the stationary distribution is given by $\pi(n) = r_n/\mu$ for all $n \geq 0$. ■

9.3.1 Stationary distribution for arbitrary chains

Now we will discuss stationary distributions of arbitrary Markov chains. We have already seen that if a stationary distribution π exists, then for any transient state $j \in \mathcal{S}$, we must have $\pi(j) = 0$.

Proposition 9.25. *Consider an arbitrary Markov chain. Assume that a stationary distribution π exists. Then any state i such that $\pi(i) > 0$ must be positive recurrent.*

Proof : Let i be such that $\pi(i) > 0$. We know that i must be recurrent. Let C be the communicating class of i . Note that C must be closed (see Proposition 7.27). Let $\pi(\cdot | C)$ be the distribution defined by restricting π to C , i.e.,

$$\pi(x | C) := \frac{\pi(x)}{\sum_{j \in C} \pi(j)} \quad (9.24)$$

for $x \in C$. Consider the restriction of the chain to C . We can easily see that the restriction chain is an irreducible recurrent Markov chain with state space C . We further claim that this restricted chain has a stationary distribution $\pi(\cdot | C)$. Indeed, for $x \in C$,

$$\pi(x | C) = \frac{\pi(x)}{\sum_{j \in C} \pi(j)} = \frac{1}{\sum_{j \in C} \pi(j)} \sum_{y \in \mathcal{S}} \pi(y) p_{yx} = \frac{1}{\sum_{j \in C} \pi(j)} \sum_{y \in \mathcal{S} \setminus T} \pi(y) p_{yx}$$

where T is the set of transient states. The last equality follows from the fact that $\pi(y) = 0$ for $y \in T$. Now, for any recurrent state y , if $y \notin C$, then $y \not\leftrightarrow x$ and hence $p_{yx} = 0$. Thus, the sum above reduces to $\sum_{y \in C} \pi(y) p_{yx}$. Hence, we have

$$\pi(x | C) = \frac{1}{\sum_{j \in C} \pi(j)} \sum_{y \in C} \pi(y) p_{yx} = \sum_{y \in C} \frac{\pi(y)}{\sum_{j \in C} \pi(j)} p_{yx} = \sum_{y \in C} \pi(y | C) p_{yx}.$$

Since the restricted chain is irreducible, we must have this is the unique stationary distribution. Therefore, we have from equation (9.19.2),

$$\frac{1}{\mathbb{E}(T_i | X_0 = i)} = \pi(i | C) > 0.$$

Thus, $\mathbb{E}(T_i \mid X_0 = i) < \infty$, i.e., i is positive recurrent. ■

The above result says that only positive recurrent states can contribute mass in the stationary distribution. Therefore, we may now strengthen Proposition 9.2.

Corollary 9.25.1. *If a chain possesses an stationary probability distribution π then $\pi(j) = 0$ for each transient or null recurrent state j .*

Example 9.26. Chains with several irreducible components may admit of several stationary solutions. In fact, if there are more than one positive recurrent classes, infinite number of stationary distributions will be possible. This is easy to see as each of the positive recurrent classes, when restricted to themselves, will admit a unique stationary distribution. Therefore, any convex combination of their stationary distribution will a stationary distribution for the Markov chain. A trite, but typical, example is presented by the random walk with two absorbing states 0 and N with state space $\mathcal{S} = \{0, 1, \dots, N\}$ (see for example, Example birth and death chains with $r_0 = r_N = 1$). Every probability distribution of the form $(\alpha, 0, 0, \dots, 0, 1 - \alpha)$, attributing positive weights only to 0 and N is stationary. ■

We show that Example 9.26 is not an aberration, rather the norm. It is now easy to show what an arbitrary stationary distribution looks like. Consider a Markov chain. Decompose it into communicating classes. Then for each positive recurrent communicating class C , there corresponds a unique stationary distribution $\pi^{[C]}$ which assigns positive probability to each state in C . This is defined by picking any state $a \in C$ and considering $\pi^{[a]}$ in C and 0 mass to states outside C (even if they are positive recurrent). In other words,

$$\pi^{[C]}(i) = \begin{cases} \pi^{[a]}(i) & \text{if } i \in C \\ 0 & \text{otherwise.} \end{cases} \quad (9.25)$$

It is obvious that $\pi^{[C]}$ is a stationary distribution for the chain. Further, any convex combination of these stationary distributions for distinct positive recurrent classes also yields a stationary distribution. An arbitrary stationary distribution π must be a convex combination of these $\pi^{[C]}$.

Proposition 9.27. *To each positive recurrent communicating class C there corresponds a stationary distribution $\pi^{[C]}$. Any stationary distribution π is necessarily of the form*

$$\pi = \sum_{\substack{C: C \text{ positive recurrent} \\ \text{communicating class}}} \alpha_C \pi^{[C]} \quad (9.26)$$

where $\alpha_C \geq 0$, such that

$$\sum_{C: C \text{ positive recurrent communicating class}} \alpha_C = 1.$$

Proof : Let $\{\pi^{[C]} : C \text{ positive recurrent communicating class}\}$ be the family of stationary distributions for positive recurrent communicating classes. Let π be an arbitrary stationary distribution. If $\pi(i) > 0$ then i belongs to some positive recurrent class C . For each such C define the distribution $\pi(\cdot | C)$ as in equation (9.24). Since C is an irreducible positive recurrent class, by the uniqueness result, this is the unique stationary distribution on C . Since $\pi^{[C]}$ is an extension of the unique stationary distribution on C , we must have

$$\pi(x | C) = \pi^{[C]}(x), \text{ for } x \in C.$$

Therefore, we have

$$\pi(x) = \sum_{C: C \text{ positive recurrent communicating class}} \pi(C) \times \pi(x | C) = \sum_{C: C \text{ positive recurrent communicating class}} \pi(C) \times \pi^{[C]}(x)$$

i.e., the above decomposition holds with $\alpha_C = \pi(C)$. Clearly $\pi(C) \geq 0$ and

$$\sum_{C: C \text{ positive recurrent communicating class}} \pi(C) = \pi\left(\bigcup_{C: C \text{ positive recurrent communicating class}} C\right) = 1$$

as π provides no mass to transient or null recurrent states. This completes the proof. ■

Example 9.28. We return to the Example 5.28 where there are two recurrent classes, viz. $\{1, 4\}$ and $\{3, 6, 8\}$ and $\{2, 5, 7\}$ as the set of transient states. Thus, restricting to attention to only to the closed class $C_1 = \{1, 4\}$ and solving the linear equations, we see that

$$\pi^{[C_1]} = (\pi^{[C_1]}(1), \pi^{[C_1]}(2), \pi^{[C_1]}(3), \pi^{[C_1]}(4), \pi^{[C_1]}(5), \pi^{[C_1]}(6), \pi^{[C_1]}(7), \pi^{[C_1]}(8))$$

is a stationary distribution where

$$\pi^{[C_1]} = (2/5, 0, 0, 3/5, 0, 0, 0, 0)$$

(see Example 9.4 also).

Another stationary distribution can be obtained by restricting our attention to the other recurrent class $C_2 = \{3, 6, 8\}$. Again, it is easy to verify that

$$\pi^{[C_2]} = (\pi^{[C_2]}(1), \pi^{[C_2]}(2), \pi^{[C_2]}(3), \pi^{[C_2]}(4), \pi^{[C_2]}(5), \pi^{[C_2]}(6), \pi^{[C_2]}(7), \pi^{[C_2]}(8))$$

where

$$\pi^{[C_2]} = (0, 0, 15/46, 0, 0, 16/46, 0, 15/46)$$

is another stationary distribution. Therefore, for any $\lambda \in [0, 1]$, $\pi = \lambda\pi^{[C_1]} + (1 - \lambda)\pi^{[C_2]}$ is a stationary distribution for this Markov chain and actually the set of all stationary distributions. ■

Chapter 10

Limit theory of Markov chains

10.1 Absolute probabilities

We have so far discussed about the conditional probabilities, i.e., assuming that we have started the Markov chain in a certain state. Now, we are going to start with the chain according to a probability distribution μ on $\mathcal{S}$, i.e., μ is a vector $(\mu(i) : i \in \mathcal{S})$ where $\mu(i) \geq 0$ and $\sum_{i \in \mathcal{S}} \mu(i) = 1$. We are going to refer this distribution as the initial distribution.

The absolute probabilities of the Markov chain, can be given in terms of the initial distribution μ and the transition matrix $\mathbf{P}$. Indeed,

$$\mathbb{P}(X_n = j) = \sum_{i \in \mathcal{S}} \mu(i) p_{ij}^{(n)} = (\mu \mathbf{P}^n)(j) \quad (10.1)$$

where $(\mu \mathbf{P}^n)(j)$ is the j -element of the row vector $\mu \mathbf{P}^n$. It is easy to check that $\mu \mathbf{P}^n$ represents a probability distribution. Thus, the distribution of X_n is given by the row vector $\mu \mathbf{P}^n$. If we take $\mu = \delta_i$, i.e., $X_0 = i$ with probability 1, we have $\mathbb{P}(X_n = j) = p_{ij}^{(n)}$.

Note that if we started with the initial distribution π which satisfies the balance equations in (9.1), then for any $n \geq 1$, the random variable X_n will have the same distribution π , i.e., $\mathbb{P}(X_n = j) = \pi(j)$ for all $j \in \mathcal{S}$ and $n \geq 1$. Thus an initial distribution satisfying (9.1) perpetuates itself for all times. For this reason it is called invariant.

Exercise 10.1. Under the stationary initial distribution, show that, for any $n \geq 1$ and $k \geq 1$,

$$\mathbb{P}(X_0 = x_0, X_1 = x_1, \dots, X_n = x_n) = \mathbb{P}(X_k = x_0, X_{k+1} = x_1, \dots, X_{k+n} = x_n)$$

for all $x_0, x_1, \dots, x_n \in \mathcal{S}$.

10.2 Limiting probabilities

We now explore the link between the existence of a stationary distribution and the limiting behaviour of the probabilities $p_{ij}^{(n)}$ as n tends to infinity. Recall our previous discussion on the physical interpretation of equilibrium of a system. The expected number of particles in state j at time n equals $N\mathbb{P}(X_n = j)$ when N is the total number of particles in the system. The physical intuition suggests that after a sufficiently long time, the system will be approximately in equilibrium so that the distribution of the number of particles at state j will not change. In fact the distribution of the particle to be in state j is given by $\pi(j)$ where π is the stationary distribution. Thus the expected number of particles at state j in equilibrium will be $N\pi(j)$ and hence, it is expected that $\mathbb{P}(X_n = j) \rightarrow \pi(j)$. Further, intuitively it is felt that the influence of the initial state should gradually wear off so that for large n the distribution (10.1) should be nearly independent of the initial distribution μ . Note that if we take the initial distribution $\mu = \delta_i$, i.e., the Markov chain starts at the state i with probability 1. If this happens for each $i \in \mathcal{S}$ for all $i \in \mathcal{S}$ and we will have $p_{ij}^{(n)} \rightarrow \pi(j)$ and hence $\mathbf{P}^n$ converges to a matrix with identical rows with each row representing the stationary distribution. It is also easy to see that $p_{ij}^{(n)} \rightarrow \pi(j)$ for any $i \in \mathcal{S}$, then by DCT, we must have for any initial distribution $\mathbb{P}(X_n = j) = \sum_{i \in \mathcal{S}} \mu(i)p_{ij}^{(n)} \rightarrow \pi(j)$ as $n \rightarrow \infty$.

First we prove a preparatory result.

Proposition 10.2. *Suppose that for some sequence $\{n_r : k \geq 1\}$, the limits of $p_{ij}^{(n_r)}$ exist for all $i, j \in \mathcal{S}$. Further assume that the limits are independent of the starting state, i.e.,*

$$\lim_{k \rightarrow \infty} p_{ij}^{(n_r)} = \pi(j)$$

for any $i \in \mathcal{S}$ and for all $j \in \mathcal{S}$. Further assume that at least one $\pi(j) > 0$. Then, $\pi = (\pi(j) : j \in \mathcal{S})$ is a stationary measure.

Proof : First note that $\sum_{j=1}^N p_{i,j}^{(n_k)} \leq 1$ for any $N \geq 1$. Therefore, letting $k \rightarrow \infty$, we have

$$\sum_{j=1}^N \pi(j) \leq 1$$

for any $N \geq 1$. Therefore we have $\sum_{j \in \mathcal{S}} \pi(j) = \sum_{j=1}^{\infty} \pi(j) \leq 1$. Thus, $0 < \alpha = \sum_{j \in \mathcal{S}} \pi(j) \leq 1$ as at least one $\pi(j) > 0$.

Furthermore, for any state $j \in \mathcal{S}$, we note

$$\sum_{k=1}^N p_{ik}^{(n_r)} p_{kj} \leq \sum_{k \in \mathcal{S}} p_{ik}^{(n_r)} p_{kj} = p_{ij}^{(n_r+1)} = \sum_{k \in \mathcal{S}} p_{ik} p_{kj}^{(n_r)}.$$

Now, we let $r \rightarrow \infty$ to deduce that the left hand side converges to $\sum_{k=1}^N \pi(k) p_{kj}$ while on the right hand side we use the dominated convergence Theorem. Since $p_{kj}^{(n_r)} \rightarrow \pi(j)$ and $p_{kj}^{(n_r)} \leq 1$ and $\sum_{k \in \mathcal{S}} p_{ik} = 1 < \infty$, we have

$$\sum_{k=1}^N \pi(k) p_{kj} \leq \sum_{k \in \mathcal{S}} p_{ik} \pi(j) = \pi(j) \sum_{k \in \mathcal{S}} p_{ik} = \pi(j).$$

Letting $N \rightarrow \infty$, for any state $j \in \mathcal{S}$, we have

$$\sum_{k \in \mathcal{S}} \pi(k) p_{kj} \leq \pi(j). \quad (10.2)$$

Next we claim that equality must hold in the above equation (10.2). Indeed, if it does not hold for some j , then we have, by summing both sides of equation (10.2)

$$0 < \alpha = \sum_{j \in \mathcal{S}} \pi(j) > \sum_{j \in \mathcal{S}} \sum_{k \in \mathcal{S}} \pi(k) p_{kj} = \sum_{k \in \mathcal{S}} \pi(k) \sum_{j \in \mathcal{S}} p_{kj} = \sum_{k \in \mathcal{S}} \pi(k) = \alpha.$$

This is a contradiction. Hence we must have equality in equation (10.2). This proves the result. $\blacksquare$

It should be noted that in such a case, the scaled limits $\{\pi(j) / \sum_{k \in \mathcal{S}} \pi(k) : j \in \mathcal{S}\}$ will be a stationary distribution. However, as we show later that we must also have $\sum_{k \in \mathcal{S}} \pi(k) = 1$, i.e., the limits (without scaling) will actually be a stationary distribution.

Note that it must be assumed that at least one $\pi(j)$ must be positive. Consider the case of random walk on integers with $p \neq 1/2$. We have already proved that the chain is transient (see Example 7.15) and hence $p_{ij}^{(n)} \rightarrow 0$ as $n \rightarrow \infty$. The same will hold for the symmetric case also which we prove subsequently.

It should also be noted that there is no guarantee that the limits will exist. The following example indicates a difficulty which arises from periodicity.

Example 10.3. Suppose $\{X_n : n \geq 0\}$ is a Markov chain on the state space $\mathcal{S} = \{1, 2\}$ and $p_{12} = p_{21} = 1$, then

$$p_{11}^{(n)} = p_{22}^{(n)} = \begin{cases} 0 & \text{if } n \text{ is odd;} \\ 1 & \text{if } n \text{ is even.} \end{cases}$$

Clearly $p_{ii}^{(n)}$ does not converge as $n \rightarrow \infty$; the reason is that both states are periodic with period 2. ■

In view of the above example, we need to impose more conditions on the chain. Indeed, it turns out that we should restrict our attentions to irreducible aperiodic chains. The principal result is the following theorem.

Theorem 10.4. *For an irreducible aperiodic chain, for all $i, j \in \mathcal{S}$, we have that*

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \frac{1}{\mathbb{E}(T_j | X_0 = j)} = \frac{1}{\mu_j}. \quad (10.3)$$

We will use a powerful probabilistic technique called *coupling*. Coupling refers to the various methods for constructing this joint law. The coupling we are interested in is a coupling of processes. Suppose that we are given the law of a stochastic process $\{X_n : n \geq 0\}$ and the law of a stochastic process $\{Y_n : n \geq 0\}$, but we are not told what the joint probabilities are. A coupling of them refers to a joint construction on the same probability space.

Suppose that we have two stochastic processes as above and that they have been coupled. Let T be a random time, often called the *meeting time* of the two processes, if

$$X_n \stackrel{d}{=} Y_n \text{ for all } n \geq T \quad (10.4)$$

Note that it makes sense to speak of a meeting time, precisely because of the assumption that the two processes have been constructed together. This T is a random variable that takes values in the set of times or it may take values $+\infty$, on the event that the two processes never meet. The following result is the so-called *coupling inequality*:

Lemma 10.5. *Let T be a meeting of two coupled stochastic processes $\{X_n : n \geq 0\}, \{Y_n : n \geq 0\}$. Then, for all $n \geq 0$*

$$\sum_{x \in \mathcal{S}} |\mathbb{P}(X_n = x) - \mathbb{P}(Y_n = x)| \leq 2\mathbb{P}(T > n). \quad (10.5)$$

If, in particular, T is finite with probability one, i.e. $\mathbb{P}(T < \infty) = 1$, then

$$\sum_{x \in \mathcal{S}} |\mathbb{P}(X_n = x) - \mathbb{P}(Y_n = x)| \rightarrow 0,$$

as $n \rightarrow \infty$.

and hence

$$\mathbb{P}(X_n = x) - \mathbb{P}(Y_n = x) \leq \mathbb{P}(X_n = x, T > n).$$

Interchanging the roles of X_n and Y_n , we obtain

$$\mathbb{P}(Y_n = x) - \mathbb{P}(X_n = x) \leq \mathbb{P}(Y_n = x, T > n).$$

Combining the two inequalities we obtain

$$|\mathbb{P}(X_n = x) - \mathbb{P}(Y_n = x)| \leq \mathbb{P}(X_n = x, T > n) + \mathbb{P}(Y_n = x, T > n)$$

for $n \geq 0, x \in \mathcal{S}$. Now, summing over all $x \in \mathcal{S}$, we obtain the first assertion.

Assume now that $\mathbb{P}(T < \infty) = 1$. Then

$$\lim_{n \rightarrow \infty} \mathbb{P}(T > n) = \mathbb{P}(T = \infty) = 0.$$

Therefore, we have

$$\sum_{x \in \mathcal{S}} |\mathbb{P}(X_n = x) - \mathbb{P}(Y_n = x)| \leq \mathbb{P}(T > n) \rightarrow 0$$

as $n \rightarrow \infty$. This completes the proof. ■

Remark. Note that,

$$\sup_{x \in \mathcal{S}} |\mathbb{P}(X_n = x) - \mathbb{P}(Y_n = x)| \leq \sum_{x \in \mathcal{S}} |\mathbb{P}(X_n = x) - \mathbb{P}(Y_n = x)|.$$

Thus, the above convergence is uniform when $T < \infty$ with probability 1. ►

Now, we return to the proof of Theorem 10.4. The idea is to couple two processes which evolves independently of each other and wait until they meet.

Proof of Theorem 10.4 : We take two Markov chains, $\{X_n : n \geq 0\}$ and $\{Y_n : n \geq 0\}$, both having the same transition matrix $\mathbf{P}$, but their initial distributions are different. At this moment, we do not put any condition on their initial distributions. Now, we couple of them as two independent processes. More precisely, let us define the bivariate process

$$\{W_n : n \geq 0\} := \{(X_n, Y_n) : n \geq 0\}$$

where $\{X_n : n \geq 0\}$ and $\{Y_n : n \geq 0\}$ are independent.

Clearly, $\{W_n : n \geq 0\}$ is a Markov chain with state space $\mathcal{S} \times \mathcal{S}$. Its transition probabilities are given by

$$\begin{aligned} q_{(x,y),(x',y')} &:= \mathbb{P}(W_{n+1} = (x', y') | W_n = (x, y)) \\ &= \mathbb{P}(X_{n+1} = x' | X_n = x) \mathbb{P}(Y_{n+1} = y' | Y_n = y) \\ &= p_{x,x'} p_{y,y'}. \end{aligned} \tag{10.6}$$

It is easy to check by induction that the n th-step transition probabilities are given by

$$q_{(x,y),(x',y')}^{(n)} = p_{x,x'}^{(n)} p_{y,y'}^{(n)}.$$

Since the chains $\{X_n : n \geq 0\}$ and $\{Y_n : n \geq 0\}$ are aperiodic, by Proposition 7.35, for any two pairs (x, x') and (y, y') , there must exist N such that $p_{x,x'}^{(n)} > 0$ and $p_{y,y'}^{(n)} > 0$ for all $n \geq N$. This implies that for $n \geq N$, $q_{(x,y),(x',y')}^{(n)} = p_{x,x'}^{(n)} p_{y,y'}^{(n)} > 0$. Hence, the chain $\{W_n : n \geq 0\}$ is aperiodic and irreducible.

Clearly it makes sense to consider the first meeting time of the above processes. More precisely, letting $D = \{(z, z) : z \in \mathcal{S}\}$ as the diagonal set, we have that T_D (the first entrance time of the set D) as

$$T_D := \inf\{n \geq 0 : W_n \in D\} = \inf\{n \geq 0 : X_n = Y_n\}$$

with the usual notation that the infimum of an empty set being $+\infty$. We note that T_D is a stopping time for the Markov chain W_n .

Suppose the chain $\{W_n : n \geq 1\}$ is recurrent and μ_1 and μ_2 are any two initial distributions on $\mathcal{S}$. Then, fix any $x_0 \in \mathcal{S}$ and note that

$$\begin{aligned} \mathbb{P}(T_D < \infty) &= \sum_{x,y \in \mathcal{S}} \mu_1(x) \mu_2(y) \mathbb{P}(W_n \in D \text{ for some } n \geq 0 | W_0 = (x, y)) \\ &\geq \sum_{x,y \in \mathcal{S}} \mu_1(x) \mu_2(y) \mathbb{P}(W_n = (x_0, x_0) \text{ for some } n \geq 0 | W_0 = (x, y)) \\ &= \sum_{x,y \in \mathcal{S}} \mu_1(x) \mu_2(y) f_{(x,y),(x_0,x_0)}^* = \sum_{x \in \mathcal{S}} \sum_{y \in \mathcal{S}} \mu_1(x) \mu_2(y) = 1 \end{aligned}$$

where we have used the fact that W_n is recurrent irreducible chain, and hence must visit (x_0, x_0) from (x, y) , i.e., $f_{(x,y),(x_0,x_0)}^* = 1$. Therefore, we have $T_D < \infty$ with probability 1 when the original chain is recurrent.

Now, we come to central idea of this proof : once the two independent chains have met, their marginal distributions after that must match, i.e.,

$$\mathbb{P}(X_n = u, T_D \leq n) = \mathbb{P}(Y_n = u, T_D \leq n).$$

Indeed, we have

$$\begin{aligned}
& \mathbb{P}(X_n = u, T_D \leq n) \\
&= \sum_{m=0}^n \sum_{x \in \mathcal{S}} \sum_{z \in \mathcal{S}} \mathbb{P}(W_n = (u, z), W_{T_D} = (x, x), T_D = m) \\
&= \sum_{m=0}^n \sum_{x \in \mathcal{S}} \mathbb{P}(W_{T_D} = (x, x), T_D = m) \sum_{z \in \mathcal{S}} \mathbb{P}(W_{n-m} = (u, z) | W_0 = (x, x)) \\
&= \sum_{m=0}^n \sum_{x \in \mathcal{S}} \mathbb{P}(W_{T_D} = (x, x), T_D = m) \sum_{z \in \mathcal{S}} p_{xu}^{(n-m)} p_{xz}^{(n-m)} \\
&= \sum_{m=0}^n \sum_{x \in \mathcal{S}} \mathbb{P}(W_{T_D} = (x, x), T_D = m) p_{xu}^{(n-m)}
\end{aligned}$$

where we have used the strong Markov property in the third line and the fact $\sum_{z \in \mathcal{S}} p_{xz}^{(n)} = 1$ in the last line. Interchanging the roles X_n and Y_n , we also have

$$\mathbb{P}(Y_n = u, T_D \leq n) = \sum_{m=0}^n \sum_{x \in \mathcal{S}} \mathbb{P}(W_{T_D} = (x, x), T_D = m) p_{xu}^{(n-m)}.$$

Thus, we note that the equation (10.4) is satisfied.

We will deal with two cases - a) the chain is positive recurrent and b) the chain is null recurrent. First for the positive recurrent case, we know that there is a unique stationary distribution, let us denote it by π . Further, we know that $\pi(j) = 1/(\mathbb{E}(T_j | X_0 = j))$ (see Corollary 9.19.2). We already know that the joint process $\{W_n : n \geq 0\}$ is an aperiodic and irreducible Markov chain on $\mathcal{S} \times \mathcal{S}$ with its transition probabilities given in equation (10.6). Now, we take

$$\sigma(x, y) = \pi(x)\pi(y) \text{ for } (x, y) \in \mathcal{S} \times \mathcal{S}.$$

Since $\pi(x) > 0$ for all $x \in \mathcal{S}$, we have that $\sigma(x, y) > 0$ for all $(x, y) \in \mathcal{S} \times \mathcal{S}$. Further, we have

$$\sum_{(x,y) \in \mathcal{S} \times \mathcal{S}} \sigma(x, y) = \sum_{x \in \mathcal{S}} \sum_{y \in \mathcal{S}} \pi(x)\pi(y) = \sum_{x \in \mathcal{S}} \pi(x) \sum_{y \in \mathcal{S}} \pi(y) = 1$$

and

$$\begin{aligned}
& \sum_{(x,y) \in \mathcal{S} \times \mathcal{S}} \sigma(x, y) q_{(x,y),(x',y')} = \sum_{x \in \mathcal{S}} \sum_{y \in \mathcal{S}} \pi(x)\pi(y) p_{x,x'} p_{y,y'} \\
&= \sum_{x \in \mathcal{S}} \pi(x) p_{x,x'} \sum_{y \in \mathcal{S}} \pi(y) p_{y,y'} = \pi(x') \pi(y') = \sigma(x', y').
\end{aligned}$$

Therefore, $\{\sigma(x, y) : x, y \in \mathcal{S}\}$ is a stationary distribution for the Markov chain. Thus the Markov chain must be positive recurrent as it is irreducible.

Now, we will specify the initial distributions: fix $i \in \mathcal{S}$ and take δ_i as a initial distribution, i.e., the Markov chain starting from the state i with probability 1 and the other initial distribution is given by π , the stationary distribution. Also, we must have

Thus, using the coupling Lemma 10.5, we have

$$\begin{aligned} \sup_{j \in \mathcal{S}} |p_{ij}^{(n)} - \frac{1}{\mathbb{E}(T_j | X_0 = j)}| &= \sup_{j \in \mathcal{S}} |p_{ij}^{(n)} - \pi(j)| \\ &= \sup_{j \in \mathcal{S}} |\mathbb{P}(X_n = j) - \mathbb{P}(Y_n = j)| \leq 2\mathbb{P}(T_D \geq n) \rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$. This proves the result for the positive recurrent case.

Now, we consider the case when the chain $\{X_n : n \geq 0\}$ is null recurrent. If the joint chain $\{W_n : n \geq 0\}$ is transient, we have that

$$\mathbb{P}(W_n = (j, j) | W_0 = (i, i)) = (p_{ij}^{(n)})^2 \rightarrow 0$$

as $n \rightarrow \infty$. This proves that $p_{ij}^{(n)} \rightarrow 0$.

Now, we assume that the joint chain $\{W_n : n \geq 0\}$ is recurrent. If the joint chain is positive recurrent, we must have for any state $i \in \mathcal{S}$, consider the return time $T_{(i,i)}$ to the state (i, i) for W_n and T_i to the state i for X_n . Then, we must have $\mathbb{E}(T_{(i,i)} | W_0 = (i, i)) < \infty$. However, it is obvious

$$T_i \leq T_{(i,i)}$$

and hence, we have $\mathbb{E}(T_i | X_0 = i) \leq \mathbb{E}(T_{(i,i)} | W_0 = (i, i)) < \infty$. Thus, the state i is positive recurrent for the chain $\{X_n : n \geq 0\}$, which contradicts the fact that the chain $\{X_n : n \geq 0\}$ is null recurrent. Therefore, the joint chain must be null-recurrent. We want to prove that

$$p_{ij}^{(n)} \rightarrow 0 \text{ as } n \rightarrow \infty$$

for all $i, j \in \mathcal{S}$.

Suppose that the above does not hold. Then, there exists a pair of states i_1, i_2 and a subsequence $\{n_k : k \geq 1\}$ such that

$$p_{i_1, i_2}^{(n_k)} \rightarrow \alpha_{i_2} > 0$$

as $k \rightarrow \infty$. Since the state space is countable, we can write down the state spaces as a sequence $\{i_k : k \geq 1\}$ starting with i_1, i_2 . Since $p_{i_k, i_{k'}}^{(n)}$ is bounded sequence, we can use

the diagonal selection argument to choose a subsequence of $\{n_{k_r} : r \geq 1\}$ such that $p_{i,j}^{(n_{k_r})}$ converges.

For any $i, j \in \mathcal{S}$, let us consider the degenerate distributions δ_i and δ_j as the initial distributions. Then, the coupling inequality will imply that

$$\sup_{k \in \mathcal{S}} |p_{ik}^{(n)} - p_{jk}^{(n)}| = \sup_{k \in \mathcal{S}} |\mathbb{P}(X_n = j) - \mathbb{P}(Y_n = j)| \leq 2\mathbb{P}(T_D \geq n) \rightarrow 0$$

as $n \rightarrow \infty$. Thus, for the sequence n_{k_r} , the limits do not depend on the starting state i . Thus, we get $\{\alpha_j : j \in \mathcal{S}\}$ such that

$$p_{i,j}^{(n_{k_r})} \rightarrow \alpha_j$$

for all $i, j \in \mathcal{S}$ as $r \rightarrow \infty$. We note that α_j 's are non-negative and at least one of them is strictly positive.

Now, we can use Proposition 10.2 to show that $\{\alpha_j : j \in \mathcal{S}\}$ is a stationary measure with $\sum_{j \in \mathcal{S}} \alpha_j \leq 1 < \infty$. Therefore, by Theorem 9.18, the chain must be positive recurrent. This is a contradiction. Therefore, we must have the limit must be 0 for any $i, j \in \mathcal{S}$. ■

10.3 Periodic chains

The periodic chain section is to be written.

10.4 Finite state chains

This section is to be written.

Chapter 11

Strong Markov Property

We know that for a Markov chain, the future process is independent of the past in the presence of the present state. We will now prove that this remains true if we replace the deterministic time by a random time, known as the *stopping time*. This stronger property is known as the *strong Markov property*.

11.1 Some measure theoretic concepts

Exercise 11.1. Show that for any family $\{\mathcal{F}_\lambda : \lambda \in \Lambda\}$, the intersection over all indices $\cap_{\lambda \in \Lambda} \mathcal{F}_\lambda$ is also a sigma algebra. ◀

This allows us to define the minimal σ -algebra containing arbitrary collection of sets $\mathcal{A}$, by considering the collection of σ -algebras containing the given collection $\mathcal{A}$ (this collection is non-empty as the power set $\mathcal{P}(\Omega)$ will always belong to this collection) and taking the intersection over this collection gives us the minimal σ -algebra. We will denote this by $\sigma(\mathcal{A})$. For example, note that when $\mathcal{A} = \{A\}$, we have $\sigma(\mathcal{A}) = (\sigma(A)) = \{\emptyset, A, A^c, \Omega\}$. For two subsets A, B , we have $\sigma(\{A, B\}) = (\sigma(A, B)) = \{\emptyset, A, A^c, B, B^c, A \cap B, A^c \cap B, A \cap B^c, A^c \cap B^c, A \cup B, A \cup B^c, A^c \cup B, A^c \cup B^c, (A \cap B) \cup (A^c \cap B^c), (A \cap B^c) \cup (A^c \cap B), \Omega\}$.

Exercise 11.2. Let $\mathcal{A} = \{A_1, A_2, \dots, A_k\}$ be a partition of Ω , i.e., $A_i \cap A_j = \emptyset$ and $\cup_{i=1}^k A_i = \Omega$. Then, the smallest σ -algebra $\sigma(\mathcal{A})$ consists of all possible finite unions of sub-collections from $\{A_1, A_2, \dots, A_k\}$, i.e.,

$$\sigma(\mathcal{A}) = \left\{ \cup_{i=1}^n B_i : B_i \text{ is either } A_i \text{ or } \emptyset \right\}.$$
 ◀

Exercise 11.3. For sets $\mathcal{A} = \{A_1, A_2, \dots, A_k\}$, find a description of $\sigma(\mathcal{A})$.

[Hint : Try to find a partition.] ◀

The collection of sets $\{\{\omega : X(\omega) \in B\} = X^{-1}(B) : B \in \mathcal{B}\}$ is a σ -algebra on Ω and is contained in $\mathcal{F}$. This is known as the σ -algebra generated / induced by X and is the minimal σ -algebra on Ω which makes X a random variable. We will denote this by $\sigma(X)$.

11.2 Stopping time

Now we introduce a special type of random times, namely, the *stopping time*. We set up a convention that the infimum of an empty set is $+\infty$. This will be followed throughout.

Definition 11.4. A sequence of σ -algebras $\{\mathcal{F}_n : n \geq 0\}$ is called a *filtration* if the σ -algebras are increasing, i.e., $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$ for all $n \geq 0$. A *random time* is a random variable assuming values in the time set $\{0, 1, 2, \dots\} \cup \{+\infty\}$ (including possibly the value $+\infty$). A random time τ is called a *stopping time* if $\{\tau = n\} \in \mathcal{F}_n$ for all $n \geq 0$.

We will generally consider a sequence of random variables and would like to consider events concerning these random variables. The intuitive explanation of a stopping time is the random time when the event that we are interested occurs. To make it more precise, for any sequence of random variables $\{X_n : n \geq 0\}$, the canonical filtration will be the sequence of σ -algebras where $\mathcal{F}_n$ is generated by the random variables $\{X_0, X_1, \dots, X_n\}$. In Markov chain theory, we will only consider the canonical filtration.

When we consider the canonical filtration, the above condition translates $\mathbb{I}(\tau = n)$ is a deterministic function of $(X_0, \dots, X_n)$ for all $n \geq 0$. In other words, the event $\{\tau = n\}$ is determined by the values of the random variables $(X_0, X_1, \dots, X_n)$, i.e., for any $n \geq 0$, there exists $C_n \subseteq \mathcal{S}^{n+1}$ such that

$$\{\tau = n\} = \{(X_0, X_1, \dots, X_n) \in C_n\}.$$

Note that any deterministic time n can always be interpreted as the stopping time which is degenerate at the value n .

Exercise 11.5. Suppose τ, τ_1, τ_2 are stopping times and $n \geq 0$. Show that

- a) $\{\tau \leq n\}$ is an event involving $X_0, X_1, \dots, X_n$.

- b) $\{\tau \geq n\}$ and $\{\tau > n\}$ are events involving $X_0, X_1, \dots, X_n$.
- c) $\tau + k$ is a stopping time for any $k \geq 1$.
- d) $\min\{\tau_1, \tau_2\}$ and $\max\{\tau_1, \tau_2\}$ are both stopping times.
- e) $\tau_1 + \tau_2$ is also a stopping time. ◀

An important example of a stopping time is the visiting time of the set A .

Definition 11.6. The visiting/hitting/entrance time of a set A is defined by

$$T_A = \inf\{n \geq 0 : X_n \in A\}. \quad (11.1)$$

If the set is a singleton, i.e., $A = \{i\}$ for some $i \in \mathcal{S}$, we will write T_i in place of $T_{\{i\}}$.

Note here that when the chain never visits the set of states in A , by our convention $T_A = +\infty$. Indeed, we note that for the event $\{T_A = n\}$ can be described by

$$\begin{aligned} \{T_A = n\} &= \{X_0 \notin A, X_1 \notin A, \dots, X_{n-1} \notin A, X_n \in A\} \\ &= \{(X_0, X_1, \dots, X_n) \in \underbrace{A^c \times A^c \times \dots \times A^c}_{n \text{ times}} \times A\}. \end{aligned}$$

Definition 11.7. Another important stopping time is the (first) return time to a set A , i.e.,

$$\tau_A = \inf\{n \geq 1 : X_n \in A\}. \quad (11.2)$$

If the set is a singleton, i.e., $A = \{i_0\}$ for some $i_0 \in \mathcal{S}$, we will write τ_{i_0} in place of $\tau_{\{i_0\}}$.

Note that if the initial state X_0 is in the set A , the the hitting time of the set T_A is 0. If the initial state is not in A , the hitting time is identical to the return time.

A set of states A may or may not be visited again. Regardless, we let $\tau_A^{(2)}$ be the time of the second return to A . (And we set $\tau_A^{(1)} = \tau_A$). Similarly, we let $\tau_A^{(3)}$ be the time of the return visit, and so on. Formally, we recursively define

$$\tau_A^{(r+1)} := \inf\{n > \tau_i^{(r)} : X_n \in A\}, \quad (11.3)$$

for $r = 1, 2, \dots$. It should be noted that if $\tau_A^{(r)} = +\infty$ for some $r \geq 1$, then obviously we are considering infimum of an empty set in the definition above and hence $\tau_i^{(r+1)} = +\infty$.

Exercise 11.8. Show that $\tau_i^{(r)}$ is a stopping time for any $r \geq 1$. ◀

Note that an example of a random time which is not a stopping time is the last visit of a specific set.

11.3 Strong Markov Property

Theorem 11.9. (Strong Markov property). *Suppose $\{X_n : n \geq 0\}$ is a Markov chain. Let τ be a stopping time. Then, conditional on the event $\tau < \infty$ and $X_\tau = j$, the future process $\{X_{\tau+n} : n \geq 1\}$ is independent of the past process $(X_0, \dots, X_\tau)$ and has the same law as the original Markov chain starting from j .*

Proof : Consider any event A determined by the $(X_0, X_1, \dots, X_\tau)$. For any such A and $i, j, j_1, j_2, \dots, j_m \in \mathcal{S}$, we need to prove that

$$\begin{aligned} & \mathbb{P}[\{X_{\tau+t} = j_t, \text{ for } 1 \leq t \leq m\} \cap A \cap \{X_\tau = j\} \cap \{\tau < \infty\} \mid X_0 = i] \\ &= p_{j,j_1} p_{j_1,j_2} \cdots p_{j_{m-1},j_m} \mathbb{P}(A \cap \{X_\tau = j\} \cap \{\tau < \infty\} \mid X_0 = i). \end{aligned}$$

Since A is determined by the $(X_0, X_1, \dots, X_\tau)$, for any $n \geq 0$, we can get $D_n \subseteq \mathcal{S}^{n+1}$ such that on the event $\{\tau = n\}$,

$$A = \{(X_0, X_1, \dots, X_n) \in D_n\}.$$

Since τ is a stopping time, we can also express in a similar way, i.e.,

$$\{\tau = n\} = \{(X_0, X_1, \dots, X_n) \in C_n\}$$

for some $C_n \subseteq \mathcal{S}^{n+1}$. Furthermore, we are also given that $\tau < \infty$. Thus, we have that

$$\begin{aligned} A \cap \{\tau < \infty\} \cap \{X_\tau = j\} &= \cup_{n=0}^{\infty} A \cap \{\tau = n\} \cap \{X_\tau = j\} \\ &= \cup_{n=0}^{\infty} \{(X_0, X_1, \dots, X_n) \in C_n \cap D_n, X_n = j\}. \end{aligned}$$

Now, for any $n \geq 0$, we have

$$\begin{aligned} & \mathbb{P}(\{X_{\tau+t} = j_t, \text{ for } 1 \leq t \leq m\} \cap A \cap \{X_\tau = j\} \cap \{\tau = n\} \mid X_0 = i) \\ &= \mathbb{P}(\{X_{n+t} = j_t, \text{ for } 1 \leq t \leq m\} \cap \{(X_0, X_1, \dots, X_n) \in C_n \cap D_n, X_n = j\} \mid X_0 = i) \\ &= \mathbb{P}(X_t = j_t, \text{ for } 1 \leq t \leq m \mid X_0 = j) \mathbb{P}((X_0, X_1, \dots, X_n) \in C_n \cap D_n, X_n = j \mid X_0 = i) \\ &= p_{j,j_1} p_{j_1,j_2} \cdots p_{j_{m-1},j_m} \mathbb{P}((X_0, X_1, \dots, X_n) \in C_n \cap D_n, X_n = j \mid X_0 = i) \end{aligned}$$

where we have used the fact the future and the past are independent given that we have the knowledge of the present state (see equation (6.2)).

Now, summing over $n \geq 0$ we have

$$\begin{aligned}
& \mathbb{P}(\{X_{\tau+t} = j_t, \text{ for } 1 \leq t \leq m\} \cap A \cap \{X_\tau = j\} \cap \{\tau < \infty\} \mid X_0 = i) \\
&= \sum_{n=0}^{\infty} \mathbb{P}(\{X_{\tau+t} = j_t, \text{ for } 1 \leq t \leq m\} \cap A \cap \{X_\tau = j\} \cap \{\tau = n\} \mid X_0 = i) \\
&= \sum_{n=0}^{\infty} p_{j,j_1} p_{j_1,j_2} \cdots p_{j_{m-1},j_m} \mathbb{P}((X_0, X_1, \dots, X_n) \in C_n \cap D_n, X_n = j \mid X_0 = i) \\
&= p_{j,j_1} p_{j_1,j_2} \cdots p_{j_{m-1},j_m} \mathbb{P}(A \cap \{\tau < \infty\} \cap \{X_\tau = j\} \mid X_0 = i).
\end{aligned}$$

This completes the proof. ■

The Strong Markov property is very useful for computing probability of events involving stopping times. Suppose that $C \subseteq \underbrace{\mathcal{S} \times \mathcal{S} \times \cdots \times \mathcal{S}}_{m \text{ times}}$. Then, summing $(j_1, \dots, j_m) \in C$, in the above relation, with A as above we see that

$$\begin{aligned}
& \mathbb{P}(\{(X_{\tau+1}, X_{\tau+2}, \dots, X_{\tau+m}) \in C\} \cap A \cap \{X_\tau = j, \tau < \infty\} \mid X_0 = i) \\
&= \mathbb{P}((X_1, X_2, \dots, X_m) \in C \mid X_0 = j) \mathbb{P}(A \cap X_\tau = j, \tau < \infty \mid X_0 = i). \tag{11.4}
\end{aligned}$$

11.4 Applications of strong Markov property

First we claim that for any left non-skipping chain, for $i \geq 1$,

$$f_{i+1,0}^* = f_{i,0}^* f_{i+1,i}^*.$$

Now to complete the proof of the claim, let T_i be the first visiting time of the state i (see equation (7.5)). For any $i \geq 1$, by left non-skipping property, starting from $i+1$, the chain must visit i first and then visit 0. So, if $T_0 < \infty$ we must have $T_i < \infty$ given $X_0 = i+1$. So, we may write the event $\{T_0 < \infty\}$ as a disjoint union $\cup_{m=1}^{\infty} \{T_0 = T_i + m\}$ and observe that the event $\{T_0 = T_i + m\}$ depends only on $\{X_{T_i+t} : t \geq 1\}$. Indeed, taking $C = \{(x_1, x_2, \dots, x_m) : x_m = 0, x_i > 0 \text{ for } t = 1, 2, \dots, m-1\}$, we observe that $\{T_0 = T_i + m\} = \{(X_{T_i+1}, X_{T_i+2}, \dots, X_{T_i+m}) \in C\}$. Also, we must have $X_{T_i} = i$. Since

T_i is a stopping time, using Theorem 11.9, we have

$$\begin{aligned}
\mathbb{P}(T_0 < \infty, X_0 = i + 1) &= \mathbb{P}(T_i < T_0 < \infty, X_{T_i} = i, X_0 = i + 1) \\
&= \sum_{m=1}^{\infty} \mathbb{P}(T_i < \infty, X_{T_i} = i, T_0 = T_i + m, X_0 = i + 1) \\
&= \sum_{m=1}^{\infty} \mathbb{P}(T_i < \infty, X_{T_i} = i, X_0 = i + 1) \mathbb{P}((X_1, X_2, \dots, X_m) \in C \mid X_0 = i) \\
&= \mathbb{P}(T_i < \infty, X_{T_i} = i, X_0 = i + 1) \sum_{m=1}^{\infty} f_{i0}^{(m)} \\
&= \mathbb{P}(T_i < \infty \mid X_0 = i + 1) \mathbb{P}(X_0 = i + 1) f_{i0}^{(*)} = f_{i+1,i}^* f_{i0}^{(*)} \mathbb{P}(X_0 = i + 1)
\end{aligned}$$

where we have used the fact that $X_{T_i} = i$ with probability 1. Now, dividing both sides by $\mathbb{P}(X_0 = i + 1)$, we have $f_{i+1,0}^* = f_{i0}^* f_{i+1,i}^*$.

11.5 Regeneration

We have observed that for a recurrent state i , $\mathbb{P}(\tau_i^{(r)} < \infty \mid X_0 = i) = 1$ for any $r \geq 1$. Suppose that a Markov chain possesses a recurrent state $i \in \mathcal{S}$ and assume that the distribution of X_0 is such that $\mathbb{P}(\tau_i < \infty) = 1$. Since $\tau_i^{(1)} = \tau_i < \infty$ with probability 1, it is the case that $\tau_i^{(r)} < \infty$ with probability 1 for all $r \geq 2$ as the state i is recurrent. Further, we must have $X_{\tau_i^{(r)}} = i$ with probability 1 for any $r \geq 1$. By the strong Markov property, we have

$$\begin{aligned}
&\mathbb{P}(\tau_i^{(k+1)} - \tau_i^{(k)} = n_k \text{ for } k = 1, 2, \dots, r) \\
&= \mathbb{P}(\tau_i^{(k+1)} - \tau_i^{(k)} = n_k \text{ for } k = 1, 2, \dots, r, \tau_i^{(r)} < \infty, X_{\tau_i^{(r)}} = i) \\
&= \mathbb{P}(X_{\tau_i^{(r)}+j} \neq i \text{ for } 1 \leq j < n_r, X_{\tau_i^{(r)}+n_r} = i, \\
&\quad \tau_i^{(k+1)} - \tau_i^{(k)} = n_k \text{ for } k = 1, 2, \dots, r-1, \tau_i^{(r)} < \infty, X_{\tau_i^{(r)}} = i) \\
&= \mathbb{P}(X_j \neq i \text{ for } 1 \leq j < n_r, X_{n_r} = i \mid X_0 = i) \\
&\quad \times \mathbb{P}(\tau_i^{(k+1)} - \tau_i^{(k)} = n_k \text{ for } k = 1, 2, \dots, r-1, \tau_i^{(r)} < \infty, X_{\tau_i^{(r)}} = i) \\
&= f_{ii}^{(n_r)} \mathbb{P}(\tau_i^{(k+1)} - \tau_i^{(k)} = n_k \text{ for } k = 1, 2, \dots, r-1) \\
&= \prod_{k=1}^r f_{ii}^{(n_k)}
\end{aligned}$$

where the last step follows by repeating the the previous step. This proves that $\{\tau_i^{(k+1)} - \tau_i^{(k)} : k \geq 1\}$ is a sequence of **iid** random variables with $\mathbb{P}(\tau_i^{(k+1)} - \tau_i^{(k)} = n) = f_{ii}^{(n)} = \mathbb{P}(\tau_i = n \mid X_0 = i)$. So, we can see that the inter-visiting times to a recurrent state are identically distributed with first visiting time conditioned on the fact that the process started at the same state. So, if the state is positive recurrent, i.e., $\mathbb{E}(\tau_i \mid X_0 = i) = \mu_i = \sum_{j=1}^{\infty} j f_{ii}^{(j)} < \infty$, we have $\mathbb{E}(\tau_i^{(k+1)} - \tau_i^{(k)}) = \mu_i < \infty$ for any $k \geq 1$. Hence, we have proved the following result.

Proposition 11.10. *Suppose that a Markov chain possesses a recurrent state $i \in \mathcal{S}$ and assume that the distribution of X_0 is such that $\mathbb{P}(\tau_i < \infty) = 1$. The sequence $\{\tau_i^{(k+1)} - \tau_i^{(k)} : k \geq 1\}$ is a sequence of **iid** random variables with the distribution $\mathbb{P}(\tau_i^{(k+1)} - \tau_i^{(k)} = n) = f_{ii}^{(n)} = \mathbb{P}(\tau_i = n \mid X_0 = i)$ for $n \geq 1$. Further, if the state i is positive recurrent, we have $\mathbb{E}(\tau_i^{(k+1)} - \tau_i^{(k)}) = \mu_i = \mathbb{E}(\tau_i \mid X_0 = i) < \infty$.*

11.6 Time average and Stationary distribution

Consider now a function $f : \mathcal{S} \rightarrow \mathbb{R}$, which we can think of as a reward received when the chain is in a state $i \in \mathcal{S}$. We are interested in the existence of the limit

$$\text{time-average reward} = \lim_{t \rightarrow \infty} \frac{1}{t} \sum_{k=0}^t f(X_k)$$

which represents the *time-average reward* received. In particular, we assume that f is either bounded or non-negative and examine those conditions which yield a nontrivial limit.

Theorem 11.11. *Suppose that a Markov chain possesses a positive recurrent state $i \in \mathcal{S}$ (i.e., $\mu_i = \mathbb{E}(T_i \mid X_0 = i) < \infty$) and assume that the distribution of X_0 is such that $\mathbb{P}(T_i < \infty) = 1$. Let $f : \mathcal{S} \rightarrow \mathbb{R}$ be a reward function. We assume that f is either bounded or non-negative. Then, with probability one, the time-average reward $\bar{f}$ exists and is a deterministic quantity:*

$$\mathbb{P}\left(\lim_{t \rightarrow \infty} \frac{1}{t} \sum_{k=0}^t f(X_k) = \bar{f}\right) = 1$$

where

$$\bar{f} = \frac{\mathbb{E}_i\left(\sum_{k=0}^{T_i-1} f(X_k)\right)}{\mathbb{E}_i(T_i)} = \frac{\mathbb{E}\left(\sum_{k=0}^{T_i-1} f(X_k) \mid X_0 = i\right)}{\mathbb{E}(T_i \mid X_0 = i)} = \sum_{i \in \mathcal{S}} f(i) \pi^{[a]}(i)$$

where $\pi^{[a]}$ is as defined in equation (9.17).

We present the proof in a way that the constant $\bar{f}$ will reveal itself. The idea is this: suppose that t is large enough and we are looking for the total reward received between 0 and t . For this, we will break up the time interval $[0, t]$ into excursions.

We look at the time points on which the chain returns to the *ground state* i . The initial excursion is defined as the part of the chain before the first visit to the ground state i . The next excursion is the part of the chain from first visit to i until the second visit to i and so on. We break the sum into the total reward received over the initial excursion, plus the total reward received over the next excursion, and so on. Thus we need to keep track of the number of complete excursions from the ground state i that occur between 0 and t .

Proof : Let us consider the visiting times of the ground state i , i.e., set $T_i^{(1)} = T_i$ and for $r \geq 1$, $T_i^{(r+1)} = \inf\{n > T_i^{(r)} : X_n = i\}$ (see equation (11.3)). Now set, for $t \geq 1$

$$N_t = \max\{r \geq 0 : T_i^{(r)} \leq t\}.$$

Clearly, N_t counts the number of completed excursions.

We first show that $\lim_{t \rightarrow \infty} N_t = \infty$ with probability 1. Observe that N_t is non-decreasing in t . Therefore, $\lim_{t \rightarrow \infty} N_t$ exists (it may be infinite and we are going to show that it is infinite) with probability 1. Suppose that $\mathbb{P}(\lim_{t \rightarrow \infty} N_t < \infty) > 0$. Consider the event $A_k = \{\lim_{t \rightarrow \infty} N_t \leq k\}$. Clearly $A_k \uparrow \{\lim_{t \rightarrow \infty} N_t < \infty\}$. Thus, $\mathbb{P}(A_k) \rightarrow \mathbb{P}(\lim_{t \rightarrow \infty} N_t < \infty)$ as $k \rightarrow \infty$. So, we may choose K large enough so that $\mathbb{P}(A_K) = \alpha > 0$. This implies that $T_i^{(K+1)} > t$ for all $t > 0$, i.e., $T_i^{(K+1)} = \infty$ on the event A_K . Hence $T_i^{(K+1)} = \infty$ with positive probability. But this contradicts the fact that i is recurrent and hence $T_i^{(r)} < \infty$ with probability 1 for all $r \geq 1$.

Now, we have

$$T_i^{(r)} = T_i^{(1)} + \sum_{k=1}^r (T_i^{(k+1)} - T_i^{(k)}).$$

Since $T_i^{(1)} < \infty$ with probability 1, we have $T_i^{(1)}/r \rightarrow 0$ as $r \rightarrow \infty$. Thus, with probability 1, we have

$$\frac{T_i^{(r)}}{r} \rightarrow \mathbb{E}(T_i^{(2)} - T_i^{(1)}) = \mathbb{E}(T_i \mid X_0 = i) = \mu_i < \infty. \quad (11.5)$$

By definition, we have that the visit with index $r = N_t$ to the ground state i is the last before t :

$$T_i^{(N_t)} \leq t < T_i^{(N_t+1)}.$$

Hence

$$\frac{T_i^{(N_t)}}{N_t} \leq \frac{t}{N_t} < \frac{T_i^{(N_t+1)}}{N_t}.$$

From (11.5) and the fact that $N_t \rightarrow \infty$ as $t \rightarrow \infty$, we have

$$\lim_{t \rightarrow \infty} \frac{T_i^{(N_t)}}{N_t} = \lim_{t \rightarrow \infty} \frac{T_i^{(N_t+1)}}{N_t + 1} \times \frac{N_t + 1}{N_t} = \mathbb{E}(T_i \mid X_0 = i).$$

Hence, we have

$$\lim_{t \rightarrow \infty} \frac{N_t}{t} = \frac{1}{\mathbb{E}(T_i \mid X_0 = i)}.$$

Let us set also, for $1 \leq i \leq N_t - 1$

$$R_i = \sum_{k=T_a^{(i)}}^{T_a^{(i+1)}-1} f(X_k).$$

Note that R_i represents the total reward of the process between i -th return and before the $(i+1)$ -th return. Let $R_{\text{first}} = \sum_{n=0}^{T_1^{(1)}-1} f(X_n)$ and $R_{\text{last}} = \sum_{n=T_{N_t}}^t f(X_n)$ be respectively the first before the first visit to the ground state and the remaining part after the last completed excursions. So, we may write

$$\sum_{k=0}^t f(X_k) = R_{\text{first}} + \sum_{i=1}^{N_t-1} R_i + R_{\text{last}}.$$

We have already seen that the sequence $\{T_i^{(k+1)} - T_i^{(k)} : k \geq 1\}$ is a sequence of iid random variables (see Proposition 11.10). For $H_1, H_2, \dots, H_r \subseteq \mathbb{R}$, we have,

$$\begin{aligned} & \mathbb{P}(R_k \in H_k \text{ for } k = 1, \dots, r) \\ &= \sum_{n=1}^{\infty} \mathbb{P}(R_k \in H_k \text{ for } k = 1, \dots, r, T_i^{(r+1)} - T_i^{(r)} = n, T_i^{(r)} < \infty, X_{T_i^{(r)}} = a) \\ &= \sum_{n=1}^{\infty} \mathbb{P}\left[f(X_{T_i^{(r)}}) + \sum_{j=1}^{n-1} f(X_{T_i^{(r)}+j}) \in H_r, X_{T_i^{(r)}+n} = a, X_{T_i^{(r)}+j} \text{ for } j = 1, \dots, n-1, \right. \\ & \quad \left. T_i^{(r)} < \infty, X_{T_i^{(r)}} = i, R_k \in H_k \text{ for } k = 1, \dots, r-1\right] \\ &= \sum_{n=1}^{\infty} \mathbb{P}\left[f(i) + \sum_{j=1}^{n-1} f(X_j) \in H_r, X_j \neq i \text{ for } 1 \leq j < n, X_n = i \mid X_0 = i\right] \\ & \quad \times \mathbb{P}\left[T_i^{(r)} < \infty, X_{T_i^{(r)}} = i, R_k \in H_k \text{ for } k = 1, \dots, r-1\right] \\ &= \mathbb{P}\left[R_k \in H_k \text{ for } k = 1, \dots, r-1\right] \\ & \quad \times \sum_{n=1}^{\infty} \mathbb{P}\left[f(i) + \sum_{j=1}^{n-1} f(X_j) \in H_r, X_j \neq i \text{ for } 1 \leq j < n, X_n = i \mid X_0 = i\right] \end{aligned}$$

using the fact that $T_i^{(r)} < \infty$, $X_{T_i^{(r)}} = i$ with probability 1. Now, using the above inductively, we have

$$\begin{aligned} & \mathbb{P}(R_k \in H_k \text{ for } k = 1, \dots, r) \\ &= \prod_{k=1}^r \sum_{n=1}^{\infty} \mathbb{P}\left[f(i) + \sum_{j=1}^{n-1} f(X_j) \in H_k, X_j \neq i \text{ for } 1 \leq j < n, X_n = i \mid X_0 = i\right]. \end{aligned}$$

Hence we have $\{R_k : k \geq 1\}$ is a sequence of **iid** random variables. Further, since f is bounded, let us assume that $|f| \leq M$. Hence, we see

$$|R_i| \leq \sum_{k=T_i^{(i)}}^{T_i^{(i+1)}-1} |f(X_k)| \leq M(T_i^{(i+1)} - T_i^{(i)}).$$

Thus, we have $\mathbb{E}(|R_k|) \leq M\mathbb{E}(T_i^{(2)} - T_i^{(1)}) = M\mu_i < \infty$. Therefore, using the SLLN, we have

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n R_i = \mathbb{E}(R_i).$$

Since $|R_{\text{first}}| \leq MT_i$ and $\mathbb{P}(T_i < \infty) = 1$, we have that $R_{\text{first}}/t \rightarrow 0$. Similarly, $R_{\text{last}}/t \rightarrow 0$. Thus, we have

$$\begin{aligned} \frac{1}{t} \sum_{k=0}^t f(X_k) &= \frac{1}{t} R_{\text{first}} + \frac{1}{t} R_{\text{last}} + \frac{N_t}{t} \times \frac{N_t - 1}{N_t} \times \frac{1}{N_t - 1} \sum_{i=1}^{N_t-1} R_i \\ &\rightarrow \frac{1}{\mathbb{E}(T_i \mid X_0 = i)} \mathbb{E}(R_1) = \frac{1}{\mathbb{E}(T_i \mid X_0 = i)} \mathbb{E}\left(\sum_{k=0}^{T_i-1} f(X_n) \mid X_0 = i\right). \end{aligned}$$

We will now identify $\bar{f}$. For any positive or bounded function f , we have

$$\begin{aligned} \bar{f} &= \frac{1}{\mathbb{E}(T_a \mid X_0 = a)} \mathbb{E}\left(\sum_{k=0}^{T_a-1} f(X_n) \mid X_0 = a\right) \\ &= \frac{1}{\mathbb{E}(T_a \mid X_0 = a)} \mathbb{E}\left(\sum_{k=0}^{T_a-1} \sum_{i \in \mathcal{S}} \mathbb{I}(X_n = i) f(i) \mid X_0 = a\right) \\ &= \frac{1}{\mathbb{E}(T_a \mid X_0 = a)} \mathbb{E}\left(\sum_{i \in \mathcal{S}} f(i) \sum_{k=0}^{T_a-1} \mathbb{I}(X_n = i) \mid X_0 = a\right) \\ &= \sum_{i \in \mathcal{S}} f(i) \frac{1}{\mathbb{E}(T_a \mid X_0 = a)} \mathbb{E}\left(\sum_{k=0}^{T_a-1} \mathbb{I}(X_n = i) \mid X_0 = a\right) \\ &= \sum_{i \in \mathcal{S}} f(i) \pi^{[a]}(i) \end{aligned} \tag{11.6}$$

where $\pi^{[a]}$ is as defined in equation (9.17). In the above interchange of summation and expectation is possible for non-negative functions or bounded functions. For unbounded function f , we will require some integrability condition. As we may observe that $\bar{f}$ is actually the expectation of the function f with respect to the stationary distribution $\pi^{[a]}$. Clearly, this is an average over the functional value over the state space, hence it is a space average. This proves the result. ■

Remark. We discuss a very important particular case of Theorem 11.11. For a state $b \in \mathcal{S}$

$$f(i) := \mathbb{I}(i = b),$$

i.e. we let f to be 1 at the state b and 0 everywhere else. Then, from Theorem 11.11, we have that for all $b \in \mathcal{S}$,

$$\lim_{t \rightarrow \infty} \frac{1}{t} \sum_{n=0}^t \mathbb{I}(X_n = b) = \frac{1}{\mathbb{E}(T_a | X_0 = a)} \mathbb{E} \left(\sum_{n=0}^{T_a-1} \mathbb{I}(X_n = b) \mid X_0 = a \right) \quad (11.7)$$

with probability 1. Letting $b = a$, we have that

$$\lim_{t \rightarrow \infty} \frac{1}{t} \sum_{n=0}^t \mathbb{I}(X_n = a) = \frac{1}{\mathbb{E}(T_a | X_0 = a)} \quad (11.8)$$

with probability 1. The physical interpretation of the above should be clear: if, on the average, it takes $\mathbb{E}_a(T_a)$ units of time between successive visits to the state a , then the long-run proportion of visits to the state a should equal $1/\mathbb{E}_a(T_a)$. ►

Remark. Suppose that two states, a, b are both positive recurrent such that both states are reached from the initial state with probability 1. We can use b as a ground state instead of a . Applying formula (11.8) with b in place of a , we find that

$$\begin{aligned} \frac{1}{\mathbb{E}(T_b | X_0 = b)} &= \lim_{t \rightarrow \infty} \frac{1}{t} \sum_{n=0}^t \mathbb{I}(X_n = b) = \frac{1}{\mathbb{E}(T_a | X_0 = a)} \mathbb{E} \left(\sum_{n=0}^{T_a-1} \mathbb{I}(X_n = b) \mid X_0 = a \right) \\ &= \frac{1}{\mathbb{E}(T_a | X_0 = a)} \mathbb{E} \left(\sum_{n=T_a^{(1)}}^{T_a^{(2)}-1} \mathbb{I}(X_n = b) \right) \end{aligned}$$

Therefore, we have

$$\left(\begin{array}{c} \text{Average no of visits to } b \\ \text{between successive visits to } a \end{array} \right) = \mathbb{E} \left(\sum_{n=T_a^{(1)}}^{T_a^{(2)}-1} \mathbb{I}(X_n = b) \right) = \frac{\mathbb{E}_a(T_a)}{\mathbb{E}_b(T_b)}. \quad \blacktriangleright$$

If we have an unique stationary distribution, we may restate Theorem 11.11.

Corollary 11.11.1. *Consider a positive recurrent irreducible Markov chain with stationary distribution π . Then for any $f : \mathcal{S} \rightarrow \mathbb{R}$ (where f is positive or bounded), we have*

$$\lim_{t \rightarrow \infty} \frac{1}{t} \sum_{k=0}^t f(X_k) \rightarrow \sum_{i \in \mathcal{S}} f(i) \pi(i)$$

with probability 1.

Chapter 12

Martingale methods in Markov chains

12.1 Introduction to Martingale

Martingales are a very powerful construct in probability theory. The theory of martingales is very rich and has applications in a wide variety of subjects. Here our main focus is Markov chain taking values in discrete set, therefore we will not get into the general theory of martingales, rather use martingales to solve some problems about Markov chain. Therefore, instead of defining martingale in the general set up, we will only consider martingales which are functions of Markov chain.

Two other related concepts are also important, namely submartingales and supermartingales, which can also be defined in a similar fashion. In general, to define martingales (as well submartingales or supermartingales), we need to consider a filtration (see Definition 11.4). For Markov chains, we generally consider the canonical filtration, i.e., consider $\{\mathcal{F}_n : n \geq 0\}$ where $\mathcal{F}_n$ is the σ -algebra generated by the Markov chain until time n . More precisely, let us assume that $\{X_n : n \geq 0\}$ is a Markov chain on the state space $\mathcal{S}$. random variables. We consider $\mathcal{F}_n = \sigma(\{X_0, X_1, \dots, X_n\})$. Henceforth, we will only consider the canonical filtration.

We will consider a sequence of random variables $\{Y_n : n \geq 0\}$ where each Y_n is a function of $(X_0, X_1, \dots, X_n)$ and n .

Definition 12.1. Suppose that the the sequence $\{Y_n : n \geq 0\}$ satisfies the following conditions:

for any $n \geq 0$,

$$\mathbb{E}(|Y_n|) < \infty.$$

(a) The sequence $\{Y_n : n \geq 0\}$ is called a supermartingale (with respect to the canonical filtration) if for any $n \geq 0$,

$$\mathbb{E}(Y_{n+1} \mid (X_1, X_2, \dots, X_n)) \leq Y_n. \quad (12.1)$$

(b) The sequence $\{Y_n : n \geq 0\}$ is called a submartingale (with respect to the canonical filtration) if for any $n \geq 0$,

$$\mathbb{E}(Y_{n+1} \mid (X_1, X_2, \dots, X_n)) \geq Y_n. \quad (12.2)$$

(c) The sequence $\{Y_n : n \geq 0\}$ is called a martingale (with respect to the canonical filtration) if for any $n \geq 0$,

$$\mathbb{E}(Y_{n+1} \mid (X_1, X_2, \dots, X_n)) = Y_n. \quad (12.3)$$

Since we are only considering the canonical filtration, we will drop the filtration part from now on. We observe that a sequence $\{Y_n : n \geq 0\}$ is a submartingale if the sequence $\{-Y_n : n \geq 0\}$ is a supermartingale. Also a sequence is a martingale if and only if it is a supermartingale as well as a submartingale.

Mostly we will restrict our attention to the case when Y_n is a function of the current state of the Markov chain X_n and current time n . In other words, let $h : \mathcal{S} \times \mathbb{N}^+ \rightarrow \mathbb{R}$ be a function. For $n \geq 0$ and set

$$Y_n = h(X_n, n).$$

For a Markov chain with transition probability $\mathbf{P}$, the main condition can be rewritten. Observe that

$$\mathbb{E}(Y_{n+1} \mid (X_1, X_2, \dots, X_n)) = \mathbb{E}(h(X_{n+1}, n) \mid (X_1, X_2, \dots, X_n)) = \mathbb{E}(h(X_{n+1}, n) \mid X_n).$$

Therefore, for any $i \in \mathcal{S}$, we have

$$\mathbb{E}(h(X_{n+1}, n) \mid X_n = i) = \sum_{j \in \mathcal{S}} h(j, n+1) p_{ij}.$$

Therefore, the main condition (in (12.1) or (12.2) or (12.3)) in Definition 12.1 can be rewritten as :

(a) for supermartingales, for all $i \in \mathcal{S}$ and $n \geq 0$,

$$\sum_{j \in \mathcal{S}} h(j, n+1) p_{ij} \leq h(i, n); \quad (12.4)$$

(b) for submartingales, for all $i \in \mathcal{S}$ and $n \geq 0$,

$$\sum_{j \in \mathcal{S}} h(j, n+1) p_{ij} \geq h(i, n). \quad (12.5)$$

(c) for martingales, for all $i \in \mathcal{S}$ and $n \geq 0$,

$$\sum_{j \in \mathcal{S}} h(j, n+1) p_{ij} = h(i, n). \quad (12.6)$$

12.2 Examples

Let us consider some examples from the Markov chains that we have seen.

Example 12.2. Let us consider the random walk example (see Example 5.10). Let $\{\xi_n : n \geq 0\}$ be the sequence of **iid** random variables taking values in $\mathbb{Z}$. Let S_0 be any random taking random variable with $\mathbb{E}(|S_0|) < \infty$. We assume that $\mathbb{E}(|\xi_n|) < \infty$ and $\mathbb{E}(\xi_n) = \mu$ (say). We define,

$$S_n = S_0 + \sum_{i=1}^n \xi_i.$$

We have observe that $\{S_n : n \geq 0\}$ is a Markov chain.

Now, we consider the function $f : \mathbb{Z} \times \mathbb{N}^+ \rightarrow \mathbb{R}$,

$$f(i, n) = i - n\mu$$

so that $Y_n = S_n - n\mu$. We note that

$$\mathbb{E}(|S_n|) \leq \mathbb{E}(|S_0| + \sum_{i=1}^n |\xi_i| + n|\mu|) = \mathbb{E}(|S_0|) + n\mathbb{E}(|\xi_1|) + n|\mu| < \infty.$$

Also, we have, for any $i \in \mathbb{Z}$,

$$\begin{aligned} \mathbb{E}(S_{n+1} - (n+1)\mu \mid S_n = i) &= \mathbb{E}(i + \xi_{n+1} \mid S_n = i) - (n+1)\mu \\ &= i + \mathbb{E}(\xi_{n+1} \mid S_n = i) - (n+1)\mu = i + \mathbb{E}(\xi_{n+1}) - (n+1)\mu = i - n\mu \end{aligned}$$

using the fact that ξ_{n+1} is independent of $\{\xi_0, \xi_1, \dots, \xi_n\}$ and hence ξ_{n+1} is independent of S_n and also $\mathbb{E}(\xi_{n+1}) = \mu$ for all $n \geq 0$.

Let us consider $\mathbb{E}(S_0^2) < \infty$ and $\mathbb{E}(\xi) = 0, \mathbb{E}(\xi_1^2) < \infty$. Now consider the function $g : \mathbb{Z} \times \mathbb{N}^+$ as $g(i, n) = i^2 - n\sigma^2$, i.e., consider

$$Z_n = S_n^2 - n\sigma^2$$

where $\sigma^2 = \text{Var}(\xi_1) = \mathbb{E}(\xi^2) - (\mathbb{E}(\xi))^2 = \mathbb{E}(\xi^2)$ since $\mathbb{E}(\xi) = 0$.

It is easy to check the conditions of martingale here also. Observe that

$$\begin{aligned} \mathbb{E}(|Z_n|) &\leq \mathbb{E}(S_n^2) + n\sigma^2 \leq (n+1)[\mathbb{E}(S_0^2) + \sum_{i=1}^n \mathbb{E}(\xi_i^2)] + n\sigma^2 \\ &= (n+1)\mathbb{E}(S_0^2) + n(n+2)\sigma^2 < \infty. \end{aligned}$$

Now, we have

$$\begin{aligned} \mathbb{E}(S_{n+1}^2 - (n+1)\sigma^2 \mid S_n = i) &= \mathbb{E}((i + \xi_{n+1})^2 \mid S_n = i) - (n+1)\sigma^2 \\ &= i^2 + 2i\mathbb{E}(\xi_{n+1} \mid S_n = i) + \mathbb{E}(\xi_{n+1}^2 \mid S_n = i) - (n+1)\sigma^2 \\ &= i^2 + 2i\mathbb{E}(\xi_{n+1}) + \mathbb{E}(\xi_{n+1}^2) - (n+1)\sigma^2 \\ &= i^2 - n\sigma^2 \end{aligned}$$

where we have used that fact that ξ_{n+1} is independent of S_n and $\mathbb{E}(\xi_{n+1}) = 0, \mathbb{E}(\xi_{n+1}^2) = \sigma^2$.

Coming to specific cases, we consider the simple random walk on one dimension (see Example 5.11) $\{S_n : n \geq 0\}$. For the symmetric case, i.e., $p = q = \frac{1}{2}$, the processes $\{S_n : n \geq 0\}$ and $\{S_n^2 - n : n \geq 0\}$ are both martingales. For the asymmetric case i.e., $p \neq q$, we note that the process $\{S_n - n(p - q) : n \geq 0\}$ is a martingale. ■

Example 12.3. a) For the Gambler's ruin problem (see Example 5.16) with $p = q = \frac{1}{2}$, the fortune $\{S_n : n \geq 0\}$ of Gambler A, forms a Markov chain on the state space $\{0, 1, \dots, N\}$ where N represents the combined capital of the gamblers. Here $\{S_n : n \geq 0\}$ is a martingale. Indeed, we see that if $i = 0$ or N , then $p_{ii} = 1$ and therefore $S_{n+1} = i$ with probability 1 so that $\mathbb{E}(S_{n+1} \mid S_n = i) = i$. For $i \in \{1, 2, \dots, N-1\}$, we note that $S_{n+1} = S_n + \xi_{n+1} = i + \xi_{n+1}$ where $\mathbb{P}(\xi_{n+1} = 1) = \mathbb{P}(\xi_{n+1} = -1) = \frac{1}{2}$. Therefore, we have $\mathbb{E}(S_{n+1} \mid S_n = i) = i$. This proves that $\{S_n : n \geq 0\}$ is a martingale. Note that the problem of gambler's ruin problem against an infinitely rich adversary, is also a martingale, by the above arguments.

b) We consider the Wright Fisher model (see Example 5.25). Here, the number of a -genes X_n will form a Markov chain on $\{0, 1, \dots, N\}$. Here again $\{X_n : n \geq 0\}$ will a

martingale. Note that here also, $p_{00} = p_{N,N} = 1$. Therefore, we have $\mathbb{E}(X_{n+1} \mid S_n = i) = i$ for $i = 0$ or N . For $i = 1, 2, \dots, N-1$, we note that $X_{n+1} \mid X_n = i$ follows a binomial distribution with parameters $2N$ and $\frac{i}{N}$. Therefore, we have $\mathbb{E}(X_{n+1} \mid X_n = i) = N \times \frac{i}{N} = i$. Therefore, $\{X_n : n \geq 0\}$ is a martingale.

c) We consider the Markov chain in cell genetics in Example 5.26. Here again we see that Note that here also, $p_{00} = p_{N,N} = 1$ and we have $\mathbb{E}(X_{n+1} \mid S_n = i) = i$ for $i = 0$ or N . For $i = 1, 2, \dots, N-1$, we note that $X_{n+1} \mid X_n = i$ follows a hypergeometric distribution with parameters $(2i, 2N - 2i, N)$. Therefore, we have $\mathbb{E}(X_{n+1} \mid X_n = i) = N \times \frac{2i}{2N} = i$. Therefore, $\{X_n : n \geq 0\}$ is a martingale.

d) We consider the branching process (see Chapter 3 and Example 5.27) with an offspring distribution ξ such that $\mathbb{P}(\xi = 1) < 1$ and $\mathbb{E}(\xi) = 1$. Then the total population at the n -th generation Z_n is a martingale. Indeed, we already know that $\{Z_n : n \geq 0\}$ is a Markov chain on $\{0, 1, 2, \dots\}$ with $p_{00} = 1$. If $Z_n = i \geq 1$, then we know that $Z_{n+1} \mid Z_n = i$ can be written as $\sum_{k=1}^i \xi_k$ where ξ_k 's are **iid**, each having the same distribution as ξ . Thus, we have $\mathbb{E}(Z_{n+1} \mid Z_n = i) = \mathbb{E}(\sum_{k=1}^i \xi_k) = i$. This proves that the branching process is a martingale under the condition given above. ■

Example 12.4. In the asymmetric case in the gambler's ruin problem, the chain is not a martingale. However, we may construct other functions which is a martingale. In fact, for the birth and death chains on finite state space $\{0, 1, 2, \dots, N\}$ where both 0 and N are absorbing states and $p_i = p, q_i = q$ and $r_i = r$ so that $p + q + r = 1$ for $i = 1, 2, \dots, N-1$, let us define the function $h : \{0, 1, \dots, N\} \times \mathbb{N}^+ \rightarrow \mathbb{R}$ by

$$h(i, n) = h(i) = \left(\frac{q}{p}\right)^i$$

We now observe that $Y_n = h(X_n)$ is a martingale. Indeed the cases when $X_n = 0$ or N are trivial. For other cases, we have

$$\begin{aligned} \mathbb{E}(Y_{n+1} \mid X_n = i) &= \sum_{j=i-1}^{i+1} h(j)p_{ij} = \left(\frac{q}{p}\right)^{i-1} \times q + \left(\frac{q}{p}\right)^i \times r + \left(\frac{q}{p}\right)^{i+1} \times p \\ &= \left(\frac{q}{p}\right)^i [p + r + q] = \left(\frac{q}{p}\right)^i = h(i). \end{aligned}$$

We should also observe that the same argument also implies that even when the state space is infinite with only 0 being the absorbing state, the process $\{Y_n : n \geq 0\}$ will remain a martingale. ■

Exercise 12.5. Consider the simple symmetric random walk in one dimension. Show that the following sequences are martingales

- (a) $\{S_n^3 - 3nS_n : n \geq 0\}$;
- (b) $\{S_n^4 - 6nS_n^2 + 2n + 3n^2 : n \geq 0\}$;
- (c) $\{S_n^5 - 10nS_n^3 + 5(2n + 3n^2)S_n : n \geq 0\}$. ◀

Exercise 12.6. Fix $\lambda \in \mathbb{R}$ and define $\phi : \mathbb{R} \rightarrow [0, \infty)$ by

$$\phi(\lambda) = p \exp(\lambda) + q \exp(-\lambda).$$

For the simple random walk in one dimension, the process $\{Z_n^{(\lambda)} = \exp(\lambda S_n) \phi(\lambda)^{-n} : n \geq 0\}$ is a martingale. ◀

Exercise 12.7. Consider a Markov chain $\{X_n : n \geq 0\}$ on the state space $\mathcal{S} = \{0, 1, 2, 3, 4, 5, 6, 7\}$ with the transition matrix given by

$$\mathbf{P} = \begin{array}{c} \begin{array}{cccccccc} & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \\ \begin{array}{r} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \end{array} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{3} & 0 & 0 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & \frac{1}{12} & \frac{1}{6} & \frac{5}{12} & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{4} & \frac{1}{8} & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{array}.$$

Show that $\{X_n : n \geq 0\}$ is a martingale. ◀

Example 12.8. Suppose $\{X_n : n \geq 0\}$ is a Markov chain with transition matrix $\mathbf{P}$ on the state space $\mathcal{S}$. Let $h : \mathcal{S} \rightarrow \mathbb{R}$ be a function. We set

$$g(i) = \mathbb{E}(h(X_1) \mid X_0 = i) = \sum_{j \in \mathcal{S}} h(j) p_{ij}.$$

Define the following sequence of random variables,

$$Y_n = \begin{cases} h(X_0) & \text{if } n = 0 \\ h(X_n) - \sum_{t=0}^{n-1} g(X_t) & \text{if } n \geq 1. \end{cases}$$

Under the assumptions of finiteness on the expectations of the function $g(X_n)$ and $h(X_n)$, show that $\{Y_n : n \geq 0\}$ is a martingale. ◀

12.3 Martingale & stopping times

Stopping times, defined earlier in Section 11.2, play a very important role in the study of martingales. Here again, we will restrict our attention to martingales / submartingales / supermartingales which are functions of a Markov chain. Let $\{X_n : n \geq 0\}$ be a Markov chain. Suppose that a random time τ is a stopping time with respect to the canonical filtration, i.e., $\{\tau = n\}$ is an event involving $(X_0, X_1, \dots, X_n)$ (see Section 11.2). More precisely, we have

$$\{\tau = n\} = \{(X_0, X_1, \dots, X_n) \in C_n\}$$

where $C_n \subseteq \mathcal{S}^{n+1}$.

Suppose that $h : \mathcal{S} \times \mathbb{N}^+ \rightarrow \mathbb{R}$ is a martingale / submartingale / supermartingale. Now we define the *stopped martingale / submartingale / supermartingale* for the stopping time τ as follows:

$$Y_{n \wedge \tau} = h(X_{n \wedge \tau}, n \wedge \tau) = \begin{cases} Y_n & \text{if } n \leq \tau \\ Y_\tau & \text{if } n > \tau. \end{cases} \quad (12.7)$$

Clearly from the definition, the stopped process tracks the original process until the stopping time τ , but as soon as the stopping time occurs, the stopped process gets fixed at the value of the process at the stopping time. Clearly, such processes are of interest for the Gambler's ruin problem and many other situations.

It is a general result that the stopped process is a martingale / supermartingale / submartingale if the original process is a martingale / supermartingale / submartingale. However, we will not be needing this. Instead, we will prove a result directly about the expectation of the stopped process which generally is a consequence of the result that the stopped process.

Proposition 12.9. *Let $\{X_n : n \geq 0\}$ be a Markov chain. Suppose that τ is a stopping time respect to the canonical filtration. Let $h : \mathcal{S} \times \mathbb{N}^+ \rightarrow \mathbb{R}$ be a function.*

(a) *If $\{Y_n = h(X_n, n) : n \geq 0\}$ is a supermartingale, for all $n \geq 0$, we have*

$$\mathbb{E}(Y_{(n+1) \wedge \tau}) \leq \mathbb{E}(Y_{n \wedge \tau}). \quad (12.8)$$

(b) *If $\{Y_n = h(X_n, n) : n \geq 0\}$ is a submartingale, for all $n \geq 0$, we have*

$$\mathbb{E}(Y_{(n+1) \wedge \tau}) \geq \mathbb{E}(Y_{n \wedge \tau}). \quad (12.9)$$

(c) *If $\{Y_n = h(X_n, n) : n \geq 0\}$ is a martingale, for all $n \geq 0$, we have*

$$\mathbb{E}(Y_{(n+1) \wedge \tau}) = \mathbb{E}(Y_{n \wedge \tau}). \quad (12.10)$$

Before we proceed, we note that $\mathbb{E}(|Y_{n \wedge \tau}|) < \infty$. Indeed, observe that $|Y_{n \wedge \tau}| \leq \sum_{j=0}^n |Y_j|$. Therefore, $\mathbb{E}(|Y_{n \wedge \tau}|) \leq \sum_{j=0}^n \mathbb{E}(|Y_j|) < \infty$.

Proof : We will prove the result for the supermartingale case. The result can be easily extended for submartingales in a straightforward way from the supermartingale case. Also combining the results of both supermartingales / submartingales, we can conclude the result for martingales.

We observe that

$$\mathbb{E}(Y_{n \wedge \tau}) = \sum_{t=0}^{n-1} \mathbb{E}(Y_{n \wedge \tau} \mathbb{I}_{\{\tau=t\}}) + \mathbb{E}(Y_{n \wedge \tau} \mathbb{I}_{\{\tau \geq n\}}) = \sum_{t=0}^{n-1} \mathbb{E}(Y_t \mathbb{I}_{\{\tau=t\}}) + \mathbb{E}(Y_n \mathbb{I}_{\{\tau \geq n\}}). \quad (12.11)$$

Also, we have

$$\begin{aligned} \mathbb{E}(Y_{(n+1) \wedge \tau}) &= \sum_{t=0}^{n-1} \mathbb{E}(Y_{(n+1) \wedge \tau} \mathbb{I}_{\{\tau=t\}}) + \mathbb{E}(Y_{(n+1) \wedge \tau} \mathbb{I}_{\{\tau=n\}}) + \mathbb{E}(Y_{(n+1) \wedge \tau} \mathbb{I}_{\{\tau \geq n+1\}}) \\ &= \sum_{t=0}^{n-1} \mathbb{E}(Y_t \mathbb{I}_{\{\tau=t\}}) + \mathbb{E}(Y_n \mathbb{I}_{\{\tau=n\}}) + \mathbb{E}(Y_{n+1} \mathbb{I}_{\{\tau \geq n+1\}}). \end{aligned} \quad (12.12)$$

Therefore, from the equations (12.11) and (12.12), we note that the result will be proved if $\mathbb{E}(Y_{n+1} \mathbb{I}_{\{\tau \geq n+1\}}) \leq \mathbb{E}(Y_n \mathbb{I}_{\{\tau \geq n+1\}})$.

Since τ is a stopping time, for any $t = 0, 1, \dots$, the event $\{\tau = t\}$ can be expressed as events in terms of $(X_0, X_1, \dots, X_t)$. More precisely, we can take events $C_t \subseteq \mathcal{S}^{t+1}$ such that $\{\tau = t\} = \{(X_0, X_1, \dots, X_t) \in C_t\}$. Now, for any n , let us take $D_t = C_t \times \mathcal{S}^{n+1-t}$ for $t = 0, 1, 2, \dots, n$. Then, we can write

$$\begin{aligned} \{\tau \leq n\} &= \cup_{t=0}^n \{\tau = t\} = \cup_{t=0}^n \{(X_0, X_1, \dots, X_t) \in C_t\} \\ &= \cup_{t=0}^n \{(X_0, X_1, \dots, X_n) \in D_t\} = \{(X_0, X_1, \dots, X_n) \in \cup_{t=0}^n D_t\}. \end{aligned}$$

Therefore, we can write

$$\begin{aligned} \{\tau \geq n+1\} &= \{\tau \leq n\}^c = \{(X_0, X_1, \dots, X_n) \in \cup_{t=0}^n D_t\}^c \\ &= \{(X_0, X_1, \dots, X_n) \in (\cup_{t=0}^n D_t)^c\} = \{(X_0, X_1, \dots, X_n) \in D\} \end{aligned}$$

where $D = (\cup_{t=0}^n D_t)^c$.

We now compute $\mathbb{E}(Y_{n+1} \mathbb{I}_{\{\tau \geq n+1\}}) = \mathbb{E}(h(X_{n+1}, n+1) \mathbb{I}_{\{(X_0, X_1, \dots, X_n) \in D\}})$. Note that

$$\begin{aligned} &\mathbb{E}(h(X_{n+1}, n+1) \mathbb{I}_{\{(X_0, X_1, \dots, X_n) \in D\}}) \\ &= \mathbb{E}[\mathbb{E}(h(X_{n+1}, n+1) \mathbb{I}_{\{(X_0, X_1, \dots, X_n) \in D\}} \mid (X_0, X_1, \dots, X_n))] \\ &= \mathbb{E}[\mathbb{I}_{\{(X_0, X_1, \dots, X_n) \in D\}} \mathbb{E}(h(X_{n+1}, n+1) \mid (X_0, X_1, \dots, X_n))] \end{aligned}$$

since the event $\{(X_0, X_1, \dots, X_n) \in D\}$ is determined by $(X_0, X_1, \dots, X_n)$.

Next, we use the Markov property of the chain $\{X_n : n \geq 0\}$. Since X_{n+1} given $(X_0, X_1, \dots, X_n)$ only depends on the current state, i.e., X_n , we must have

$$\mathbb{E}(h(X_{n+1}, n+1) \mid (X_0, X_1, \dots, X_n)) = \mathbb{E}(h(X_{n+1}, n+1) \mid X_n) \leq h(X_n, n) = Y_n$$

where the last step follows from the supermartingale property. Thus, we have

$$\begin{aligned} \mathbb{E}(Y_{n+1} \mathbb{I}_{\{\tau \geq n+1\}}) &= \mathbb{E}[\mathbb{I}_{\{(X_0, X_1, \dots, X_n) \in D\}} \mathbb{E}(h(X_{n+1}, n+1) \mid (X_0, X_1, \dots, X_n))] \\ &\leq \mathbb{E}[\mathbb{I}_{\{(X_0, X_1, \dots, X_n) \in D\}} Y_n] = \mathbb{E}[\mathbb{I}_{\{\tau \geq n+1\}} Y_n]. \end{aligned}$$

This completes the proof. ■

We will often try to compute the expected value of the process at the stopping time, i.e., compute $\mathbb{E}(Y_\tau)$. From Proposition 12.9, for $n \geq 0$, noting that $\tau \geq 0$, we may conclude by induction that

(a) for supermartingales,

$$\mathbb{E}(Y_{n \wedge \tau}) \leq \mathbb{E}(Y_{0 \wedge \tau}) = \mathbb{E}(Y_0);$$

(b) for submartingales,

$$\mathbb{E}(Y_{n \wedge \tau}) \geq \mathbb{E}(Y_{0 \wedge \tau}) = \mathbb{E}(Y_0);$$

(c) for martingales,

$$\mathbb{E}(Y_{n \wedge \tau}) = \mathbb{E}(Y_{0 \wedge \tau}) = \mathbb{E}(Y_0).$$

If we further have $\tau < \infty$ with probability 1, then we have

$$\lim_{n \rightarrow \infty} Y_{n \wedge \tau} = Y_\tau \text{ with probability 1.}$$

Therefore, if we can take limit as $n \rightarrow \infty$ and interchange the limit and expectation, we should get

$$\mathbb{E}(Y_\tau) = \mathbb{E}\left(\lim_{n \rightarrow \infty} Y_{n \wedge \tau}\right) = \lim_{n \rightarrow \infty} \mathbb{E}(Y_{n \wedge \tau}) = \lim_{n \rightarrow \infty} \mathbb{E}(Y_0) = \mathbb{E}(Y_0). \quad (12.13)$$

This is what is known as Doob's *Optional stopping theorem*.

Theorem 12.10. (Optional Stopping Theorem) *Let $\{Y_n : n \geq 0\}$ be a function of a Markov chain $\{X_n : n \geq 0\}$. Suppose that τ is a stopping time respect to the canonical filtration. Suppose that any of the following conditions hold:*

(a) The stopping time τ is finite and the process is bounded with probability 1, i.e., there exists a constant $M > 0$ such that for all $n \geq 1$, $|Y_n| \leq M$ with probability 1.

(b) The stopping time τ bounded with probability 1, i.e., there exists a constant $M \geq 0$ such that $\tau \leq M$ with probability 1.

(c) The stopping time τ has finite expectation. Furthermore, we assume that the conditional expectations of $|Y_{n+1} - Y_n|$ with respect to X_n are bounded with probability 1 for all $n \geq 0$ on the event $\{\tau \geq n\}$. In other words, we assume that $\mathbb{E}(\tau) < \infty$ and there exists $M > 0$ such that for all $n \geq 0$, $\mathbb{E}(|Y_{n+1} - Y_n| \mid X_n) \leq M$ on the event $\{\tau \geq n\}$.

Under the above conditions, X_τ is a well defined random variable. Further,

- (i) if $\{Y_n : n \geq 0\}$ is supermartingale, then $\mathbb{E}(Y_\tau) \leq \mathbb{E}(Y_0)$;
- (ii) if $\{Y_n : n \geq 0\}$ is submartingale, then $\mathbb{E}(Y_\tau) \geq \mathbb{E}(Y_0)$;
- (iii) if $\{Y_n : n \geq 0\}$ is martingale, then $\mathbb{E}(Y_\tau) = \mathbb{E}(Y_0)$.

Before we embark on the proof of the optional stopping theorem, we will apply it for the Markov chain case. We have encountered examples of Markov chains on the state space $\{0, 1, \dots, N\}$ where the states 0 and N are absorbing states and rest of the states $1, 2, \dots, N$ are all transient (see Example 12.3). We will now use the optional stopping theorem to compute the expressions for the probability of absorption at N (see also Theorem 8.1).

Proposition 12.11. Let $\{X_n : n \geq 0\}$ be a Markov chain on $\{0, 1, \dots, N\}$. Assume that the states 0 and N are absorbing and rest of the states are transient. Suppose that $h : \{0, 1, \dots, N\} \rightarrow \mathbb{R}$ is a function such that $\{Y_n = h(X_n) : n \geq 0\}$ is a martingale. Then, we have

$$\mathbb{P}(\text{Absorption at } N \mid X_0 = i) = \frac{h(i) - h(0)}{h(N) - h(0)}.$$

Proof : Consider the hitting time $\tau = T_{\{0, N\}}$ of the set $\{0, N\}$, i.e., the random time when the chain enters one of the states 0 or N . We have already seen that τ is a stopping time (see Definition 11.6). Further, $\tau < \infty$ with probability 1. Hence, we must have $X_\tau = 0$ or N with probability 1. We are interested in computing $\mathbb{P}(X_\tau = N \mid X_0 = i)$. Since we are interested in the conditional probability given $X_0 = i$, we may assume that $X_0 = i$ with probability 1 so that we don't have to write the conditional probability.

Consider the function $g : \{0, 1, \dots, N\} \rightarrow \mathbb{R}$ defined by $g(i) = h(i) - h(0)$ so that $g(0) = 0$. Therefore, we need to show that $\mathbb{P}(X_\tau = N) = g(i)/g(N)$. Note that we must

have $g(X_\tau) = g(0) = 0$ or $g(N) = h(N) - h(0)$. Therefore, we have

$$\mathbb{E}(g(X_\tau)) = g(0)\mathbb{P}(X_\tau = 0) + g(N)\mathbb{P}(X_\tau = N) = g(N)\mathbb{P}(X_\tau = N).$$

Next, we take $Z_n = g(X_n) = Y_n - h(0) = h(X_n) - h(0)$. Since the conditional expectation is linear and the conditional expectation of a constant is that constant, we also have that $\{Z_n : n \geq 0\}$ is a martingale. Therefore for the stopped process $\{Z_{n \wedge \tau} : n \geq 0\}$, by Proposition 12.9, we have $\mathbb{E}(Z_{n \wedge \tau}) = \mathbb{E}(Z_0) = g(X_0) = g(i) = h(i) - h(0)$. Furthermore, $|g(X_{n \wedge \tau})| \leq 2 \max\{|h(j)| : 0 \leq j \leq N\} < \infty$. Therefore, we may apply the *optional stopping theorem*. Therefore, we have

$$\mathbb{E}(g(X_\tau)) = \mathbb{E}(g(X_0)) = g(i).$$

This shows that $\mathbb{P}(X_\tau = N) = g(i)/g(N)$ which proves the result. ■

Example 12.12. We now return to Example 12.3. For cases (a), (b) and (c), i.e., the symmetric case of the gambler's run problem, the Wright Fisher model and the Markov chain in cell genetics, we have noted that the chain is a martingale, i.e., we have take $h(i) = i$. Therefore, the probability of absorption at N given the initial state is i , is given by $(h(i) - h(0))/(h(N) - h(0)) = i/N$.

For the asymmetric case in the gambler's run problem, we have observed in Example 12.4, the process $\{(\frac{q}{p})^{X_n} : n \geq 0\}$ is a martingale. In other words, we can take $h : \mathcal{S} \rightarrow \mathbb{R}$ where $h(i) = (\frac{q}{p})^i$. Therefore, we have the probability of absorption at N given the initial state is i , is given by $(h(i) - h(0))/(h(N) - h(0)) = (1 - (\frac{q}{p})^i)/(1 - (\frac{q}{p})^N)$. ■

Exercise 12.13. For Exercise 12.7, find the probability of absorption at 7 given the initial state is i , for $i = 1, 2, \dots, 6$. ◀

Example 12.14. We consider the simple random walk starting from 1. We want to study that the hitting time of the state 0, i.e., we consider

$$\tau = \tau_0 = \inf\{n \geq 0 : S_n = 0\}.$$

For $\lambda \in \mathbb{R}$, we have seen $\{Z_n^{(\lambda)} = \exp(-\lambda S_n)\phi(\lambda)^n : n \geq 0\}$ is a martingale (see Exercise 12.6) where

$$\phi(\lambda) = p \exp(-\lambda) + q \exp(\lambda).$$

Now, specialize to the case of $\lambda > 0$. Therefore, using Proposition 12.9, we obtain

$$\mathbb{E}(Z_{n \wedge \tau}^{(\lambda)}) = \mathbb{E}(Z_{0 \wedge \tau}^{(\lambda)}) = \mathbb{E}(Z_0^{(\lambda)}) = \exp(-\lambda).$$

Hence, $\phi(\lambda) > 1$ for any λ . Further, by continuity of ϕ , we have $\phi(\lambda) \rightarrow 1$ as $\lambda \downarrow 0$.

Now, let D_n be the difference of the random walks at time n . We have $D_0 = 2$. Then,

$$M_n = \exp(-\lambda D_n) (\phi(\lambda))^{-n}$$

is a martingale with respect to standard σ -algebra. Note $M_0 = \exp(-\lambda D_0) = \exp(-2\lambda)$.

Now, you need to use DCT on the LHS in the above to show that

$$\mathbb{E}[\mathbb{I}(\tau < \infty) (\phi(\lambda))^{-\tau}] = \exp(-2\lambda)$$

for $\lambda > 0$. Now letting λ decrease to 0, observe that $\mathbb{E}[\mathbb{I}(\tau < \infty)] = \mathbb{P}(\tau < \infty) = 1$. Thus, we obtain that

$$\mathbb{E}[(\phi(\lambda))^{-\tau}] = \exp(-2\lambda).$$

Setting $s = 1/\phi(\lambda)$ and denoting the inverse function by $\lambda = \psi(s)$, the above equation can be rewritten as

$$\mathbb{E}(s^\tau) = \exp(-2\psi(s))$$

which provides the generating function of the stopping time. You can get all information from this. For example, to see that the expectation is infinite you just need to differentiate the RHS and let $s \rightarrow 1$ and see that the expression goes to infinity. In fact, you may find the tail behaviour and it behaves as inverse of square root of n .

12.4 Convergence of Martingale

This section is to be written.

12.5 Sufficient Criteria for Recurrence

We return to the question of recurrence of irreducible chains on an infinite state space. In Theorem 8.13 which involves solving a system of linear equations (see equations (8.21)). However, in many cases, the system of equations in (8.21) may not be solvable. The following theorem provides a sufficient condition which is often useful in many cases.

Theorem 12.15. *For an irreducible Markov chain on the state space $\{0, 1, 2, \dots\}$, suppose that there exists a function $h : \mathcal{S} \rightarrow \mathbb{R}$ such that*

$$(a) \ h(i) \rightarrow \infty \text{ as } i \rightarrow \infty;$$

(b) for $i \geq 1$,

$$\sum_{j=0}^{\infty} p_{ij} h(j) \leq h(i).$$

Then, the Markov chain is recurrent.

Note that, we may restate the second condition in terms of supermartingale. Let us set τ be the hitting time of the state 0, i.e., $\tau = \tau_0$. Now, we look at the stopped process: for $n \geq 1$,

$$Y_n = h(X_{n \wedge \tau}).$$

The condition (b) now implies that the process $\{Y_n : n \geq 1\}$ is a supermartingale.

Proof : We will define a new transition matrix $\mathbf{R} = (r_{ij})$ where

$$r_{ij} = \begin{cases} p_{ij} & \text{if } i \geq 1 \\ 1 & \text{if } i = 0, j = 0 \\ 0 & \text{if } i = 0, j \geq 1 \end{cases}, \quad \text{i.e., } \mathbf{R} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & \dots \\ p_{10} & p_{11} & p_{12} & p_{13} & p_{14} & \dots \\ p_{20} & p_{21} & p_{22} & p_{23} & p_{24} & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \dots \end{bmatrix}.$$

Clearly the state 0 is converted to an absorbing state in the modified chain. Since for $i \geq 1$, $r_{ij} = p_{ij}$ and hence, for $i \geq 1$,

$$\sum_{j=0}^{\infty} r_{ij} x_j = \sum_{j=0}^{\infty} p_{ij} x_j \leq x_i.$$

Next, since $x_i \rightarrow \infty$, we have $x_i \geq -M$ for all $i \geq 0$. Now, taking $z_i = x_i + M$, we see that $z_i \geq 0$ for all $i \geq 0$. Further, for $i \geq 1$,

$$\sum_{j=0}^{\infty} r_{ij} z_j = \sum_{j=0}^{\infty} p_{ij} (x_j + M) = \sum_{j=0}^{\infty} p_{ij} x_j + M \sum_{j=0}^{\infty} p_{ij} \leq x_i + M = z_i.$$

So, without loss of generality, we may take $x_i \geq 0$ for all $i \geq 1$.

Now, for any $n \geq 1$, we have

$$\begin{aligned} \sum_{j=0}^{\infty} r_{ij}^{(n+1)} x_j &= \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} r_{ik}^{(n)} r_{kj} x_j = \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} r_{ik}^{(n)} r_{kj} x_j = \sum_{k=0}^{\infty} r_{ik}^{(n)} \sum_{j=0}^{\infty} r_{kj} x_j \\ &= \sum_{k=1}^{\infty} r_{ik}^{(n)} \sum_{j=0}^{\infty} r_{kj} x_j + r_{i0}^{(n)} \sum_{j=0}^{\infty} r_{0j} x_j \\ &\leq \sum_{k=1}^{\infty} r_{ik}^{(n)} x_k + r_{i0}^{(n)} [1 \times x_0] = \sum_{k=0}^{\infty} r_{ik}^{(n)} x_k \end{aligned}$$

where we have used the fact that $r_{00} = 1$ and $r_{0j} = 0$ for all $j \geq 1$. Using induction, we now conclude that, for any $n \geq 1$ and for all $i \geq 1$,

$$\sum_{j=0}^{\infty} r_{ij}^{(n)} x_j \leq x_i.$$

Next, we show that all states except 0 are transient in the modified chain $\{\tilde{X}_n : n \geq 0\}$. Since 0 is an absorbing state, it is enough to show that $i \rightsquigarrow 0$ for any $i \geq 1$, in the modified chain. Since the original chain was irreducible, for any $i \geq 1$, $i \rightsquigarrow 0$ (in the original chain) and hence $p_{i0}^{(n)} > 0$ for some $n \geq 1$. Choose the minimum such n . Then, we have, by Chapman-Kolmogorov equations,

$$p_{i0}^{(n)} = \sum_{i_1, i_2, \dots, i_{n-1}} p_{i, i_1} p_{i_1, i_2} \cdots p_{i_{n-1}, 0} > 0.$$

So, we can choose $i_1, i_2, \dots, i_{n-1}$ such that $p_{i, i_1} p_{i_1, i_2} \cdots p_{i_{n-1}, 0} > 0$. Further, we must have $i_1, i_2, \dots, i_{n-1} \geq 1$. If we have some $i_l = 0$, we observe that $p_{i0}^{(l)} \geq p_{i, i_1} p_{i_1, i_2} \cdots p_{i_{l-1}, 0} > 0$, contradicting the fact that n is the minimum positive integer for which $p_{i0}^{(n)} > 0$. Then, we note that

$$r_{i0}^{(n)} \geq r_{i, i_1} r_{i_1, i_2} \cdots r_{i_{n-1}, 0} = p_{i, i_1} p_{i_1, i_2} \cdots p_{i_{n-1}, 0} > 0.$$

Thus, $i \rightsquigarrow 0$ in the modified chain.

Now, fix $i \geq 1$. Given $\epsilon > 0$, we choose N so large that $x_k > x_i/\epsilon$ for all $k \geq N$ and hence $\min\{x_k : k \geq N\} > x_i/\epsilon$. We have

$$x_i \geq \sum_{j=0}^{\infty} r_{ij}^{(n)} x_j = \sum_{j=0}^{N-1} r_{ij}^{(n)} x_j + \sum_{j=N}^{\infty} r_{ij}^{(n)} x_j \geq \min\{x_k : k \geq N\} \sum_{j=N}^{\infty} r_{ij}^{(n)}.$$

Let A_n denote the event that the modified chain does not visit the state 0 till time n and A_{∞} be the event that the modified chain never visits state 0. In other words, $A_n := \{\tilde{X}_k \neq 0, \text{ for } 1 \leq k \leq n\}$ and $A_{\infty} := \{\tilde{X}_k \neq 0, \text{ for } k \geq 1\}$. Now we have

$$\begin{aligned} \mathbb{P}(A_{\infty} | X_0 = i) &= \lim_{n \rightarrow \infty} \mathbb{P}(A_n | X_0) = \lim_{n \rightarrow \infty} \sum_{j=1}^{\infty} r_{ij}^{(n)} \\ &\leq \lim_{n \rightarrow \infty} \left[\sum_{j=1}^{N-1} r_{ij}^{(n)} + \frac{x_i}{\min\{x_k : k \geq N\}} \right] \leq \epsilon. \end{aligned}$$

We have used the fact that the state j is transient and hence $\lim_{n \rightarrow \infty} r_{ij}^{(n)} = 0$. Thus, $\mathbb{P}(A_{\infty} | X_0 = i) = 0$. However, note that if the events that the modified chain never visits

the state 0 depends only on the transition probabilities of the non-zero states. Since the transition probability of the non-zero states remain the same in both the original chain as well as the modified chain, we have that the probability of the original chain never visits the state 0 is also 0. So, we have obtained that $1 - f_{i0}^* = 0$, i.e., $f_{i0}^* = 1$ for any $i \geq 1$ for the original chain. Using (??), we now conclude that $f_{00}^* = p_{00} + \sum_{i=1}^{\infty} p_{0i} f_{i0}^* = p_{00} + \sum_{i=1}^{\infty} p_{0i} = 1$. This proves that the state 0 is recurrent and hence the original chain must be recurrent. ■

Example 12.16. Now we return to the queueing model in Example ??. Let us consider the random variable ξ defined there. First consider the case $\mathbb{E}(\xi) \leq 1$. In this case, we show that the sequence $\{x_i : i \geq 0\}$ where $x_i = i$ satisfy the above criteria. Indeed, we have seen that we can write

$$X_{n+1} = f(X_n, \xi_{n+1})$$

where $f(i, j) = i - 1 + j$ if $i \geq 1$. We take $x_i = i$ for $i \geq 0$. Therefore, we have, for $i \geq 1$,

$$\sum_{j=0}^{\infty} x_j p_{ij} = \sum_{j=0}^{\infty} j p_{ij} = \mathbb{E}(f(i, \xi)) = i - 1 + \mathbb{E}(\xi) \leq i = x_i$$

as $\mathbb{E}(\xi) \leq 1$.

If $\mathbb{E}(\xi) > 1$, consider the probability generating function $P_{\xi}(s)$ of ξ . We know that $P_{\xi}(s) = s$ has a positive solution in $(0, 1)$. Let σ denote the solution of $P(s) = s$ in $(0, 1)$. We verify that $\{1 - \sigma^i : i \geq 1\}$ is a solution of system of linear equations in (8.21). For $i = 1$

$$\begin{aligned} \sum_{k=1}^{\infty} p_{1k}(1 - \sigma^k) &= \sum_{k=1}^{\infty} \mathbb{P}(\xi = k)(1 - \sigma^k) = \sum_{k=1}^{\infty} \mathbb{P}(\xi = k) - \sum_{k=1}^{\infty} \mathbb{P}(\xi = k)\sigma^k \\ &= 1 - \mathbb{P}(\xi = 0) - (P_{\xi}(\sigma) - \mathbb{P}(\xi = 0)) = 1 - \sigma \end{aligned}$$

and for $i \geq 2$,

$$\begin{aligned} \sum_{k=1}^{\infty} p_{ik}(1 - \sigma^k) &= \sum_{k=i-1}^{\infty} \mathbb{P}(\xi = k - (i - 1))(1 - \sigma^k) \\ &= \sum_{k=i-1}^{\infty} \mathbb{P}(\xi = k - (i - 1)) - \sum_{k=i-1}^{\infty} \mathbb{P}(\xi = k - (i - 1))\sigma^{k-(i-1)+(i-1)} \\ &= \sum_{k=0}^{\infty} \mathbb{P}(\xi = k) - \sigma^{i-1} \sum_{k=0}^{\infty} \mathbb{P}(\xi = k)\sigma^k = 1 - \sigma^{i-1}P_{\xi}(\sigma) = 1 - \sigma^i. \end{aligned}$$

Thus, we have established a bounded non-trivial solution of (8.21) and hence the chain is transient. However, since we do not have the general solutions of (8.21), we are unable to claim that this is the maximal solution. For that we will use a more direct computation.

Further in this model, we have, for $i \geq 1$,

$$f_{i+1,i}^* = f_{1,0}^*.$$

By induction, this will imply $f_{i,0}^* = (f_{1,0}^*)^i$ for $i \geq 1$. Now, using the equation (??), we have that

$$f_{10}^* = p_{10} + \sum_{k=1}^{\infty} p_{1k} f_{k,0}^* = \alpha_0 + \sum_{k=1}^{\infty} \alpha_k (f_{10}^*)^k = P_{\xi}(f_{10}^*).$$

Hence, f_{10}^* must satisfy the equation $P_{\xi}(s) = s$. Since we know that the chain is transient, we have $f_{10}^* < 1$. Thus, it is given by the smaller solution of $P_{\xi}(s) = s$ (the other solution being 1), i.e., $f_{10}^* = \sigma$.

To conclude our result, we know for this model, using left non-skipping,

$$\begin{aligned} f_{i+1,i}^{(n)} &= \sum_{i_1, i_2, \dots, i_{n-1} \neq i} \mathbb{P}(X_n = i, X_j = i_j \text{ for } 1 \leq j < n | X_0 = i + 1) \\ &= \sum_{i_1, i_2, \dots, i_{n-1} \neq i} p_{i+1, i_1} p_{i_1, i_2} \cdots p_{i_{n-2}, i_{n-1}} p_{i_{n-1}, i} \\ &= \sum_{i_1, i_2, \dots, i_{n-1} \geq i+1} p_{i+1, i_1} p_{i_1, i_2} \cdots p_{i_{n-2}, i_{n-1}} p_{i_{n-1}, i} \\ &= \sum_{i_1, i_2, \dots, i_{n-1} \geq i+1} p_{1, i_1-i} p_{i_1-i, i_2-i} \cdots p_{i_{n-2}-i, i_{n-1}-i} p_{i_{n-1}-i, 0} \\ &= \sum_{j_1, j_2, \dots, j_{n-1} \geq 1} p_{1, j_1} p_{j_1, j_2} \cdots p_{j_{n-2}, j_{n-1}} p_{j_{n-1}, 0} \\ &= f_{1,0}^{(n)}. \end{aligned}$$

Here we have used the fact that for $i, j \geq 1$, we have $p_{i, i+k} = p_{j, j+k}$ for all $k \geq -1$. Now, summing both sides over n , we conclude that $f_{i+1,i}^* = f_{1,0}^*$. ■

Chapter 13

Simulation

13.1 Simulation Basics

For constructing examples where random experiments may result in a continuous region, discrete sets, as considered above, are not sufficient. We have already discussed that there are σ -algebras defined on $\mathbb{R}$ or $(0, 1]$. Now we state a result (without proof) about the existence of a probability on $\mathcal{B}(0, 1]$.

Theorem 13.1. *For $\Omega = (0, 1]$ and $\mathcal{F} = \mathcal{B}((0, 1])$, there exists an unique probability ℓ such that for any $0 \leq a \leq b \leq 1$, we have*

$$\ell((a, b]) = b - a. \quad (13.1)$$

This is known as the Lebesgue measure on the unit interval. In probability theory, this probability holds a very important position, as we will discuss later.

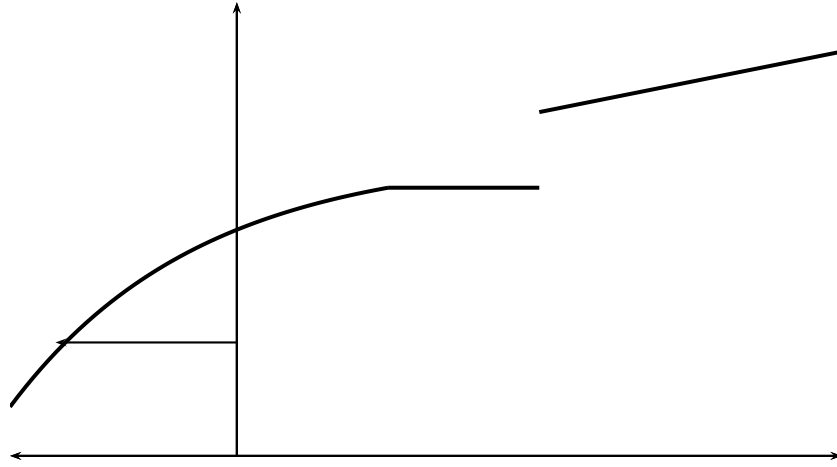
Quantile Function

For any distribution function F , we define the **quantile function**, $F^{\leftarrow} : (0, 1) \rightarrow \mathbb{R}$ as follows: for $0 < p < 1$

$$F^{\leftarrow}(p) = \inf\{x : F(x) \geq p\}.$$

This is also called the p -th quantile or fractile of F . For example, $F^{\leftarrow}(1/2)$ is the *median* of F .

The following figure shows the plot of the quantile function:



The following exercise lists some of the properties of the *quantile function*.

Exercise 13.2. The quantile function $F^{\leftarrow}$ has the following properties:

1. The set $I_p = \{y : F(y) \geq p\}$ is an infinite right-open interval, i.e., if $u \in I_p$ and $v > u$, then $v \in I_p$.
2. $F^{\leftarrow}$ is a non-decreasing.
3. $F^{\leftarrow}(F(x)) \leq x$ for any $x \in \mathbb{R}$.
4. $p \leq F(F^{\leftarrow}(p))$ for all $p \in (0, 1)$.
5. $F(F^{\leftarrow}(p-)) \geq p$ for all $p \in (0, 1)$. Conclude that $F^{\leftarrow}$ is left continuous.
6. If F is continuous and strictly increasing, then F has an inverse function (which is continuous and strictly increasing). Show that the quantile function is the inverse function in this case. ◀

Next we prove a very important property of the quantile function.

Proposition 13.3. For $u \in (0, 1)$ and $x \in \mathbb{R}$, we have

$$F^{\leftarrow}(u) \leq x \iff u \leq F(x).$$

Proof : Assume that $F^{\leftarrow}(u) \leq x$, i.e., $\inf\{z : F(z) \geq u\} \leq x$. For any $\epsilon > 0$, we must have $x + \epsilon \in \{z : F(z) \geq u\}$ and hence $F(x + \epsilon) \geq u$. Using the right continuity of F ,

we have $F(x) \geq u$. Conversely assume that $u \leq F(x)$. So, $x \in \{z : F(z) \geq u\}$. Since the infimum of a set must be smaller or equal to any member, we have $F^{\leftarrow}(u) = \inf\{z : F(z) \geq u\} \leq x$. ■

Now we show that every distribution function is actually a (cumulative) distribution function of a random variable. In other words, given any distribution function, we may construct a random variable having the given (cumulative) distribution function. Note that $F^{\leftarrow}$ is a non-decreasing function and hence for any random variable X , $Y = F^{\leftarrow}(X)$ is also a random variable.

Theorem 13.4. *For any distribution function F and $U \sim U(0,1)$, $F^{\leftarrow}(U) \sim F(\cdot)$, i.e.,*

$$\mathbb{P}(F^{\leftarrow}(U) \leq z) = F(z).$$

Proof : From Proposition 13.3, we have $\{\omega : F^{\leftarrow}(U(\omega)) \leq z\} = \{\omega : U(\omega) \leq F(z)\}$. So, we have $\mathbb{P}(F^{\leftarrow}(U) \leq z) = \mathbb{P}(U \leq F(z)) = F(z)$ by equation (1.19). This completes the proof. ■

13.2 Simulate Markov chains

Natural question that should be asked here is that given a distribution μ on $\mathcal{S}$ and a transition matrix $\mathbf{P}$ on $\mathcal{S}$, is there a Markov chain with the initial distribution μ and transition matrix $\mathbf{P}$? The answer is true.

Proposition 13.5. *Given a distribution μ on $\mathcal{S}$ and a transition matrix $\mathbf{P}$ on $\mathcal{S}$, there exists a Markov chain with the initial distribution μ and transition matrix $\mathbf{P}$.*

Proof : We start with a probability space on which we define independent random variables X_0 and a sequence of random variables $\{U_t : t \geq 1\}$. Further, we assume that X_0 has the distribution μ and U_t is uniformly distributed on $(0, 1]$.

Let us also assume that we have ordered the states $\mathcal{S} = \{j_1, j_2, \dots\}$. Now, we define the function $f : \mathcal{S} \times (0, 1] \rightarrow \mathcal{S}$ as follows:

$$f(i, u) = j_k \text{ if } \sum_{m=1}^{k-1} p_{i,j_m} < u \leq \sum_{m=1}^k p_{i,j_m}$$

where we are using the convention that the sum over empty indices is 0. Now, for $t \geq 0$, set

$$X_{t+1} = f(X_t, U_{t+1}).$$

By the Proposition 5.7, $\{X_n : n \geq 0\}$ is a Markov chain. Further, the initial distribution is given by μ . Now, the transition probability for a state $j = j_k$, we have

$$\mathbb{P}(f(i, U_1) = j_k) = \mathbb{P}\left(\sum_{m=1}^{k-1} p_{i,j_m} < U_1 \leq \sum_{m=1}^k p_{i,j_m}\right) = p_{i,j_k} = p_{i,j}.$$

This completes the proof. ■

13.3 Computation on Markov chain

This section is to be written.

13.4 Python Code

This section is to be written.

Index

- Abel's Lemma, 57–59, 76
- absorbing boundary, 132
- Bayes theorem, 16
- Bernoulli distribution
 - pgf of, 56
- Bernoulli-Laplace model, 135
- Binomial distribution
 - pgf of, 56
- birth and death chain, 131
 - finite state, 131
 - infinite state, 131
- Bonferroni's inequality, 8
- Boole's inequality, 10
- Borel σ -algebra, 6
- Branching Process
 - critical, 91
 - definition, 81
 - extinction, 85
 - extinction probability, 86
 - extinction time, 89
 - immigration, 97
 - subcritical, 90
 - supercritical, 93
 - total progeny, 97
- branching process, 142
- coin toss example, 7
- conditional probability, 15
- Convolution, 71
- dice roll example, 7
- diffusion model, 134
- discrete queue, 138
- doubly stochastic matrix, 130
- Ehrenfest model, 134
 - modified, 135
- elastic barrier, 134
- event, 5
- Expectation
 - as sum of tail probabilities, 60
 - relation to pgf, 55
- Exponential function
 - as generating function, 56
- Fibonacci sequence, 65
- finite state Markov chain, 143
- gambler's ruin, 132
- Generating Function
 - computation, 61
 - convexity, 58, 75
 - definition, 55
 - monotonicity, 57
 - uniqueness of distribution, 57
- genetics models, 140
- Geometric distribution
 - pgf of, 56

- independence, 10
- independent events, 10
 - finite collection, 10
 - infinite collection, 10
- independent random variables, 125
- initial distribution, 122
- Kounias's inequality, 9
- Laplace transform, 57
- lazy chain, 146
- left non-skipping chain, 139
- Markov chain
 - birth and death, 131
 - branching process, 142
 - canonical representation, 123
 - circular state space, 129
 - definition, 121
 - examples, 125
 - gambler's ruin, 132
 - genetics, 140
 - graph random walk, 136
 - homogeneous, 122
 - independent case, 125
 - queueing, 138
 - random walk, 127
 - renewal events, 139
 - two-state, 125
- Markov condition, 121
- mitosis model, 142
- Moment
 - factorial, 59
 - first, 59
- Negative Binomial distribution
 - pgf of, 57
- non-Markov process, 143
- Offspring distribution, 82
- PageRank algorithm, 137
- pairwise independent events, 11
- Poisson distribution
 - pgf of, 56
- probability, 6
- Probability Generating Function, *see* Generating Function
- product chain, 146
- Pólya urn, 143
- queueing model, 138
- Radius of convergence, 55, 56
- Random sum, 74
- Random variable
 - non-negative integer valued, 55
- Random Walk, 101
- random walk, 127
 - chess pieces, 138
 - general, 127
 - multi-dimensional, 128
 - on circle, 129
 - on graphs, 136
 - one-dimensional, 127
 - reflecting barrier, 133
- Recurrence relation, 63
- reflecting barrier, 133
- renewal process, 139
- right non-skipping chain, 139
- sample space, 5

sigma-algebra
 Borel, 6
 σ -algebra, 5
simple random walk, 127
snake chain, 146
state space, 121
 countable, 121
stochastic matrix, 122
stochastic process, 121
success run chain, 140
symmetric random walk, 128

time-homogeneous, 122
Total progeny, 98
transition matrix, 122
two-state chain, 125

Uniform distribution (discrete)
 pgf of, 56
urn model, 15

Wright-Fisher model, 140

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